

SARKARI PATH

sarkaripath.onl

MONTHLY COMPENDIUM

CURRENT AFFAIRS

AUGUST 2026 EDITION

Coverage: 25th July 2026 to 24th August 2026

- [+] Comprehensive Coverage of National & International Events
- [+] Polity, Economy, Science-Tech, Environment & Prelims Facts
- [+] SPECIAL ADDITION: 200 Practice MCQs & FAQs Included
- [+] Official Verified Answer Key (Q1 to Q200) Included at the End

TARGETED FOR ALL COMPETITIVE EXAMS

- UPSC & State PSCs: CSE, UPPSC, BPSC, RAS, MPPSC, WBPSC, etc.
- SSC Exams: CGL, CHSL, CPO, MTS, GD Constable, Stenographer
- Railway Exams: RRB NTPC, Group D, ALP, Technician, JE
- Banking & Insurance: IBPS PO / Clerk, SBI PO / Clerk, RBI, LIC
- Defence & Teaching: CDS, NDA, AFCAT, CAPF, CTET, State TET

Download Free Study Notes, Mock Tests & Daily Current Affairs

VISIT: [SARKARIPATH.ONL](https://sarkaripath.onl)

(Coverage from 25th July 2026 to 24th August 2026)

INDEX

Polity & Governance

1-25

• SC Eases 3-Year Practice Rule for Judicial Services	1
• SHANTI Act, 2025 and Nuclear Liability Concerns	1
• Foreign Contribution (Regulation) Amendment Bill, 2026	3
• Supreme Court on Electoral Black Money	3
• 80th Independence Day: 'Sapt Dhara' Framework for Viksit Bharat 2047	4
• Relevance of Regional Parties in India	5
• Deepening Inclusive Development in Rural India	7
• PM CARES Fund Under Scrutiny Over Transparency	8
• Tribunals Reforms Bill, 2026	9
• Supreme Court's New Handbook on Gender Sensitivity	10
• 10 Years of Government e-Marketplace (GeM)	11
• Supreme Court Directs Measures to Curb 'Digital Arrest' Scams	12
• Digital Transformation of Indian Judiciary	13
• GOBARdhan: National Circular Bioenergy Scheme	13
• SC Extends Protection Against Domestic Cruelty to Live-in Relationships	14
• Indian Statistical Institute Bill, 2026	15
• Pradhan Mantri Viksit Bharat Rojgar Yojana	16
• PM-KISAN Scheme	16
• Pradhan Mantri Surya Sarovar Yojana	16
• Jan Aushadhi Nutraceuticals	17
• Revamped Khelo India Scheme	18
• Prevention of Insults to National Honour (Amendment) Act, 2026	18
• Tackling Non-Revenue Water for a Water-Secure Viksit Bharat	19
• Gen Z Protests: Redefining Democratic Participation in India	19
• Public Examinations Amendment Bill, 2026	21
• Right to Protest and Excessive Policing	22
• Amrit Bharat Station Scheme	23
• PM Announces Fast-Track Courts to Tackle Public Examination Paper Leaks	24

• Smart Border Architecture to Strengthen India's Border Security	25
• Paternity Test in Matrimonial Dispute	25

Nation & States

26-33

• Strengthening India's Defence through Self-Reliance	26
• Emerging Terror Threats from Social Media	27
• Parliament Passes Bill to Rename Kerala as Keralam	28
• National Security Guidelines for Renewable Energy Projects Near International Borders	29
• India Formally Names 27 Places in Arunachal Pradesh	29
• Naxal-Free India Through Integrated Strategies	30
• Cauvery Water Dispute	32
• India at Commonwealth Games 2026	32
• Kargil Vijay Diwas 2026	33

Economic Scenario

34-55

• Surge in India's Merchandise Exports	34
• Electronics Component Manufacturing Scheme	35
• Foreign Assets of Small Taxpayers Disclosure Scheme, 2026	35
• India's Makhana Sector	36
• Building a Business-Ready India	37
• NITI Aayog Report on India's Regulatory Regime in Professional Services	38
• Greenium and Sovereign Green Bonds	39
• National Cooperative Development Corporation (Amendment) Bill, 2026	40
• Mines and Minerals (Development and Regulation) Amendment Bill, 2026	40
• Container Manufacturing Assistance Scheme (CMAS)	42
• Ship Recycling in India	43
• Taxation and Other Laws (Amendment) Act, 2026	43
• National Handloom Day 2026	45

● RBI to Resume Licensing of Urban Cooperative Banks	46
● Geographical Indications in India	46
● Critical Minerals: The Foundation for Strategic Resource Security	47
● India's Solar Growth Enters a New Era	49
● Indian Rupee: From Overvaluation to Undervaluation	50
● Minimum Support Price and the Need for Alternative Farmer Support	51
● India as a Tourism Hub	52
● ILO Convention No. 193 on Decent Work in the Platform Economy	53
● India Allows FDI in Inventory-Based E-Commerce Model for Exports	55

International Relations 56-64

● India-Japan Maritime Security Pact	56
● India's Trade Agreements in Action	56
● The Makkah Joint Defence Agreement	57
● A Climate Resilience Pathway Between India and China	59
● India's Trade Balancing Act: Navigating US, China and Strategic Autonomy	60
● US-Saudi Arabia Nuclear Deal	61
● President's State Visit to Romania	62
● Rome Declaration for an Unarmed and Disarming Peace	63
● Bay of Bengal as India's SHANTI Anchor	63

Environment & Ecology 65-75

● World Elephant Day 2026	65
● Thinning-Induced Deceleration of Zaskar Glaciers	66
● Agasthyamalai Evictions and the Forest Rights Act, 2006	67
● World Lion Day	68
● SC Extends 10 km Mining Buffer Rule to Wetland Reserves Across India	69
● India Launches Clouded Leopard Conservation Action Plan (CAP)	69
● Ken-Betwa River Linking Project	70
● SC Verdict on Ex Post Facto Environmental Clearances	70

● Cloudbursts in India	71
● Western Ghats Ecologically Sensitive Area: 7th Draft Notification	72
● Teesta Tunnel Disaster: Methane Risks and Himalayan Infrastructure Vulnerability	73
● Soil Management for Sustainable Agriculture	74

Science & Technology 76-82

● Generative AI in Synthetic Biology	76
● Transforming India's Space Ecosystem	77
● Rotating Detonation Engines	78
● Total Solar Eclipse	78
● Physicist Deepak Dhar Awarded the 2026 Dirac Medal	79
● NASA Invites ISRO to Join Moon Base Programme	80
● Lab-Grown Diamonds	80
● World Hepatitis Day (WHD) 2026	81
● Mega Data Centres and the Sustainability Challenge	82

History, Art & Culture 83-84

● Royal Indian Navy Mutiny of 1946	83
● Indian Independence Act, 1947	83
● Veeragallu Discovery Sheds Light on Ruler Vishnuvardhana's Death	84

Social Issues 85-89

● Centre's Opposition to Creamy Layer for SC/STs	85
● Debate on Alcohol Prohibition	86
● The State of Food Security and Nutrition in the World 2026	87
● NITI Aayog Report on India's Caregiving Ecosystem	88

Facts for Prelims 96-115

■ Report & Indices	90
■ Awards & Honours	91
■ Important Days	92
■ Defence & Security	93
■ Summits & Conferences	94
■ Places in News	94
■ Portals & Apps	98
■ Species in News	98
■ Miscellaneous	99

Polity & Governance

SC Eases 3-Year Practice Rule for Judicial Services

SC has modified the operational framework of its May 2025 judgment mandating a **three-year legal practice requirement** for entry into the judicial services by introducing a **transitional structured training mechanism** for fresh law graduates.

- **May 2025 SC ruling:** Restored **3-year legal practice requirement** for entry-level judicial services, reversing the **2002 All India Judges' Association case** that allowed fresh graduates to directly appear.
- **Transition Relief Window (20 May 2025 – 31 March 2027):**
 - Fresh law graduates can appear without prior practice.
 - They will be considered to have completed **1 year of practice**.
- **Trainee Judicial Officer Framework:** Selected candidates will initially be appointed as Trainee Judicial Officers and receive half the remuneration of a Judicial Magistrate First Class (JMFC) during training.
- **Training as Practice Equivalence:**
 - One year of State Judicial Academy training will count as one year of Bar practice.
 - One year of supervised judicial clerkship (six months under a District/Sessions Judge and six months under a High Court Judge) will count as another one year of Bar practice.
- **Regular Appointment:** Full judicial rank, regular pay scales and service benefits will be granted after satisfactory completion and evaluation of the two-year academy and clerkship programme.
- **Long-Term Rule (From 1st April 2027):** Candidates appearing for the Civil Judge (Junior Division) examination must have at least one year of actual certified practice in effective court proceedings. The remaining requirement will be fulfilled through structured training after selection.
- **Implementation & Review:** High Courts and State Governments must amend subordinate judicial service rules within three months. The scheme will remain operational for five years and will be reviewed based on empirical data.
- **Rationale:** SC held that Civil Judges handle issues of life, liberty, property and personal rights from day one, making familiarity with courtroom procedures and Bench–Bar dynamics essential.

Constitutional Provisions Governing Subordinate Judiciary

Article	Provision
Article 234	Recruitment of persons other than District Judges: Appointments are made by the Governor according to rules framed after consultation with the State Public Service Commission and High Court .
Article 235	Vests administrative, disciplinary and operational control of district and subordinate courts in the High Court .
Article 233	Governor appoints, posts and promotes District Judges in consultation with the High Court. A person from outside government service must have 7 years' experience as an advocate/pleader and be recommended by the High Court.

SHANTI Act, 2025 and Nuclear Liability Concerns

SC is examining whether liability caps under the **Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, 2025** can restrict constitutional courts from awarding **fair and adequate compensation** to nuclear-disaster victims.

- SC is also examining whether **Section 17(4)** on appointments to the **Atomic Energy Regulatory Board (AERB)** creates a **conflict of interest**.
- A **three-judge Bench headed by the CJI** issued notices to the **Union Government and AERB** on petitions alleging inadequate victim compensation under statutory caps.

SHANTI Act, 2025

- **About:** Replaced the **Civil Liability for Nuclear Damage (CLND) Act, 2010** to facilitate **private-sector participation** by easing liability norms and creating a predictable civil-liability framework.
 - “**Company**” follows **Section 2(20), Companies Act, 2013**, but excludes **foreign-incorporated companies**.
 - Gives **AERB statutory status**.
- **Strategic State Control:** Uranium enrichment, spent-fuel reprocessing, high-level waste management and heavy-water production remain reserved for the **Central Government/wholly owned institutions**.
- **Dispute Redressal:** Creates an **Atomic Energy Redressal Advisory Council**; **Appellate Tribunal for Electricity** is the appellate authority.
- **Nuclear Liability under Previous Framework:** CLND Act, 2010

- Established a **no-fault regime**: victims had to prove damage, not negligence.
- Operator liability for large reactors: **₹1,500 crore**; overall maximum liability: **300 million SDR** per nuclear incident.
- Operators had a statutory **right of recourse against suppliers** for accidents caused by **defective equipment, patent or latent defects**.

- **Nuclear Liability under SHANTI Act**: Based on the thermal capacity and risk profile of the nuclear installation:

Nuclear Installation	Operator Liability
Above 3,600 MW thermal power	₹3,000 crore
Above 1,500 MW–3,600 MW	₹1,500 crore
Above 750 MW–1,500 MW	₹750 crore
Above 150 MW–750 MW	₹300 crore
Up to 150 MW, specified fuel-cycle facilities and nuclear-material transport	₹100 crore

- **State Absorption**: After the operator limit is exhausted, the **Central Government** pays remaining compensation.
 - ❖ Total liability per incident is capped at the rupee equivalent of **300 million SDR**, or a **higher amount notified by the Central Government**.
 - ❖ Beyond this, the Government may take additional measures, including seeking funds under the **Convention on Supplementary Compensation (CSC)**.
- **Supplier Recourse Diluted**: Domestic and foreign suppliers are largely shielded unless liability is **expressly contractual** or the incident is caused **intentionally**.

- **Adjudication & Information Control**: Claims handled by specialised bodies such as a **Claims Commissioner/Nuclear Damage Claims Commission**; ordinary civil courts are excluded.

- Claims must be filed within **10 years for property damage** and **20 years for personal injury**.
- Union Government may classify certain nuclear operational information as **restricted**, limiting disclosure under the **RTI framework**.

- India is **not a party to the Paris Convention or Vienna Convention**.
- India ratified **Convention on Supplementary Compensation (CSC)**, 1997 in 2016 and is a **State Party**.
 - India is **not party to the 2026 Amendment**, which revises the public-fund contribution formula for States without nuclear reactors.

Nuclear Liability Concerns Regarding the SHANTI Act, 2025

- **Inadequate Liability Ceiling**: **300 million SDR**, about **₹3,900–4,000 crore**, may be insufficient for catastrophic accidents.

- **Chernobyl**: Estimated losses of **USD 235–700 billion**.
- **Fukushima**: Around **USD 400–445 billion** in clean-up, decommissioning and compensation.
- Nuclear disasters can cause **multi-generational health damage, environmental contamination, displacement, livelihood losses and long-term land-use/economic restrictions**.

- **Restrictive Time Limits**: **10 years** for property and **20 years** for personal injury may exclude **radiation-induced cancers, genetic damage and congenital anomalies** appearing decades later.

- **Weaker Safety Incentives**: Liability caps may reduce incentives for **advanced safety systems, rigorous maintenance and operational redundancy**; excessive supplier protection may further weaken accountability.

- **Constitutional and Legal Fault Lines**:

- **Absolute Liability, Article 21**: *M.C. Mehta (1987)* held enterprises engaged in inherently hazardous activities **absolutely liable for all damage without exception**; statutory caps may conflict with this principle.
- **Polluter Pays**: *Indian Council for Enviro-Legal Action (1996)* recognised that polluters must bear environmental costs.
 - ❖ Under SHANTI, Government assumption of residual liability may shift costs of **private nuclear activities to taxpayers**.
- **Article 14**: Reactor-size-based liability, including only **₹100 crore for small reactors**, is challenged as arbitrary because even smaller failures may cause severe radiation hazards.
- **Article 19(1)(a)**: **Section 39** restricting information under the **RTI Act, 2005** may weaken safety transparency and victims' ability to obtain evidence for claims.
- **AERB Independence**: Government control over appointments may create conflict between **promotion and regulation of nuclear energy**.
 - ❖ Effective regulation requires **institutional, financial and technical autonomy** from executive, political and commercial influence.

Measures to Strengthen the Nuclear Liability Framework

- **Independent Nuclear Safety Regulator**: CAG reported an **85% shortfall in regulatory inspections of hazardous sites** and absence of a **National Nuclear Safety Policy**.
 - Revive a modern **Nuclear Safety Regulatory Authority (NSRA)** framework based on the **Raja Ramanna Committee, 1997** and **NSRA Bill, 2011**.
 - **Raja Ramanna Committee, 1997**: Recommended amending the **Atomic Energy Act, 1962** to strengthen nuclear regulation.

- **NSRA Bill, 2011:** Proposed a stronger regulator after the **Mayapuri radiation accident, 2010** and **Fukushima disaster, 2011**; it lapsed and did not establish a truly autonomous regulator.
- Would address the structural conflict arising from the **Department of Atomic Energy (DAE)** being involved in both nuclear promotion and regulation.
- **Use CSC Compensation Mechanism:** Develop mechanisms to access the **CSC international supplementary compensation pool** when domestic liability limits are exceeded, reducing pressure on the Indian exchequer.
- **Strengthen India Nuclear Insurance Pool (INIP):** Expand the **INIP**, managed by **General Insurance Corporation of India**, and supplement it with **international reinsurance and Sovereign Catastrophe Bonds (Cat-Bonds)**.
- **Contractual Supplier Liability:** Since statutory recourse has weakened, **NPCIL and private operators** should make the **contractual right of recourse mandatory**.

- Procurement contracts should expressly impose supplier liability for **latent design defects**, while remaining **CSC-compliant**.

The SHANTI Act, 2025 seeks to promote private participation in nuclear energy while restructuring nuclear liability. Examine whether its liability framework adequately balances investment interests with citizens' right to life under Article 21.

Drishti Mains Question

Foreign Contribution (Regulation) Amendment Bill, 2026

The **FCRA Bill, 2026** has been referred to a **Joint Parliamentary Committee (JPC)** for detailed examination and stakeholder consultations.

- The move follows notification of the **FCRA Amendment Rules, 2026** aimed at improving **compliance, transparency and accountability** in regulation of foreign contributions received by NGOs and associations.

Key Provisions of the FCRA Bill, 2026	Details
Management of Foreign Contribution Assets	Provides framework for supervision, management and disposal of assets after an organisation loses FCRA registration
Cessation of FCRA Registration	Registration may end due to cancellation, voluntary surrender, or non-submission/rejection of renewal
Designated Authority	Created to take custody, manage and dispose of foreign contribution assets after registration ceases
Provisional Vesting of Assets	Assets remain under Authority's custody and are returned if registration is restored
Permanent Vesting of Assets	If registration is not restored, assets vest with Authority, used for public purposes and sale proceeds transferred to the Consolidated Fund of India
Protection of Religious Sites	Authority must preserve religious character of places of worship and cannot convert or repurpose them
Appeal Mechanism	Revision against Authority's order within 90 days ; appeal allowed before District Judge
Automatic Cessation (Section 14B)	Registration automatically ends if renewal is not completed before expiry
Rationalised Punishment	Maximum imprisonment reduced from 5 years to 1 year
Coordinated Investigations	State governments and agencies require Central Government approval before FCRA investigations

Supreme Court on Electoral Black Money

Sc observed that removing **black money from elections** is a responsibility of the **ECI**.

- Illicit money can distort **free voter choice** and undermine **democracy, rule of law and the electoral process**.
- Court directed **time-bound investigations, seizure reporting and speedy trials** in election-related black money cases.

Key Directives Issued by the SC

- **Seizure Reporting:** Cash/assets seized must be reported within **24 hours** to the District Magistrate, Additional District Magistrate or competent court, with a **prima facie link to an electoral offence**.
- **Time-Bound Investigation:** Election-related FIR investigations should be completed within **1 year**; delays must be justified to the ECI.

- **Quarterly status reports** must be submitted to the ECI.

- **Income Tax Coordination: Static Surveillance Teams** must report cash above **₹10 lakh** to Income Tax authorities.
- **Speedy Trials:** High Courts must ensure **expeditious disposal**, including through designated courts where required.
- **Withdrawal of Cases:** Withdrawal of election-cycle criminal cases against candidates requires **HC approval**.
- **Compliance:** **ECI and State Governments** must submit compliance reports by **18th Nov 2026**.

Black Money as a Threat to Electoral Democracy

- **Subversion of Constitutional Rights:** Monetary inducements undermine free voter choice and the principle of **Universal Adult Suffrage under Article 326**.
 - **PUCL v. Union of India (2013):** Right to cast a **negative vote (NOTA)** is important for free democratic choice.

- **Destruction of the Level Playing Field (Article 14):** Unaccounted wealth disadvantages **resource-poor candidates** and promotes plutocratic influence.
 - **Kanwar Lal Gupta v. Amar Nath Chawla (1974):** Financial disparity can distort electoral competition.
 - **Indira Nehru Gandhi v. Raj Narain (1975):** Free and fair elections are part of the **Basic Structure**.
- **Policy Capture and Crony Capitalism:** Illicit funding can produce **quid pro quo**, biased procurement and policies favouring financiers.
 - **ADR v. Union of India (2024):** Electoral Bonds Scheme struck down; anonymous funding violated citizens' **Right to Information under Article 19(1)(a)** and could facilitate quid pro quo.
- **Criminal-Political Nexus: Vohra Committee (1993)** highlighted links between criminal networks and political financing.
 - **Public Interest Foundation v. Union of India (2018):** Court stressed **decriminalisation of politics** and electoral transparency.

Legal and Institutional Framework to Curb Black Money in Elections

- **Section 123(1), RPA 1951:** Bribery is a **corrupt practice**.
- **Conduct of Election Rules, 1961:** Framework for **candidate expenditure limits**.
- **Common Cause v. Union of India (1996):** Political parties must maintain and submit **election-related financial statements**.
- **Union of India v. ADR (2002):** Voters have a right to know candidates' **criminal records, assets, liabilities and qualifications**.
- **ECI monitoring:** Expenditure Observers, Static Surveillance Teams, Flying Squads and Video Surveillance Teams.
- **cVIGIL App:** Enables real-time citizen reporting of **MCC violations and voter inducements**.
- **Income Tax Act, 2025 and PMLA, 2002:** Used to investigate unaccounted money and proceeds of crime.

Persisting Challenges in

Curbing Black Money in Elections

- **Expenditure Loophole:** Section 77, RPA 1951 caps candidate expenditure, but there is **no overall ceiling on political-party expenditure**.
- **Anonymous Small Donations:** Section 29C, RPA 1951 does not require donor disclosure for contributions below **₹20,000**, enabling donation splitting.
- **Limited ECI Enforcement Capacity:** ECI depends on State police and administration and lacks an **independent financial-forensics wing**.

- **Inter-Agency Coordination:** Tracking **hawala, shell companies and complex financial flows** requires multiple agencies.
- **MCC Enforcement Gap:** **Model Code of Conduct** is largely non-statutory.
- **Cash-Based Economy:** Makes it difficult to establish the source and electoral link of seized cash.
- **Last-Mile Inducements:** Cash, liquor and gifts are often distributed **24–48 hours before polling**.
- **Detection–Conviction Gap:** Seizures and FIRs often do not translate into convictions.

Measures Needed to Ensure the Integrity of the Electoral Process

- **Statutory Capping of Party Expenditure:** Introduce enforceable expenditure limits for political parties and empower the ECI to **deregister non-serious/shell parties**.
- **Complete Digitization of Political Funding:** Phase out cash donations and route contributions through **traceable digital banking channels** with independent auditing.
- **Empowering ECI with Financial Intelligence:** Create an autonomous financial enforcement wing and integrate real-time data from **Financial Intelligence Unit (FIU-IND)** and CBDT.
- **Partial State Funding of Elections:** As recommended by the **Indrajit Gupta Committee (1998)**, provide state support in kind.
- **Judicial Fast-Tracking and Decriminalisation:** Create specialised fast-track courts, enforce the SC's **1-year investigation timeline**, and strengthen disqualification for serious electoral offences.

Black money in elections compromises not merely electoral finance but the very autonomy of the voter. Discuss.

Drishiti Mains Question

80th Independence Day: 'Sapt Dhara' Framework for Viksit Bharat 2047

On India's 80th Independence Day, PM outlined the '**Sapt Dhara**' (**Seven Streams of Strength**) framework for **Viksit Bharat by 2047**. The full version of '**Vande Mataram**' was rendered at the Red Fort for the first time during Independence Day celebrations, marking its **150th anniversary**.

- The development draws attention to **Panch Pran (Five Resolves)**, outlined in the **2022 Independence Day address**, to guide India's journey towards **Viksit Bharat 2047**.

PM's 'Sapt Dhara' Framework for Viksit Bharat by 2047

Stream	Key Focus
Manufacturing	■ Develop the entire domestic value chain, zero-defect precision manufacturing and position India as a trusted global supply-chain hub through Make in India, Swadeshi and Vocal for Local. India has had FTAs with nearly 40 countries since 2014.
Agriculture and Food Production	■ Promote food processing, value addition, farm-to-export integration and chemical-free farming; maintain access to affordable Urea and Di-ammonium Phosphate (DAP).
Technology and Innovation	■ Build capabilities in AI, quantum technology, space, robotics, data centres, next-generation communication and semiconductors. Initiatives include UPI/DPI, Bharat 6G Vision, National Drone Policy, Atal Tinkering Labs and RDI Scheme. Target: AI skilling of 1 crore youth within one year; nationwide sports talent hunt for 5–15-year-olds for 2036 Olympic ambitions.
Gati Shakti	■ Expand high-speed rail, highways, inland waterways, airports and multimodal logistics. Modern railway coach production increased 21-fold; electrification rose from about 30% to near-total. Hydrogen-powered trains, port-led development and blue economy were also highlighted.
Defence Power	■ Strengthen defence self-reliance and exports, particularly drones, counter-drone systems, hypersonic technologies and cybersecurity. Mission Sudarshan Chakra and stronger civil defence highlighted.
Green and Blue Economy	■ Leadership in green hydrogen, renewable energy, storage, green mobility and manufacturing. SHANTI Act 2025 supports a target of 100 GW nuclear capacity by 2047; Nuclear Energy Mission targets at least 5 indigenous SMRs by 2033. Focus also on Fast Breeder Reactors, critical minerals, fisheries, coastal tourism and ocean technology. ■ Piped gas distribution expanded from 70 to 700 cities, reaching around 1.75 crore households; EV sales reached 25 lakh in 2025–26, compared with 1.5 lakh in 2009–10. ■ National Critical Minerals Mission and Critical Minerals Corridor seek to strengthen mineral security for clean-energy and technology transitions.
Soft Power	■ Leverage Yoga, Ayurveda, culture, tourism, handicrafts, films, VFX, animation, gaming and digital content. Heal in India and WAVES Summit highlighted.

Panch Pran (Five Resolves) for Viksit Bharat by 2047

- **Developed India:** Transform India into a prosperous global power with high living standards.
- **Removal of Colonial Mindset:** Eliminate colonial mentality from institutions, systems and society.
- **Pride in Heritage:** Promote pride in India's culture, traditions and heritage.
- **Unity and Integrity:** Strengthen national unity beyond caste, region and language divisions.
- **Sense of Duty:** Emphasise Kartavya and responsible citizenship in nation-building.

India's journey towards Viksit Bharat requires a balance between cultural confidence, civic responsibility and economic competitiveness. Discuss in the light of Panch Pran and Sapt Dhara.

Drishti Mains Question

Relevance of Regional Parties in India

The relevance of regional parties has come under debate following electoral setbacks, defections and concerns that the proposed **One Nation, One Election (ONOE)** framework may strengthen dominant national narratives at the expense of State-specific politics.

Classification of the Party System in India

- **Political Party:** An association/group of Indian citizens registered with the ECI under Section 29A of the Representation of the People Act, 1951.
 - India follows a **multi-party system**.
 - ECI classifies parties as recognised and unrecognised.
 - ❖ **Recognised Parties:** National Parties and State Parties under the **Election Symbols (Reservation and Allotment) Order, 1968**.
- State Parties are commonly called **regional parties** because their influence is concentrated in particular States.
- The term political party was not expressly mentioned in the original Constitution; it entered the constitutional framework through the **52nd Constitutional Amendment Act, 1985**, which inserted the Tenth Schedule.

Why is the Political Space for**Regional Parties Shrinking in India?**

- **Dominance of National Parties:** As of August 2026, regional parties independently headed governments in only **Andhra Pradesh, Meghalaya, Nagaland and Puducherry**, the lowest level in nearly 2.5 decades, compared with nine States during 2015–20.

■ Changing Political Phases:

- **1952–67:** One-party dominance.
- **1967–89:** Rise of linguistic, cultural and regional aspirations.
- **1989–2014:** Coalition politics, with regional parties acting as **kingmakers**.
- Renewed single-party dominance has reduced their role in national policymaking.

■ **FPTP Effect:** Even a small decline in vote share can cause disproportionate seat losses, while concentrated gains can yield large seat increases; therefore, seat losses may exaggerate actual erosion of voter support.

■ **Dynastic/Leader-Centric Politics:** Weak internal democracy and dependence on dominant leaders or families can produce **succession crises, factionalism, splits and loss of emerging leaders**.

■ **Resource Asymmetry:** Candidate expenditure limits are ₹95 lakh/₹40 lakh for Lok Sabha/Assembly seats in larger States and ₹75 lakh/₹28 lakh in smaller States, but parties themselves have no overall election-expenditure ceiling, benefiting larger national parties.

■ **Defections:** Concerns over misuse of the **Tenth Schedule's merger provision**, party splits and defections can weaken regional parties.

■ **Political Pressure:** Allegations concerning selective use of central investigative agencies, corruption and money power may further affect organisational stability and bargaining capacity.

■ **Reduced State Policy Space:** Governor-related disputes, delayed assent to State laws, dependence on **Centrally Sponsored Schemes**, and greater use of **cesses and surcharges outside the divisible pool under Article 271** can constrain States' fiscal autonomy and their ability to pursue region-specific development policies.

■ **ONOE Concern:** Simultaneous elections may allow national-security and macroeconomic narratives to overshadow State-specific issues, potentially favouring better-resourced national parties.

■ **Identity to Aspirational Politics:** Voters increasingly emphasise **development, infrastructure, employment and national security** alongside regional identity; regional parties must combine identity politics with credible governance.

Evidence of Resilience

- **Stable Vote Share:** Regional parties secured about **33.52% of total votes in the 2024 Lok Sabha election**.
- **Assembly Strength:** Their performance remains stronger in State Assembly elections, indicating continued regional political preferences.

■ **Coalition Relevance:** They remain dominant coalition partners in **Andhra Pradesh, Meghalaya, Nagaland and Puducherry** and participate in governments in **Bihar, UP, Assam, Goa, Maharashtra, Tripura, J&K & Kerala**.

Significance of Regional Parties in Indian Democracy

■ **Checks Centralisation:** Act as a counterweight to excessive centralisation and demand greater administrative and fiscal devolution.

■ **Electoral Federalism:** During **1989–2014**, coalition politics “federalised” the party system by compelling the Centre to accommodate State aspirations.

■ **Welfare Innovation:** State-level models can influence national policy; the **Midday Meal Scheme in Tamil Nadu** is a key example.

■ **Democratic Choice:** Regional parties widen electoral competition and participation.

- Tamil Nadu recorded **84.69% voter turnout in the 2026 Assembly election**, its highest ever.

■ **Voice to Regional Interests:** Bring linguistic, ethnic and cultural concerns into national politics.

■ **National Agenda Setting:** Highlight issues such as **inter-State river disputes, State-specific financial needs, agricultural concerns and regional infrastructure** in Parliament and national policymaking.

Measures to Strengthen Regional Parties

■ **State Funding of Elections:** Partial State funding, as recommended by the **Indrajit Gupta Committee (1998)**, can reduce resource asymmetry.

■ **Curb Wholesale Defections:** Tighten/remove the two-thirds merger exemption under the Tenth Schedule.

- **Dinesh Goswami Committee (1990):** Disqualification by President/Governor on ECI advice.
- **Law Commission, 170th Report (1999):** Transfer decision-making to an independent tribunal.
- Reiterated by the **National Commission to Review the Working of the Constitution (2002)**.

■ **Intra-Party Democracy:** Build a second rung of leadership and reduce personality-centric organisation.

■ **Re-evaluate ONOE:** Safeguard State-level issues and regional political narratives during simultaneous elections.

■ **Fair Delimitation:** Ensure Southern States are not politically marginalised due to demographic differences.

■ **Developmental Regionalism:** Shift from purely caste/language mobilisation towards State-specific governance focused on **infrastructure, health and fiscal prudence**.

“Strong regional parties are essential for maintaining political competition and preventing excessive centralisation.” Critically examine.

Drishiti Mains Question

Deepening Inclusive Development in Rural India

As India marks its 80th Independence Day, the Government highlighted initiatives for rural and tribal welfare, inclusion, livelihoods, infrastructure, education and financial security.

India's Shift from Welfare Delivery to Decentralised and Inclusive Development

From Welfare Delivery to Decentralised Partnerships

- **Direct Fiscal Transfers:** Panchayat transfers increased from ₹2.36 lakh crore under 15th Finance Commission (2021–26) to ₹4.35 lakh crore under 16th Finance Commission (2026–31).
- **Rural Development Budget:** ₹87,765 crore (2016–17) → ₹2.73 lakh crore (2026–27).
- **Tribal Affairs Budget:** ₹4,826 crore (2016–17) → ₹15,421.97 crore (2026–27).

Foundations of Dignity: Universal Access to Basic Needs

- **Jal Jeevan Mission:** Tap-water coverage rose from 3.23 crore households (2019) to 15.91+ crore (August 2026); 2.89 lakh villages reported 100% household coverage.
- **Swachh Bharat Abhiyan-Grameen:** 12.19+ crore household toilets constructed; India declared ODF in 2019. 1,288+ GOBARDhan biogas plants operational by August 2026.
- **Pradhan Mantri Ujjwala Yojana:** 10.57+ crore LPG connections provided to poor women; total LPG connections rose from 14.52 crore (2014) to 33.39 crore (2026).
- **Universal Village Electrification:** 100% willing household electrification achieved by March 2019 under SAUBHAGYA; 100% village electrification achieved by 2025 under DDUGJY.
- **Ayushman Bharat:** 45.51+ crore Ayushman Cards, 96.43 crore ABHA accounts and 1.86+ lakh Ayushman Arogya Mandirs by August 2026.
- **Food Security for All: PMGKAY** covers 81.35+ crore beneficiaries.
 - 99.8% Fair Price Shops automated for Aadhaar-based distribution.
 - Ration cards 100% digitised across States/UTs.
 - 221+ crore ONORC portability transactions recorded.
- **Samagra Shiksha Abhiyan:** During 2018–19 to 2025–26, 4,073 schools upgraded, 1.49 lakh ICT/digital initiatives implemented and 25,000 schools covered under skill education.
 - 97.3% schools had functional sanitation facilities for girls.
 - Female dropout: Primary 4.6% → 0.1%; Secondary 14.5% → 8% between 2013–14 and 2025–26.
- **EWS Reservation and Improved Enrolment:** 103rd Constitutional Amendment Act, 2019 enabled up to 10% EWS reservation in education and public employment.

- **GER 2025–26:** Primary 89.4, secondary 81.5; higher-secondary increased from 46.4 (2014–15) to 61.7.
- **Beti Bachao Beti Padhao:** Girls' secondary GER increased from 75.5% (2014–15) to 83.4% (2025–26).
- **Kasturba Gandhi Balika Vidyalyas and PM eVidya:** Sanctioned KGBVs increased from 3,609 to 5,639; 5,133 operational in 2024–25.
 - Enrolment increased from 3.52 lakh to 7.11 lakh.
 - **SWAYAM:** 6.5+ crore enrolments.
 - **DIKSHA:** 2.35 crore registered users.

Building the Backbone: Infrastructure and Connectivity

- **Pradhan Mantri Awaas Yojana-Grameen:** 3.91+ crore houses sanctioned and 3.13 crore completed by August 2026.
- **Pradhan Mantri Gram Sadak Yojana:** 99.6% eligible habitations connected by all-weather roads by June 2026.
 - Rural roads completed: 3.86 lakh km (2000–14) vs 4.13 lakh km (2014–2026).
 - Bridges completed: 484 → 10,404 over the same periods.
- **Digitalisation of Gram Panchayats:**
 - BharatNet: 2.19 lakh Gram Panchayats connected.
 - eGramSwaraj: ₹3.16+ lakh crore online transactions processed.
 - SabhaSaar: Used by 1.28+ lakh Gram Panchayats for AI-enabled meeting summaries.

Pathways to Prosperity:

Livelihoods, Skills and Financial Security

- **DAY-NRLM:** 10.19 crore households mobilised into 94.46 lakh SHGs.
 - Lakhpati Didi: 3.46 crore created by July 2026.
- **Community Resource Persons for Last-Mile Delivery:** 9+ lakh CRPs, including 50,548 Bank Sakhis, 1.91 lakh Krishi Sakhis, 2,012 Matsya Sakhis and 1.70 lakh Pashu Sakhis.
- **Self-Help Entrepreneur Mart (SHE-Mart):** Announced in Union Budget 2026–27; aims to benefit 1 crore women through community-owned retail spaces.
- **Deendayal Upadhyaya Grameen Kaushalya Yojana:** 18.45 lakh trained and 12.35 lakh placed by August 2026.
- **Pradhan Mantri Kaushal Vikas Yojana:** 1.64+ crore youth trained up to June 2026.
- **Rural Employment Guarantee:** Under the predecessor MGNREGA framework, person-days increased from 1,660 crore (2006–14) to 3,036.7 crore (2014–25); women's participation rose from 48% to 58.19%.
 - From 1 July 2026, MGNREGA was replaced by the **Viksit Bharat-Guarantee for Rozgar and Ajeevika Mission**

(Gramin) Act, 2025, guaranteeing 125 days of wage employment per rural household.

- **e-Shram Portal:** Registrations rose from 14.40 crore (2021) to 31.87 crore (August 2026).
- **PM Vishwakarma Yojana:** 30+ lakh artisans registered; 24.32 lakh trained and 17.70 lakh received toolkits.
- **PM SVANidhi:** Beneficiaries increased from 26.37 lakh (2021) to 78.17 lakh (2026); ₹18,971.18 crore loans disbursed; 46% beneficiaries are women.
- **PM MUDRA Yojana:** Loans sanctioned increased from 3.49 crore (2016) to 59.14+ crore (July 2026).
- **Pradhan Mantri Jan Dhan Yojana:** Beneficiaries increased from 17.9 crore (2015) to 58.90 crore (2026).
- **Expanding Social Security Coverage:**
 - **NPS:** 1.05 crore → 2.17 crore beneficiaries.
 - **Atal Pension Yojana:** 48.80 lakh → 9.29 crore beneficiaries.
 - **PMJJBY:** 27.90 crore people covered.
 - **PMSBY:** 58.78 crore people covered.
 - **PM Shram Yogi Maan-dhan:** 54+ lakh workers registered.

Accelerating Tribal Development through Integrated Interventions

- **PM-JANMAN:** 540 Van Dhan Vikas Kendras sanctioned, benefiting 46,042 PVTG persons.
- **Pradhan Mantri Janjatiya Vikas Mission (PM-JVM):** 4,172 VDVKS sanctioned, benefiting about 12.48 lakh tribal people.
- **Pradhan Mantri Janjatiya Unnat Gram Abhiyan (PM-JUGA):** Integrates 17 Ministries for tribal-majority villages and PVTG habitations.
 - 7.60 lakh tribal houses completed; 28,303 villages gained drinking-water access.
 - 59,812 households/public institutions electrified; 657 Anganwadi Centres operationalised.
 - 692 hostels sanctioned for tribal students.
- **Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY):** 4.16+ lakh SC houses, 4.17+ lakh SC households under health protection and 3.57+ lakh households with individual toilets.
 - 17,578 villages declared Adarsh Grams.
- **Eklavya Model Residential Schools (EMRSs):** Operational schools increased from 129 (2014–15) to 511 (2026); enrolment around 1.68 lakh.
- **PM-YASASVI:** In 2024–25, 20.61 lakh pre-matric and 24.53 lakh post-matric scholarships provided.
- **Scholarships for Scheduled Tribe Students:**
 - **Pre-Matric Scholarship:** Benefited 28.99+ lakh tribal students between April 2024 and August 2026.

- **National Fellowship for ST Students:** Supported 2,961 research scholars in 2025–26; 56% women.

- **Scholarships for Higher Education for Young Achievers Scheme (SHREYAS):** 8,571 SC students studied in institutions including IITs, IIMs, NITs and AIIMS in 2025–26.
- **National Action for Mechanised Sanitation Ecosystem (NAMASTE):** 89,915 sewer/septic-tank workers and 2.81+ lakh waste pickers validated.
- **Scheme for Economic Empowerment of De-Notified Tribes (SEED):**
 - ₹26.75 crore disbursed for education of 4,485 DNT students in 2025–26.
 - 73,569 Ayushman Bharat cards issued to DNT families.

India's rural and tribal development strategy is moving from welfare delivery to empowerment and decentralised partnerships. Discuss.

Drishti Mains Question

PM CARES Fund Under Scrutiny Over Transparency

The **PM CARES Fund's** transparency has come under scrutiny, as no audited financial statements have been publicly available since FY 2022-23.

Major Concerns

- **Public Authority Ambiguity:**
 - Despite PM/Union Ministers as trustees and PMO administrative support, government maintains PM CARES is neither a government fund nor a "public authority" under RTI Act, 2005; RTI status remains under judicial consideration.
- **Limited Parliamentary Scrutiny:**
 - In 2026, PMO told Lok Sabha Secretariat that PM CARES questions were inadmissible as it is not primarily a Government of India concern; critics see weaker accountability, though the Speaker has final say on admissibility.
- **No CAG Audit:** Audited by an independent chartered accountant, not CAG; critics seek stronger public audit/disclosure due to public purpose, official association and significant PSU contributions.
- **PSU Contributions:** CSR contributions by government-owned PSUs strengthen the debate over public accountability and disclosure.
- **Institutional Overlap:** Questions arise over separate PM CARES alongside PMNRF and statutory NDRF under Section 46, Disaster Management Act, 2005; Supreme Court (2020) held PM CARES and NDRF are distinct with different objects/purposes.

Key Facts	
Aspect	Quick Revision
Nature & Origin	Prime Minister's Citizen Assistance and Relief in Emergency Situations Fund; Public Charitable Trust established in March 2020 for emergencies, calamities and distress situations such as COVID-19.
Objectives	Relief during public-health emergencies and natural/man-made calamities; healthcare/pharmaceutical facilities, infrastructure, research and other assistance.
Trust Structure	Prime Minister: ex-officio Chairperson; Defence, Home and Finance Ministers: ex-officio Trustees; PM may nominate other pro bono Trustees.
Funding	Entirely voluntary contributions; no budgetary support; eligible CSR contributions from companies and PSUs accepted.
Tax/CSR Status	Donations qualify for 100% deduction under Section 133, Income-tax Act, 2025 (earlier Section 80G, 1961); company contributions qualify as CSR under Companies Act, 2013.
Foreign Contributions	Exempt from FCRA, 2010 provisions; may receive overseas donations through a separate foreign-contribution account.
FY 2022-23	Receipts -₹6,723 crore; payments -₹439 crore; closing balance -₹6,284 crore as on 31 March 2023.

Steps to Enhance Accountability

- **Clarify Legal/RTI Status:** Government or Parliament should define whether PM CARES is a public authority under RTI Act, 2005 through a clear statutory framework.
- **Stronger Audit/Oversight:** Adopt stronger public-audit standards, detailed auditor observations, utilisation certificates and periodic independent scrutiny, especially given PSU contributions.
- **Time-Bound Disclosure:** Publish audited accounts, receipts, expenditure, project-wise utilisation and closing balances annually within a fixed timeline.
- **Parliamentary Accountability:** Consider questions case-by-case under Lok Sabha rules, with admissibility decided by the Speaker rather than blanket exclusion.
- **Outcome-Based Disclosure:** Publish major donor categories, project-wise allocations, beneficiary criteria, implementing agencies and measurable outcomes while protecting legitimate privacy.

Transparency and accountability are essential for public trust in emergency relief funds. Discuss in the context of the PM CARES Fund and the debate over its legal and institutional status.

Drishti Mains Question

Tribunals Reforms Bill, 2026

The Lok Sabha passed the **Tribunals Reforms Bill, 2026** by voice vote without debate. It seeks to **replace the Tribunals Reforms Act, 2021** and establish a National Tribunals Commission (NTC) to oversee tribunal appointments and service conditions.

Key Provisions of the Bill

- **Repeal of 2021 Act:** Repeals the **Tribunals Reforms Act, 2021** and aligns tribunal appointments/service conditions

with **SC directions on judicial independence and separation of powers.**

- **Madras Bar Association v. Union of India** rulings (2010–2025) restricted executive overreach in tribunals.
- **National Tribunals Commission:** Establishes the **NTC** for appointments, performance review, complaints, and the **National Tribunals Data Grid**.
 - Composition: **Chairperson** (who has been a Judge of SC or a Chief Justice of HC), **2 Judicial Members** and **2 Technical Members**.
 - Technical members require **25 years' relevant experience**.
 - Tenure: **5 years** or up to **70 years** for Chairperson and **67 years** for members, whichever is earlier.
 - Appointments by **Central Government**; **CJI consultation** for Chairperson and judicial members; Secretariat headed by a **Central Government Secretary**.
- **Selection of Tribunal Members:** Search-cum-selection committee recommends **one candidate plus one waitlisted name**; appointment must be finalised within **3 months**.
- **Reappointment:** Based on **previous performance**, with consultation of the concerned Tribunal Chairperson for members.
- **Removal:** Grounds include **insolvency, moral-turpitude conviction, incapacity, abuse of position and conflict of interest**; tribunal members may also be removed for **incompetence, inefficiency or paid assignments**.
- **Service Conditions:** **Central Government Rules** will govern qualifications, selection, salaries, allowances, resignation, removal and other service conditions.

2021 Act Provisions Struck Down by Supreme Court

Provision	Supreme Court Position
50-year minimum age	Arbitrary; excluded younger advocates/experts despite eligibility based on 10 years' legal practice.
Four-year tenure	Invalid; undermined judicial independence and continuity. Minimum five-year tenure restored.
Two names per vacancy	Invalid; increased executive discretion. Committee should recommend one name per post.
Re-enacting invalid provisions	Parliament cannot re-enact struck-down provisions without curing defects; judicial review is part of the Basic Structure.

Tribunals in India: Background

- **Initial Establishment:** Tribunals developed parallel to courts to reduce workload and provide expertise; **Income Tax Appellate Tribunal (ITAT) (1941)** was the first tribunal.
- **Constitutional Incorporation (1976):** Tribunals were absent from the original Constitution. Based on **Swaran Singh Committee (1976)** recommendations, **Part XIV-A** and **Articles 323A & 323B** were added by the **42nd Constitutional Amendment Act, 1976**.
- **Part XIV-A:**
 - **Article 323-A:** Parliament can establish **Administrative Tribunals** for recruitment and service conditions of public servants.
 - **Article 323-B:** Parliament and State Legislatures can establish tribunals for matters such as **taxation, foreign exchange, labour disputes and land reforms**.
- **Expansion and Consolidation:** Sector-specific tribunals expanded from the 1980s; the **Finance Act, 2017** merged several tribunals. In **2010**, the SC held that legislatures may create tribunals for any **Seventh Schedule** subject, not only Article 323B matters.
- **Judicial Review:** In **S.P. Sampath Kumar (1986)**, the SC allowed alternatives to High Courts if equally effective.
- In **L. Chandra Kumar (1997)**, the SC held tribunal decisions subject to **High Court and Supreme Court writ jurisdiction**; tribunals cannot fully substitute higher constitutional courts.

Major Issues

- **Executive Dominance:** Government controls **appointments, salaries, service conditions and removals** despite being a major litigant. **Roger Mathew v. South Indian Bank (2019)** stressed **judicial dominance in selection**.
- **Vacancies and Pendency:** Chronic vacancies cause large backlogs; **ITAT had ₹6.7 trillion in disputes in 2024**, affecting revenue and investor confidence.

- **Short Tenures and Reappointment:** Short terms limit experience; reappointment may increase **executive dependence** and affect impartiality.
- **Administrative Dependence:** Reliance on parent ministries for **finance, infrastructure and staff** weakens autonomy.
- **Additional Litigation:** **L. Chandra Kumar v. Union of India (1997)** upheld **High Court judicial review**, creating a possible additional litigation layer.
- **Inadequate Infrastructure:** Tribunals face shortages in **infrastructure, digital facilities and staff**, along with non-uniform procedures and service conditions.

Reforms Needed

- **Independent Administration:** Shift financial and administrative control from litigating parent ministries to an independent nodal authority with separate budgetary allocations.
- **Streamline Appeals:** Create dedicated Tribunal Appellate Benches in High Courts; consider a **UK-style two-tier tribunal system** with **First-tier and Upper Tribunals**, supported by specialised **subject-specific Chambers**.
- **Judicial Impact Assessment:** Conduct Judicial Impact Assessments (JIA) before expanding tribunal jurisdiction to estimate required infrastructure, judges and resources.
- **Capacity Building:** Cross-train technical members in law and judicial members in domain expertise through the National Judicial Academy.
- **Pre-Institution Mediation:** Mandate mediation for suitable disputes to reduce frivolous litigation and pendency.
- **Performance Audits:** Independent annual audits of disposal rates, adjournments, pendency and judicial quality, with transparent publication.

"Tribunalisation was intended to deliver specialised and speedy justice, but institutional weaknesses have often undermined this objective." Critically examine.

Drishti Mains Question

Supreme Court's New Handbook on Gender Sensitivity

SC's 2026 report, 'Judgments and Gender: Sensitivity and Compassion in Writing Judgments', seeks trauma-informed, survivor-centric judicial language and courtroom practices, building on its 2023 Handbook on Combating Gender Stereotypes.

Key Highlights of SC's New Handbook on Gender Sensitivity

- **Evolution of Gender Jurisprudence:** Judicial jurisprudence evolved from **protective discrimination** to stronger safeguards for **dignity, autonomy and protection against gender discrimination**.

Case	Key Principle
Yusuf Abdul Aziz v. State of Bombay (1954)	Upheld protective discrimination (affirmative action) under Article 15(3).
State of Punjab v. Gurmit Singh (1996)	Supported in-camera trials and cautioned against doubting survivor testimony without corroboration.
Vishaka v. State of Rajasthan (1997)	Established guidelines against workplace sexual harassment , paving the way for the POSH Act, 2013
NALSA v. Union of India (2014)	Recognised transgender identity, dignity and constitutional rights .
Joseph Shine v. Union of India (2019)	Struck down Section 497 IPC on adultery as unconstitutional.
Aparna Bhat v. State of MP (2021)	Prohibited gender stereotyping and suggestions of compromise in sexual-offence cases.
State of Jharkhand v. Shailendra Kumar Rai (2022)	Banned the unscientific two-finger test in rape examinations.
X v. Principal Secretary, Health, Delhi (2023)	Affirmed reproductive autonomy of unmarried women .
Vaneeta Patnaik v. Nirmal Kanti Chakrabarti (2025)	A hostile work environment can constitute continuing sexual harassment , requiring a survivor-sensitive interpretation under the POSH Act, 2013 .

Key Recommendations

- **No Stereotyping:** Building on *Aparna Bhat v. State of Madhya Pradesh (2021)*, the Court directed judges to avoid gender stereotypes and marriage/compromise suggestions in rape cases; clothing, marital status, sexual history or lifestyle must not determine consent or credibility (Section 146, Indian Evidence Act, 1872; Section 48, BSA, 2023).
- **Rights-Based Language:** Replace stereotypical expressions such as “prosecutrix”, “modesty”, “lust” and “chastity” with terms such as “survivor”, “sexual assault” and “bodily autonomy violated”; use inclusive LGBTQIA+ terminology, including SOGIESC (Sexual Orientation, Gender Identity, Expression and Sex Characteristics).
- **Victim/Witness Protection:** Strengthen Witness Protection Scheme, 2018 (upheld in *Mahender Chawla v. Union of India (2018)*) and provide mandatory legal aid from the FIR stage (*Delhi Domestic Working Women’s Forum v. Union of India (1995)*).
- **In-Camera & Child-Sensitive Trials:** Use in-camera rape/POCSO trials under Section 366(2), BNSS, 2023; video-link/screen testimony (*Sakshi v. Union of India, 2004*), support persons for children (*Bachpan Bachao Andolan v. Union of India, 2013*) and time-bound POCSO trials.

- **Witness as Guest:** Treat witnesses with dignity; avoid repeated adjournments and long waiting (*State of U.P. v. Shambhunath Singh, 2001*).
- **Pre-Trial Counselling:** Provide counselling to reduce anxiety and secondary victimisation.

Significance	
Constitutional	Advances Article 21 (dignity, fair trial), Articles 14 & 15 (substantive equality), and aligns with Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), 1979 .
Reduces Secondary Victimisation	Insensitive language and hostile cross-examination deter reporting. Rights-based judicial language can reduce secondary victimisation and promote survivor support.
Institutional Accountability	Promotes a survivor-sensitive, trauma-informed justice process beyond an exclusively adversarial approach.
Legal Consistency	Complements BNSS 2023 (in-camera trials), Bharatiya Sakshya Adhiniyam, 2023 (BSA) (rape-shield provisions), POCSO Act 2012 , and POSH Act 2013 .

Related Concerns	
Non-binding Nature	Guidelines are recommendatory ; effectiveness depends on judicial training and internalisation.
Linguistic Diversity	Uniform glossary is difficult to implement in vernacular trial-court proceedings.
Low Awareness	80.7% of respondents were unaware of the Witness Protection Scheme .
Continuous Sensitisation	Sensitisation is an “ongoing process” ; language and social attitudes keep evolving.
Deep-rooted Gender Bias	Judicial sensitisation alone cannot remove gender bias without broader societal behavioural and attitudinal change.

How can gender stereotypes embedded in judicial reasoning undermine the constitutional guarantees of equality and dignity?

Drishti Mains Question

10 Years of Government e-Marketplace (GeM)

GeM completed **10 years of operations** on **9th August 2026**.

Key Features of GeM

- **About GeM:** Unified digital platform for **paperless, cashless, contactless, transparent and competitive** government procurement.
- **Purpose:** Uses digital technology and standardised processes to simplify public procurement.
- **Coverage:** Serves **Central/State Ministries, Departments, PSUs, Autonomous Bodies, local bodies, Panchayats and Cooperatives**.

- **End-to-End Procurement:** Covers seller registration, product listing, bidding, contract award, order fulfilment and payment.
- **Transparency & Competition:** Supports **direct purchase, comparison, e-bidding and reverse auctions** with digital records and real-time monitoring.
- **Data-Driven Procurement:** Uses **GeM Availability Report, Past Transaction Summary, Last Purchase Price and price-trend analytics** for informed purchasing.
- **Integration:** Being integrated with **Central Public Procurement Portal (CPPP), Indian Railway Electronic Procurement System (IREPS) and Defence Public Procurement Portal (DPPP).**
- **Scale:** Department of Military Affairs crossed **₹1.35 lakh crore cumulative GMV** as of **6 Aug 2026.**

GeM's Role in Expanding the Scale and Inclusiveness of Public Procurement

- **Expanding Procurement Scale:** Connects **1.37 lakh+ government buyer organisations** with nearly **25 lakh sellers/service providers**; cumulative procurement exceeds **₹20 lakh crore.**
- **Broad Marketplace Coverage:** Offers **10,660+ product and 350 service categories.**
- **Digital Procurement Ecosystem:** Supports **Digital India, Make in India, Aatmanirbhar Bharat, Vocal for Local and Ease of Doing Business** through multiple procurement modes, integrated payments and online contracts.
- **Ease-of-Business Reforms:** **Rate Contracts** and removal of **Caution Money** reduce seller compliance burden.
- **Innovation Partnerships:** Partnered with **IN-SPACe** for space-tech and **Drone Federation India** for drone procurement in **April 2025.**
- **Seller Financing through GeMSahay:** Provides eligible proprietorship sellers **instant, collateral-free, paperless working-capital loans** without buyer consent.
- **Capacity and Compliance Support:** Partnerships with **Arun Jaitley National Institute of Financial Management (AJNIFM), CareEdge Ratings, National Centre for Good Governance (NCGG) and Indian Institute of Public Administration (IIPA)** support training, research and policy; **EPFO integration** strengthens manpower-service compliance.
- **Last-Mile Support through Suvridha Kendras:** **50 GeM Suvridha Kendras** being piloted under the **June 2026 CSC-SPV partnership** for registration, vendor assessment, catalogue creation, capacity building and grievance redressal.
- **MSE Participation:** In **2025–26, 11 lakh+ MSEs** executed **51 lakh+ orders worth ₹2.36 lakh crore+.**
- **Women-Led Enterprises:** **2.1 lakh+ women-led MSEs** secured **₹28,000 crore+ procurement**, supported by **Womaniya** and **UN Women** partnerships.
- **SC/ST and Startup Participation:** SC/ST enterprise procurement exceeded **₹6,000 crore** with nearly **28% annual growth**; startup procurement crossed **₹19,000 crore** with over **36% annual growth.**
- **Inclusive Procurement through SWAYATT:** Promotes participation of **startups, women entrepreneurs, MSEs and SHGs.**
- **Dedicated Market Platforms:** **Womaniya** supports women entrepreneurs/SHGs, **SARAS Collection** links rural SHGs with buyers, and **Startup Runway 2.0** showcases startup innovations.

Supreme Court Directs Measures to Curb 'Digital Arrest' Scams

SC directed **coordinated action by the Centre, States/UTs, RBI, telecom authorities and investigators to curb digital-arrest fraud** and speed up recovery of victims' money.

- **SC Directions:** RBI to issue an SOP within four weeks for temporary debit holds on mule accounts linked to cyber fraud/money laundering; States/UTs to establish grievance-redressal and money-restoration mechanisms aligned with MHA SOPs for the National Cyber Crime Reporting Portal and Cyber RMS.
- **Digital Arrest:** Fraudsters impersonate law-enforcement agencies, use spoofed numbers/fake documents, falsely link victims to criminal cases and threaten arrest or account freezing to extract fake fines or security deposits.
- **Complaints Declined:** I4C data: 1,23,672 complaints in 2024 → 58,249 in 2025 → 16,377 by June 2026.
- **Related Initiatives:**
 - **Money Restoration Mechanism:** Data-sharing MoU between RBI Innovation Hub and I4C; covers 1,23,590 branches of 69 banks, with 57 banks participating; **₹18.05 crore** restored in 36,290 cases.
 - **e-Zero FIR & Cyber Centres:** e-Zero FIR operational in 19 States; 14 States have Cyber Crime Coordination Centres. SC directed remaining States/UTs to implement both within four weeks.
 - **Telecom Rules:** Radio Equipment Possession Authorisation Rules, 2025 notified; User Identification Rules, 2025 in final notification stage.
 - **Committee Recommendations:** Consider lowering the **₹10-crore CBI threshold** and include multiple frauds by the same network that cumulatively exceed it.

Digital Transformation of Indian Judiciary

The **e-Courts Mission Mode Project** has transformed India's paper-based judiciary into a digitally enabled, transparent and trackable system through e-filing, virtual courts, digitised records, e-Sewa Kendras, video conferencing, AI tools and integrated criminal-justice platforms.

Key Features of e-Courts Mission Mode Project

- **Phase I – Basic Computerisation (2011–2015):** Computerised **14,000+ courts** and created basic network infrastructure.
- **Phase II – Citizen-Centric Digital Services (2015–2023):** Introduced **NJDG, eFiling and eSewa Kendras** and expanded video-conferencing infrastructure **over fivefold**.
- **Phase III – Digital and Intelligent Courts (2023–2027):**
 - Focuses on a **digital, paperless and technology-driven judiciary**.
 - **4,519 courts** have e-filing and **474 courts** have paperless infrastructure.
 - **538 courts** support virtual hearings, **1,806 eSewa Kendras** are functional, and **VC Video Conferencing (VC)** facilities cover **7,553 courts, jails and hospitals**.

- **National Service and Tracking of Electronic Processes (NSTEP):** 6,895 courts; **Inter-operable Criminal Justice System (ICJS):** all High Courts; **734 court websites** migrated to **Secure, Scalable and Sugamya Website as a Service (S3WAAS)** platform; **Software-Defined Wide Area Network (SD-WAN):** 174 sites.
- Uses **AI, OCR and analytics**; **IIT Madras** is developing the **Learning Management System (LMS)**; solar facilities cover **1,626 court complexes**.

AI-Assisted Judicial Processes

- **Funding:** ₹53.57 crore of the ₹7,210 crore Phase III outlay is earmarked for AI/ML applications in HCs through 2027.
- **Tools:** LegRAA supports legal research/document analysis; Digital Courts 2.1 enables paperless case management; ASR-SHRUTI provides voice-to-text; PANINI assists translation and dictation of orders/judgments.
- **Multilingual Judgments:** eSCR portal translates judgments into 18 Indian languages; 83,000+ translations, including 36,344 in Hindi, were completed by March 2025.
- **AI Filing Scrutiny:** An IIT Madras tool detects defects and extracts case metadata from e-filed petitions; prototype access given to 200 Advocates-on-Record, with e-filing/ICMIS integration underway.

Major Benefits and Digital Platforms	
Platform / Area	Quick Revision
National Judicial Data Grid (NJDG)	Public dashboard with real-time orders, judgments, case details and pendency; separate dashboards for SC, HCs and District Courts; tracks case stage, disposal and delay reasons.
e-Filing & e-Payments	Online filing in English/regional languages across District Courts, HCs & SCs. By 30 June 2026: 1.25 crore+ e-filed cases; about ₹1,404 crore court fees and ₹75 crore fines processed.
Video / Virtual Courts	4.18 crore+ remote hearings; livestreaming in 11 HCs; 31 Virtual Courts handled 11.33 crore traffic challans worth ₹1,135.79 crore.
Nyaya Shruti	Launched in 2024 under ICJS; enables accused, witnesses, police, prosecutors, experts and prisoners to appear/testify virtually.
Case Tracking	e-Courts portal/apps/SMS/email provide case status and cause lists; daily: 4 lakh+ SMSs, 6 lakh+ emails and ~35 lakh portal hits. e-Courts Services App: ~3.84 crore downloads; JustIS app: 22,594.
National Service and Tracking of Electronic Processes (NSTEP)	GPS-enabled electronic service and tracking of summons/notices across districts and States.
e-Sewa Kendras	Assist with case information, e-filing, e-signatures, e-stamps, e-payments, VC hearings, judgments, Virtual Courts, legal aid and eMulakat. As of 30 June 2026, the document reports 49 in High Courts and 2,535 in District Courts.

The e-Courts Mission Mode Project has transformed India's judiciary from a paper-based system to a digital justice delivery framework. Discuss.

Drishiti Mains Question

GOBARDHAN: National Circular Bioenergy Scheme

Recently, the Union Cabinet approved **GOBARDHAN, National Circular Bioenergy Scheme**, to convert organic waste into clean energy and promote rural prosperity.

Key Facts about the Scheme

- **About: GOBARDhan, or the National Circular Bioenergy Scheme**, aims to convert organic waste such as **animal waste, crop residue, kitchen waste and market waste** into compressed biogas and bio-slurry.
- **Implementation:** Approved on **6th August 2026**, the scheme will be implemented from **FY 2026–27 to FY 2035–36** under the **Ministry of Petroleum and Natural Gas**.
- **Financial Outlay:** The scheme has a total outlay of ₹23,731 crore and consolidates the previously **fragmented CBG ecosystem** under a unified national framework.
- **Support Framework:** The scheme is **structured around six growth engines** covering assured offtake, pricing, capital support, pipelines, credit guarantees and ecosystem development.
- **Objectives and Vision:** The scheme seeks to **increase domestic CBG production nearly ten-fold, attract private investment and develop a circular bioeconomy** in which waste supports growth, villages become clean-energy centres and farmers contribute to national energy security.
- **Need:** India imports **nearly half of its natural gas requirements, while disruptions around the Strait of Hormuz—through which about 55–60% of its LNG imports pass**—highlight the urgent need to strengthen domestic energy production and security.
- **GOBARDhan Unified Registration Portal:**
 - The **GOBARDhan Unified Registration Portal** enables existing and proposed CBG/Bio-CNG plants to obtain the registration number required for accessing MoPNG support.
 - As of **6th August 2026**, **1,908 plants** were registered, including **217 commissioned plants** producing **0.4 MMSCMD** and **339 plants under construction**.

Six Growth Engines Under the GOBARDhan Scheme

Component	Key Provisions	Expected Impact
Assured CBG Offtake	City Gas Distribution entities will procure CBG in line with the notified obligation of 3% in FY 2026–27, 4% in FY 2027–28 and 5% from FY 2028–29 onwards in the CNG-Transport and PNG-Domestic segments.	Creates a predictable long-term market, improves project bankability and capacity utilisation, and encourages private investment.
Stable CBG Pricing Framework	Introduces a government-backed administered price of ₹2,110 per MMBTU , equivalent to ₹105 per kg of CBG , with a minimum 10-year horizon and a government-supported, market-based cost-sharing mechanism.	Provides revenue certainty, remunerative returns and consumer affordability while supporting investment and capacity expansion.
Capital Assistance	Eligible greenfield projects will receive assistance of up to ₹2 crore per tonne per day of installed CBG capacity . Support also covers feedstock aggregation, organic-manure processing and value-addition assets. Capacity-expanding brownfield projects are also eligible.	Reduces initial capital costs, accelerates financial closure and expands participation by private developers, MSMEs, cooperatives and rural entrepreneurs.
Pipeline Infrastructure Development	Supports cluster-based and standalone pipelines connecting CBG plants with trunk pipelines and City Gas Distribution networks.	Reduces evacuation costs, improves supply reliability, expands market reach and increases utilisation of domestically produced CBG.
Credit Guarantee Support	Provides credit-guarantee coverage of up to 85% on eligible loans for MSME-based CBG projects.	Improves lender confidence, reduces collateral requirements, enables affordable institutional finance and encourages MSMEs, women entrepreneurs and first-time developers.
CBG Ecosystem Challenge Fund	Supports feedstock assessment and mapping, aggregation infrastructure, district-level CBG planning, technology adoption, process improvement, organic-manure value addition, capacity building and stakeholder awareness.	Accelerates district-level implementation and strengthens local CBG value chains and ecosystem capabilities.

SC Extends Protection Against Domestic Cruelty to Live-in Relationships

In **Dr. Lokesh B.H. & Ors. v. State of Karnataka & Anr. (2026)**, SC held that **Section 498A IPC (now Section 85 BNS)** may apply to a **live-in relationship** that is in the nature of marriage, provided intent to marry is established.

- **Purposive Interpretation of Law:** The Court held that **Section 498A IPC** aims to prevent domestic cruelty, irrespective of a formal marriage. Though it originally protected only legally married women, courts later extended it to women deceived into **void or voidable marriages**.
 - Denying protection to women in eligible live-in relationships while protecting married women violates **Article 14**, as the distinction has no rational nexus with preventing domestic violence.
 - The Court adopted the DV Act concept of a “**relationship in the nature of marriage**”, identified through shared finances, domestic life, relationship stability, and public recognition as spouses.
- **Initial Burden of Proof:** The female live-in partner must initially establish the **intent to marry**. No fixed parameters were prescribed, but a **higher standard of proof** applies because criminal liability is involved.
- **DV Act vs Criminal Law:** The **Protection of Women from Domestic Violence Act, 2005** mainly provides civil remedies such as maintenance and protection orders, while **Section 498A IPC/Section 85 BNS** provides criminal sanctions and deterrence; therefore, both laws are distinct.
- **Safeguard Against Arrest:** The Court directed strict compliance with **Arnesh Kumar v. State of Bihar (2014)**, requiring a mandatory preliminary inquiry before arresting the live-in partner or their relatives.

Indian Statistical Institute Bill, 2026

The **Bill introduced in Lok Sabha** seeks to reform **ISI governance, strengthen research and interdisciplinary collaboration**, and develop it as a global centre of excellence, but has raised concerns over Central control and institutional autonomy.

Key Provisions

- **Legal Status:** Repeals the Indian Statistical Institute Act, 1959 and converts ISI from a registered society into a not-for-profit statutory body corporate.
- **Visitor:** The President of India will review functioning, order inquiries and issue binding directions based on reviews.
- **Board of Governors:** Replaces the 33-member Council with an 11-member Board comprising a Visitor-nominated Chairperson, two Central officials, four eminent persons and four ISI representatives.
- **Board Functions:** Administration, budgets, departments and centres, degrees, regulations, posts and periodic performance reviews.
- **Councils and Director:** Creates an Academic Council, Finance Committee and centre-level Management Councils; the Director is appointed by the Board with the Visitor's prior approval and serves as Chief Executive Officer.
- **Regulations:** Central approval is required only for the first set of regulations, unlike the 1959 Act's requirement for every amendment.

Associated Concerns	
Institutional Autonomy	An 11-member Board may reduce faculty participation compared with the existing representative 33-member Council.
Executive Oversight	Visitor and Central Government roles in appointments, reviews and directions may increase executive influence.
Academic Authority	The Academic Council remains the principal academic body, but final authority rests with the Board.
Research Priorities	Industry collaboration and revenue generation may shift focus from fundamental research to market-driven work.

Historical Evolution of ISI

- **Historical Status:** Founded in 1931 by **P.C. Mahalanobis** as a laboratory at Presidency College; declared an Institution of National Importance under the Indian Statistical Institute Act, 1959.
- **Current Reach:** Headquartered in Kolkata with centres in Delhi, Bangalore, Chennai and Tezpur.
- **Apex Authority:** A **33-member council** with an elected chairman, independent scientists, and **six Central representatives** from **MoSPI, the Ministry of Finance, and RBI**.
- **Executive Independence:** The council directly appoints the institute's **director**, who is the academic and administrative head.

Measures for Effective Reform

- **Balanced Governance:** Ensure representation of government nominees, elected faculty and independent experts to combine efficiency with academic independence.
- **Protect Academic Freedom:** Define limits on the Visitor's powers and safeguard research, curricula and publication of statistical findings from executive interference.
- **Merit-Based Leadership:** Use an independent Search-cum-Selection Committee with eminent academicians and peer institutions for appointing the Director.
- **Research Funding:** Maintain adequate public funding for fundamental statistical and socio-economic research while encouraging industry collaboration.

Discuss the importance of institutional autonomy in Institutions of National Importance. How can governance reforms balance accountability with academic freedom?

Drishiti Mains Question

Pradhan Mantri Viksit Bharat Rojgar Yojana

PM-VBRY completed one year on **1st Aug 2026**, advancing formal employment and social security coverage.

Key Points

- **About:** PM-VBRY incentivises first-time employment and additional job creation by supporting employees and employers.
- **Implementation:** Administered by the **Employees' Provident Fund Organisation (EPFO)**, it expands formal employment and social security coverage.
- **Implementation Period:** Operational from **1st August 2025 to 31st July 2027**, with an outlay of **₹99,446 crore**.
- **Employment Target:** Aims to create over **3.5 crore jobs**, including **1.92 crore first-time employees**.
- **Social Security Expansion:** Promotes EPFO-linked employment across sectors, with emphasis on **labour-intensive manufacturing** and emerging industries.
- **Support to National Manufacturing Mission:** Strengthens the industrial workforce by encouraging formal job creation.
- **Two-Part Incentive Framework:** Provides separate incentives to first-time employees and employers.
- **Incentive to First-Time Employee:** Employees earning up to **₹1 lakh per month** receive one month's **EPF wage**, capped at **₹15,000**. The first instalment is paid after **six months**, and the second after **12 months of continuous service** and completion of a **financial literacy programme**. Part of the incentive is placed in a designated savings instrument.
- **Support to Employers:** Employers receive up to **₹3,000 per month for two years** for each eligible additional employee earning up to **₹1 lakh** and retained for at least **six months**. In manufacturing, support continues for the **third and fourth years**.

Impact of PM-VBRY

- **Employment and Workforce Formalisation:** In its first year, PM-VBRY promoted new and first-time hiring, expanded **EPFO-based social security** and strengthened the organised workforce.
- **Beneficiary Reach:** Since **Aug 2025**, over **72 lakh first-time employees** joined the formal workforce, while more than **15 lakh beneficiaries** received employment-linked incentives.
- **Women's Participation:** Women account for nearly **30% of beneficiaries**.
- **Digital Implementation:** The scheme uses an **EPFO-linked digital platform**, **Aadhaar authentication**, **UANs**, **ECR filings** and **Face Authentication Technology** through the **UMANG App**.

- **Transparent Benefit Delivery:** Incentives are transferred through **DBT** to Aadhaar-linked bank accounts, while the **PM-VBRY portal** enables real-time beneficiary and sector-wise monitoring.
- **Outreach and Support:** Nationwide campaigns and joint teams of **EPFO**, **ESIC** and the **Chief Labour Commissioner** support grassroots participation and implementation.

PM-KISAN Scheme

The Union Cabinet approved PM-KISAN's continuation for **2026–27 to 2030–31** with an outlay of **₹3.15 lakh crore**.

- The Budget 2026–27 allocated ₹60,000 crore.

Key Points

- **About:** Central Sector Scheme launched in **2019** to provide assured income support to landholding farmer families with cultivable land.
- **Assistance:** ₹6,000 annually in three instalments of ₹2,000.
- **Transfer:** Directly to Aadhaar-seeded bank accounts through DBT.
- **Objectives:** Support farm inputs, income, productivity and reduce informal credit dependence.
- **Identification:** Eligible farmer households identified by State Governments.
- **Monitoring:** Multi-tier national, State and district framework; national review chaired by the Cabinet Secretary.
- **Digital Delivery:** PM-KISAN portal and app enable registration, verification, tracking and e-KYC.
- **Key Achievements:**
 - Over **₹4.47 lakh crore** transferred through **23 instalments**.
 - Over **₹1.71 lakh crore** disbursed during Covid-19.
 - Women farmers received over **₹1.06 lakh crore** and form nearly one-fourth of beneficiaries.
- **Impact:** Over **92% of beneficiaries** used PM-KISAN assistance for **agricultural activities and investment (NITI Aayog)**.
 - 85% of beneficiaries **reported higher agricultural income** and reduced dependence on informal credit.
 - **Contributed to agricultural growth** between 2020–21 and 2025–26 - **cultivated area increased by about 9.65%**, productivity by 10.53% and total foodgrain production by 21.18%.

Pradhan Mantri Surya Sarovar Yojana

The Union Cabinet approved **PM-SSY** to boost India's **renewable energy capacity and energy security**.

PM-SSY

- **About:** A Central Sector Scheme promoting **Floating Solar Photovoltaic (FSPV) projects** integrated with **Energy Storage Systems (ESS)**.
 - **FSPV:** Solar panels are mounted on buoyant platforms over inland water bodies, expanding renewable capacity without competing for land.
- **Outlay and Target:** ₹5,070 crore to develop **5,000 MW** of FSPV capacity between **FY 2026–27 and FY 2030–31**.
- **Implementing Basis:** NISE estimates India's floating solar potential at **102.18 GWp** across reservoirs and inland water bodies, against about **700 MW** installed capacity.
- **Storage Mandate:** Co-located ESS with at least **2-hour storage**, totalling **10,000 MWh**.
- **Financial Support:**
 - **Central Financial Assistance:** ₹1 crore per MW after successful commissioning of eligible projects.
 - **Project De-risking:** Up to ₹50 lakh per project for bathymetry, hydrography, feasibility and environmental studies.
- **Significance:**
 - **Capacity Addition:** Increase floating solar capacity from about **700 MW to 5,700 MW**.
 - **Climate Goals:** Reduce nearly **10 million tonnes of CO₂ emissions annually**.
 - **Employment:** Generate about **16,000–17,000 full-time equivalent jobs**.
 - **Aatmanirbhar Bharat:** Promote domestic production of floatation systems, PV cells, modules and energy storage systems.
 - **Grid Stability:** **2-hour BESS** will improve grid reliability and manage peak demand.
- **Omkareshwar Floating Solar Park:** India's largest operational floating solar project at **278 MW** on the **Narmada River, Madhya Pradesh**; planned capacity **600 MW**.
- **Global Status:** Floating solar capacity reached about **9.6 GW in 2024**, with **Asia accounting for nearly 90%**.

Status of Solar Energy in India

- **Capacity:** Increased from **3 GW in 2014** to about **129 GW in 2025**; supports the **500 GW non-fossil capacity target by 2030**.
- **Leading States:** Rajasthan, Gujarat, Maharashtra and Tamil Nadu.
- **Grid Integration:** **Green Energy Corridor** strengthens transmission and renewable energy evacuation.

- **Rooftop and Agricultural Solar:** **PM Surya Ghar Yojana** enabled around **24 lakh rooftop installations by December 2025**; **PM-KUSUM** promotes grid-connected solar pumps.
- **BESS:** ₹5,400 crore VGF scheme supports **30 GWh** battery storage to improve renewable energy reliability.
- **Domestic Manufacturing:** India achieved **100 GW** solar PV module manufacturing capacity under **ALMM**.
- **International Solar Alliance:** India-led initiative headquartered in **Gurugram**, mobilising solar investment and demand among sun-rich countries between the **Tropics of Cancer and Capricorn**.

Jan Aushadhi Nutraceuticals

The Union government highlighted affordable nutraceutical products under the **Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)**, available through Janaushadhi Kendras.

- **Aim:** Make quality nutritional supplements affordable and accessible, especially for low-income families. To advance the vision of **"Quality Healthcare and Nutrition for All"**.
- **Coverage:** Includes protein supplements, immunity products, nutritional foods and specialised formulations for children, women, senior citizens, fitness enthusiasts and people with specific health conditions.

Nutraceuticals

- Biologically active products derived from food that combine nutritional and therapeutic benefits.
- The term combines "nutrition" and "pharmaceutical" and was coined by Dr Stephen De Felice in 1989.
- Include vitamins, minerals, proteins, probiotics, functional foods, herbal products and foods designed for specific medical needs.
- Support **health promotion, disease prevention** and management of conditions such as **cardiovascular diseases, diabetes, cancer**, skin disorders and immune-related problems.
- Challenges include low bioavailability, instability, poor taste and formulation difficulties for children and old adults.
- Nanoformulations, liposomes, nanoemulsions and lipid-based carriers can improve stability, absorption, controlled release and effectiveness.
- Require strict regulation to prevent contamination, adulteration and harmful drug interactions.

PMBJP

- **Launched:** November 2008 by the Department of Pharmaceuticals, Ministry of Chemicals and Fertilisers.
- **Implemented by:** Pharmaceuticals and Medical Devices Bureau of India.

- **Objectives:** Affordable generic medicines, lower out-of-pocket expenditure, menstrual hygiene and employment generation by encouraging individual entrepreneurs to establish Jan Aushadhi Kendras.
- **Coverage:** Over **19,000 Kendras** by March 2026; target of **25,000 by March 2027**.
- **Products:** Around **2,100 medicines** and **300 surgical and medical items**.
- **Savings:** Medicines are generally **50–80% cheaper** than branded equivalents.
- **Quality Assurance:** WHO-GMP manufacturers and NABL-accredited laboratory testing.
- **Special Incentive:** A one-time special incentive of ₹2 lakh is provided for Kendras opened by women, ex-servicemen, Divyangjan and Scheduled Caste or Scheduled Tribe entrepreneurs, as well as those established in Aspirational Districts, Himalayan regions, island territories and North-Eastern States.
- **Janaushadhi Suvidha:** Oxo-biodegradable sanitary pads at ₹1 (menstrual hygiene); sales crossed **100 crore pads by March 2026**.
- **Jan Aushadhi Sugam App:** Helps in locating Kendras, medicine availability and comparison of prices of generic and branded medicines.

Revamped Khelo India Scheme

Union Cabinet approved an expanded Khelo India Scheme and enhanced Assistance to National Sports Federations (ANSFs) with a combined financial outlay of ₹36,441 crore for the period 2026–27 to 2030–31.

- **Objective:** Promote youth development and nation-building through opportunities from grassroots to elite sports.
- **Policy Alignment:** Aligned with **Khelo Bharat Niti 2025** and **NEP 2020**, integrating sports with education, fitness, technology and high-performance training.
- **Athlete Development Pathway:** Talent identification → scientific training → competitions → international exposure → Olympic preparation through Khelo India and **TOPS**.
- **Institutions:** National and Khelo India Centres of Excellence, SAI Training Centres, accredited academies, Khelo India Centres and Armed Forces' Youth Sports Companies.
- **School and Talent Initiatives:** **Khelo India Feeder Schools**, **Khelo India Utkrishta Vidyalayas** and **Emerging Khelo India Athletes (E-KIAs)** to expand the athlete pool nearly ten-fold.
- **Coaching and Technology:** National Coach Accreditation Board, unified digital sports platform and partnerships with private and corporate institutions.

Prevention of Insults to National Honour (Amendment) Act, 2026

The Prevention of Insults to National Honour (Amendment) Bill, 2026, passed by Parliament, became an Act after receiving Presidential assent.

Key Provisions

- Amends the **Prevention of Insults to National Honour Act, 1971** to include Vande Mataram within its scope.
- **Section 3 of the Act, 1971:** It prohibits intentionally preventing the singing of the National Anthem (Jana Gana Mana) or causing disturbance to an assembly during its singing.
 - The offence is punishable with imprisonment up to 3 years, a fine, or both, while repeat offenders face a minimum imprisonment of 1 year.
- Seeks to give the National Song **statutory protection similar to the National Anthem**.
- Based on Dr. Rajendra Prasad's **24 January 1950** statement according equal honour to Vande Mataram and Jana Gana Mana.

Vande Mataram

- Composed by **Bankim Chandra Chattopadhyay** in Sanskritised Bengali.
- Published in **Bangadarshan** on **7th November 1875** and included in **Anandamath (1882)**.
- Set to music by **Rabindranath Tagore**.
- First two stanzas adopted as the National Song by the Congress Working Committee in **1937**.
- Accorded equal honour with Jana Gana Mana on **24 January 1950**.
- The Constitution does not expressly recognise a National Song.
 - **Article 51A(a)** makes it a Fundamental Duty to abide by the Constitution and respect its ideals and institutions, the National Flag, and the National Anthem.
- **Role in Freedom Struggle:**
 - Rallying cry of the Swadeshi and anti-colonial struggle.
 - First sung by Tagore at the **1896 Calcutta Congress Session**.
 - Became a political slogan during the **1905 Partition of Bengal** agitation.
- **Nationalist Organisations & Press:** **Bande Mataram Sampradaya (1905)** promoted patriotic awareness, while the **English daily Bande Mataram (1906)**, led by **Bipin Chandra Pal** and **Sri Aurobindo**, became an influential nationalist publication.

- **Hyderabad-Karnataka Movement (1938):** Students defied restrictions by singing Vande Mataram, leading to arrests and expulsions.
- **Global Influence: Madam Bhikaji Cama** unfurled India's tricolour bearing "Vande Mataram" at **Stuttgart (1907)**, **Madan Lal Dhingra** reportedly uttered "Bande Mataram" before his execution in **1909**, and nationalist publications carrying the same name were circulated from **Geneva**.

Tackling Non-Revenue

Water for a Water-Secure Viksit Bharat

India's **Viksit Bharat** goal depends on water-secure cities, but **groundwater over-exploitation, rapid urbanisation and climate change** are intensifying urban water stress.

- Urban utilities lose over **30% of supplied water** as **Non-Revenue Water (NRW)**, weakening municipal finances, service efficiency and long-term water security.

Non-Revenue Water (NRW)

- **NRW:** NRW is treated water supplied into the distribution network but lost before delivery or billing.
 - **Physical Losses:** Leakage from old pipelines, broken valves and overflowing reservoirs.
 - **Commercial Losses:** Faulty meters, illegal connections and unbilled public use.
- NRW is below **10%** in advanced global cities but around **40–50%** in many Indian metropolitan areas.
 - Nearly half the water treated at a **high cost yields zero municipal revenue and never reaches the tap**.

Challenges in Urban Water Governance

- **Financial Unsustainability: Urban Local Bodies (ULBs)** recover less than **55%** of operation and maintenance costs, limiting infrastructure repair and expansion.
- **Dependence on Distant Sources:** Reliance on distant rivers and deep aquifers increases pumping costs, drought vulnerability and inter-state disputes.
- **Infrastructure and Treatment Gaps:** Only about **28% of urban wastewater** is treated, while wastewater reuse remains limited.
 - **Weak Monitoring:** Lack of **GIS mapping, SCADA systems and District Metered Areas (DMAs)** prevents early detection of leaks and efficient network management.
- **Inequitable Distribution:** Poor households often depend on expensive tankers, while connected households receive subsidised piped water.
- **Institutional Fragmentation:** State agencies develop infrastructure, while financially and technically weak ULBs operate it, creating accountability gaps.

- **Ecological Degradation:** Encroachment of wetlands, lakes, floodplains and recharge zones increases both urban flooding and summer water scarcity.

Measures Needed for Effective Urban Water Governance

- **Mandate Sponge City Infrastructure:** Integrate **permeable pavements, bioswales and restored peri-urban wetlands** into master plans for aquifer recharge and flood mitigation.
- **Decouple Water Supply:** Legally separate **potable and non-potable networks** and enforce **Zero Liquid Discharge (ZLD)** for heavy industries and large commercial complexes using recycled municipal effluent.
- **Price Groundwater Extraction:** Introduce **volumetric charges on private borewells** in affluent and commercial areas to curb groundwater depletion.
- **Integrate GIS and SCADA:** Digitise underground assets through **GIS** and use **SCADA** for real-time pressure monitoring and predictive leak management.
- **Ring-Fence NRW Budgets through DMAs:** Divide networks into **District Metered Areas (DMAs)** and link state fund devolution to verified reductions in **Non-Revenue Water (NRW)**.
- **Conduct Concurrent Sewer Audits:** Audit sewer networks alongside water systems to prevent sewage leakage, groundwater contamination and disease outbreaks.
- **Regulate the Tanker Market:** Mandate **municipal licensing, GPS tracking and water-quality testing** for private tankers.
- **Redefine Service Benchmarks:** Measure success through **equitable water security**, prioritising pressure management and smart standposts in informal settlements.
- **Create Water Regulatory Authorities:** Establish statutory **State Urban Water Regulatory Authorities** to set NRW targets, standardise pumping-energy audits and rationalise tariffs.
- **Use Hybrid Annuity Models:** Link private concessionaire payments under **HAM-based PPPs** to verifiable long-term reductions in NRW.

What is Non-Revenue Water (NRW)? Discuss its implications for urban governance and suggest measures to reduce NRW in Indian cities.

Drishiti Mains Question

Gen Z Protests: Redefining Democratic Participation in India

Generation Z-led protests, spearheaded by the satirical **CJP**, have emerged over alleged **NEET irregularities**, rising **graduate unemployment**, and the increasing **cost of education**.

- The demonstrations, compared with youth movements in **Bangladesh, Nepal, Indonesia, and Peru**.

How are Gen Z Protests Reshaping Global Politics?

- **Gen Z:** Gen Z, or Zoomers, generally includes people born from the **mid-1990s to early 2010s**.
 - It is marked by **digital nativity, social media use, and strong civic awareness**.

■ Global Examples

Country	Trigger	Outcome
Bangladesh	Student protests over job quotas	Government change and formation of an interim administration
Nepal	Anti-corruption protests	Political change and rise of youth-backed leadership
Indonesia	Lawmakers' privileges amid economic hardship	Withdrawal of several controversial measures
Peru	Crime and political instability	Further political instability and leadership changes
India	Alleged NEET paper leak and demand for exam reforms	Greater pressure for accountability and examination reforms

Evolution of Youth Movements

- **Ideological Politics (1950s–1960s):** Campus-based movements led by organisations such as **SFI** and **ABVP**.
 - Focused on **language, education, and ideological mobilisation**.
 - **Key Developments:** **Anti-Hindi Agitation (1965)** in Tamil Nadu & **Naxalbari Movement (1967)** with significant student participation.
- **Anti-Establishment Mobilisation (1970s):** Youth movements expanded beyond campuses to oppose **corruption, inflation, and authoritarianism**.
 - **Key Developments:** **Nav Nirman Andolan (1974)** in Gujarat & **JP Movement (1974–75)** against corruption and misuse of power.
 - Student activism influenced developments preceding the **Emergency (1975–77)**.
- **Identity and Social Justice (1980s–1990s):** Movements centred on **regional identity, migration, and reservation policies**.
 - **Key Developments:** **Assam Agitation (1979–85)** led by **AASU**, resulting in the **Assam Accord (1985)**.
 - **Anti-Mandal Protests (1990)** against reservation in education and public employment.
- **Rights-Based Activism (2000s–2010s):** Internet and social media strengthened campaigns for **transparency, accountability, civil rights, and gender justice**.
 - **Key Developments:**
 - ❖ **India Against Corruption Movement (2011)** demanding the **Jan Lokpal Bill**.
 - ❖ **Nirbhaya Protests (2012)**, contributing to reforms in criminal laws concerning women's safety.

■ Role in Politics:

- Gen Z has become a major political force against **corruption, unemployment, inequality, and governance failures**.
- Its movements are usually **decentralised, digitally coordinated, and youth-led**.
- Sustained mobilisation has contributed to **policy reversals, leadership changes, and political reforms**.

- **Gen Z and Issue-Based Activism (2020s–Present):** Movements are increasingly **digital, decentralised, and issue-specific**, focusing on education, employment, governance, and social justice.

- **Key Developments:** **Anti-CAA/NRC protests (2019–20); Agnipath protests (2022)**.
- **CJP movement (2026)** over alleged **NEET irregularities, unemployment, and rising education costs**.

Major Shifts in Youth Movements

- **From Charismatic Leaders to Decentralised Networks:** Movements have shifted from leaders like **JP Narayan** and **Anna Hazare** to **leaderless, digitally coordinated networks**.
- **From Ideology to Issue-Based Activism:** Focus has moved from ideological politics to issues such as **education, unemployment, climate action, examination integrity, and gig workers' rights**.
- **From Party-Linked to Independent Mobilisation:** Recent movements often function outside political parties, using **memes, satire, and online campaigns**.
- **From Physical to Hybrid Protests:** Modern movements combine **social media mobilisation** with **street demonstrations**, enabling faster and wider participation.
- **From Campus Issues to National Policy Debates:** Youth movements increasingly shape debates on **education, employment, governance, accountability, and institutional reforms**.

Underlying Causes of Youth Disillusionment

Macroeconomic Asymmetries: The “Educated Unemployed”

- **Jobless Growth and Skill Mismatch:** Economic growth has not generated enough secure jobs for educated youth.

- The **ILO Report 2024** states that youth constitute **83% of India's unemployed**, while graduate unemployment is **29.1%**, compared with **3.4%** among those without schooling. **AI and automation** may further displace routine jobs, making large-scale reskilling essential.
- **Informalisation of Employment:** Many new opportunities are concentrated in the **gig economy**, which often lacks social security, stable incomes, and career progression.
- **Educational Inflation:** Expansion and privatisation of higher education have increased the number of degrees without matching improvements in employable skills, reducing the economic value of qualifications.

Institutional Decay and Compromised Meritocracy

- **Shattered Social Contract:** Paper leaks, proxy candidates, exam cancellations, and recruitment delays weaken trust in competitive examinations as fair routes to social mobility.
- **Regulatory Deficit:** Weak oversight of examination systems and the commercialisation of coaching have made education increasingly costly and exclusionary.

Sociological and Psychological Dimensions

- **Anomie and Relative Deprivation:** The gap between aspirations and available opportunities creates frustration, while exposure to global lifestyles through digital media intensifies feelings of deprivation.
- **Student Suicides:** Financial pressure, education loans, and examination failure have increased psychological distress. The **NCRB ADSI 2024** reported that those aged **18–45** accounted for **65.6% of suicides**, while students formed **8.5% of total cases**.
- **Brain Drain:** Repeated institutional failures are pushing skilled and middle-class youth to seek education and employment abroad, causing loss of human and financial capital.

Political Economy and Democratic Alienation

- **Failure of Interest Aggregation:** Mainstream politics often gives limited attention to Gen Z concerns such as **housing, inflation, employment, and economic security**.
- **Shrinking Space for Dissent:** Use of **Section 163 of the BNSS**, internet shutdowns, surveillance, and prohibitory orders against protests deepens youth distrust and democratic alienation.

Measures Needed to Address Youth Discontent

- **Adopt a Youth-Centric Policy Framework:** Introduce **Youth in All Policies** (like Finland's Youth Policy) so that major policies assess their impact on young people.
- **Expand New-Age Employment:** Promote the **creative, creator, sports, care, space, and green economies** through finance, incubation, and intellectual property support.

- **Institutionalise Youth Participation:** Establish a **National Youth Council** and strengthen **Youth Parliaments** for structured involvement in policymaking.
- **Restore Trust in Examinations:** Strengthen fast-track courts under the **Public Examinations Act, 2024**, reform the **NTA**, and use secure technologies such as blockchain verification and encrypted testing.
- **Reform Education and Skills:** Implement the **NEP 2020** by integrating AI, entrepreneurship, climate literacy, financial literacy, and apprenticeships into higher education.
- **Strengthen Youth Mental Health:** Create a **National Youth Well-being Mission** providing counselling, career guidance, and suicide-prevention services in educational and coaching institutions.
- **Create Multiple Success Pathways:** Reduce dependence on single high-stakes examinations through multiple testing windows, **micro-credentials**, **PARAKH**, and skill-based recruitment.
- **Promote Digital Democracy:** Develop **MyGov** into an interactive platform for youth participation in draft policies, budgets, and legislation.
- **Support Civic Innovation:** Establish **Youth Innovation Missions** and civic-tech incubators aligned with **Atal Innovation Mission** and **Startup India**.
- **Prepare a National Youth Strategy 2047:** Set measurable targets for education, employment, entrepreneurship, mental health, civic participation, and digital inclusion.
- **Generate Employment-Intensive Growth:** Support labour-intensive sectors such as **textiles, tourism, leather, footwear, and food processing**, alongside the **PLI Scheme**.
 - Improve affordable credit through **OCEN**, integrate MSMEs into **Global Value Chains**, and implement the **Code on Social Security, 2020** for gig-worker welfare and insurance.

Youth aspirations increasingly shape democratic governance in India. Examine the structural factors driving youth disillusionment and suggest policy reforms.

Drishti Mains Question

Public Examinations Amendment Bill, 2026

The Union Government dismissed **47 NTA officials**, introducing the **Public Examinations (Prevention of Unfair Means) Amendment Bill, 2026**, and forming a high-powered task force chaired by former **UIDAI chief Nandan Nilekani** to strengthen public examination integrity amid concerns over **NEET** and **CJP protests**.

Key Reforms to Ensure Integrity of Public Examinations

- **Public Examinations (Prevention of Unfair Means) Amendment Bill, 2026**

- **Harsher Imprisonment Terms:** General offences, including paper leaks and OMR tampering: **5–10 years' imprisonment**, up from a minimum of 3 years.
- **Organised examination fraud:** Minimum imprisonment increased from **5 to 7 years**.
- **Steep Financial Penalties:** General offences: Maximum fine increased from **₹10 lakh to ₹50 lakh**.
 - **Organised crime networks:** Minimum fine increased from **₹1 crore to ₹10 crore**.
- **Accountability of Service Providers:** Complicit private IT vendors or printers may face fines up to **₹5 crore**, against ₹1 crore earlier.
 - ❖ Debarment from public examinations increased from **4 to 8 years**.

Right to Protest and Excessive Policing

Violent clashes during the **youth-led 'Chalo Sansad' march** in New Delhi have reignited debate over the **constitutional limits of policing public protests** and the State's power to restrict civil and digital liberties.

Constitutional Basis for the Right to Protest

- **Right to Protest:** The Constitution guarantees the right to protest through FR (Part III) but does not explicitly mention the term "protest".
 - **Article 19(1)(a):** Freedom of speech and expression.
 - **Article 19(1)(b):** Right to assemble peacefully and without arms.
 - **Article 19(1)(c):** Right to form associations or unions.
 - **Article 21:** Right to life and liberty, including right to live with dignity and voice grievances against the State.
 - Essential to hold the **executive accountable, ventilate public grievances & foster informed public discourse**.
- **Limitations: It is not absolute.** State can impose "reasonable restrictions".
 - **Under Article 19(2):** Restrictions on freedom of speech in the interests of the sovereignty and integrity of India, the security of the State, friendly relations with foreign States, **public order**, decency or morality, or in relation to contempt of court, defamation, or incitement to an offence.
 - **Under Article 19(3):** Restrictions on the freedom of assembly in the interests of the **sovereignty and integrity of India or public order**.
- **Statutory Framework:**
 - **BNSS 2023, Section 163:** Empowers Executive Magistrates to issue prohibitory orders in urgent situations.
 - **Delhi Police Act, 1978:** Authorise law enforcement to regulate assembly routes, timings, and conditions.

- **Unlawful Assembly:** Under the **BNS 2023**, a gathering of 5 or more individuals becomes unlawful, only if its common object involves using criminal force, resisting the law, committing an offence, or compelling others by threat. A lawful assembly can turn unlawful if its conduct changes during the event.

Standards and Laws Govern Police Action and Crowd Control

- **Democratic Policing Framework:** Police are guardians of citizen rights and the rule of law (NHRC Manual). Excessive force undermines public trust and may escalate unrest.
- **International Guidelines:** UN Basic Principles on the Use of Force and Firearms (1990) mandate that force be lawful, necessary, and proportionate.
- **Principle of Minimum Force:** Under the **BNSS, 2023** and police code, warnings and persuasion must precede only the **irreducible minimum force** necessary.
- **Identification and Accountability:** **BNSS, 2023** and **D.K. Basu (1997)** requires arresting officers to wear visible name tags, but no similar rule exists for **crowd-control personnel**, creating accountability gaps.

Supreme Court Judgments on Protests and Police Excess

- **Ram Manohar Lohia v. Bihar (1965):** SC distinguished between "law and order" and "public order." Protests can be restricted only if public order is genuinely threatened.
- **Himat Lal K. Shah v. Commissioner (1973):** State may regulate but cannot impose arbitrary restrictions on right to protest. Assemblies cannot be arbitrarily restricted.
- **Re: Ramlila Maidan (2012):** Peaceful protest is a constitutional right; force only as last resort.
- **Anita Thakur v. J&K (2016):** Excessive force violates fundamental rights.
- **Mazdoor Kisan Shakti Sangathan v. Union of India (2018):** Right to assemble cannot be extinguished.
- **Amit Sahni v. Commissioner (2020):** Protests cannot indefinitely occupy public spaces or block traffic.

Concerns Regarding Excessive Policing

- **Political Interference:** Influence over transfers, postings, and investigations undermines police independence and impartiality.
- **Training and Legal Gaps:** Inadequate police training in **human rights, de-escalation, forensic investigation and community policing** increases the risk of excessive force. Although India signed the **UN Convention Against Torture (UNCAT) in 1997**, it has neither ratified it nor enacted a specific anti-torture law.

- **Manpower Shortages & Stress:** With 155 personnel per lakh (Jan 2024) against the UN norm of 222 and 21% vacancies, long hours cause high stress (83.8% of personnel), leading to coercive practices.
- **Disproportionate Force:** Tear gas, lathi charges, preventive detention, and mass arrests against peaceful protesters often violate the **Doctrine of Proportionality** and **UN Basic Principles on Use of Force**, which mandate minimum necessary force.
- **Accountability Issues:** Masked or unidentified officers and absent name badges violate transparency norms (D.K. Basu v. West Bengal, 1997).
- **Gender-Sensitive Safeguards:** Inadequate deployment of women police and reports of harassment breach Article 21 and statutory protections.
- **Digital Restrictions:** Internet shutdowns and communication blackouts during protests can unduly limit Articles 19(1)(a) & 19(1)(g). Courts require legality, necessity, proportionality, and periodic review (Anuradha Bhasin v. India, 2020).
- **Weak Civilian Oversight:** Poor implementation of **Prakash Singh (2006) directives**, including ineffective Police Complaints Authorities and State Security Commissions, fosters impunity.
- **Colonial Structure:** Most States retain the **Police Act, 1861**, making the force executive-controlled rather than public-accountable, despite the **Model Police Act, 2006**.

Way Forward

- **SOPs & Training:** Implement BNSS/BNS provisions; focus on non-lethal crowd control and negotiation.
- **Independent Oversight:** Enforce Prakash Singh directives by operationalising empowered State and District Police Complaints Authorities.
- **Separate Investigation & Law & Order:** Adopt 2nd ARC and Malimath Committee recommendations to enhance forensic-based investigations and reduce coerced confessions.
- **Modernisation & Capacity Building:** Fill 21% manpower gap via the MPF scheme; improve officer-to-citizen ratio; integrate technology like body-worn cameras and CCTV.
- **SMART Policing:** Transform policing to be **Strict & Sensitive, Modern & Mobile, Alert & Accountable, Reliable & Responsive, Tech-savvy & Trained**.

"The right to protest is fundamental but not absolute." Examine the constitutional and statutory framework governing public protests in India.

Drishti Mains Question

Amrit Bharat Station Scheme

Recently, the Prime Minister inaugurated **75 redeveloped railway stations across 20 States** under the **Amrit Bharat Station Scheme**, as part of efforts to modernise railway infrastructure and transform stations into passenger-centric transit hubs.

- This is the **second major nationwide dedication** of redeveloped stations under the scheme. Earlier, on **22 May 2025**, the Prime Minister dedicated **103 redeveloped Amrit Bharat Stations across 18 States and Union Territories**.

Amrit Bharat Station Scheme

- **About:** Launched by the **Ministry of Railways** in December 2022, the Amrit Bharat Station Scheme (ABSS) is a **long-term, phased initiative** to transform railway stations into modern city centres instead of undertaking one-time facelifts.
 - As of July 2026, redevelopment work had been completed at **261 of the 1,340 stations** covered under the scheme, with station-specific plans prepared according to local needs, passenger footfall and future growth.
 - The scheme focuses on improving passenger experience, integrating regional architecture, enhancing accessibility, and supporting urban development.
- **Key Features:**
 - **Customised & Culturally Integrated Redevelopment:** Stations are redeveloped with location-specific designs combining modern facilities with local heritage, culture, and architectural themes.
 - **Passenger Amenities & Modern Facilities:** Provides improved entry and exit points, modern waiting areas, covered platforms, clean restrooms, lifts, escalators, free Wi-Fi, executive lounges, business areas, food courts, kiosks, and enhanced passenger information systems.
 - ❖ Includes aesthetically designed façades, landscaping, roof plazas, improved lighting, parking facilities, and better circulation areas.
 - **Inclusive & Accessible Design:** Aligned with the **Accessible India Campaign (Sugamya Bharat Abhiyan)**, the scheme ensures facilities for **Divyangjan and passengers with reduced mobility** through ramps, accessible parking, low-height counters, Braille signage, tactile pathways, accessible toilets, and lift-equipped subways/foot overbridges.
 - **Local Economic Promotion (One Station One Product):** Promotes indigenous and regional products by providing kiosks and outlets at railway stations for local artisans, weavers, tribal communities, handicrafts, handlooms, and region-specific food products.

- **Urban Connectivity & Sustainability:**

- ❖ Develops stations as **multimodal urban centres** by integrating railways with bus and metro networks and connecting both sides of cities through improved station planning.
- ❖ Incorporates eco-friendly measures such as green spaces, sustainable infrastructure, and improved environmental planning.

PM Announces Fast-Track Courts to Tackle Public Examination Paper Leaks

The Prime Minister announced the **establishment of FTCs** for swift punishment of individuals and syndicates involved in public examination paper leaks.

- The move follows protests over **irregularities in NEET-UG 2026 examination**.
- Delhi HC constituted a **Special Fast-Track Court** under the **Public Examinations (Prevention of Unfair Means) Act, 2024**, transferring pending cases for speedy disposal.

FTCs

- **About:** Specialized judicial mechanisms to expedite trial and disposal of specific, high-priority or long-pending cases, reducing burden on regular judiciary.
- **Constitutional Backing:** SC has read the right to speedy trial into Article 21 (*Hussainara Khatoon v. State of Bihar*, 1979).
- **No Single Central Act:** FTCs are created through Finance Commission recommendations, central schemes or High Court notifications.
- **Targeted Categories of Cases:**
 - **Heinous Crimes & Long-Pending Disputes:** Eleventh Finance Commission (2000–2005) recommended FTCs for pending criminal cases; Fourteenth Finance Commission (2015–2020) expanded scope to heinous crimes, old civil disputes and vulnerable groups.
 - **Vulnerable Demographics:** Cases involving women, children, senior citizens, persons with disabilities and terminally ill individuals.
 - **Fast-Track Special Courts (FTSCs):** Established under Centrally Sponsored Scheme in October 2019 for rape and POCSO Act, 2012 cases; funded partly through Nirbhaya Fund.
 - In 2017, Supreme Court directed setting up special FTCs for criminal trials against MPs and MLAs.
- **Constitutional Scrutiny:** Under Article 14, classification must have rational basis; **speed alone cannot be an arbitrary criterion** (*State of West Bengal vs Anwar Ali Sarkar*, 1952).

- **Efficiency vs. Structural Challenges:** FTSCs dispose of about **9.5 cases/month** compared to regular trial courts' **3.3 cases/month**, but over **2.4 lakh cases** remained pending by end-2023.

- **Landmark Judicial Precedents Governing FTCs:**

- **Brij Mohan Lal v. Union of India (2012):** States are primarily responsible for FTC continuation; permanent judicial infrastructure is needed.
- **Anita Kushwaha v. Pushap Sudan (2016):** Access to justice is a Fundamental Right under Articles 14 and 21.
- **P. Ramachandra Rao v. State of Karnataka (2002):** No fixed time limit can be prescribed for criminal trials.

Significance of FTCs in Public Examination Paper Leak Cases

- **Restoring Institutional Trust:** Reinforces meritocracy and confidence in examination systems.
- **Strengthening Deterrence:** Swift punishment discourages organised examination fraud.
- **Protecting India's Demographic Dividend:** Reduces exam disruptions and protects career prospects of aspirants.

Limitations of FTCs in Tackling Public Examination Paper Leaks

- **Post-Facto Remedy:** FTCs act after leaks occur and cannot prevent damage caused by cancellations and loss of trust.
- **Inter-State Jurisdiction Challenges:** Paper leak syndicates operate across States; localised courts need coordinated investigations.
- **Digital Forensics Bottleneck:** Encrypted apps, dark web, IP logs and crypto transactions require strong cyber-forensic capacity.
- **Investigation–Trial Gap:** FTCs function only after charge sheets; multi-agency investigations may delay trials.
- **Structural Deficiencies:** Issues include ad hoc funding, inadequate infrastructure, shortage of judges, prosecutors and support staff.
- **Funding Friction:** State responsibility after central grants creates uneven performance among States.
- **Witness & Prosecution Challenges:** Weak witness protection and lack of cyber experts affect conviction rates.

Way Forward

- **Strengthen NTA:** Make NTA an autonomous statutory body with accountability like UPSC.
- **Dedicated Investigation SOP:** Ensure time-bound investigations and charge sheets through specialised cyber cells.

- **Technological Fortification:** Use **Quantum Key Distribution (QKD)** and blockchain-based audit trails for examination security.
- **Zero-Tolerance:** Introduce **Command-Failure Liability Clause** under the Public Examinations (Prevention of Unfair Means) Act, 2024.

Addressing Institutional Gaps in Public Examinations: The Dr. K. Radhakrishnan Committee Reforms

- FTCs provide post-facto remedies, while prevention requires systemic reforms.
- **Dr. K. Radhakrishnan High-Level Committee (2024):**
 - **Examination Format:** Shift from single-day OMR exams to multi-shift CBTs.
 - **Examination Logistics:** Reduce outsourcing of sensitive logistics.
 - **Candidate Identity Verification:** Implement **DIGI-EXAM** biometric identity layer.

Fast-Track Courts can ensure speedy justice but cannot substitute systemic reforms in preventing public examination paper leaks. Discuss.

Drishti Mains Question

Smart Border Architecture to Strengthen India's Border Security

MHA is spearheading a paradigm shift in internal security by developing a **technology-driven, integrated command-and-control system** to establish a “**smart border**” architecture across India's land frontiers.

- **Strategic Focus Area:**
 - The initiative specifically targets regions where physical fencing is not feasible, such as the **3,488-km undemarcated border with China (Ladakh to Arunachal Pradesh)** and flood-prone **riverine stretches along the Bangladesh border**.
- **Pilot Project:** The initial pilot project will commence along the **Pakistan border**, with a major focus on neutralizing the growing threat of **cross-border drones used for smuggling and reconnaissance**.
- **Four-Pronged Security Strategy:**
 - The architecture integrates the **border guarding forces, local population, State police, and technological solutions** to strengthen monitoring and response capabilities.
- **Technological Integration:** The system creates a common operational picture by linking **AI, sensors, surveillance cameras, and image-processing tools** to local command centres, which feed live data to field and New Delhi headquarters.

Customised Border Solutions:

- Recognizing that “**no two borders are the same,**” the deployment will avoid a uniform approach, instead utilizing **terrain-specific technological solutions** for deserts, mountains, and forests, thereby reducing the need for constant physical patrolling.
- **Quadrangular Security Grid:** Following the first-ever **Land Border Districts' Superintendents of Police Conference–2026**, a localized security grid involving border communities and police across **119 border districts (17 States/UTs)** will be finalized within the next 1.5 years.

Paternity Test in Matrimonial Dispute

SC, in a matrimonial dispute involving allegations of **unchastity and disputed paternity**, upheld the orders of a **lower Court** directing a **DNA test of the child**.

Legitimacy of a Child and Paternity test

- **Presumption of Legitimacy of a Child:** **Section 116 of the Bharatiya Sakshya Adhinyam (BSA), 2023** a child born during a **valid marriage** or within **280 days after its dissolution** is conclusively presumed to be the legitimate child of the husband.
 - Section 116 of the **BSA, 2023** corresponds to **Section 112 of the Indian Evidence Act, 1872**.
- **Paternity Test:** **DNA tests cannot be ordered merely on disputed paternity claims** but courts may direct DNA testing only in **exceptional cases where it is necessary to resolve the dispute** and other evidence is inadequate.

Supreme Court Jurisprudence on DNA Testing

- **Goutam Kundu v. State of West Bengal (1993):** The Supreme Court held that **DNA/medical tests cannot be ordered as a first resort** and require a strong **prima facie case of non-access**.
- **Banarsi Dass v. Teeku Dutta (2005):** The Court ruled that **DNA testing should not be directed routinely** and must be limited to exceptional cases.
- **Nandlal Wasudeo Badwaik v. Lata Nandlal Badwaik (2014):** The Court held that where **scientific evidence conflicts with legal presumptions**, scientific truth may prevail.
- **Rohit Shekhar v. Narayan Dutt Tiwari (2014):** The Court allowed DNA testing, prioritising the **child's right to know biological identity** over privacy concerns.
- **Ivan Rathinam v. Milan Joseph (2025):** The Court held that DNA tests require an “**eminent need**” and courts must balance privacy, social stigma, and the child's interest.

Nation & States

Strengthening India's Defence through Self-Reliance

On India's 80th Independence Day, the Government highlighted the transformation of India's defence sector towards greater **self-reliance, technological capability and global credibility**.

Strengthening India's Defence Capabilities and Industrial Base

Defence Investment in Strengthening Military Capabilities

- **Defence Budget:** Increased from ₹2.53 lakh crore in 2013–14 to ₹7.85 lakh crore in 2026–27, nearly 3.1-fold growth.
- **Capital Expenditure:** Increased from ₹94,587.95 crore in 2014–15 to ₹2.19 lakh crore in 2026–27, around 2.3-fold growth, supporting military modernisation.

Investment Driving Indigenous Defence Production

- **Defence Production:** Reached a record ₹1.78 lakh crore in FY 2025–26 under Atmanirbhar Bharat.
- **Private-Sector Participation:** Public sector accounted for 76% of defence production in FY 2025–26 and private sector 24% of defence production.

Domestic Industrial Base Expanding Defence Exports

- **Defence Exports:** Increased from ₹686 crore in FY 2013–14 to ₹38,424 crore in FY 2025–26, nearly 56-fold growth.
- Indian defence products are exported to **80+ countries**; export value rose by over **5,500% in 12 years**.
- **FY 2025–26 exports:** Private sector ₹17,353 crore (45.16%); DPSUs ₹21,071 crore (54.84%).
- **2029 Targets:** ₹3 lakh crore defence production and ₹50,000 crore defence exports.

Innovation Strengthening the Technological Foundation

- **Defence R&D allocation:** Increased from ₹13,716.14 crore in 2014–15 to ₹29,100.25 crore in 2026–27, over 112% growth.
- Since 2022–23, 25% of defence R&D budget has been opened to industry, start-ups and academia.
- Testing facilities of **24 DRDO laboratories** are accessible to domestic industry through the **Defence Testing Portal (DTP)**.

Strengthening India's Defence Industrial Base

Defence Acquisition Reforms

- **Defence Acquisition Council (DAC):** Granted **Acceptance of Necessity** worth over ₹6 lakh crore for DRDO-designed systems to be manufactured by Indian industry.
- **Major approvals include:**
 - 97 Tejas Mk-1A fighters: ₹62,000 crore.
 - 156 LCH Prachand helicopters: ₹62,700 crore.
- **Defence Acquisition Procedure 2020:** Strengthened indigenous procurement and domestic manufacturing.
- **Defence Procurement Procedure 2016:** Streamlined acquisitions and promoted **Make in India**.
- **Defence Procurement Manual 2025:** Streamlined nearly ₹1 lakh crore revenue procurement, including faster approvals and relaxed penalties for indigenous projects.
- **Draft DAP 2026:** Proposes simplified acquisition categories and indigenous-content requirements of up to 60%.

Boosting Defence Innovation through iDEX, ADITI and TDF

Initiative	Key Facts
iDEX	Outlay ₹498.78 crore; engaged 676 startups, MSMEs and innovators and led to 551 contracts by March 2026.
ADITI Scheme	Outlay ₹750 crore for 2023–24 to 2025–26 for critical defence technologies.
Technology Development Fund (TDF)	Grants up to ₹50 crore; 80 projects worth ₹334 crore under implementation by June 2026; additional ₹500 crore corpus for deep-tech and emerging technologies.

Technology Transfer and Industry Participation

- **Development-cum-Production Partner framework:** Technologies transferred to **134 partner companies**.
- Around **2,180 technology-transfer agreements** signed.
- Over **2,780 DRDO intellectual property rights** opened for industry use by March 2026.

Indigenisation and Domestic Manufacturing

- **Positive Indigenisation Lists:** 10 lists covering 5,521 items notified by May 2026.
- **Srijan Defence Equipment Empowerment Platform:** Listed over **41,000 vendors** and **2.7 lakh products** by May 2026.
- **Uttar Pradesh Defence Industrial Corridor:** ₹42,057 crore investment commitments; around ₹4,409 crore grounded.

- **Tamil Nadu Defence Industrial Corridor:** Around ₹32,699 crore committed; about ₹6,446 crore realised.
- **BrahMos Integration and Testing Facility, Lucknow:** Began manufacturing missiles.

Ease of Doing Business and Defence Trade

- **Industrial Licence:** Validity extended to up to **18 years**; lifetime validity available under the **Arms Act**.
- **Revamped Defence Exim Portal:** Enables end-to-end processing, automated verification, tracking and secure payments.
- **FDI in Defence:** Up to **74% under Automatic Route** and **100% under Government Route**; cumulative inflows reached around ₹6,670.59 crore by March 2026.

Advancing Defence Self-Reliance in India in FY 2025–26

Indigenisation Strengthening Operational Self-Reliance

- **Ammunition Self-Sufficiency:** Army achieved around **91% self-sufficiency**, indigenising **159 of 175 ammunition variants**.
- **Strategic Missiles:** Successful tests of **Agni-4 IRBM** and **Agni-5 long-range surface-to-surface ballistic missile**.
- **Precision Strike:** **TARA Glide Weapon** and **RudraM-II air-to-surface missile** demonstrated indigenous strike capability.
- **Air Defence:** Progress on **Kusha Long Range Air Defence Missile** and **VSHORADS**.
- **Hypersonic Technology:** DRDO conducted a **12-minute ground test of an actively cooled scramjet combustor**.
- **Integrated Air Defence:** Indigenous system combining missile interceptors, short-range weapons and laser technologies was successfully tested.

Advancing Maritime Defence and Strike Capabilities

- **Naval Fleet:** Navy commissioned **12 warships and submarines**, including **INS Surat, INS Nilgiri and INS Vaghsheer**.
- **INS Mahendragiri:** Stealth frigate with over **75% indigenous content**.
- **SHACHI:** First of **11 Next Generation Offshore Patrol Vessels** launched.
- **INS Dunagiri, INS Sanshodhak and INS Agray:** Expanded indigenous naval, survey and surveillance capabilities.
- **INS Sanshodhak:** Contributed to completion of the indigenous **Sandhayak-class survey fleet**.
- **Naval Anti-Ship Missile–Medium Range:** Successfully tested by DRDO to strengthen indigenous maritime strike capability.

India's defence self-reliance is being driven by indigenous production, innovation, private-sector participation and acquisition reforms. Discuss.

Drishti Mains Question

Emerging Terror Threats from Social Media

Indian security agencies dismantled the **Shahzad Bhatti Network (SBN)**, an **ISI-backed terror syndicate**.

- Highlights growing use of social media for **extremist propaganda, radicalisation and recruitment**, bringing the intersection of social media, terrorism and India's internal security into sharper focus.

Social Media: A Force-Multiplier for Security Threats

- **Online Radicalisation:**
 - Extremist outfits exploit algorithms of platforms such as Instagram, X and YouTube Shorts.
 - Emotionally charged propaganda targets vulnerable youth, including in **J&K and Punjab**; engagement-driven algorithms may trap them in extremist **echo chambers**.
- **Low Cost and Decentralized Recruitment Funnel:**
 - Terrorist handlers identify users engaging with extremist content on open platforms → contact established → selected people shifted to **encrypted/private deep-web platforms**, often without physical meetings.
- **Advanced Digital Tradecraft:**
 - **Dead-Drop Emails:** Credentials of a single, anonymous email account are shared; messages kept in the **Drafts folder** without being sent, leaving minimal digital footprint.
 - **Metadata-less Apps:** Decentralised **E2EE platforms** such as **Threema**, without stored metadata/phone numbers, hinder conventional cyber-forensics.
 - **Multi-layered VPNs:** VPNs and Dark Web used to mask IP addresses and coordinate logistics.
- **Lone-Wolf Threat:** Extremist content and DIY tactical manuals can enable **self-radicalised lone-wolf attackers**, who are difficult to detect due to absence of formal command structures.
- **Digital Terror Financing:** Shift from hawala towards online financing complicates tracking by **FIU-IND**.
 - Deceptive crowdfunding and cryptocurrencies such as **Monero/Bitcoin** may bypass AML mechanisms.

Implications of Social Media for India's Internal Security

- **Amplification of Polarisation:** **PsyOps** can manipulate emotions/opinions, deepen social fault lines and destabilise society.

- **Misinformation & Public Order:** Fake news can trigger **panic and communal tension**, sometimes forcing the government to rely on harsh countermeasures like internet shutdowns.
 - Rapid spread of unverified rumors on platforms like WhatsApp has historically led to mob lynchings and localized riots.
- **Election Interference: AI-generated deepfakes and bot-driven campaigns** may distort voter perception and electoral narratives.

Challenges for Law Enforcement in Curbing Social Media-Driven Threats

- **Jurisdictional Barriers:** Overseas servers require use of **MLATs**, causing delays in time-sensitive investigations.
- **Volume & Decentralisation:** Huge content volumes and migration to fringe/decentralised platforms complicate monitoring.
- **Language & Context Gaps:** Automated moderation struggles with **local languages, dialects, coded terms and cultural context**, risking both under-detection and excessive censorship.
- **Security vs Speech & Privacy:** National security measures must respect free speech and the **Right to Privacy** recognised in **K.S. Puttaswamy (2017)**.
- **First-Responder Limitations:** Local police often lack advanced digital-forensic tools, specialised **OSINT (Open Source Intelligence) training** and technical expertise though the central agencies such as **I4C** are highly capable.

India's Initiatives

- **IT Act, 2000 — Section 69A:** Allows blocking of online content threatening India's **sovereignty, integrity, defence and security**.
- **IT Rules, 2021:** Due-diligence obligations on significant social-media intermediaries, including mechanisms for addressing unlawful content and identifying the first originator in specified cases.
- **I4C & CERT-In:** I4C coordinates cybercrime/cyber-enabled terrorism response; CERT-In handles cybersecurity incidents.
- **Intelligence Fusion: MAC, NATGRID and CCTNS** facilitate information sharing between Central agencies and State police.
- **UAPA, 1967:** Principal framework against terrorism, including **online recruitment, radicalisation, financing and facilitation**.

Suggestions

- **Bilateral Executive Agreements:** Direct arrangements similar to the **US CLOUD Act framework** could provide faster access to overseas digital evidence than MLATs.
- **Traceability Rules:** Enforce first-originator provisions under the **IT Rules, 2021** without necessarily breaking E2EE for all users.

- Explore **Client-Side Scanning (CSS)** against databases of known terror hashes before encryption.
- **NATGRID & NETRA:** Optimise intelligence systems and use **AI/NLP** for real-time surface-web monitoring, coded dialect detection and deepfake identification.
- **FATF Travel Rule:** Enforce PMLA-linked crypto regulations requiring exchanges to collect/share originator and beneficiary information with **FIU-IND**.
- **State-Level Cyber-OSINT Cells:** Establish specialised OSINT cells in every State police headquarters.
- **Public-Private Partnerships:** I4C, CERT-In and technology firms can conduct **red-teaming** of platforms/algorithms.
- **Digital De-radicalisation:** Identify vulnerable youth and use structured counselling, following models such as **Maharashtra ATS**.
 - Develop targeted counter-narratives against extremist messaging.
- **Content Regulation:** Tech companies must be mandated to conduct independent audits of their recommendation engines.
 - Platforms can use digital hashes to prevent reappearance of removed terror content.
- **Digital Literacy:** Improve youth capacity to identify **disinformation, manipulative narratives and extremist recruitment attempts**.

Parliament Passes Bill to Rename Kerala as Keralam

Parliament passed the **Kerala (Alteration of Name) Bill, 2026**, paving the way for Kerala to be officially renamed **"Keralam"**, following a resolution of the Kerala Legislative Assembly.

- **Meaning:** "Keralam" reflects the historical demand for a united Malayalam-speaking region associated with the **Aikya Kerala Movement of the 1920s**.
- **Constitutional Changes:** The Bill amends the **First and Fourth Schedules**, with consequential changes to **Articles 31A and 290A**.
- **Constitutional Procedure:** Renaming a State is governed by **Article 3**, which empowers Parliament to alter the area, boundaries or name of a State.
 - The Bill requires the **President's prior recommendation** for introduction.
 - The President refers it to the concerned **State Legislature for its views**, which are **not binding on Parliament**.
 - Under **Article 4**, a law under Article 3 is not treated as a constitutional amendment under Article 368.
 - After **Presidential assent**, the change becomes law and the First Schedule is amended.

- **Examples:** Uttaranchal → Uttarakhand (2007); Orissa → Odisha (2011).
- **Historical Roots:** Ashoka's Rock Edict II (257 BCE) refers to "Keralaputra," often linked to the Chera dynasty.
 - Scholar Herman Gundert linked "Keralam" to "Cheram/Cheralam," with *cher* meaning "to join" and *alam* "land/region."
 - The **Aikya Kerala Movement** strengthened the demand for a unified Malayalam-speaking region.
 - Following the **States Reorganisation Commission, 1956**, modern Kerala was formed on **1 November 1956**, celebrated as Kerala Piravi Day; the Constitution, however, used the anglicised name "Kerala."

National Security Guidelines for Renewable Energy Projects Near International Borders

MHA issued **national security guidelines for solar, wind and hybrid renewable energy projects** located near **India's International Border, LoC and LAC**.

- **Objective:** MHA guidelines seek to balance **renewable-energy expansion with border security, surveillance, military mobility, foreign participation and protection of sensitive infrastructure**.

Key Security Norms

Zone/Provision	Requirement
Within 1 km of IB/LoC/LAC	No solar, wind or hybrid renewable-energy activity permitted
1–50 km	MHA security clearance mandatory , case-by-case
1–20 km	Additional Ministry of Defence NOC required
1–8 km	Civil/operational infrastructure height capped at 3 m
8–20 km	Height cap 5 m
20–50 km	Height cap 15 m

- **Core Infrastructure Exception:** Height limits do **not apply to essential project infrastructure such as wind turbines**.
- **Centralised Approval:** Proposals must be routed through **MNRE**, after State government's in-principle land-allotment approval and submission of project coordinates; developers/States cannot directly approach MHA or Defence Ministry.
 - MHA should, as far as possible, decide applications within **60 days**.
- **Foreign Personnel:** Workers/engineers from **land-border-sharing countries**, particularly Pakistan, Bangladesh and China, require prior Central approval; foreign nationals in sensitive areas are subject to monitoring.

- **Land Safeguard:** Project land cannot be transferred/ allotted to a **foreign company without Central approval and MHA security clearance**; foreign investment must comply with **FDI and DPIIT rules**.
- **Anti-Drone Security:** Developers must provide **anti-drone systems**, operated by **CISF or State police**, with costs borne by the developer.
- **Security-Oriented Design:** Avoid long project stretches parallel to the border; project roads should permit use by **armed forces/border-security personnel during emergencies**.

India Formally Names 27 Places in Arunachal Pradesh

India formally identified 27 locations in Arunachal Pradesh on the official Survey of India map in response to China's repeated attempts to assign alternative names to places in the State.

- **China's "Cartographic Aggression":** China issued renaming lists in 2017, 2021 and 2023 and refers to Arunachal Pradesh as "Zangnan" (South Tibet) to reinforce its territorial claims.
- **Sovereignty Assertion:** Official identification strengthens India's cartographic and administrative claim over Arunachal Pradesh and improves public awareness of border geography.

Key Locations

Category	Places / Significance
Border Flashpoints	Long Ju: early India-China border flashpoint in 1959; Thag La: fighting began here in the 1962 India-China War.
Strategic Passes	Bisa, Dzo La, Riza La, Pukur La and Thag La.
Connectivity	Jairampur in Changlang district is an important logistics hub for movement towards eastern Arunachal Pradesh and the Myanmar border.
Cultural/Historical	Jaswant Garh commemorates Jaswant Singh Rawat; Sher-e-Thapa Memorial honours 1962 war hero Trilok Singh Thapa; Sambho Sarovar is a high-altitude lake.

Strategic Significance of Arunachal Pradesh

- **Borders:** Shares borders with China (Tibet Autonomous Region) to the north, Bhutan to the west and Myanmar to the east; Assam and Nagaland lie to the south.
- **Geopolitical Buffer:** As the largest northeastern State, it acts as a shield for the Brahmaputra Valley.
- **Water Security:** It is the upper catchment for major Brahmaputra tributaries such as Subansiri, Kameng and

Lohit; Chinese control would threaten water and food security in Northeast India and Bangladesh.

- **Military Geography:** Control of high-altitude passes such as Thag La and Bum La shapes military dominance in the region.
- **Tawang Factor:** Tawang is the birthplace of the 6th Dalai Lama; China fears Indian control may give New Delhi influence in Tibetan Buddhist geopolitics and future Dalai Lama succession.
- **McMahon Line:** India follows the McMahon Line drawn in the 1914 Simla Convention; China rejects it as a legacy of British imperialism.

Naxal-Free India Through Integrated Strategies

India became effectively free from **Left-Wing Extremism** after nearly six decades, due to sustained security operations, technology-based intelligence, development, welfare and rehabilitation under **Vishwaas, Nirman and Jan Kalyan**.

India's Efforts to Counter Left-Wing Extremism

- **Background:** For nearly six decades, **LWE** affected tribal and remote regions marked by violence, weak connectivity and poor public services.
- **Historic Milestone:** An integrated strategy addressing security and structural causes made India effectively free from **Naxalism by 31st March 2026**.
- **Strategic Framework:**
 - **Vishwaas:** Stronger security operations, inter-agency coordination, surrender and rehabilitation programmes, and community engagement.
 - **Nirman:** Expansion of physical and digital connectivity, governance, infrastructure and livelihoods.
 - **Jan Kalyan:** Focus on welfare delivery, dignity, cultural inclusion and mainstream development.
- **Integrated Approach:** Security enabled development, development built trust, and trust improved welfare delivery.
- **Shrinking LWE Footprint:** LWE-affected districts declined from **126 in 2014 to 2 in 2026**; most-affected districts fell from **35 to zero**.
- **Decline in Violence:** Incidents declined from **870 in 2014 to 234 in 2025**, while fatalities fell from **310 to 100**. In **2010**, incidents and deaths had peaked at **1,936** and **1,005**, respectively.

Role of Vishwaas in Restoring

Faith in the State in LWE-Affected Areas

- **About:** **Vishwaas** rebuilt trust through sustained state presence, better security-governance coordination and stronger engagement with remote communities.

- **Origins:** **LWE** began with the **1967 Naxalbari uprising**. Several extremist groups merged in **2004** to form the **CPI (Maoist)**.
- **Peak of Violence:** Violence peaked in **2010** with **1,936 incidents** and **720 civilian deaths**.
- **Decadal Impact:** Between **2004 and 2014**, **17,542 incidents** killed **1,913 security personnel** and **5,019 civilians**. Nearly **92% of Naxalite weapons** were reportedly looted from police armouries.
- **Policy Gaps:** India lacked a consistent national strategy, while States acted with limited coordination. In **2009**, Naxalism was acknowledged as India's gravest internal security challenge, with a wider geographical spread than insurgencies in Kashmir and the Northeast.
- **Transition to Nirman:** Fortified police stations, **CAPF camps** and cadre surrenders expanded governance into inaccessible areas, enabling roads, schools, healthcare, banking and digital connectivity.

National Policy and Action Plan

- **Policy Shift:** The **2015 National Policy and Action Plan** replaced ad hoc responses with a structured, whole-of-government framework.
- **Security and Development Measures:** It combined **CAPF deployment**, specialised training and funding through **Security Related Expenditure, Special Infrastructure Scheme** and **Special Central Assistance** with socio-economic development.
- **Three-Pronged Approach:** The strategy focused on **Dialogue, Security and Coordination**, with the **Ministry of Home Affairs** providing central direction.
- **Time-Bound Achievement:** The target announced on **24th August 2024** to make India Naxal-free by **31st March 2026** was pursued through intensified operations, coordination, infrastructure and development outreach.

Expanded Security Grid

- **Strengthening the Security Grid:** Security infrastructure, specialised forces and operational coordination improved state presence, mobility, intelligence and counter-insurgency capacity.
- **Police and CAPF Expansion:** **597 fortified police stations** were built, compared with **66 before 2014**. Police stations reporting Naxal incidents fell from **333 to 16**, while **408 CAPF camps** were established.
- **Operational and Welfare Support:** **68 night-landing helipads**, **400 bullet- and blast-proof vehicles** and **five hospitals** supported troop movement, evacuation and welfare.

- **Specialised Security Architecture:** The framework included CoBRA, CRPF, District Reserve Guard, Jharkhand Jaguar, Chhattisgarh Police units and Greyhounds.
- **Integrated Operations:** Joint training of the DRG, STF, CRPF and CoBRA improved interoperability, readiness and command coordination.

Technology-Led Operations

- **Technology-Driven Operations:** After 2014, UAVs, drones, satellite imagery and AI-based analysis enabled real-time monitoring and precise surveillance.
- **Intelligence and Tracking:** Location tracking, call-log analysis, mobile-data analytics and social-media monitoring improved movement tracking and operational efficiency.

Trace Target Neutralise Doctrine

- **Operational Doctrine:** The “Trace, Target, Neutralise” strategy combined sustained security action with intelligence-led operations to reclaim areas under LWE influence for over three decades.
- **Major Operations:** Operation Black Forest, Octopus, Double Bull, Thunderstorm, Bhimbarg and Chakrabandha dismantled Maoist strongholds and restored state presence.
- **Major Operational Gains:** Operation Black Forest eliminated over 30 Maoists and increased arrests and surrenders. Operation Double Bull helped make Gumla, Lohardaga and Latehar Naxal-free.

Disrupting Financial Networks

- **Integrated Strategy:** An All-Agency approach targeted armed cadres and the wider LWE support ecosystem, combining development and security efforts.
- **Disrupting Financial Networks:** By December 2025, the NIA seized assets worth over ₹40 crore, the ED attached ₹12 crore, and State agencies confiscated another ₹40 crore.
- **Investigation and Prosecution:** By Dec 2025, 111 cases were investigated and 98 charge sheets filed. By June 2026, the figures rose to 112 cases and 100 charge sheets.

Surrender and Rehabilitation

- **Financial Assistance:** Surrendering cadres received ₹5 lakh for higher ranks and ₹2.5 lakh for others, a ₹10,000 monthly stipend for 36 months, an additional ₹50,000 incentive—doubled for group surrenders—and compensation for deposited weapons.
- **Rising Surrenders:** The policy led to 2,337 surrenders in 2025 and 3,927 between 2024 and March 2026, aided by civil society members, journalists, community leaders and public representatives.

Role of NIRMAN in Building

a New Future in LWE-Affected Areas

- **Focus of Nirman:** Expanded roads, digital connectivity, public services and governance in remote areas, integrating them with mainstream development.
- **Development Vision:** Ensured tribal communities received essential facilities and opportunities comparable to major cities.
- **Road Connectivity:** Over 12,249 km of roads were built after 2014. Projects covering 17,319 km were approved with ₹20,557 crore.
- **Mobile and Digital Connectivity:** Over 9,600 mobile towers were installed. Connectivity reached 44,728 of 46,592 villages, covering nearly 96%.
- **Financial Inclusion:** Between April 2015 and March 2026, 1,804 bank branches, 1,321 ATMs, 6,025 post offices and 74,720 banking correspondents were added.
- **Educational Infrastructure:** 259 Eklaya Model Residential Schools were sanctioned, of which 179 were developed.
- **Skill Development:** 46 ITIs and 49 Skill Development Centres were established with nearly ₹800 crore. Over 90,000 youth and women received training.
- **Civic Action Programmes:** Works worth ₹212 crore supported health camps, medicines and community welfare.
- **Community Engagement:** Tribal Youth Exchange Programmes and ₹50.45 crore under the Media Plan Scheme strengthened awareness and State-community engagement.

Role of JAN KALYAN in Promoting

Welfare and Dignity in LWE-Affected Areas

- **Focus of Jan Kalyan:** Ensured dignified, inclusive and last-mile delivery of essential public services to marginalised and remote communities.
- **Strengthening Tribal Livelihoods:** Improved access to education, healthcare, housing, infrastructure and livelihoods reduced LWE influence and strengthened trust in democratic institutions.
- **Rehabilitation of Surrendered Cadres:** Former cadres received skill training, livelihood support, self-employment finance and housing under Pradhan Mantri Awas Yojana. Their children received free education up to Class XII.
- **Incentives for Democratic Governance:** Development grants were given to villages that ended Naxal influence and formed elected panchayats.

Targeted Initiatives Accelerating the Transformation

- Special Central Assistance
- Aspirational Districts Programme
- PM-JANMAN
- Dharti Aaba Janjatiya Gram Utkarsh Abhiyan

India's success against Left-Wing Extremism shows that internal security challenges require both security operations and inclusive development. Discuss.

Drishti Mains Question**Cauvery Water Dispute**

Cauvery Water Management Authority (CWMA) upheld the **CWRC (Cauvery Water Regulation Committee)** order directing Karnataka to release 3,500 cusecs (around 4.5 tmc ft) of water to Tamil Nadu despite a 60% shortfall in reservoir inflows.

- Karnataka cited inadequate rainfall; Tamil Nadu demanded **9.8 TMC ft**.
- Tamil Nadu also opposed Karnataka's **Mekedatu Reservoir Project**.

Cauvery River

- Called the **"Ganga of the South"**; east - flowing.
- Originates at **Talakaveri, Brahmagiri Hills, Karnataka (Kodagu)**, and drains into the Bay of Bengal near **Poompuhar**.
- Basin: **81,155 sq. km** across Karnataka, Tamil Nadu, Kerala and Puducherry.
- Perennial due to both Southwest and Northeast monsoons.
- **Left-bank tributaries:** Harangi, Hemavati, Shimsha, Arkavati.
- **Right-bank tributaries:** Lakshmana Tirtha, Kabini, Suvarnavathi, Bhavani, Noyyal, Amaravati.
- **Major dams:** Krishna Raja Sagara (Karnataka) and Mettur (Tamil Nadu).

Cauvery River Water Dispute

- Originated in the **1892 and 1924 agreements** between the princely state of Mysore and Madras presidency.
- Karnataka (upper riparian) seeks water for Bengaluru and irrigation; Tamil Nadu (lower riparian) seeks assured releases for Cauvery delta agriculture (particularly Kuruvai crop).
- **CWDT constituted:** 1990 under Article 262 and the 1956 Act.
- **CWDT Award, 2007:** Tamil Nadu 419 TMC, Karnataka 270 TMC, Kerala 30 TMC, Puducherry 7 TMC and environment 14 TMC.

- **Supreme Court, 2018:** Karnataka increased to **284.75 TMC** and Tamil Nadu reduced to **404.25 TMC**.
- Held inter-state rivers to be national assets and ordered creation of CWMA and CWRC.

Mekedatu Project

- Proposed **48 TMC balancing reservoir** at Mekedatu, Karnataka.
- Objectives: Bengaluru's drinking water, groundwater recharge and hydropower.
- Karnataka says it will not alter existing allocations.
- Tamil Nadu fears upstream storage may reduce downstream releases, especially during distress years.
 - It cites CWDT Award, which prohibits upstream projects that adversely affect scheduled water deliveries without mutual agreement and consultation with the regulatory authority.

Inter-state river water disputes reflect the challenges of balancing federalism, water security, and climate resilience. Discuss with reference to the Cauvery dispute.

Drishti Mains Question**India at Commonwealth Games 2026**

India finished 4th at the 2026 CWG in Glasgow achieving several landmark firsts.

- At the closing ceremony, India formally took over as host of the centenary CWG 2030 in Ahmedabad. Neeraj Chopra received the Commonwealth Flag, while boxing gold medallist Jaismine Lamboria served as India's flag bearer.

Anahat Singh became the first Indian to win the **girls' singles title at the World Junior Squash Championships**.

India's Performance

- **Medal tally:** India won 39 medals—13 gold, 17 silver and 9 bronze—finishing behind Australia, England and Canada.
 - India achieved this despite fielding only 122 athletes due to doping-related restrictions and the exclusion of traditional medal-rich sports such as **shooting, wrestling, badminton, hockey, and table tennis** that had contributed 25 of its 61 medals at Birmingham 2022.
- **Better conversion rate:** Despite winning fewer medals than in 2022, India's medal success rate improved to 31%, compared with 29% in Birmingham.
- **India's Major Achievements:**
 - **Top Medal Winners:** Boxing emerged as India's most successful discipline with **10 medals (7 Gold, 3 Silver)** - the highest boxing gold tally by any nation at a single CWG.
 - **Historic Para Campaign:** India recorded its **best-ever CWG para campaign**, winning **7 medals**, equalling the combined tally from all previous editions.

Commonwealth Games

- World's second-largest multi-sport event after the Olympics, involving 71 nations and territories.
- Brainchild of Canadian sportswriter **Melville Marks Robinson**.
- Idea originated from the **1911 Inter-Empire Championships** in London.
- First held as the **British Empire Games** in Hamilton, Canada, in 1930.
- Canadian **Gordon Smallacombe** won the first CWG gold in triple jump.
- Held every four years, except in 1942 and 1946; renamed CWG in 1978.
- Governed by the **Commonwealth Games Federation (CGF)**.

India and CWG

- Debut: **1934 London Games**.
- First medal: **Rashid Anwar**, wrestling bronze.
- First individual gold: **Milkha Singh**, 440 yards, 1958 Cardiff.
- Best performance: **Delhi 2010**—101 medals and 2nd position.
- **Host the 2030 CWG**.

India's Doping Challenge

- India recorded a **3.6% positivity rate** in WADA's 2024 report—the highest among countries conducting over 5,000 tests for the third consecutive year.
- Anti-doping rule violations cost India one weightlifting quota at CWG 2026.
- Persistent doping cases undermine India's sporting credibility as it prepares to host the **2030 CWG** and bids for the **2036 Olympic Games**.
- **Need for Institutional Reforms**: Strengthen NADA, intelligence-led enforcement, athlete education and supplement awareness, with stricter action against complicit coaches, doctors and suppliers.

Kargil Vijay Diwas 2026

India observed the **27th Kargil Vijay Diwas** on **26 July 2026** to honour soldiers who recaptured the **Kargil heights** during the **1999 war**. The **Kargil War** is regarded as the only conventional conflict fought between **two nuclear-armed nations**.

Key Facts about the Kargil War

- **Background**: In 1999, shortly after the **Lahore Declaration**, Pakistani forces occupied winter-vacated Indian posts along strategic Kargil ridges.
- **Operation Badr**:
 - Pakistani soldiers, disguised as armed militants, crossed the **Line of Control (LoC)** and occupied positions overlooking the **Srinagar–Leh Highway**.
- **Pakistan's Objectives**: To disrupt connectivity between **Kashmir and Ladakh**, isolate Indian troops in **Siachen**, and force negotiations on Kashmir.
- **Detection**:
 - Suspicious movement was reported on **3 May 1999**; patrols and aerial surveillance later confirmed large-scale intrusion.
- **Challenging Terrain**: Fighting occurred at heights above **5,000 metres**, amid freezing temperatures, thin air, steep slopes, and enemy fire.
- **Operation Vijay**: India launched **Operation Vijay** to evict the intruders, while directing its forces not to cross the LoC.
- **Tri-Service Response**: The Indian Air Force began **Operation Safed Sagar** on **26 May 1999**, while the Navy deployed forces to demonstrate strategic readiness.
- **Military Strategy**: Indian forces sought to contain the infiltrators, recapture occupied posts, and prevent reinforcements.
- **Tololing**: India fully recaptured **Tololing and Point 4590** on **13 June 1999**, marking a major breakthrough.
- **Tiger Hill**: **Tiger Hill** was recaptured on **4 July 1999**, decisively shifting the war's momentum.
- **Other Major Battles**: Intense fighting occurred at **Point 4875, Batalik, Mushkoh Valley, and Kaksar**.
- **Successful Completion**: India recaptured all occupied positions without crossing the LoC, demonstrating military professionalism and restraint.
- **Declaration of Victory**: On **14 July 1999**, Prime Minister **Atal Bihari Vajpayee** declared Operation Vijay successful.

Economic Scenario

Surge in India's Merchandise Exports

India's merchandise exports increased by nearly 20% YoY to USD 44.2 billion in July 2026, according to the Ministry of Commerce and Industry.

- Growth occurred despite **West Asia geopolitical tensions**, supported by market diversification and maritime trade rerouting.

Key Trends in India's Exports

- **Merchandise Trade Growth:** Goods exports grew **19.6% YoY**, while imports increased **17.5% to USD 76.2 billion**.
- **Services Trade:** Services exports grew **6.4% to USD 35.9 billion**, while imports increased **9.5% to USD 18.9 billion**.
- **Trade Deficit:** Combined trade deficit widened to **USD 15 billion (from USD 11.4 billion)** due to faster growth in services imports.
- **Factors Driving Export Growth:**
 - **Petroleum Products:** Contributed around **39% of export growth**, driven by high global prices.
 - **Electronics & Engineering:** Electronics exports grew **30.7%** (April–July 2026), while engineering goods grew **18.2%**.
 - **Agri & Allied Sectors:** Growth in marine products, meat, dairy, poultry and handicrafts.

Strategic Rerouting & Market Diversification

- **Alternative Shipping Routes:** Exporters reduced dependence on the **Strait of Hormuz** by shifting cargo through ports in **Oman, Fujairah and Khor Fakkan (UAE)**.
- **New Markets:**
 - **China:** Exports grew by **65%**; Singapore's share increased from **2.7% to 4.7%**.
 - **Africa:** Exports to Tanzania increased by **130%**, with growth in Kenya and SACU region.
- **Traditional Market:** The **US remained the largest export destination**, accounting for about one-fifth of exports.

Challenges to India's Export Competitiveness

- **Middle-Technology Trap (Dual Competitiveness Challenge):** India faces competition from low-cost economies like Bangladesh and Vietnam while lagging behind China, South Korea and Taiwan in high-tech manufacturing.
- **Low R&D Spending:** GERD remains around **0.65% of GDP**, limiting innovation and movement from assembly to high-value manufacturing.

- **Weaponisation of Trade Standards:** Non-Tariff Barriers like **EU's CBAM**, deforestation laws act as green protectionism, directly threatening India's carbon-intensive exports like steel, cement, and aluminum.
 - US's increasing use of "forced labor" investigations and stringent ESG compliance demands create massive compliance costs and market-entry bottlenecks for Indian MSMEs.
- **Export Basket Vulnerability:** Heavy dependence on refined petroleum exports makes growth vulnerable to crude prices.
- **Geopolitical Risks:** West Asia and Red Sea disruptions increase freight costs and transit time.
- **Inverted Duty Structure:** Higher duties on raw materials than finished goods reduce domestic value addition and weaken India's competitiveness in Global Value Chains (GVCs).
- **Chinese Input Dependence:** Pharmaceuticals and electronics remain dependent on Chinese intermediates.
- **Low FTA Utilisation (25%):** Indian exporters have low utilisation of FTAs due to complex **Rules of Origin (ROO)** and limited awareness.
 - Conversely, partner countries often leverage these FTAs more effectively to capture the Indian domestic market.

Measures to Strengthen Export Competitiveness

- **High-Tech GVC Integration:** Move beyond assembly-based PLI schemes towards deep-tier manufacturing in semiconductors, green technologies and precision engineering.
 - Strengthen **Electronics Component Manufacturing Scheme (ECMS)**, India Semiconductor Mission 2.0 and domestic manufacturing of APIs and telecom components.
- **ESG Compliance Support:** Develop domestic carbon markets and assist MSMEs through the **TRACE (Trade Regulations, Accreditation & Compliance Enablement)** framework.
- **FTA 2.0 Strategy:** Must be negotiated with a dual-track strategy. Focus on greater market access for services, Mode 4 mobility and stronger ROO to prevent dumping.
- **Logistics Improvement:** Implement **National Logistics Policy** and LIFT scheme to reduce logistics costs towards global benchmark of **7–8% of GDP**.
- **Correct Inverted Duty Structure:** Ensure inputs are not taxed higher than finished products.

- **Strengthen MSME Exports:** Provide affordable trade finance, export credit, insurance, branding, certification and market intelligence through initiatives like **Export Promotion Mission (Niryat Protsahan and Niryat Disha)**.

Electronics Component Manufacturing Scheme

MeitY approved **31 additional ECMS proposals** involving **₹6,844 crore investment**.

- New components include **electrolyte additives, hermetic terminals, metalised capacitor films and coils**, being manufactured in India for the **first time**.

Key Facts About the Scheme

- **ECMS:** Launched in **April 2025** by MeitY to build a **self-sustaining electronics component ecosystem** and integrate India with **Global Value Chains (GVCs)**.
- **Incentives:** Provides **turnover-linked, capex-linked or hybrid incentives**, partly linked to **employment generation**.
 - Benefits are given on a **first-come, first-served** basis to firms ready for early production.
- **Tenure:** **6 years**; turnover-linked incentives have **1-year gestation**, while capex incentives are available for **5 years**.
- **Target Segments:** Covers **sub-assemblies, bare components and capital equipment**, together accounting for nearly **90% of mobile-phone Bill of Materials (BoM)**.
- **Investment Commitments:** By **December 2025**, ECMS attracted **₹1.15 lakh crore**, nearly double the initial **₹59,350 crore target**.
- **Projected Outcomes:** Expected to generate **₹10.34 lakh crore production over 6 years** and create **1.41 lakh direct jobs**.
- **Synergy with ISM 2.0:** Complements the **India Semiconductor Mission 2.0** across materials, equipment and advanced packaging.

Significance

- **Self-Reliance:** India is self-reliant in **enclosures, relays, anode material and optical transceivers**.
 - Domestic production meets about **80% of laminate, 75% connector, 60% lithium-ion cell** and **55% transducer demand**.
- **Geographical Spread:** Latest projects span **10 States**—Himachal Pradesh, Uttarakhand, UP, Haryana, Gujarat, Maharashtra, Goa, Karnataka, Tamil Nadu and Telangana; **Tamil Nadu leads with 7 approvals**.

- **Expected Output and Jobs:** New projects are expected to generate about **₹82,243 crore production** and nearly **10,000 jobs**.

Supporting Policy Framework

- **PLI Scheme:** Covers **14 sectors**; nearly **70% of USD 4 billion electronics FDI since 2020** is attributed to PLI beneficiaries.
- **100% FDI** permitted in electronics manufacturing, subject to applicable laws.
- **Modified Electronics Manufacturing Clusters (EMC 2.0) Scheme**
- **Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS)**
- **National Policy on Electronics (NPE) 2019**
- **Union Budget 2026–27:** Announced **Basic Customs Duty exemptions for microwave-oven inputs** and **Social Welfare Surcharge exemptions for electronic toy parts**.

Foreign Assets of Small

Taxpayers Disclosure Scheme, 2026

CBDT notified the **Foreign Assets of Small Taxpayers–Disclosure Scheme (FAST-DS) 2026**, a one-time voluntary disclosure mechanism announced in **Union Budget 2026–27**.

Key Facts About FAST-DS 2026

- **About:** One-time scheme under the **Finance Act, 2026** for declaring specified **undisclosed foreign assets/income or undeclared foreign assets**.
 - Effective from **16 August 2026**.
 - Declaration deadline: **31 December 2026**.
 - Asset valuation date: **31 March 2026**.
- **Eligibility:** Available to persons who were **resident in India in the relevant previous year**.
 - A current **Non-Resident or Resident but Not Ordinarily Resident (RNOR)** can also use it if resident in India when the relevant foreign income arose or asset was acquired.
- **Grounds for Declaration:** Applicable where the taxpayer:
 - Did not file an income-tax return.
 - Omitted foreign asset/income from a return filed before commencement of the Scheme.
 - Had foreign asset/income that escaped assessment.

Category	Coverage	Limit
Category I– Undisclosed foreign asset/income	Foreign asset/income not previously offered to tax	₹1 crore
Category II– Undeclared foreign asset	Foreign asset already taxed or acquired while non-resident , but not disclosed in return	₹5 crore

- **Tax/Fee Payable:**
 - **Category I:** 30% tax + additional amount equal to 100% of tax = effective 60% payment.
 - ❖ **Example:** ₹80 lakh declaration = ₹48 lakh payable.
 - **Category II:** Flat ₹1 lakh fee.
- **Valuation of Foreign Assets:** Fair Market Value (FMV) generally equals the higher of **cost of acquisition** or **open-market value on 31 March 2026**.
 - Values must be converted into **Indian Rupees** using prescribed forex rules.
 - Separate valuation rules apply to **jewellery, securities, unquoted shares, immovable property, foreign bank accounts and interests in foreign partnerships/Limited Liability Partnerships (LLPs)**.
- **Benefits:** Valid declaration and payment provide immunity from further **tax, penalty and prosecution under the Black Money Act, 2015** for the declared asset/income.
 - Declared income/investment is not included in total income under the **Income-tax Act, 1961** or **Black Money Act, 2015**.
 - Declarant cannot later seek **rectification, revision, set-off or relief** regarding the declared amount.
- **Exclusions:** Income/assets representing **proceeds of crime** where proceedings under the **PMLA, 2002** have been initiated or are pending.
 - Assessment years where proceedings have already been completed under the **Black Money Act, 2015**.
- **Significance:** Promotes **voluntary compliance** for small-value **foreign assets/income** while providing certainty against further **penalties and prosecution**.

India's Makhana Sector

Government highlighted India's growing **makhana sector** after launch of the **National Makhana Board in Bihar on 15th September 2025** and approval of the **Central Sector Scheme for Development of Makhana** with **₹476.03 crore** outlay for **2025–26 to 2030–31**.

Current Status of India's Makhana Sector

- **Global Leadership:** India is the **world's largest producer**; Bihar contributes nearly **75% of India's production** and around **80–85% of global supply**.
- **Production Hub:** Concentrated in **North Bihar**, especially **Supaul, Saharsa, Madhepura, Mithila and Seemanchal**.
- **Production:** Increased from **63,910 MT in 2024–25** to **80,590 MT in 2025–26**; productivity rose from **2.03 MT/ha** to **2.34 MT/ha**.
 - Bihar produced around **60,000 MT** in 2025–26.

- **Domestic Market:** Grew around **17–18% annually** during **2021–22 to 2024–25**; projected at **₹11,000–12,000 crore** by 2029–30.
- **Prices:** Rose from about **₹500/kg** in 2020–22 to nearly **₹1,250/kg** in 2025.
- **Consumption:** **3,000–3,500 MT/month**, rising to around **5,000 MT** during festivals.
- **Exports:** After separate **HSN codes in 2025**, India exported **7,264.89 MT** worth **₹19,296.08 lakh** in 2025–26.
- **Major Markets:** **US 40%, Canada 20%, UAE 17%**, together accounting for around **77% of exports**, while markets like **Germany, Nepal and Australia** offer higher unit-price realisations.

Key Facts about Makhana

- **Makhana (*Euryale ferox*):** Aquatic crop grown in **shallow ponds, lakes and wetlands**; edible seeds are processed and popped.
- **Major States:** Bihar, West Bengal and Assam.
- **GI Tag:** Mithila Makhana.
- **Cultivation:** Increasing shift from traditional pond cultivation to **field-system farming**.
- **Improved Varieties:** **Swarna Vaidehi** and **Sabour Makhana-1**.
- **Nutrition:** Rich in **carbohydrates, protein, fibre, magnesium, potassium, phosphorus and iron**; low in fat and sodium and cholesterol-free.
 - Growing preference for **healthy, plant-based snacks** has increased its premium-food demand.

Government Initiatives to Strengthen the Makhana Sector

- **National Makhana Board:** Announced in **Union Budget 2025–26** and launched in Bihar on **15 September 2025**.
- **Central Sector Scheme:** ₹476.03 crore for 2025–26 to 2030–31.
- **Focus Areas:** Research, quality seeds, farmer training, harvesting, post-harvest management, value addition, branding, marketing, exports and quality control.
- **Allocation:** ₹30 crore in 2025–26 and ₹90 crore in 2026–27.
- Around **8,159 farmers** benefited in 2025–26.
- **1,010 additional hectares** brought under cultivation and **73 hectares** under seed production.
- **89 Front Line Demonstrations** conducted through **State Agriculture Universities (SAUs), Central Agricultural Universities (CAUs)** and **Indian Council of Agricultural Research (ICAR)** institutions.

Strengthening the Makhana Value Chain through Processing, Research and Technology

- **Primary Processing:** Seeds undergo **drying, cleaning, roasting, popping, polishing, sorting and grading**.

- **Value Addition:** Expansion into **flavoured makhana, ready-to-eat snacks and makhana flour**.
- **Start-ups and FPOs:** Supporting **processing, branding, value addition and market access**.
- **National Research Centre for Makhana (NRCM):**
 - Promotes **water-efficient and integrated farming**, including **makhana-cum-fish farming**.
 - Distributed around **15,824.1 kg of high-yielding seeds**.
 - Trained over **3,000 farmers between 2012 and 2023**.
 - Supported **24 enterprises**.
 - Developed **seed washers, graders, roasting and popping machines** for commercial use.
- **Overall Potential:** Can improve **farmer incomes, rural employment, value addition and agri-food exports**.

Makhana has the potential to transform wetland-based agriculture into a high-value agri-food economy. Discuss.

Drishti Mains Question

Building a Business-Ready India

On India's 80th Independence Day, the Government highlighted reforms to improve **Ease of Doing Business, digital governance, compliance simplification, market access and enterprise financing**.

Key Initiatives Launched by the Government for Reforming India's Business Environment

India's Business Environment

- **National Single Window System (NSWS):** Launched in 2021; integrates **32 Central Ministries and 34 States**, covering **686+ Central and 7,498 State approvals**. Granted **8.29 lakh+ approvals by November 2025**.
- **Startup India:** Launched in 2016; DPIIT-recognised startups rose from **502 in 2016 to 2.47 lakh+ by 12 August 2026**.
- **SPICe+ Form:** Introduced in 2020; integrates **11 services from 3 Central Ministries/Departments**, including incorporation, DIN, PAN, TAN, EPFO, ESIC, GSTIN and bank account opening.
- **MCA21 V3:** AI-driven corporate registry platform for companies and Limited Liability Partnerships (LLPs).
 - Processed **3.84 crore filings during 2021-25**, with **3.33 crore approved through Straight Through Process**.
- **Udyam Registration:** Introduced in 2020 as a **free, paperless, self-declaration-based MSME registration system**.
 - Registrations increased from **0.10 lakh in October 2020 to 9.27 lakh+ by 13 August 2026**.
- **Business Reform Action Plan (BRAP):** Launched in 2015 to promote State/UT reforms in **single-window clearances, land, labour, inspections, taxation and environmental approvals**.

Unlocking Finance for Enterprise Growth

- **Pradhan Mantri MUDRA Yojana (PMMY):** Launched in 2015; collateral-free loans up to **₹20 lakh** for non-corporate, non-farm activities.
 - By **27 March 2026**, **57.79 crore loans worth ₹40.07 lakh crore** were sanctioned.
- **Credit Guarantee Scheme:** Enables collateral-free credit up to **₹10 crore** for Micro and Small Enterprises.
 - By **March 2026**, **Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE)** approved **₹13.67 lakh crore** coverage across **1.41 crore** guarantees.
- **Credit Assessment Model (CAM):** Introduced by **Public Sector Banks in 2025** to automate MSME loan appraisal using digital data.
 - **3.96 lakh+ applications worth ₹52,300 crore+** sanctioned during **April-December 2025**.
- **Trade Receivables Discounting System (TReDS):** Enables MSMEs to finance trade receivables through multiple financiers.
 - Invoice discounting rose from **~₹40,000 crore in 2021-22 to ₹3.47 lakh crore in 2025-26**.
 - **Union Budget 2026-27** proposed credit guarantee for invoice discounting and wider TReDS use by CPSEs.

From Compliance to Trust-Based Governance

- **Jan Vishwas (Amendment of Provisions) Act, 2026:** Decriminalises **717 provisions** and amends **784 provisions** across **79 Central Acts and 23 Ministries**.
- **Reduction in Compliance Burden:** By **November 2025**, **47,000+ compliances** reduced:
 - 16,108 simplified; 22,287 digitised; 4,458 decriminalised; 4,270 redundant compliances removed
- **GST 2.0:** Reforms announced in **September 2025** moved towards a **simplified two-rate structure**.
 - GST taxpayers increased from **~60 lakh in 2017 to 1.68 crore+ by 31 July 2026**.
- **Faceless Assessment Scheme:** Uses **automated case selection, random allocation and centralised digital communication** to make tax administration more objective, standardised and transparent.

Expanding Markets for Indian Businesses

- **Government e-Marketplace (GeM):** Launched in 2016.
 - By **6 August 2026**, connected **1.37 lakh+ government buyers** with **~25 lakh sellers/service providers** and facilitated procurement worth **₹20 lakh crore+**.
 - **75 lakh+ orders** executed in **FY 2025-26**.
- **PM GatiShakti:** Launched in 2021; integrates **58 Ministries/Departments, 36 States/UTs and 3,291 data layers**.

- By **February 2026**, **352 projects worth ₹16.10 lakh crore** evaluated; **201 sanctioned** and **167 under implementation**.
- **Open Network for Digital Commerce (ONDC)**: Launched in **2022** to promote interoperable digital commerce.
 - Has **7.64 lakh+ sellers** across **616+ cities**.

Strengthening India's Global Business Footprint

- **Export Promotion Mission**: Launched in **2025**; focuses on **MSMEs, first-time exporters, labour-intensive sectors and low-export-intensity regions**.
- **Trade Connect e-Platform**: Provides trade information, market opportunities and global-buyer access.
 - By **August 2026**, crossed **20 lakh registered users** and facilitated **40 lakh+ Certificates of Origin**.
- **Indian Customs Electronic Gateway (ICEGATE)**: Enables electronic customs filing, duty payment, amendments, query resolution and refunds.
 - Bills of Entry increased from **1.43 lakh+ in April 2020** to **6.14 lakh+ in June 2026**.

India's Ease of Doing Business reforms are shifting from compliance reduction to building an enterprise-led growth ecosystem. Discuss.

Drishti Mains Question

NITI Aayog Report on India's Regulatory Regime in Professional Services

NITI Aayog released "*India's Services Sector: Insights on Regulatory Regime in Professional Services*" to identify regulatory bottlenecks in knowledge-intensive professional services and improve global competitiveness.

Key Findings of NITI Aayog's Report

- **Macro Significance**: Services contributed **~55% of GDP/GVA in FY24**, grew at **~7% CAGR (2011-12 to 2023-24)**, account for **~30% of employment**, and remain India's top FDI-recipient sector.
 - Professional and management consulting contributed **~20% of services exports in 2024-25**; exports grew at **18% CAGR (FY15-FY25)**.
 - India's **Revealed Comparative Advantage (RCA)** in professional and management consulting rose from **0.95 (2005) to 3.0 (2024)**.
 - India is the **7th-largest services exporter**, with **4.3% global share in 2024**, up from **2% in 2005**.
- **Regulatory Landscape**: Governed by laws including **Companies Act, 2013; LLP Act, 2008; Partnership Act, 1932; and FEMA/RBI regulations**.
- **OECD's Services Trade Restrictiveness Index (STRI)**: Measures regulatory barriers to services trade across 22 sectors and 51 countries, updated annually.

Sector	India's Rank	Regime
Architectural Services	2 nd most restrictive	Highly restrictive
Legal Services	3 rd most restrictive	Highly restrictive
Accounting Services	4 th most restrictive	Highly restrictive
Engineering Services	38 th most restrictive	Comparatively liberal

- **India and Global Trade Rules Context**: Under **General Agreement on Trade in Services (GATS)**, services trade occurs through-
 - **Mode 1**: Cross-border supply
 - **Mode 2**: Consumption abroad
 - **Mode 3**: Commercial presence
 - **Mode 4**: Presence of natural persons
 - India has **no GATS commitments in legal and architectural services**; engineering has limited commitments, with **Mode 3 foreign equity capped at 51%**.
 - India is **not a signatory** to the WTO's **2017 Joint Initiative on Services Domestic Regulation**, though it benefits through the **Most-Favoured-Nation (MFN)** principle.

Challenges Facing India's Professional Services Sector

- **Legal Services**: **Advocates Act, 1961** mainly recognises advocates; **in-house counsels, Legal Process Outsourcing (LPO) professionals and legal consultants** lack a clear framework.
 - India has **no Mutual Recognition Agreements (MRAs) with foreign bar bodies**.
 - Restrictions on **corporate structures and Multi-Disciplinary Practices (MDPs)** limit scalability and global competitiveness.
- **Accounting and Auditing**: Dual oversight by **Institute of Chartered Accountants of India (ICAI)** and the **National Financial Reporting Authority (NFRA)** creates jurisdictional concerns.
 - Foreign accountants face practice restrictions; **limited MRAs** constrain mobility.
 - **Bookkeeping** lacks a dedicated framework for qualifications, ethics and record-keeping.
- **Architectural and Engineering Services**: Architecture regulated by **Architects Act, 1972**; engineering lacks a comprehensive statutory framework.
 - **Engineers Bill** remains pending; **AICTE** mainly regulates education.
 - Overlap between architects and engineers creates accountability gaps.

- Architects generally cannot practise through **LLPs or companies, unlike in countries such as the UK, New Zealand and Singapore.**
- Multiple State requirements overlap with **Council of Architecture registration**; India has **no MRAs for architects.**
- **Healthcare & Allied Healthcare Services:** Jurisdictional overlap in **AYUSH** among the Ministry, **National Commission for Indian System of Medicine (NCISM)** and **State bodies.**
 - Interstate mobility requires **NOCs between State Medical Councils.** Delayed **National Exit Test (NExT)** and partial implementation of the **NCAHP Act, 2021** hinder standardisation.

Recommendations of NITI Aayog

- **Continuous Professional Development (CPD) for Professionals:** Introduce a **uniform and enforceable Continuous Professional Development (CPD)-hour regime** across professions.
- **Adoption of Best (Global Regulatory) Practices:** Ensure transparent **qualification/licensing criteria, appeal mechanisms and processing timelines.**
 - **Models:** Malaysia's Unified Public Consultation portal and Australia's DFAT Professional Services and Mutual Recognition Unit.
- **Elevate Professional Services within the Global Value Chain (GVC):** Shift towards **R&D, strategic technology, engineering design and complex advisory.**
 - Leverage India's **~1,700 GCCs employing over 1.9 million professionals.**
- **Advance Emerging Trends:** Build capability in **AI/automation, carbon accounting, climate-risk advisory, sustainable finance, supply-chain restructuring and compliance advisory.**

India's services sector has emerged as a major pillar of economic growth, but regulatory fragmentation is limiting its potential in high-value professional services. Critically examine.

Drishti Mains Question

Greenium and Sovereign Green Bonds

India's Sovereign Green Bonds (SGRBs) are trading at a persistent **greenium** over comparable conventional government securities, reflecting stronger investor demand.

- The **average greenium in H1 FY27** reached its highest level since India began issuing SGRBs in **FY2022-23**, indicating scope for a larger issuance of green bonds in **H2 FY27.**

Greenium

- **Greenium (green premium):** Lower yield (interest rate) accepted by investors on green bonds compared with conventional government securities.

- Lower yield reflects willingness to pay a **higher price** for environmentally sustainable investments.

■ Significance of a Stable Greenium:

- **Cost-Effective Climate Action:** Reduces government borrowing costs for **renewable energy, electric mobility and adaptation projects**, supporting **Net Zero by 2070.**
- **Market Maturation:** Indicates a maturing green finance ecosystem and investor confidence in **green taxonomies and ESG project evaluation.**

India's Sovereign Green Bond Framework

- **SGRBs:** Government debt instruments introduced in **Union Budget 2022-23** to finance India's transition to a low-carbon economy.
 - Proceeds are exclusively earmarked for **eligible green projects.**
 - Framework aligned with **ICMA Green Bond Principles, 2021.**
 - Issued through **uniform price auctions**, eligible for **repo**, tradable in secondary markets and eligible for **Statutory Liquidity Ratio (SLR).**
- **Sovereign Green Bonds Framework:** Released by the **Finance Ministry in 2022.**
 - Validated by **CICERO** with '**Medium Green**' and '**Good Governance**' ratings.
- **Management Sovereign Green Bonds:**
 - Proceeds deposited in the **Consolidated Fund of India** and managed by the **Public Debt Management Cell** through the **Green Register.**
 - Project selection and evaluation by the **Green Finance Working Committee (GFWC)**, chaired by the **Chief Economic Adviser.**
 - Allocation and utilisation audited by the **CAG.**
- **Current Status of India's SGRB Market:**
 - **₹877 billion** SGRBs outstanding.
 - **30-year bonds** dominate, with outstanding issuance above **₹500 billion.**
 - Earlier issuances were cancelled/reduced due to weak demand at acceptable yields; demand has since improved, especially for longer maturities.
- **Drivers of Demand:**
 - **Insurers' Investment Needs:** Long-term assets match long-term liabilities.
 - **Infrastructure Status:** Green bonds are treated as infrastructure investments.
 - **Asset-Liability Management (ALM) Requirements:** Long-duration bonds suit insurers and pension funds, supporting demand for **30-year maturities.**

Sustainable Finance Instruments

Instrument	Use of Proceeds	Key Feature	Example
Green Bonds	Ring-fenced for environmental projects	Exclusively finances green/environmental objectives	India's SGrBs
Social Bonds	Ring-fenced for social projects	Targets vulnerable or underserved groups	Educate Girls Development Impact Bond, Rajasthan (2015-18) ; world's first education Development Impact Bond.
Sustainability Bonds	Ring-fenced for both green and social projects	Combines environmental and social objectives	Mexico's BONDESG/Bondes G , aligned with SDGs
Sustainability-Linked Bonds (SLBs)	Can be used for general purposes	Bond terms are linked to achievement of specified sustainability performance targets	UltraTech Cement, 2021 ; targeted 22.2% reduction in carbon intensity by 2030

National Cooperative Development Corporation (Amendment) Bill, 2026

Parliament passed the **National Cooperative Development Corporation (Amendment) Bill, 2026** to amend the **NCDC Act, 1962** and expand the mandate and financing powers of **NCDC**.

Key Highlights of the Bill

- **Direct Financial Assistance:** Earlier, NCDC could directly provide loans and grants mainly to **national-level and multi-State cooperatives**.
 - The Bill extends direct loans and grants to **any cooperative society or entity engaged in cooperative development**, subject to prescribed conditions.
 - NCDC can continue financing cooperatives through **State Governments**, while gaining an additional direct-financing channel.
- **Expanded Definition of "Foodstuffs":** Includes **processed food and other food items notified by the Central Government**, expanding eligibility for NCDC assistance.
- **Industrial Goods:** Removes the requirement that eligible industrial cooperatives, cottage/village industries and allied industries must be **located in rural areas**.
- **Wider Investment Powers:** With prior Central Government approval, NCDC may participate in the **share capital of cooperatives/entities engaged in cooperative development**.
- **Information Sharing:** NCDC may collect and furnish **credit and other information** and share it with the **Central Government, RBI, banks and notified financial institutions**.

Concerns Raised Regarding the Bill

- **Federal Friction:** Cooperatives confined to one State fall under **Entry 32 of the State List**. Direct NCDC funding may bypass State Governments, raising concerns over **centralisation and federal balance**.
- **Implementation Challenges:** Since States perform important oversight and auditing functions, direct funding

without State coordination may cause **overlapping jurisdiction and monitoring problems**.

NCDC

- **About:** Statutory organisation established in **1963** under the **NCDC Act, 1962**.
 - The Act was earlier amended in **1973, 1974 and 2002**, progressively expanding NCDC's scope and direct-financing powers.
- **Nodal Ministry:** Ministry of Cooperation.
- **Mandate:** Promote and develop farmer cooperatives to improve **production, productivity and post-harvest facilities**.
 - Finance **processing, storage, cold chain and marketing** of agricultural produce.
 - Support dairy, livestock, handloom, poultry, fisheries and cooperatives led by **women and SC/ST communities**.
- **Financial Growth:** Disbursements increased from **₹5,735 crore in 2014–15 to ₹95,182 crore in 2024–25**.
- **National Cooperation Policy (NCP) 2025:**
 - **Vision:** Build a supportive legal, economic and institutional environment for cooperatives aligned with **Viksit Bharat 2047**.
 - **Implementation:** Focuses on transparent, technology-driven and professionally managed cooperatives, involving **NCDC, IFFCO, NAFED, Amul, KRIBHCO, NDDDB, NCEL and NABARD**.

Mines and Minerals (Development and Regulation) Amendment Bill, 2026

Parliament passed the **Mines and Minerals Amendment Bill, 2026** to create a more uniform and predictable fiscal framework by restricting State taxes and levies on **mineral rights and mineral-bearing lands**.

- The Bill addresses **taxation challenges** while raising concerns over **mineral security, fiscal federalism and Centre-State relations**, especially after the **Supreme Court's 2024 judgment**.

Key Features of the MMDR Amendment Bill, 2026

- **Union Control over Mineral-Bearing Lands:** Amends Section 2 of the **MMDR Act, 1957** to explicitly bring **mineral-bearing lands under Union regulatory control**.
- **Capping State Levies — New Section 9D:** States cannot impose tax, cess or other levies on mineral rights or mineral-bearing lands—whether based on quantity, value or royalty—except under **conditions prescribed by the Centre**.
- **Past Dues:** Unpaid or unrecovered levies before the amendment become invalid; amounts already paid will not be refunded.
- **Rule-Making:** Section 13 is amended to empower the Central Government to prescribe parameters for future State levies.

Significance of the MMDR Amendment Bill, 2026

- **Tax Certainty:** Reduces multiple and unpredictable levies.
- **Uniformity:** Promotes more uniform taxation across States and lowers compliance costs.
- **Investment & Production:** Reduces uncertainty and mining costs, potentially encouraging domestic mineral production and investment.

Concerns Regarding the MMDR Amendment Bill, 2026

- **Federalism:** Restrictions on State taxation of mineral rights and mineral-bearing land may affect powers under **Entries 49 and 50 of the State List**.
 - Parliament can limit mineral-right taxation under Entry 50, but questions remain over restrictions on land taxation under Entry 49.
- **Supreme Court's 2024 Judgment:** In *Mineral Area Development Authority v. Steel Authority of India*, an **8:1 majority of a 9-judge Constitution Bench** upheld State power to tax mineral rights under Entry 50 and mineral-bearing lands under Entry 49.
 - The Court also permitted recovery of past tax dues retrospectively from **1 April 2005**.
 - By invalidating certain unpaid/unrecovered past levies, the Bill seeks to alter the legal consequences of this ruling, raising questions of **separation of powers and legislative competence**.
- **Article 14:** Invalidating unpaid dues while not refunding paid levies could treat compliant and non-compliant companies differently.
- **Excessive Delegation:** Key conditions governing State taxation are left to Central rules, raising concerns that Parliament should itself define clearer principles.
- **State Revenues:** Restrictions may reduce the fiscal autonomy and revenue base of mineral-rich States.

Mines and Minerals**(Development and Regulation) Act, 1957**

- **About:** Enacted under **Union List Entry 54** to provide central legislative control over mineral regulation and development.
- **2015 Amendment:** Introduced mandatory **auction-based mineral concessions**, created the **District Mineral Foundation (DMF)** and **National Mineral Exploration Trust (NMET)**, and strengthened penalties for illegal mining.
- **2016:** Defined “leased area” and allowed transfer of captive mining leases granted otherwise than through auction, subject to prescribed conditions and charges.
- **2020:** Allowed companies without prior coal-mining experience to participate in coal/lignite auctions, widening participation and FDI.
- **2021 Amendment:** Removed the distinction between captive and merchant mines; captive mines could sell up to **50% of annual production** after meeting end-use requirements, while auction-based allocation was reinforced.
- **2023 Amendment:** Removed **six critical minerals**—lithium, titanium, beryllium, niobium, tantalum and zirconium-bearing minerals/ores—from the list of 12 atomic minerals reserved for State agencies.
 - Empowered the Centre to exclusively auction concessions for critical minerals and introduced **Exploration Licences** to attract FDI and junior mining companies.
 - Supports access to deep-seated minerals required for future technologies and India's **net-zero 2070** goal.
- **2025:** Introduced a **Mineral Exchange** for transparent electronic trading of minerals, concentrates and processed metals.
 - Removed the **50% sale cap on captive mines**, allowing open-market sale after meeting end-use requirements.

Measures Needed to Strengthen Mineral and Mining Regulation

- **Centre-State Coordination:** Create a permanent mechanism to coordinate taxation, leases, auctions, production and mineral development.
- **Improve DMF Outcomes:** Ensure transparent, outcome-based use of DMF funds for health, education, livelihoods and local infrastructure in mining-affected areas.
- **Deep Ocean Mission:** Expand exploration of polymetallic nodules and sulphides containing **nickel, cobalt, copper and manganese**, while developing indigenous deep-sea mining technology with strong environmental safeguards.
- **National Critical Mineral Mission:** Expand domestic exploration, acquire overseas assets, promote recycling, and build beneficiation, processing and refining capacity across the critical-mineral value chain.

- **Mine Closure & Restoration:** Make progressive mine closure, land reclamation and ecological restoration enforceable throughout the mining lifecycle.
- **Curb Illegal Mining:** Strengthen satellite monitoring, drone surveillance, GPS-based mineral tracking and e-permits.

Container Manufacturing Assistance Scheme (CMAS)

India's rising trade and manufacturing needs have increased demand for a strong domestic container ecosystem and reduced import dependence.

- **Government has proposed CMAS** to strengthen maritime, shipping and logistics capabilities.

Container Manufacturing Assistance Scheme

- **Budget:** Announced in Union Budget 2026–27.
- **Outlay:** ₹10,000 crore over 5 years.
- **Objective:** Develop a globally competitive domestic container-manufacturing ecosystem and strengthen supply-chain resilience.
- **Capacity Target:** Raise annual capacity to about **7.5 lakh Twenty-foot Equivalent Units (TEUs)**, nearly 10 times the existing capacity.
- **Financial Support:** Assistance for new greenfield units, expansion of brownfield units and operational competitiveness.
- **Ecosystem Development:** Supports testing infrastructure, skilling and capacity building.
- **Policy Alignment:** Complements Make in India, Maritime Amrit Kaal Vision 2047 and multimodal logistics development.

Shipping Containers

- Standardised steel freight boxes enabling seamless movement across ships, railways and roads without repacking.
- Manufactured according to international **ISO standards**.
- Common sizes: **20-foot and 40-foot** containers.
- Capacity is generally measured in **TEUs**.

Other Government Initiatives Supporting Container Manufacturing and Maritime Development

Building an Integrated Maritime Ecosystem

- Established through an MoU in **Feb 2026**.
- Partners: **Shipping Corporation of India (SCI), Container Corporation of India (CONCOR), Jawaharlal Nehru Port Authority, V.O. Chidambaranar Port Authority and Sagarmala Finance Corporation Limited.**
- Envisages about **₹99,149 crore** investment for **51 container vessels** and domestically manufactured containers.

Integrated Maritime Ecosystem

- **PM Gati Shakti:** Improves port-rail-road-industrial connectivity.
- **National Logistics Policy:** Promotes digital integration, efficiency and lower logistics costs.
- **Sagarmala Programme:** Sagarmala, along with CMAS and BCSL, supports port-led development and maritime infrastructure.
- **Shipbuilding Support:** ₹70,000 crore Shipbuilding Financial Assistance Package.
- **Digital Platforms:** One Nation One Port Process, Maritime Single Window and e-Samudra.
- **Modern Laws:** Merchant Shipping Act, 2025; Coastal Shipping Act, 2025; Indian Ports Act, 2025.
- **Major Projects:** VadHAVAN Port, Galathea Bay transshipment port, Tuna Tekra Container Terminal and Outer Harbour Container Terminal at V.O. Chidambaranar Port.
- **Port Competitiveness:** Three Indian ports ranked among the world's top 30 in the Container Port Performance Index 2025.

Recent Progress in Container Manufacturing

- In **July 2026**, India rolled out its first domestically manufactured export-import (EXIM) shipping container for **A.P. Moller–Maersk** at the Maersk–CONCOR ICD, Dadri, UP.
- Manufactured according to **ISO standards and International Convention for Safe Containers (CSC)**.
- Maersk placed an order for **1,000 additional Made-in-India containers** with DCM Shriram Group.

Why Does India Need Domestic Container Manufacturing?

- Around **80% of global merchandise trade by volume** moves by sea.
- Containerised cargo accounts for nearly **two-thirds of international trade by value**.
- Geopolitical tensions and shipping disruptions have exposed supply-chain vulnerabilities and freight-rate volatility.
- India's expanding manufacturing and EXIM trade are increasing domestic container demand.
- India imports nearly **2 million empty containers annually**, creating dependence on global prices and availability.
- Domestic manufacturing can improve **container availability, supply-chain resilience and trade competitiveness**.

Domestic container manufacturing is essential for strengthening India's maritime logistics, supply-chain resilience and trade competitiveness. Discuss.

Drishiti Mains Question

Ship Recycling in India

India became the **world’s leading ship-recycling nation in 2025**, with global share rising from **30.1% (2024) to 35.4% (2025) (UNCTAD)**.

- Recycling volume increased from **1.86 million GT to 2.99 million GT**, nearly **60% growth**.
- India achieved the **Maritime India Vision 2030** target of global leadership **five years early**.

How Has India Emerged as a Global Leader?

- **Alang-Sosiya, Gujarat**: Backbone of the sector, accounting for about **98% of India’s ship recycling**, with annual capacity of around **6 million GT**.
- **Capacity Expansion**: India plans to raise recycling capacity to about **9 million light displacement tons (LDT)** through expansion of Alang.
- **Steel Recovery**: Alang recovers around **3.50 MMT of steel annually**, supporting resource conservation and the **circular economy**.
- **Integrated Ecosystem**: Includes authorised recycling yards, steel re-rolling mills, waste-management facilities and allied infrastructure.
- **Worker Training**: Alang’s **Safety Training and Labour Welfare Institute** has trained **1.5 lakh+ workers** in safety, hazardous-material handling and emergency response.
- **Other Facilities**: Ship recycling also takes place at **Amar Iron Udyog, Kolkata** and **Steel Industrials Kerala Limited (SILK), Kerala**.
- **Modernisation**: Yards now use engineered plots, impermeable surfaces, drainage systems, mechanised handling, segregated hazardous-waste storage and authorised disposal systems.

Regulatory Framework

Legal and Institutional Architecture

- **Ship Recycling Act, 2019**: It provides the **principal statutory framework for ship recycling in India**, designates the **Directorate General of Shipping as the National Authority**, and provides for **authorisation, certification, inspection and enforcement mechanism**.
- **Ship Recycling Rules, 2021**: The Rules operationalise the Act by prescribing requirements relating to **facility authorisation, ship-recycling plans, hazardous-material management, worker safety, training and reporting**.
- **Ship Recycling Regulations, 2026**: These Regulations strengthen implementation through more detailed **operational, environmental and safety standards**, promoting uniform compliance across recycling facilities.

Provision	Detail
Ship Recycling Act, 2019	Principal law; Directorate General of Shipping is the National Authority; covers authorisation, certification, inspection and enforcement.
Ship Recycling Rules, 2021	Covers facility authorisation, recycling plans, hazardous-material management, worker safety, training and reporting.
Ship Recycling Regulations, 2026	Strengthen operational, environmental and safety standards and uniform compliance.

Hong Kong Convention Compliance

- **Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships** became legally binding on **26 June 2025**.
- India aligned its **Ship Recycling Act, Rules and Regulations** with Convention standards.
- Government provided **₹53.5 crore** for modernisation, helping **115 facilities** achieve compliance.
- **EU Green Listing– 35 Alang yards** have applied; **Shree Ram Vessel Scrap** and **Y. S. Investments** were included in the draft **16th European List**.

Inventory of Hazardous Materials (IHM)

Identifies hazardous substances aboard ships and facilitates their **safe handling, traceability, transparency and risk management** throughout the ship lifecycle.

Ship-breaking Credit Note Scheme

- Introduced under **Ship Building Financial Assistance Scheme (SBFA) 2.0**.
- Recycling a vessel at a compliant Indian yard generates a **Credit Note equal to 40% of the vessel’s scrap value**.
- Credit Note can finance up to **5% of the value of a new Indian-built vessel**.
- **Objective**: Promote both **sustainable ship recycling and domestic shipbuilding**.

Ship recycling can strengthen India’s maritime economy, resource security and circular economy goals. Discuss.

Drishti Mains Question

Taxation and Other Laws (Amendment) Act, 2026

The Taxation and Other Laws (Amendment) Bill, 2026, passed by Parliament, became an Act after receiving Presidential assent, replacing the **Income-tax (Amendment) Ordinance, 2026** and **amending the Income-tax Act, 2025, Finance Act, 2026, and Payment and Settlement Systems Act, 2007**.

Key Provisions of the Act

- **Push for Electronics Manufacturing:** Under the **Income Tax Act 2025**, foreign companies supplying capital goods or utilizing Indian factories for contract manufacturing of electronics were **exempt from income tax for five years**.
 - The new Act explicitly defines eligible goods (mobile phones, laptops, servers, wearables) and extends this **tax holiday** by another ten years (up to **2040-41**).
- **Support for Electronics and Diamond Sectors:** Grants **tax exemptions** to eligible **foreign diamond companies** and **electronics manufacturers**, including income from **rough diamond sales** and **storage of electronic components** in customs bonded warehouses.
- **Tax Exemption for FIIs and BIS:** Exempts **Foreign Institutional Investors (FIIs)** and the **Bank for International Settlements (BIS)** from income tax on **interest and capital gains** from government securities.
- **Easing Rules for Foreign Fund Managers:** The Act simplifies tax rules to allow **foreign fund managers to relocate to India** without making their foreign funds taxable, while retaining safeguards **against tax avoidance and round-tripping**.
- **Boost for Data Centres:** It removes complex, multi-layered government approval requirements previously mandated for foreign cloud companies to claim tax exemptions when using **Indian data centers**.
 - It also allows Indian data centers to operate on a leased basis rather than strictly under direct ownership, fostering a flexible ecosystem.
- **Tax Benefits for Investment Trusts (REITs & InvITs):** Restores tax-free dividend income for investors in **Real Estate Investment Trusts (REITs)** and **Infrastructure Investment Trusts (InvITs)** by exempting unit holders, while increasing the surcharge on Special Purpose Vehicles (SPVs) of business trusts from 10% to 25%.
- **UPI Transactions:** Amends the Payment and Settlement Systems Act, 2007 to **empower the Central Government to permit banks and payment service providers to levy charges on Unified Payments Interface (UPI) and other notified electronic payment modes**, replacing the existing prohibition on such charges.
 - The measure aims to create a **sustainable revenue model** for banks, **payment service providers (PSPs)**, and payment infrastructure firms while supporting the growth of the digital payments ecosystem.
 - The amendments would allow the Central Government to enable the **Merchant Discount Rate (MDR)** on UPI and other notified electronic payment modes.
 - Currently, banks and payment system providers are prohibited from levying any charges, directly or

indirectly, on notified electronic payment modes under **Section 269SU of the Income-tax Act, 1961**.

Significance

- **Encourages foreign direct investment (FDI)** and global capital inflows.
- **Strengthens the Make in India initiative** by supporting electronics manufacturing.
- Promotes India as a **global hub for data centres and cloud infrastructure**.
- Positions India as an attractive destination for **global investment fund management**.
- Encourages greater investment in **real estate and infrastructure** through REITs and InvITs.
- Improves **ease of doing business** by reducing regulatory hurdles and providing tax certainty.

Concerns Associated with the Act

- **Impact on Digital Payments:** The removal of the **Zero- MDR** provision may allow charges on **UPI** transactions, potentially affecting digital payment adoption, especially among **small merchants and low-income users**.
 - However, the move comes amid concerns that the zero-MDR regime has made the UPI ecosystem **financially unsustainable**. With UPI expected to add 600 million users and process 100–150 billion monthly transactions, sustained investment beyond government support will be essential.
- **Revenue Implications:** **Long-term tax exemptions** for foreign companies, **FIIs**, and other entities could reduce government tax revenues and affect **fiscal consolidation**.
- **Risk of Round-Tripping:** Relaxed rules for foreign fund managers may increase the risk of **tax avoidance and round-tripping**, despite anti-abuse safeguards.
- **Uneven Competition:** Targeted tax incentives for **foreign companies** may place **domestic Micro, Small and Medium Enterprises (MSMEs)** at a competitive disadvantage.

Way Forward

- **Link Tax Incentives to Performance:** Make long-term tax exemptions **performance-based**, with **periodic reviews and sunset clauses** linked to outcomes such as **job creation, exports, and technology transfer**.
- **Ensure Data Security:** Align incentives for **foreign cloud companies and data centres** with the **Digital Personal Data Protection (DPDP) Act, 2023** to balance investment promotion with **data protection and cybersecurity**.
- **Strengthen Domestic Manufacturing:** Complement tax incentives with expanded **Production Linked Incentive (PLI) schemes** and **R&D support for MSMEs** to build a resilient domestic electronics supply chain.

- **Prevent Tax Abuse:** Strengthen SEBI and Central Board of Direct Taxes (CBDT) oversight through AI-based monitoring and strict enforcement of the General Anti-Avoidance Rules (GAAR) to curb tax avoidance and round-tripping.

Relaxation of tax rules for foreign investment funds can attract capital but also raises concerns regarding tax avoidance. Examine.

Drishti Mains Question

National Handloom Day 2026

India observed the 12th National Handloom Day on 7th August 2026. Instituted in 2015, it commemorates the launch of the Swadeshi Movement in 1905 and recognises handloom weavers and allied workers.

Key Features of India's Handloom Sector	
Legacy	Dates to the Indus Valley Civilisation; spinning, weaving and Indian cotton/silk are mentioned in the Rigveda, Ramayana, Mahabharata and Kautilya's works.
Nature & Market	Manual, household-based and region-specific craftsmanship; contemporary design, digital platforms and exhibitions are widening market reach.
Census & Registration	Four censuses: 1987-88, 1995-96, 2009-10 and 2019-20. Pehchan/e-Pehchan identifies workers; the portal launched in January 2025 and approved about 84,000 additional workers by July 2026.
Employment	Fourth All India Handloom Census 2019-20: 31.45 lakh handloom households and 35.22 lakh weavers/allied workers; 8.48 lakh in pre- and post-loom work.
Women & Inclusion	Women: over 72% (about 25.46 lakh). Workforce includes 4.84 lakh SC, 6.28 lakh ST and 12.67 lakh OBC workers.
Rural Base	Of 28.20 lakh looms, 25.30 lakh were rural and 2.90 lakh urban - nearly nine in ten looms are in rural India.

Government Initiatives Related to the Handloom Sector

National Handloom Development Programme: NHDP (2021-22 to 2025-26) supports welfare, infrastructure, production, credit and marketing through public bodies, cooperatives, producer companies and banks. About 200 Handloom Producer Companies were formed during 2020-21 to 2026-27.

Key components

- **Small Cluster Development Programme:** Up to ₹2 crore per cluster for looms, lighting, worksheds, design and

marketing. 357 clusters sanctioned from 2021-22 to June 2026-27; ₹94 crore released in 2025-26.

- Up to 100% workshed subsidy; daily output +53%, working hours +60%, monthly working days +28% and daily earnings +42% (₹328 to ₹465).
- **Electronic Jacquard Impact:** Raised daily production by 55%, wages by 56% (₹360 to ₹560) and monthly income from ₹9,600 to ₹15,000; 96% reported productivity gains.
- ❖ **Handloom Marketing Assistance:** Over 960 expos, craft melas and exhibitions held during 2021-22 to June 2026-27. Sant Kabir Award: ₹3.5 lakh plus honours; National Handloom Award: ₹2 lakh plus honours. The 2024 award cycle honoured 24 recipients.
- ❖ **Concessional Credit:** Weaver MUDRA loans at 6% for three years, with interest subvention up to 7%, margin money and guarantee-fee support; over 40,000 loans worth more than ₹280 crore. **PMJJBY:** ₹2 lakh life cover; **PMSBY:** ₹2 lakh for accidental death/total disability and ₹1 lakh for partial disability.
- ❖ **Expansion of Social Security:** Insurance enrolment rose 80.8%, from 1,11,957 (2021-22) to 2,02,393 by May 2026-27; released funds rose 63.6%, from ₹204.83 lakh to ₹335 lakh. Scholarships up to ₹2 lakh annually for two children; 531 applications sanctioned by June 2026.
- ❖ **Mega Cluster Development:** Each covers at least 10,000 handlooms and can receive up to ₹30 crore over five years. Eight functional and four under implementation; ₹370.72 crore released for 1,62,682 weavers.
- ❖ **Special Infrastructure:** Need-based projects receive up to ₹12 crore for product development, quality, productivity, value addition and marketing. Six Craft Handloom Villages are operational and three under implementation.

- **Raw-Material Supply Scheme (RMSS):** Freight reimbursement on all yarn and 15% price subsidy on specified yarns. NHDP and RMSS continued in 2026-27 with allocations of ₹205 crore and ₹200 crore, respectively.

Union Budget 2026-27 and GST Rationalisation

- **GST Rationalisation:** GST on 36 handicraft items, cotton handloom rugs, handwoven carpets, and sewing machines reduced from 12% to 5% after the 56th GST Council meeting (September 2025).

- **National Fibre Scheme:** Strengthen the textile raw-material base and self-reliance in **natural, man-made, and new-age fibres**.
- **Textile Expansion and Employment Scheme:** Modernise traditional clusters through machinery support, technology upgrades, and common testing and certification centres.
- **National Handloom and Handicraft Programme:** Integrate existing schemes for coordinated and targeted sectoral support.
- **Mahatma Gandhi Gram Swaraj Initiative:** Support khadi, handloom, handicrafts, village industries, rural youth, and **ODOP products** through branding, skilling, quality improvement, and global market linkages.
- **ODOP Integration:** Promote district-specific products, local entrepreneurship, balanced regional development, branding, and wider market access.

Strengthening Market Access and Buyer Confidence

- **Growing Export Presence:** Handloom exports reached **₹1,330.96 crore in 2025–26**; major destinations were the **UAE, US, France, UK, and Spain**.
- **Digital Market Access:** Around **1.50 lakh weavers and handloom entities** joined **GeM** to supply directly to government buyers.
- **Direct-to-Consumer Platform:** **Indiahandmade** enables verified weavers and producer groups to sell authentic and **GI-tagged products** directly to consumers; **4,186 weavers and artisans** were onboarded between **2023–24 and June 2026**.
- **Handloom Mark:** Launched in **2006** to certify genuinely handwoven products and identify manufacturers; **29,402 registrations** and **815 retail outlets** by **30 June 2026**.
- **India Handloom Brand:** Launched on **7 Aug 2015** for premium natural-fibre products meeting standards on design, colourfastness, skin-friendly dyes, social compliance, zero defects, and environmental impact; **2,305 registrations across 184 categories** by **June 2026**.
- **GI Protection:** Under the **Geographical Indications of Goods Act, 1999**, GI registration prevents unauthorised use and enables legal action against counterfeit production and marketing.
 - **GI Coverage:** By **Aug 2026**, registrations covered **106 handloom products, six product logos, and 227 handicraft products**.
- **Legal Reservation for Handlooms:** The **Handlooms (Reservation of Articles for Production) Act, 1985** reserves **11 textile articles** exclusively for handloom production, protecting weavers from powerlooms and mills.

RBI to Resume Licensing of Urban Cooperative Banks

RBI will resume 'on-tap' UCB licensing after over two decades; new licences had been paused in 2004 following governance and financial instability under the 1993-2001 liberalised regime.

- The Union Cooperation Ministry has urged UCBs to join the **National Urban Cooperative Finance and Development Corporation (NUCFDC)** to strengthen technology, cybersecurity, employee training, and depositor confidence.

Draft Eligibility Conditions

Condition	Requirement
Operational Track Record	Credit cooperative society must have at least 10 years of active operations.
Capital Base	Minimum deposits of ₹10,000 crore and net worth of at least ₹300 crore.
Financial Health	Capital to Risk-Weighted Assets Ratio (CRAR) of at least 12% and net NPA ratio not above 3% at the time the license is granted.
Geographical Footprint	Initial priority to societies registered under Multi-State Co-operative Societies Act, 2002 with operations across multiple States.
Interest-Rate Reform	RBI also proposes rationalising Marginal Cost of Funds Based Lending Rate (MCLR) and External Benchmark Lending Rate (EBLR) to standardise interest calculation and benchmark resets, improve transparency, transmission and consumer protection.

NUCFDC

- It is a specialised **NBFC** registered with the RBI and the apex umbrella organisation for UCBs.
- It supports UCBs through shared digital banking technology, cybersecurity, liquidity assistance, regulatory compliance and capacity building, helping smaller banks reduce costs and improve governance.
- NUCFDC aims to secure official Self-Regulatory Organisation (SRO) status after meeting the required Rs 300 crore paid-up capital threshold, strengthening self-regulation, governance and compliance among UCBs.
- NUCFDC is currently headquartered in **New Delhi** and is proposed to be shifted to **Maharashtra**, as **Maharashtra, Gujarat, Goa and Karnataka** account for nearly **70% of UCBs**.

Geographical Indications in India

GI tags link distinctive products to their geographical origin, protect cultural heritage and misuse, and **improve market recognition for artisans, farmers, weavers and producer groups**.

- Government support in finance, exports, tourism, and marketing is strengthening rural incomes, consumer trust, and global visibility.

Geographical Indication

- **Meaning:** A designation for products whose quality, reputation or other characteristics are essentially attributable to a specific geographical origin.
- **International Framework:** Article 22(1) of TRIPS defines GIs; they are recognised under the Paris Convention **for the Protection of Industrial Property** and the WTO’s TRIPS Agreement.

Indian GI Framework

Aspect	Key Point
Law	Geographical Indications of Goods (Registration and Protection) Act, 1999 and Rules, 2002.
Administration	GI Registry administers registration and protection; registration lasts 10 years and is renewable for successive 10-year periods.
Cancellation	Section 27 permits cancellation, variation or rectification where registration was improperly obtained, misleading, generic or legally non-compliant.
2025 Amendment	Application and related fees reduced by 80%; renewal fee lowered from ₹3,000 to ₹500.
Growth	Over 800 registered GI products; 607 granted since 2014; authorised users increased from 365 to about 29,000 by Jan 2025.
Leading States	As of Dec 2025– UP (81), Tamil Nadu (76) and Maharashtra (55) GI-tagged products.
First Registrations	Darjeeling Tea was India’s first registered GI; early registrations included Aranmula Kannadi and Pochampally Ikat.

Significance

- **Heritage and Authenticity:** Preserve traditional knowledge and craftsmanship while assuring consumers of origin, quality and cultural value.
- **Livelihoods:** Greater visibility, bargaining power and premium-market access **can raise artisan incomes by 20-30%** and support local livelihoods.
- **Exports:** GI recognition improves global access; 30 metric tonnes of Muzaffarnagar jaggery were exported to Bangladesh in 2025, benefiting 545 farmer-members.

Government Initiatives to Strengthen India’s GI Ecosystem

Initiative	Key Provision / Outcome
PM Ekta Malls	₹5,000 crore allocated for dedicated retail spaces for GI products, ODOP items and handicrafts across States.
One Station One Product	Railway stalls enable artisans, farmers, weavers and producer groups to directly sell regional products.
GI-Based Tourism	Weaving-village and tea-estate trails; Assam proposed the Muga Silk Trail, Silk Tourism Park and Muga Utsav.
MSME Innovative Scheme	IPR component reimburses up to 100% of GI-registration expenses, capped at ₹2 lakh per registered GI.
Legal Support	Development Commissioner for Handicrafts provides up to ₹1.5 lakh for GI enforcement and legal expenses.
Handloom Promotion	By June 2026, 104 GI-registered handloom products and 53 registrations were supported under Handloom Marketing Assistance.
APEDA & Export Markets	Supports GI agricultural exports; examples include Karnataka GI fruits to Maldives, Salem Sago to Canada, Joha Rice to UK and Italy, and Tezpur Litchi to Dubai.
Trade & Promotion	GI protection included in trade talks; Indi Lime exported to Oman after India-Oman CEPA, while exhibitions and events promote GI goods.

Geographical Indication tags can transform India’s traditional products into instruments of cultural preservation and rural economic growth. Discuss.

Drishti Mains Question

Critical Minerals: The Foundation for Strategic Resource Security

The IEA’s **Global Critical Minerals Outlook 2026** warns that concentrated refining capacity, supply disruptions, and China’s 2025 rare earth export controls are increasing global mineral supply risks. In response, India is accelerating the **National Critical Mineral Mission**, overseas acquisitions, and global partnerships to reduce import dependence and support clean energy and manufacturing.

Critical Minerals and Their Importance

- **Meaning:** Non-fuel metallic or non-metallic elements vital for advanced technologies; countries classify them by economic importance, substitutability and supply risk.

- **India's List:** 30 minerals identified by the Ministry of Mines, including Lithium, Cobalt, Nickel, Graphite, Copper and Rare Earth Elements (REEs).
- **Clean Energy:** Essential for EVs, batteries, solar cells and wind turbines—Silicon/Tellurium/Indium/Gallium for solar; Neodymium/Dysprosium for wind; Lithium/Cobalt/Nickel for storage.
 - **NITI Aayog's Net Zero Scenario** projects India's cumulative demand for **Critical Energy Transition Minerals (CETMs)** at **~169 Mt by 2070**, nearly **51% higher** than the **Current Policy Scenario**.
- **Strategic Uses:** Inputs for semiconductors, AI, electronics, telecom, missiles, jet engines and rare-earth magnets; support **SDGs 7** (Affordable & Clean Energy), **9** (Industry, Innovation & Infrastructure), and **13** (Climate Action).
- **Security Dimension:** Export controls and resource nationalism (e.g., China's **2025 rare earth export controls**) have turned critical minerals into the "new oil" of the clean-energy era; access is vital for energy, industrial, technological and defence security.

Major Reserves/Occurrence in India

Mineral	Major Occurrence
Lithium	Karnataka and J&K.
Cobalt & Nickel	Associated with copper ores; small deposits in Odisha and Jharkhand.
REEs	Andhra Pradesh, Karnataka, Odisha and Kerala; Kerala monazite sands are rich in REEs. Gujarat and Rajasthan also hold 1.29 Mt of in-situ rare-earth oxide (REO) in hard-rock deposits.
Graphite	Arunachal Pradesh, Jharkhand and Tamil Nadu.

How Geopolitics Is Reshaping Supply Chains

- **Weaponised Controls:** China's licensing restrictions exposed global vulnerabilities; European prices of gallium, dysprosium and terbium reached nearly five times Chinese prices (**IEA's Global Critical Minerals Outlook 2026**).
- **Friendshoring:** US-led Minerals Security Partnership, US–Australia Compact and EU Critical Raw Materials Act seek trusted parallel supply chains.
- **Chokepoint Risks:** China's May 2026 sulphuric-acid restrictions and Strait of Hormuz disruptions affected copper, lithium, cobalt, sulphur and aluminium flows.
- **China+1:** Countries are diversifying sourcing, domestic mining/refining and partnerships with trusted suppliers.

Global Approaches

Country/Bloc	Approach
China	Dominance across mining, processing, refining and exports; uses minerals for geo-economic influence.
US	Friendshoring, alliances, domestic processing and overseas investment to de-risk from China.
EU	Strategic autonomy through extraction, processing, recycling and supplier diversification.
Australia	Trusted supplier using reserves, downstream processing and partnerships.
Japan	Resource diplomacy, overseas assets, long-term contracts and strategic stockpiles.
India	Integrated "Mine-to-Market" strategy: domestic exploration, overseas acquisitions and global partnerships.

Major Steps Taken by India in this Regard

- **NCMM (2025):** Implemented from FY 2024–25 to FY 2030–31; covers exploration, mining, processing, recycling, R&D, skills and stockpiling. Targets 1,200 exploration projects, production of 15 critical minerals, 270 kt annual recycling capacity to recover 40 kt, 1,000 patents and 7 Centres of Excellence.
- **MMDR Amendment, 2023:** Empowers the Centre to auction 24 of India's 30 identified critical minerals.
 - **Khanij Bidesh India Limited (KABIL):** Secured 15,703 hectares in Catamarca, Argentina, for lithium exploration; target of 50 overseas mining assets by 2030–31.
- **International Partnerships:** India–US Critical Minerals and Rare Earths Framework (May 2026), Pax Silica Initiative (2026), and Quad Critical Minerals Initiative (2025), which plans to mobilise up to USD 20 billion.
- **Rare Earth Corridors:** Proposed in Odisha, Kerala, Andhra Pradesh and Tamil Nadu under Budget 2026–27.
- **Scheme to Promote Manufacturing of Sintered Rare Earth Permanent Magnets (REPM):** Seeks a 6,000 Metric Tonnes Per Annum (MTPA) integrated ecosystem from rare-earth oxides to finished permanent magnets.
- **PLI Schemes:** Support ACC batteries, solar PV modules, electronics and semiconductors.

Key Challenges

- **Heavy Import Dependence:** Lithium is almost fully imported; cobalt mainly from DRC (processed mainly in China), nickel from Indonesia and rare-earth magnets from China.

- **“Missing Middle”:** India lacks battery-grade chemicals, high-purity graphite, rare-earth separation, advanced smelting and refining, where 70–90% of global value addition is concentrated.
- **China’s Midstream Monopoly:** Refines over 90% of REEs and graphite, around 75% of cobalt and nearly 70% of lithium chemicals.
- **Inadequate Exploration:** Current surveys are largely shallow, while deep-seated minerals require geophysical, geochemical, hyperspectral and airborne techniques.
- **Weak Project Economics:** Small/low-grade or remote/LWE-affected deposits and weak logistics, power and water reduce viability.
- **Limited Private Capital:** High-risk, patient funding is constrained by weak junior-mining and risk-sharing ecosystems.
- **Weak Circular Economy:** Limited recovery from e-waste, batteries, tailings, fly ash and red mud due to fragmented collection and technology gaps.
- **Resource Nationalism:** Export restrictions by producers such as Indonesia (Nickel) and Latin American countries increase overseas-sourcing uncertainty.
- **No Strategic Stockpile:** India lacks comprehensive critical-mineral reserves comparable to Strategic Petroleum Reserves.

Measures to Secure Supply

- **Explore Deep-Seated Deposits:** Use AI, airborne geophysics, hyperspectral mapping and 3D geological modelling under NCMM.
- **Build Processing Capacity:** Create battery-grade lithium, nickel, cobalt and rare-earth refining, mineral parks and integrated value chains through incentives and PPPs.
- **Diversify Overseas Sources:** Use KABIL and long-term acquisitions in Australia, Argentina, Chile, Namibia and African countries.
- **Create Strategic Reserves:** Establish Strategic Critical Mineral Reserves to cushion geopolitical and supply shocks.
- **Strengthen Circular Economy:** Scale urban mining and recycling of e-waste, batteries, end-of-life vehicles, tailings, fly ash and red mud through EPR.
- **Improve Ease of Mining:** Streamline clearances, improve geological data and introduce risk-sharing to attract private investment.
 - **Adopt a National Critical Minerals Strategy:** Set mineral-specific risk thresholds, strengthen inter-ministerial coordination, monitor supply chains and integrate minerals into industrial, energy and national-security planning.

“Critical minerals are the new oil of the 21st century.” Examine the strategic significance of critical minerals in the context of global energy transition and geopolitical competition.

Drishti Mains Question

India’s Solar Growth Enters a New Era

India’s installed solar capacity rose more than fiftyfold—from about 3 GW in 2014 to 162.15 GW on 30 June 2026—making solar the largest component of renewable-energy capacity.

- Growth has been driven by policy support, infrastructure, domestic manufacturing and public participation, with PM Surya Sarovar Yojana as the latest push.

Solar Capacity Snapshot

Indicator	Status
Global Position	Added 37 GW in 2025, surpassing the US in annual additions and becoming the world’s second-largest solar growth market.
Policy Foundation	National Solar Mission, Solar Parks Scheme, Green Energy Corridor Programme and competitive bidding.
Regional Expansion	Rajasthan, Gujarat, Karnataka, Tamil Nadu, Andhra Pradesh and Maharashtra account for a major share.
Utility-Scale Solar	About 118.79 GW, including Bhadla, Pavagada and Rewa solar parks.
Rooftop Solar	Grid-connected rooftop systems contribute 27.88 GW; the balance comes from hybrid and off-grid systems.
Record Addition	44.61 GW added in FY 2025–26, nearly twice the 23.83 GW added in FY 2024–25.
Manufacturing	Domestic module manufacturing rose from 2.3 GW in 2014 to nearly 172 GW by March 2026.
Non-Fossil Capacity	283.46 GW total, including 274.68 GW renewable capacity; FY 2025–26 added a record 55.29 GW.
Renewable Share	Renewables met 51.5% of electricity demand on 29 July 2025, the highest recorded during the period.

Policy Architecture Driving Expansion

- **Vision and Targets:** Panchamrit goals announced at COP26 include 500 GW non-fossil capacity by 2030 and net-zero emissions by 2070, reflected in India’s updated NDCs. **Mission LiFE promotes sustainable consumption.**
- **Demand Creation:** National Solar Mission created sustained demand; Renewable Purchase Obligations provide market certainty, with the 2026–27 target fixed at

35.95%. PM Surya Ghar and PM-KUSUM broaden household and farmer participation.

- **Lower Costs:** e-Reverse Auctions and SECI-led procurement reduced tariffs from about ₹18/unit in 2010 to ₹2.44/unit at Bhadla in 2017. In 2025, utility-scale solar LCOE was USD 35/MWh versus the global average of USD 44/MWh.
- **Enabling Infrastructure**
 - **Solar Parks:** Target raised from 20 GW to 40 GW in 2017. By February 2026, 54 parks with 39,188 MW sanctioned capacity had been approved.
 - **State Leaders:** Gujarat—12,150 MW across 7 parks; Rajasthan—10,726 MW across 11 parks; Madhya Pradesh—4,208 MW across 9 parks.
 - **Major Parks:** Bhadla (Rajasthan), Pavagada (Karnataka) and Rewa (Madhya Pradesh). Bhadla’s 2,245 MW capacity shows how common land and transmission infrastructure can attract investment and lower tariffs.
 - **Grid Integration:** By February 2026, the Green Energy Corridor enabled evacuation of 24,567.8 MW; planned projects in 10 States aim to support around 44 GW.
- **Supporting Manufacturing**
 - **Manufacturing PLI:** Launched in 2021 with ₹4,500 crore and expanded in 2022 by ₹19,500 crore to promote

high-efficiency PV modules and reduce imports. By October 2024, it attracted about ₹35,000 crore investment and generated around 10,000 direct jobs.

- **ALMM:** Introduced in 2019 to ensure prescribed quality and performance standards for modules and cells used in specified projects while supporting domestic manufacturing.

India’s Solar Growth Enters a New Era

- **Related Initiatives:** PM Surya Ghar Muft Bijli Yojana, PM-KUSUM
- **Global:** International Solar Alliance, One Sun One World One Grid (OSOWOG), Green Grids Initiative (GGI)–OSOWOG– launched by India and the UK at COP26.
 - During its **G20 Presidency**, India helped build consensus on tripling global renewable-energy capacity by 2030.

Indian Rupee: From Overvaluation to Undervaluation

RBI Governor stated that the rupee is undervalued in nominal and REER terms despite nearly 5.8% depreciation against the US dollar in 2026. The RBI attributes the weakness mainly to global financial headwinds, not deterioration in India’s macroeconomic fundamentals.

NEER and REER

Metric	Meaning / Interpretation	Key Details / Limitations
NEER (Nominal Effective Exchange Rate)	■ A weighted average index measuring the nominal (face-value) strength of the Indian rupee against a basket of currencies of India’s major trading partners.	■ RBI uses 6-currency basket (USD, Euro, Yuan, Pound, Yen, Hong Kong Dollar) and 40-currency basket covering nearly 88% of merchandise trade. ■ Base: 2015-16 = 100. Does not adjust for inflation or price-level differences b/w countries.
REER (Real Effective Exchange Rate)	■ Inflation-adjusted version of NEER - a trade-weighted exchange rate index that measures the value of the Indian rupee against the currencies of its major trading partners after adjusting for inflation differentials. ■ A stronger indicator of external competitiveness and currency valuation, and a proxy for assessing currency misalignment relative to PPP (Purchasing Power Parity).	■ Above 100 = overvalued and exports become costlier; below 100 = undervalued and may improve export competitiveness. ■ Ignores non-price factors such as quality, logistics and technology, and is less suited to India’s services-led exports because it is merchandise-trade weighted.

Why Does the RBI Consider the Rupee Undervalued?

- **Correction in External Valuation:**
 - Average exchange rate moved from Rs 84.4 per US dollar (November 2024) to about Rs 95.5 (May 2026). NEER fell from 91.68 to 77.19; REER declined from 108.03 to 89.08 and recovered slightly to 91.26 in June 2026, indicating a shift from overvaluation to undervaluation (below 100).
- **Strong Fundamentals:**
 - GDP growth above 6%, moderating inflation, a stable external sector and forex reserves covering over 11

months of imports suggest depreciation does not reflect underlying economic weakness.

- **Global Headwinds:**
 - Elevated crude oil prices, a stronger US dollar, geopolitical tensions and FPI outflows increased dollar demand and placed temporary downward pressure on the rupee.
- **REER-Based Assessment:**
 - The rupee is trading below its equilibrium value, implying depreciation beyond what India’s macroeconomic fundamentals would ordinarily justify.

Impact of an Undervalued Rupee

Potential Benefits	Associated Risks
■ Makes Indian exports relatively cheaper and improves export competitiveness. ■ Strengthens domestic manufacturing against imports and supports Make in India. ■ May reduce the trade deficit if export growth exceeds import growth. ■ However, exchange-rate advantage alone is insufficient without higher productivity, efficient logistics and stronger manufacturing.	■ Raises the cost of crude oil, fertilisers, electronics and other imports, causing imported inflation. ■ Increases input costs for import-dependent industries. ■ Raises servicing costs of External Commercial Borrowings (ECBs). ■ May widen the Current Account Deficit (CAD) when oil prices are high.

What is the Real Effective Exchange Rate (REER)? Examine its significance in assessing currency valuation and external competitiveness.

Drishti Mains Question

Minimum Support Price and the Need for Alternative Farmer Support

Madhya Pradesh farmers ended protests after the State agreed to procure **60% of estimated summer moong output**, against the earlier **25% limit**, reviving debate over MSP sustainability and the need for structural reforms in agricultural support.

Why MSP is Difficult to Sustain

- **Fiscal Burden:** Extending MSP-backed procurement to all 23 notified crops could cost ₹10-17 lakh crore, reducing funds for irrigation, R&D, storage and rural infrastructure.
- **Market Distortion:** A legally binding price floor means no transaction can legally occur below the MSP. This artificially inflates prices, discourages private sector participation, and forces the government into becoming the sole buyer (monopsony).
- **Cropping Pattern Distortion:** Assured Rice–wheat procurement incentivises monoculture, discourages diversification towards pulses, oilseeds and horticulture and undermines nutritional and import security.
- **Ecological Damage:** Paddy cultivation in Punjab and Haryana contributes to groundwater depletion, soil degradation and stubble burning.
- **Limited & Unequal Benefits:** The Shanta Kumar Committee (2015) found that only about **6% of farmers** benefit from MSP procurement, mainly in Punjab, Haryana and among large surplus-producing farmers, excluding most small and marginal farmers.

- **Storage Burden:** Excess procurement leaves FCI with rice and wheat stocks beyond PDS and buffer needs, increasing storage costs and wastage.
- **Outdated Costs:** Lagged estimates may not reflect sudden rises in fertiliser, diesel, labour and other input costs.
- **WTO Compliance Concerns:** For developing countries, the WTO AoA (Agreement on Agriculture) caps trade-distorting Amber Box support at the 10% de minimis limit; aggressive MSP procurement risks breaching this threshold and triggering trade disputes.

Minimum Support Price

- **Meaning:** Pre-announced price at which the government procures specified crops, protecting farmers from price crashes and distress sales.
- **Announcement:** Declared at the beginning of the sowing season on recommendations of the **Commission for Agricultural Costs and Prices (CACP)**.
 - CACP is an attached office of the **Ministry of Agriculture & Farmers Welfare**.
 - Final MSP approved by the **Cabinet Committee on Economic Affairs (CCEA)**.
- **Coverage:** MSP for 22 mandated crops and Fair and Remunerative Price for sugarcane.
- **Procurement Agencies:** FCI, NAFED and Cotton Corporation of India.

Cost Formula

- **A2:** Paid-out expenses such as seeds, fertilisers, hired labour, fuel and irrigation.
- **A2+FL:** A2 plus imputed family labour; current basis for MSP.
- Since 2018–19, MSP is fixed at least **1.5 times** the all-India weighted average cost of production.
- **C2+50%:** The **National Commission on Farmers (NCF)**, chaired by **Dr. M.S. Swaminathan** recommended fixing the MSP at least 50% above the comprehensive C2 cost (including A2+FL, imputed rent on owned land, and interest on fixed capital).
 - Demanded by farmers but not accepted due to fiscal and market concerns.

Alternatives Beyond MSP

Measure	Purpose
Minimum Income Support (MIP)	Per-acre transfers and enhanced PM-KISAN support without market distortion.
Price Deficiency Payment Systems (PDPS)	Compensate the MSP–market price gap under PM-AASHA without physical procurement. A combination of PM-AASHA (Price Deficiency Payments) , MIP , crop diversification , FPOs , and market reforms offers a more sustainable approach to protecting farmer incomes.

Market Infrastructure	Expand AIF, warehouses, cold chains and e-NAM for transparent and competitive price discovery.
FPOs and Cooperatives	Improve bargaining power of small farmers through FPOs, enabling better market access and price realisation.
Crop Diversification	Promote pulses, oilseeds and millets through agro-climatic planning - for import reduction and ecological sustainability.
Contract Farming	Enable farmer-buyer agreements with legal safeguards to ensure assured markets, stable prices and technology transfer.
R&D and Extension	Promote climate-resilient seeds, precision farming, Soil Health Cards and Per Drop More Crop.
Targeted MSP	Restrict assured procurement to strategic crops and regions. Allow States to design context-specific procurement models based on local agricultural conditions.
WTO-Compatible Support	Shift from Amber Box price support to Green Box income and infrastructure support.

Discuss the limitations of the existing Minimum Support Price (MSP)-based agricultural support system. Suggest policy measures to protect farmers beyond MSP.

Drishti Mains Question

India as a Tourism Hub

Govt. highlighted India's potential as a **major tourism hub**, driven by better **infrastructure, connectivity, digital services, travel facilitation, global promotion**, and sustainable tourism.

Tourism as a National Growth Engine

- **Economic Linkages:** Supports **hospitality, transport, handicrafts, food services, wellness, culture**, and local enterprises.
- **Domestic Tourism:** Around **2.9 billion visits in 2024**, exceeding pre-pandemic levels.
- **International Arrivals:** Nearly **2.02 crore in 2025**.
- **GDP Contribution:** About **₹15.73 lakh crore**, or **5.22% of GDP**, in 2023–24.
- **Employment:** Supported around **84.6 million jobs in 2023–24**.
- **Foreign Exchange Earnings:** Nearly **USD 31.7 billion in 2025**.
- **Regional Development:** Generates local livelihoods, promotes small businesses, and improves infrastructure in less-developed regions.

- **Global Potential:** India's **heritage, landscapes, spirituality, wellness traditions**, and experiential tourism strengthen its global tourism prospects.

- **Heritage:** **45 UNESCO World Heritage Sites** showcase India's cultural and natural richness.
- **Nature:** **106 National Parks, 18 Biosphere Reserves**, forests, beaches, islands and diverse landscapes support nature tourism.

Major Tourism Segments

- **Spirituality:** India, the birthplace of **Hinduism, Buddhism, Jainism and Sikhism**, hosts major pilgrimage centres such as **Char Dham, Kashi, Ayodhya, Bodh Gaya, Amritsar, Tirupati and Puri**.
- **Buddhist Tourism:** **Bodh Gaya, Sarnath and Kushinagar** promote India's civilisational links with Asia.
- **Heritage and Cultural Tourism:** **Forts, temples, museums, festivals, handicrafts, dance, music** and regional cuisines showcase India's cultural heritage.
- **Wellness and Medical Tourism:** **Yoga, Ayurveda**, affordable treatment and advanced medical facilities attract international visitors.
- **Eco-tourism and Wildlife Tourism:** Parks, reserves, wetlands and Himalayan regions support conservation tourism and local livelihoods.
- **Rural and Community Tourism:** **Homestays, crafts, cuisine** and village experiences generate income for rural communities.
- **MICE Tourism:** Modern convention infrastructure supports **meetings, incentives, conferences and exhibitions**, creating year-round activity.
- **Coastal Cruise and Adventure Tourism:** India's **7,500-km coastline**, islands, rivers, deserts and mountains support cruises, water sports, trekking, rafting and safaris.

Role of Infrastructure-Led Development in Transforming Tourism Destinations

- **Swadesh Darshan:** **76 projects** worth **₹5,295.24 crore** sanctioned across **14 thematic circuits; 75 completed**.
- **Swadesh Darshan 2.0:** **53 destination-centric projects** worth **₹2,207.08 crore** sanctioned for sustainable infrastructure and better visitor experience.
- **PRASHAD Scheme:** **54 projects** worth **₹1,726.24 crore** sanctioned for pilgrimage and heritage destinations.
- **Capital Support to States:** **40 projects** across **23 States**, worth **₹3,295.76 crore**, approved for globally competitive tourist centres.
- **Challenge-Based Destination Development:** **37 projects** worth **₹687.99 crore** sanctioned, mainly for **spiritual and eco-tourism** destinations.

- **Improved Connectivity:** Highways, modern railway stations, **Vande Bharat**, **UDAN**, e-buses and last-mile transport are improving tourist access.

- **Viksit UDAN:** Over **₹28,840 crore** allocated for more than **100 aerodromes** and **200 helipads** in Tier-2, Tier-3 and remote regions.

India's Efforts to Improve Ease of Travel and Digital Tourism

- **e-Tourist Visa:** Available to nationals of **175 countries** through **33 airports**, **19 seaports** and **4 land ports**, simplifying international travel.
- **Visa Facilitation:** Expanded e-Visa categories, digital authorisation and designated entry points have eased tourist access.
- **NIDHI+:** Unified platform for registration, recognition and classification of tourism service providers, with online applications, payments and transparent approvals.
- **Tourism Startups:** **1,497 DPIIT-recognised startups** across **262 districts** generated nearly **13,919 jobs** as of **April 2023**.
- **Regional Entrepreneurship:** About **58%** of tourism startups are located in **Tier-II and Tier-III cities**.
- **Technology-Driven Services:** Startups support online booking, homestays, rural tourism, smart mobility and AI-based travel planning.
 - **Notable Innovations:** **Highway Delite** connects highway amenities, **VilloTale** promotes rural homestays and **udChalo** serves defence personnel and veterans.

Role of Sustainable and Inclusive Tourism in Long-Term Growth

- **Sustainable Tourism Model:** Integrates environmental protection, community participation, economic inclusion and quality visitor experiences.
- **Travel for LiFE:** Promotes responsible travel, resource conservation and eco-friendly choices.
- **Destination Diversification:** Nature-based activities reduce pressure on crowded destinations and support regional tourism.
- **Community Benefits:** Creates jobs and income for guides, artisans, homestays and local enterprises.
- **Protection:** Preserves local culture, heritage, biodiversity and natural resources.
- **SDG Contribution:** Supports **SDG 8.9**, **SDG 12.B** and **SDG 14.7**.

Tourism can act as a driver of economic growth, employment and cultural diplomacy in India. Discuss.

Drishti Mains Question

ILO Convention No. 193 on Decent Work in the Platform Economy

At the **114th International Labour Conference, Geneva**, the ILO adopted **Convention No. 193 (Decent Work in the Platform Economy)**, the world's first legally binding treaty for app-based gig and platform workers.

- Under the ILO's tripartite voting framework (comprising government, employer, and worker representatives), India's government delegates abstained, whereas both its employer and worker delegates voted in favor of the treaty.

Key Provisions

- Adopted in **June 2026**.
- Ensures minimum wages, timely payment, social security and occupational safety.
- Applies irrespective of classification as employees, contractors or platform partners.
- Mandates transparency in algorithms governing work allocation, wages, ratings and account deactivation.
- Requires human oversight of significant automated decisions.
- Directs governments to classify workers **based on the actual nature of work rather than contractual labels**, preventing the misclassification of employees as independent contractors.

Why India Abstained

- **Domestic Law:** Code on Social Security, 2020 already covers gig and platform workers.
- **Conformity Principle:** India generally ratifies ILO Conventions only after ensuring **full alignment with domestic laws**.
- **Federal Concerns:** Labour is in the Concurrent List; the Centre prefers retaining flexibility for States to frame their own labour regulations.
- **Economic Concerns:** Provisions on **worker reclassification** and **algorithmic transparency** could increase compliance costs, affect the platform business model, and discourage digital innovation.
- **Cost and Consequences of Abstention:**
 - May allow worker misclassification and opaque algorithmic control.
 - Denies workers an international legal baseline against platform abuses.
 - Widens the protection gap with ratifying countries.
 - May affect India's image as the voice of the Global South.

Gig Economy in India

- **Workforce:** 7.7 million in 2020–21, projected to reach 23.5 million by 2029–30 (NITI Aayog)—about 6.7% of the non-agricultural workforce.
- **Growth drivers:** Expansion of internet connectivity (96.96 crore connections in 2024), widespread smartphone adoption (85.5% of households), and app-based platforms under Digital India.
 - Rapid growth of e-commerce, quick commerce, ride-hailing, and food delivery platforms.
 - Urban demand for on-demand services.
 - High unemployment, underemployment, and a large semi-skilled workforce have made gig work an accessible source of income.
 - Preference among youth for flexible work and multiple incomes.

Indian Legal Framework

- **Code on Social Security, 2020:** Defines gig and platform workers.
 - Aggregators must contribute 1–2% of annual turnover, capped at 5% of worker payouts, to a social security fund.
 - Benefits include life and disability cover, accident insurance, health, maternity, pension and crèche facilities.
- **State Initiatives:**
 - **Rajasthan:** Platform-Based Gig Workers Act, 2023.
 - **Karnataka and Telangana:** Drafted welfare boards and fund mechanisms.

Electronic Gold Receipt

India's **EGR ecosystem** is promoting **transparent, standardised, and exchange-based trading** of physical gold. By digitising gold ownership while allowing **physical delivery**, EGRs are modernising the bullion market and improving **investor confidence**.

Electronic Gold Receipt (EGR)

- **Gold Exchange Ecosystem:** The **SEBI-regulated Gold Exchange** provides a national platform for standardised trading of physical gold. It formalises India's fragmented bullion market and enables a **transparent, real-time domestic spot price**.
- **EGR:** An **EGR** is a SEBI-regulated electronic security representing ownership of physical gold stored in **SEBI-accredited vaults**.
 - EGRs are held in **demat form** and traded on recognised stock exchanges. Investors can convert them into physical gold when required.
 - Each EGR is **100% backed** by equivalent physical gold, generally of **995 or 999 fineness**, stored with a **SEBI-registered Vault Manager**.
 - The holder becomes the **beneficial owner** of the gold without storing or transporting it.

- **Need for EGR:** SEBI introduced the **Gold Exchange Framework, 2022** to address **opaque pricing, inconsistent purity, high transaction costs, fragmented trade, and dominance of the unorganised sector**.
- The framework aims to ensure **transparent price discovery, standardised quality, investor protection**, and easy conversion between dematerialised receipts and physical gold.
- **Key Participants in the EGR Ecosystem:**
 - **Vault Manager:** Stores, verifies, assays, and delivers physical gold under the SEBI Regulations, 2021.
 - **Depository (NSDL/CDSL):** Holds EGRs electronically in investors' demat accounts.
 - **Stock Exchanges (NSE/BSE):** Provide platforms for EGR trading.
 - **Stock Brokers:** Facilitate the purchase, sale, and redemption of EGRs.
 - **Investors:** Individuals, HUFs, NRIs, trusts, and institutions holding EGRs through demat accounts.
- **Working of EGR:**
 - **Stage I—Creation:** **995/999 purity gold** is deposited with a **SEBI-registered Vault Manager** for verification and storage. Equivalent EGRs are issued by the Depository and credited to the depositor's **demat account**.
 - **Stage II—Trading:** EGRs are traded on the **NSE and BSE Gold Exchanges** at prices linked to the domestic spot market. Ownership transfers electronically on a **T+1 settlement basis**, while the gold remains in the vault.
 - **Stage III—Redemption (Withdrawal):** Investors request physical gold through a broker or **Depository Participant (DP)**. The Vault Manager delivers the gold and extinguishes the corresponding EGR.
- **Significance of EGRs**
 - **Unified Price Discovery:** Creates a transparent **"One Nation, One Price"** domestic spot price based on real-time demand and supply.
 - **Reduced Dependence on Foreign Benchmarks:** Develops an Indian reference price, reducing reliance on **LBMA** and **Dubai benchmarks**.
 - **Storage and Security:** Eliminates individual locker costs and theft risks, as the gold is stored in secure, insured vaults.
 - **Quality Assurance:** Mandatory purity verification reduces adulteration risks in the unorganised market.
 - **Tax Efficiency:** **Zero GST** applies to EGR trading; **3% GST** applies only when EGRs are converted into physical gold.
- **Limitations of EGRs:** EGR adoption is limited by **low awareness, low trading volumes, developing vault infrastructure, storage and redemption charges, GST on physical withdrawal**, and competition from **Gold ETFs, digital gold, and Sovereign Gold Bonds**.

EGR and India International Bullion Exchange (IIBX)

- **IIBX:** Located in **GIFT City, Gujarat**, it is India's first international bullion exchange, regulated by the **IFSCA**. It facilitates bullion imports and international trading.
- **Link with EGRs:** IIBX supplies standardised bullion, while EGRs enable its domestic electronic trading.
- Gold imported through IIBX can be deposited in **SEBI-accredited vaults** and converted into EGRs after meeting prescribed purity and quality standards.

India Allows FDI in Inventory-Based E-Commerce Model for Exports

The government has revised the **Foreign Direct Investment policy** to permit foreign-funded e-commerce entities to maintain inventories exclusively for exporting goods manufactured or produced in India.

Key Facts About Revised Foreign Direct Investment (FDI) Policy

- **About:** Foreign-owned e-commerce companies can **now own and manage inventory in India**, provided the goods are domestically produced and sold only in overseas markets. The **Department for Promotion of Industry and Internal Trade** has created an **export-specific exception** to the restrictions on inventory-based e-commerce.
 - **However, foreign-funded inventory-based e-commerce remains prohibited for domestic retail sales**, preserving safeguards for small traders and the ban on FDI in multi-brand retail.
- **Objective:** The policy aims to improve **Indian sellers' access to global markets**, promote exports and reduce the compliance burden on MSMEs. The measure could **help Tier-2 and Tier-3 manufacturers, artisans and small businesses** export products such as handicrafts, garments, books, art, gems and jewellery.
- **Beneficiaries:** Global platforms such as Amazon and Walmart-owned Flipkart can source products from Indian manufacturers, manage inventories and handle export-

related paperwork, testing, labelling and traceability requirements.

- **Implementation:** The decision will take effect after the relevant notification under the **Foreign Exchange Management Act, 1999**.
- **Existing Policy:** India earlier permitted **100% FDI under the automatic route only in marketplace-based e-commerce and B2B e-commerce**, while prohibiting FDI in inventory-based models.
- **Concerns:** Critics argue that **separating export and domestic inventories may be difficult to monitor** and could eventually create pressure to permit foreign-funded inventory-based e-commerce in the domestic market.

FDI policy evolution for ecom

- | | |
|---|--|
| <p>2000: Foreign direct investments allowed in business-to-business e-commerce under automatic route</p> <p>2016: FDI permitted in marketplace model under automatic route</p> <p>■ FDI prohibited in inventory-based B2C e-commerce</p> <p>2018: Marketplace entities barred from exercising ownership or</p> | <p>control over inventory</p> <p>■ Curbs tightened on vendor purchases, exclusive arrangements</p> <p>2026: FDI allowed in inventory-based ecom entities for exclusively exporting domestically manufactured goods</p> <p>■ Ban on FDI in inventory-based domestic B2C ecom unchanged</p> |
|---|--|

Significance

- **Export Promotion:** Provides Indian sellers with an additional e-commerce channel to reach global consumers and expand merchandise exports.
- **Support for SMEs:** Benefits small and medium enterprises in sectors such as apparel, jewellery, handicrafts, home products and organic wellness by easing access to international markets.
- **Opportunities for Regional Manufacturers:** Enables manufacturers in tier-2 and tier-3 cities to widen their overseas customer base and participate more actively in global trade.
- **Long-Term Economic Impact:** Supports India's manufacturing and export-growth objectives, including the target of achieving \$80 billion in cumulative e-commerce exports by 2030.

Difference Between Major E-Commerce Models

Feature	Inventory-Based Model	Marketplace Model
Ownership of Goods	Owned by e-commerce entity	Owned by sellers
Role of Platform	An e-commerce entity owns the inventory and sells goods directly to consumers.	The platform acts as a facilitator between buyers and sellers.
FDI Status in India	Allowed only for exports under new policy	Up to 100% FDI is permitted under the automatic route
Domestic Retail	Foreign-funded inventory-based B2C e-commerce remains prohibited for domestic sales.	Foreign-funded platforms may facilitate domestic sales under marketplace rules but cannot own or control sellers' inventory
Export Potential	Direct export facilitation	Seller-driven exports

International Relations

India-Japan Maritime Security Pact

India and Japan signed a **Memorandum of Arrangement (MoA) on Maritime Security Cooperation** in New Delhi to deepen **defence and maritime security cooperation**.

- **About the MoA:** Provides a framework for cooperation between the **Indian Navy and Japan Maritime Self-Defense Force (JMSDF)** in **Maritime Domain Awareness, Search and Rescue, and HADR**.
- **Naval Shipbuilding and Repair:** Both sides will explore **joint naval shipbuilding and design** and reciprocal ship-repair arrangements, including greater use of Indian shipyards under **Make in India**.
- **Military Exercises:** Cooperation will expand through **Dharma Guardian, JAIMEX and Veer Guardian 26**.
 - **Veer Guardian 26** will mark the **first participation of Japanese fighter aircraft in an air exercise in India**.
 - Future drills may include **short-notice and unmanned-system-enabled exercises**.
- **Defence Technology Cooperation:** **DRDO and Japan's Acquisition, Technology & Logistics Agency (ATLA)** will deepen advanced technology cooperation; a **Defence Industry Forum** will promote industrial collaboration.
- **UNICORN Project:** The shipborne **UNICORN integrated communications antenna system** is being pursued as the **first defence equipment transfer project between India and Japan**.
- **Special Operations and Theatre Commands:** Both sides will promote **Special Operations Forces exchanges** and explore cooperation with India's future **integrated theatre commands**.
- **Indo-Pacific Cooperation:** Reaffirmed support for a **free and open Indo-Pacific**, freedom of navigation and opposition to unilateral alteration of the status quo by force.
 - Cooperation with third countries will expand through exercises and the **Indo-Pacific Logistics Network**.
- **Institutional Mechanism:** A **Director-General/Joint Secretary-level Working Group** will coordinate operational, intelligence, equipment, technology and industrial cooperation.
 - Both sides will also prepare for the **4th India-Japan 2+2 Foreign and Defence Ministerial Dialogue in Tokyo**.

India's Trade Agreements in Action

The government highlighted India's growing use of **FTAs**, reflected in higher exports, greater use of **preferential Certificates of Origin (CoOs)** and wider product diversification across partner markets.

Expanding Export Opportunities and Global Market Access through India's FTAs

- **Expanding Trade Network:** India has widened its FTA network across major markets to deepen global economic integration.
- **Focus:** Convert negotiated market access into **higher exports, investment and employment**, while protecting sensitive domestic sectors.
- **Record Exports:** Merchandise and services exports reached **USD 863.1 billion in FY 2025–26**, including **USD 441.8 billion merchandise exports**.
- **April–June 2026:** Combined exports grew **11.37%** year-on-year.

India-UAE CEPA: Early Trade Gains

- **Implemented:** **1 May 2022**, after negotiations completed in **88 days**.
- India exported **USD 37.36 billion** of merchandise to UAE in FY 2025–26, making it India's **largest individual FTA export market**.
- Bilateral trade crossed **USD 100.06 billion in FY 2024–25**, growing **19.6%**.
- **Target:** **USD 200 billion** bilateral trade by 2032.

Export Facilitation through Certificates of Origin

- **Preferential CoO:** Certifies compliance with **rules of origin** and enables concessional or zero tariffs under FTAs.
- **e-CoO 2.0:** Provides end-to-end digital CoO issuance with **Aadhaar-based e-signatures and QR verification**.
- **Simplified Rules:**
 - **India-EFTA TEPA and India-UK CETA:** Permit **self-declaration of origin**. UK agreement also permits **importer's knowledge** and waives documentation below **£1,000**.
 - **Australia ECTA:** Allows multiple qualifying products under a **single CoO**.
- **Trade Connect:** Helps exporters, especially **MSMEs**, access tariff information, FTA opportunities and CoO support.

India-Australia ECTA: Expanding Trade Partnership

- **Implemented:** **29 December 2022**.
- India's exports to Australia rose from about **USD 4 billion in FY 2020–21** to **USD 7.28 billion in FY 2025–26**, over **80% growth**.
- Australia initially gave zero-duty access on **98.3% tariff lines**; from **2026**, all Indian exports became eligible for **zero-duty access**.

- ECTA is the basis for negotiations on a broader **India-Australia CECA**.

Connecting Indian Goods and Skills to Global Markets through FTAs

Expanded Market Access for Merchandise Goods	
Agreement	Market Access
India-UK CETA	Nearly 99% of Indian exports duty-free
India-Oman CEPA	Concessions on about 98% of tariff lines
India-EFTA TEPA	92.2% tariff lines, covering about 99.6% of Indian exports
India-Mauritius CECPA	Preferential access for 310 Indian product lines
India-UAE CEPA	Around 97% of tariff lines, covering nearly 99% of Indian exports
India-Australia ECTA	100% of Australian tariff lines duty-free

- **Awaiting Implementation:**
 - **India-EU FTA:** Preferential treatment across about **97%** of EU tariff lines, covering nearly **99.5%** of bilateral trade value.
 - **India-New Zealand FTA:** Duty-free access for **100%** of Indian exports from entry into force.
- Major beneficiaries include **textiles, agriculture and processed food, leather, footwear, marine products, gems and jewellery, carpets and handicrafts.**
- Tariff calibration and transition arrangements protect farmers and sensitive industries.

New Opportunities for Indian Services and Professionals

- **Services exports:** Around **USD 421.3 billion** in FY 2025–26; services account for nearly **30%** of employment.
- **India-New Zealand FTA:** Up to **5,000** skilled Indians can stay for up to **3 years**.
 - Commitments in **118 services sectors** and MFN treatment in **139 sectors**.
- **India-EU FTA:** Covers **144 services sub-sectors** and facilitates mobility of business visitors and intra-corporate transferees, along with dependant work rights and social-security cooperation.
- **India-UK CETA:** Facilitates mobility of **business visitors, intra-corporate transferees, contractual suppliers and independent professionals**.
 - **Double Contribution Convention** expected to save over **₹4,000 crore** in dual social-security payments.
- **India-EFTA TEPA:** Access to **128 sub-sectors** in Switzerland, **114** in Norway, **110** in Iceland and **107** in Liechtenstein.

- Provides for **MRAs** in nursing, chartered accountancy and architecture.

- **India-Oman CEPA:** Facilitates temporary entry of **intra-corporate transferees, contractual suppliers, business visitors and independent professionals**.
- **India-Australia ECTA:** Commitments across about **135 services sub-sectors** and MFN treatment across **120 sub-sectors**.
- **India-UAE CEPA:** Access to about **111 sub-sectors across 11 broad areas**.
- **India-Mauritius CECPA:** Access to about **115 services sub-sectors across 11 broad areas**.

India's Free Trade Agreement strategy is shifting from market access creation to effective utilisation. Discuss.

Drishti Mains Question

The Makkah Joint Defence Agreement

Saudi Arabia, Türkiye and Pakistan signed the Makkah Joint Defence Agreement (MJDA) at Al-Safa Palace, Makkah. **An armed attack on one member will be treated as an attack on all three, creating a new "triple alliance"/"Sunni security triangle"** amid shifting West Asian security dynamics.

Key Features

- **Background:** Builds on the Saudi-Pakistan Strategic Mutual Defence Agreement (Sept 2025); Türkiye joined in August 2026, converting it into a trilateral arrangement.
- **Collective Defence:** An attack on one is treated as an attack on all three. It resembles NATO Article 5 in principle, but operational commitments and enforcement mechanisms remain unclear.
- **Military Coordination:** Regular joint training and combined military exercises to improve interoperability and readiness.
- **Intelligence Sharing:** Formal channels for strategic intelligence, security data and threat assessments.

Why Was the Pact Formed?

- **Saudi Arabia:** Seeks reduced dependence on the US security umbrella; faces Houthi, Iranian missile/drone/proxy threats and risks around Hormuz and Bab-el-Mandeb, with implications for oil exports and Vision 2030.
- **Türkiye:** Seeks greater strategic leverage amid concerns over Israel's military presence in Gaza and Syria.
- **Pakistan:** Gains strategic relevance in West Asia and stronger defence, economic and energy ties with Saudi Arabia.

Strategic Contributions and Gains		
Country	Strategic Contribution	Expected Gains
Saudi Arabia	Financial heft, energy resources, Gulf location and defence-spending capacity.	Collective deterrence; access to Pakistani military capability and Turkish defence technology; stronger security for Vision 2030.
Türkiye	NATO-standard military-industrial capability, drones, aerospace technology and defence industry.	Saudi capital and defence markets, higher exports and stronger West Asian influence.
Pakistan	Nuclear and conventional military capabilities, manpower and long defence ties with Saudi Arabia.	Greater regional relevance, stronger economic/energy ties and diplomatic leverage.

Earlier Pan-Islamic/Pan-Arab Defence Arrangements	
Arrangement	Key Feature
Treaty of Joint Defense and Economic Cooperation between the States of the Arab League (1950)	Pan-Arab mutual-defence framework; limited by political divisions and weak military coordination.
Baghdad Pact/CENTO (1955)	Iraq, Turkey, Pakistan, Iran and UK; aimed to contain Soviet expansion. Renamed Central Treaty Organization (CENTO) after Iraq's 1958 withdrawal; dissolved in 1979.
Peninsula Shield Force (1984)	GCC joint military force created during the Iran-Iraq War; notably deployed in Bahrain in 2011.
Islamic Military Counter Terrorism Coalition (IMCTC) (2015)	Saudi-led coalition of 40+ countries against terrorism through military, financial, ideological and media cooperation; criticised as primarily anti-Iran.

Potential Implications for India

- **Pakistan-Türkiye-China Axis:** Deeper Saudi ties with Pakistan and Türkiye, both close to China, **could strengthen a network along India's western flank**; Türkiye's Gulf defence push may compete with Indian systems such as BrahMos and Akash.
- **India-Saudi Relations:** Pakistan and Türkiye have **opposed India on Kashmir in multilateral forums**; greater Saudi dependence on Pakistan could give Islamabad leverage. However, India-Saudi economic, energy, investment and strategic ties remain institutionalised through the Strategic Partnership Council.
- **Western Indian Ocean:** The pact could extend the three countries' strategic presence into an area where India seeks to remain a net security provider.
- **Conflict Threshold:** The **"attack on one is an attack on all" clause creates uncertainty** over possible Saudi or Turkish political/military support in an India-Pakistan conflict.
- **Pakistan Military Modernisation:** Saudi finance plus Turkish technology could accelerate Pakistani acquisition of advanced fighters, drones and air-defence systems.
- **Counter-Terrorism Diplomacy:** Closer Saudi-Pakistan security ties may provide Islamabad greater diplomatic space and complicate India's efforts against cross-border terrorism.
- **Nuclear Ambiguity:** Pakistan's nuclear capability adds strategic weight, though no formal Pakistani nuclear guarantee to Saudi Arabia has been publicly confirmed.

Way Forward

- **Deepen Minilaterals:** Accelerate **IMEC and I2U2** to strengthen ties with Saudi Arabia, UAE and other Gulf states.
- **Use Iran as a Balancer:** Strengthen Chabahar Port and International North-South Transport Corridor (INSTC) to preserve connectivity with Central Asia and Europe.
- **Diversify Gulf Partnerships:** Deepen defence, maritime-security and technology cooperation with UAE, Oman and Qatar; strengthen ties with Greece, Cyprus, Armenia and Israel.
- **Maintain UAE Defence Autonomy:** Expand the India-UAE Strategic Defence Partnership without a mutual-defence commitment to avoid automatic involvement in regional conflicts.
- **Build Indigenous Capability:** Accelerate AI, drones, counter-drone systems, electronic warfare, hypersonic weapons and advanced air defence under Atmanirbhar Bharat.
- **Strengthen Indian Ocean Security:** Use India's IORA chairmanship to promote transparency and inclusive, open regional security cooperation rather than bloc-based arrangements.
- **Defence Forces Vision 2047:** Pursue **"autonomy through capability and power"** by strengthening military, economic and technological capacity and treating **West Asia as a strategic hub** for energy, trade and diaspora interests.

How could the Makkah Joint Defence Agreement affect India's strategic and diplomatic interests in the Gulf?

Drishiti Mains Question

A Climate Resilience

Pathway Between India and China

Experts have highlighted that shared climate vulnerabilities could serve as a **pragmatic, low-risk avenue for India-China climate cooperation** following El Niño-delayed monsoons and extreme flooding in Assam, Odisha, Shaanxi and Gansu.

- The **April 2026 visit of China's Special Envoy for Climate Change to New Delhi** reinforced **climate resilience and disaster management** as potential areas of bilateral engagement despite geopolitical tensions.

Shared Climate Vulnerabilities of India and China

- **The Urban "Concrete Trap" & Flooding:** Rapid urbanisation has paved over **wetlands and natural floodplains**, causing drainage failures and urban flooding in places like **Mumbai and Guangxi**.
- **The Third Pole (Himalayan Ecosystem) Risks:** Accelerating glacial retreat in the shared **Hindu Kush Himalaya (HKH) region** increases risks of **Glacial Lake Outburst Floods (GLOFs)** and threatens transboundary rivers like **Yarlung Tsangpo (Brahmaputra) and Sutlej**.
- **Inland Heatwaves vs. Coastal Cyclones:** Both countries face prolonged **heatwaves, flash floods, storm surges, extreme precipitation and rising sea levels** affecting densely populated regions.
- **Public Health and Economic Cascades:** Climate disasters disrupt **supply chains, agricultural productivity and public health**, threatening developmental gains.

Historical Precedent for India-China Climate Engagement

- **Institutional Cooperation (1990s–2020):** India and China adopted **Joint Statements, Joint Declarations and MoUs** on climate change and disaster management.
 - The **2015 Joint Climate Statement** supported an **equitable global climate agreement** and cooperation in **renewable energy, electric mobility, low-carbon urbanisation, climate adaptation, joint R&D, data sharing and disaster governance**.
 - As founding members of the **BASIC Group (Brazil, South Africa, India and China)**, both coordinated positions in global climate negotiations.
- **Strategic Economic Dialogue (SED):** Facilitated cooperation in **sustainable urban planning, waste management, sewage treatment, water-use efficiency and capacity building**.
- **Sister City Agreements:** **Delhi–Beijing (2013), Mumbai–Shanghai (2014) and Chennai–Chongqing (2015)** promoted collaboration in **urban governance, environmental management and climate resilience**.

Challenges Constraining India-China Climate Cooperation

- **Strategic Mistrust:** Border tensions since the **2020 clashes** disrupted climate cooperation and **transboundary hydrological data sharing**, especially on the **Brahmaputra River**.
- **Economic Competition:** China's dominance in **solar manufacturing, batteries and critical minerals**, along with India's **Atmanirbhar Bharat strategy** and **USD 112.6 billion trade deficit (2025–26)**, limits green technology cooperation.
 - India has gradually eased restrictions on Chinese firms for selective economic cooperation.
- **Regulatory Barriers:** India's **2020 Press Note 3** restrictions and China's export controls on **critical minerals and processing technologies** constrain collaboration.
- **Weak Implementation:** Sister City initiatives saw limited progress due to diplomatic tensions and weak institutional partnerships.
- **Carbon Market Cooperation:** In 2026, China, Brazil and the EU became founding members of the **Open Coalition for Carbon Markets**; India was not a founding member.
- **Divergent Climate Leadership:** India has not joined China's **BRI International Green Development Coalition**, while China is not part of India's **International Solar Alliance (ISA)** and **Global Biofuels Alliance (GBA)**.

India's Major Bilateral Climate Partnerships Across the World

- Under the **India–US Climate and Clean Energy Agenda 2030**, cooperation exists in **clean energy, climate finance, hydrogen, EVs and clean technology innovation**.
- Through **Green Growth Equity Fund (GGEF)**, India and UK mobilise investment in **renewable energy, energy efficiency and EVs** and lead **GGI–OSOWOG**.
- **Japan–India Clean Energy Partnership (CEP)** and **Joint Crediting Mechanism (JCM)** covers **EVs, hydrogen, ammonia technologies, battery storage, energy efficiency and carbon credits**.
- Through **Green and Sustainable Development Partnership (GSDP)** and **International Climate Initiative (IKI)**, Germany supports **renewable energy, climate-resilient infrastructure, biodiversity conservation and adaptation**.
- **India–Denmark Green Strategic Partnership (2020)**
- India and France launched **International Solar Alliance (ISA)** at **COP21 (2015)**.
- Through **India–EU Clean Energy and Climate Partnership**, cooperation covers **clean energy transition, green finance, low-carbon development, carbon markets and climate resilience**.

Strategic Significance for India

- **Bridging the Climate Finance Gap:** Partnerships provide **concessional loans, blended finance models and green bonds** to support India's NDC goals.

- **Critical Technology Transfer:** Cooperation with countries like **Denmark and Germany** enables access to technologies in **offshore wind, green hydrogen and battery storage**.
- **Geopolitical Leverage:** Green alliances strengthen India's role as a **responsible global climate leader** and support **Common but Differentiated Responsibilities (CBDR)**.

Measures for India and China to Strengthen Cooperation on Climate Resilience

- **Focusing on "Low-Politics" Entry Points:** Separate disaster management cooperation from border disputes and begin with areas like **sustainable agriculture, urban waste management and early warning systems**.
- **Synergizing Subnational Best Practices:** Promote city-to-city cooperation.
 - India can adopt China's **"Sponge Cities" model**, while China can learn from India's **community-led Heat Action Plans (Ahmedabad model)**.
- **Leveraging Minilateral Platforms:** Platforms like **BRICS and ADB** can provide neutral spaces for technical and financial cooperation.
- **Institutionalizing Blended Climate Finance:** Develop frameworks for **blended finance and green municipal bonds** to de-risk private investment and support Global South climate finance.
- **Himalayan Hydrological Surveillance:** Establish **continuous, year-round joint glaciological monitoring** of the **HKH region** to predict GLOFs and manage transboundary rivers.

Examine the role of climate diplomacy in strengthening India's bilateral relations. How can climate resilience contribute to confidence-building between India and China despite geopolitical tensions?

Drishti Mains Question

India's Trade Balancing Act: Navigating US, China and Strategic Autonomy

India eased **FDI, anti-dumping and e-commerce norms** to attract investment, strengthen manufacturing and boost exports while balancing ties with the **US and China**.

Recalibration of India's Trade and Investment Policy

- **The "China Plus One" Capital Constraint:** **Press Note 3 (2020)** restricted Chinese equity but did not curb the trade deficit, which crossed **USD 112 billion in FY25-26**. India requires Chinese capital and intermediate goods to diversify supply chains.
- **The PLI Competitiveness Paradox:** Anti-dumping duties on Chinese **APIs, electronic components and specialty chemicals** raised input costs, weakening the competitiveness of **PLI-supported products** in **Global Value Chains**.

- **Bureaucratic Friction in Capital Formation:** Inter-ministerial screening under **Press Note 3** delayed investments, including global venture capital funds with minor passive Chinese investors.
- **Mitigating US Trade Coercion:** Easing export-oriented e-commerce FDI and banning forced-labour imports aim to protect US market access and reduce proposed tariffs on Indian exports.

Balancing Trade Liberalization with Strategic Interests in India

- **"Small Yard, High Fence" Doctrine:** The **March 2026 FDI relaxation** permits up to **10% beneficial ownership** from land-bordering countries through the automatic route, while retaining scrutiny over controlling stakes and critical infrastructure.
- **Prioritizing Domestic Value Addition (DVA):** India favours conditional local manufacturing by Chinese firms to increase **DVA** and exports rather than relying only on imported intermediates.
- **Selective Trade Defense Application:** Anti-dumping duties are eased on essential industrial inputs but retained on finished goods. China-related imports formed nearly **72% of rejected duty recommendations between 2000 and Dec 2025**.
- **Multi-alignment in Digital Trade:** Inventory-based e-commerce FDI is allowed only for exports, supporting global integration without opening domestic retail to foreign predatory pricing.

Concerns Regarding India's Trade Strategy

- **The "Trojan Horse" Dilemma (Assembly vs. Manufacturing):** Cheap intermediate imports and Chinese joint ventures may deepen supply-chain dependence and reduce India to a low-margin assembly hub.
- **Institutional Opacity and Erosion of DGTR:** Unreasoned rejection of **DGTR** recommendations weakens institutional credibility, creates regulatory uncertainty and exposes domestic producers to subsidised dumping.
- **Weaponisation of Bilateral Dependencies:** Dependence on the **US for export markets** and **China for intermediate goods** leaves India vulnerable to sanctions, tariffs and supply disruptions.
- **Ideological and Domestic Friction:** Diluting WTO-compliant trade defense mechanisms (anti-dumping) may weaken **Atmanirbhar Bharat** and reduce investment, capacity utilisation and employment among affected domestic producers.

Way Forward

- **Enact a Statutory "Public Interest Test":** Strengthen the **Customs Tariff Act, 1975** to require transparent reasons

for rejecting **DGTR** recommendations, balancing exporter gains with **MSME** protection.

- **Transition to Value-Addition Linked Incentives (VALI):** Link incentives to verified growth in **Domestic Value Addition** and mandate time-bound technology transfer in joint ventures with land-bordering countries.
- **Strengthen Customs Rules of Origin:** Use blockchain-enabled audits under **Customs (Administration of Rules of Origin under Trade Agreements) Rules, 2020 (CAROTAR, 2020)** to prevent tariff circumvention through third countries without genuine value addition.
- **Leverage:** Use the **US–India TRUST Initiative** and deepen semiconductor and deep-tech partnerships with **Japan and South Korea** to diversify supply chains.

‘What introduces friction into the ties between India and the United States is that Washington is still unable to find for India a position in its global strategy, which would satisfy India’s National self-esteem and ambitions’. Explain with suitable examples. (2019)

Drishti Mains Question

US–Saudi Arabia Nuclear Deal

The US and Saudi Arabia signed a **30-year Civil Nuclear Cooperation Agreement** under **Section 123 of the US Atomic Energy Act, 1954**, along with a **Bilateral Nuclear Safeguards Agreement**. The US linked its implementation to Saudi Arabia joining the **Abraham Accords** and normalising relations with Israel.

- **Abraham Accords:** US-brokered agreements signed in **2020** to normalise Israel’s relations with Arab countries. **Signatories:** UAE, Bahrain and Morocco (2020); Sudan (2021). Strengthened India’s ties with Israel and Gulf countries and supported the **I2U2 Group—India, Israel, UAE and US**—for cooperation in energy, food security, water, health, transport, space and emerging technologies.

Key Features

- **Civil Cooperation:** Transfer of US nuclear materials, equipment, technology and expertise; US companies to

receive priority in developing Saudi Arabia’s civilian nuclear infrastructure.

- **Fuel Supply:** Saudi Arabia will initially import nuclear fuel.
- **Enrichment:** A two-year feasibility study will assess uranium enrichment in Saudi Arabia under a possible **“black box” arrangement**, keeping sensitive enrichment technology under US control.
- **Safeguards:** Unlike the **US–UAE Civil Nuclear Agreement, 2009**, Saudi Arabia is **not required to permanently renounce uranium enrichment or spent fuel reprocessing** (the so-called ‘**Gold Standard**’).
 - Adoption of the **IAEA Additional Protocol** is **not mandatory**, limiting intrusive inspections.
 - **Concern:** The deal may weaken global non-proliferation norms by legitimising dual-use nuclear technologies.

Strategic Significance

- **Saudi Arabia:** Supports **Vision 2030**, energy diversification, rising electricity demand, AI data centres, industries and mega projects. Uranium reserves, Iran’s nuclear programme and regional instability also drive its search for energy security and strategic deterrence.
- **United States:** Secures contracts for US nuclear companies and counters Chinese and Russian influence in West Asia.
- **Iran:** Iran has advanced centrifuges and uranium enriched up to 60%. Saudi enrichment rights may harden Iran’s position on retaining enrichment, complicating nuclear negotiations and regional non-proliferation.

Implications for India

- **Energy Security:** Saudi–Iran rivalry may intensify and threaten Strait of Hormuz oil supplies and raise India’s import bill; nearly **40% of India’s crude imports** come from West Asia.
- Tests **India’s multi-alignment policy** as it has strategic partnerships with Saudi Arabia, Iran, Israel and the US.
- Regional instability and delayed Saudi–Israel normalisation may slow the **India–Middle East–Europe Economic Corridor**.
- **Economic Opportunities:** Indian engineering and infrastructure firms may participate in non-sensitive Saudi nuclear infrastructure projects.

India–US Deal vs US–Saudi Deal

Parameter	India–US Deal, 2008	US–Saudi Deal, 2026
Status under NPT	India is not a signatory to the Nuclear Non-Proliferation Treaty (NPT), considering its discriminatory nature. It received a 2008 NSG waiver, becoming the only non-NPT state permitted to engage in global civil nuclear commerce.	Saudi Arabia is an NPT Non-Nuclear Weapon State and has the right to pursue peaceful nuclear energy under IAEA safeguards.
Uranium Enrichment and Reprocessing (ENR)	Granted blanket consent for ENR. India separated its civilian and military nuclear facilities and can reprocess spent fuel under IAEA safeguards .	Domestic uranium enrichment is not prohibited. Proposes a two-year feasibility study for a US-operated enrichment facility in Saudi Arabia, marking a departure from the US-UAE ‘ Gold Standard ’ model.

Contd...

India–US Deal vs US–Saudi Deal		
Parameter	India–US Deal, 2008	US–Saudi Deal, 2026
IAEA Safeguards	India signed an India-specific IAEA Safeguards Agreement and Protocol , covering only its declared civilian facilities while military facilities remain outside international inspection.	Saudi Arabia is covered by an IAEA Comprehensive Safeguards Agreement (CSA) , but the agreement does not require adoption of the IAEA Additional Protocol , limiting the scope of intrusive inspections.
Geopolitical Aim	Driven by the US objective of integrating India into the global non-proliferation framework while strengthening the India-US strategic partnership in the Indo-Pacific amid China's rise.	Driven by Saudi Vision 2030 , energy diversification, and the US effort to prevent Riyadh from relying on China or Russia for nuclear technology. US has additionally linked the agreement to Saudi Arabia joining the Abraham Accords .

How does the US-Saudi Arabia Nuclear Agreement differ from the Indo-US Civil Nuclear Deal (2008)? Discuss

Drishti Mains Question

President's State Visit to Romania

President **Droupadi Murmu** visited **Romania** during her State Visits to **Moldova, North Macedonia and Romania**, marking the **first State Visit by an Indian President to Romania in nearly three decades**.

- India and Romania reaffirmed their **70-year partnership** and agreed to enhance cooperation in **trade, investment and resilient supply chains**, with a target to **double bilateral trade within three years**.
 - India has a **trade surplus** with Romania. In **FY 2024–25**, Romania accounted for **0.24% of India's exports** and was India's **56th largest export destination**.
 - Major Indian exports: **petroleum products, engineering goods and pharmaceuticals**. Major imports: **machinery, vegetable oils and iron & steel**.
- **Importance of India–EU FTA**: Romania's role as an **EU partner** was highlighted, with the **India–EU FTA** expected to create opportunities for **industries and skilled professionals**.
- **Strengthening Education & Cultural Ties**: Three agreements were signed on **education, culture and people-to-people exchanges**.
- **Science & Technology Cooperation**: Both countries agreed to expand **joint R&D** under the **Programme of Cooperation in Science and Technology**.
- **Promotion of Indian Culture**: An **Indian Studies Chair** will be established at the **University of Bucharest** with support from **ICCR**.
- **Capacity Building**: India will **double training opportunities for Romania** under the **ITEC Programme**.
- **India–Romania Year of Innovation**: Both countries will celebrate **2028 as the "India–Romania Year of Innovation"** to mark the **80th anniversary of diplomatic relations**.

Romania

- **Location**: A **South-Eastern European** country bordered by **Ukraine, Moldova, Black Sea, Bulgaria, Serbia and Hungary**. Capital: **Bucharest**.
- **Political Profile**: A **democratic republic** and member of **EU and NATO**.
- **Physical Features**: Characterised by **one-third mountains, one-third hills and one-third plains**.
 - Major features: **Carpathian Mountains, Transylvanian Basin, Danube Delta (UNESCO World Heritage Site), and Chernozem plains**.
- **Economy**: A **high-income economy** and among the **fastest-growing economies in the EU**.
- **Strategic Significance**: Located on the **Black Sea**, Romania hosts the **Port of Constanța**, a key **Europe–Asia trade gateway**.
 - It is an important partner for India in **IT, pharmaceuticals, automobiles, renewable energy** and strengthening **India–EU economic cooperation**.



Rome Declaration for an Unarmed and Disarming Peace

The **Rome Declaration for an Unarmed and Disarming Peace**, signed in **July 2026** by **Nobel laureates, AI scientists, religious leaders and global luminaries**, raised concerns over the **“convergent peril”** of delegating moral and ethical decisions to **Artificial Intelligence (AI)** systems.

Rome Declaration for an Unarmed and Disarming Peace

- **About:** A strategic and moral appeal calling for an **international treaty prohibiting autonomous AI systems from accessing or making nuclear launch decisions**, emphasising that **human ethics, accountability and judgment must not be outsourced to AI**.
- **Current Status:** A normative **“nudge”** rather than a binding legal framework due to lack of global consensus on autonomous AI regulation in warfare.
- **Principles of the Rome Declaration:**
 - **Disarming the Next Arms Race:** Prevent an AI-driven military and nuclear arms race.
 - **Responsible Development:** Ensure transparent and ethical AI development.
 - **Responsible Use of AI:** Restrict autonomous AI decisions in high-risk military domains.
 - **Responsible Governance:** Promote international oversight, regulation and accountability.
 - **Responsible Leadership:** Prioritise ethics over technological competition.
 - **Nuclear Disarmament:** Renew commitment towards complete, irreversible and verifiable elimination of nuclear weapons.
- **Ethical & Operational Mandates:**
 - **Mandatory Meaningful Human Control:** AI must not make final decisions on nuclear weapon deployment.
 - **The ‘Digital Commons’ Model:** Democratise data access for independent analysis of AI risks.
 - **Responsible Development:** Ensure AI systems remain transparent, auditable and controllable.
 - **Internal Arsenal Vulnerability Audits:** Nuclear states must assess AI-driven cyber threats to command-and-control systems.
 - **Time-Bound, Verifiable Disarmament:** Renew negotiations for complete elimination of nuclear arsenals.

The Need for the Declaration

- **Compression of Decision-Making Windows:** AI-based military systems reduce decision time from hours to seconds, removing the human hesitation and moral pause that prevented disasters like the **1962 Cuban Missile Crisis** and **1983 Soviet false alarm**.

- **Algorithmic Failure Modes:** AI systems face risks of **hallucinations and data poisoning**, which may trigger false threat perceptions and catastrophic retaliation.
- **Collapse of Global Security Infrastructure:** Weakening arms control treaties like **Open Skies** and **New START** creates incentives for states to automate defence systems.
- **Real-World Triggers (The Iran Scenario):** In sensitive environments like the **Iranian nuclear programme**, digital intrusions or attacks could be misinterpreted by automated systems, escalating conflicts.

“Convergent Perils” in AI

- **Convergent Peril:** A situation where multiple risks interact and amplify each other, creating systemic threats.
 - In AI, it represents the intersection of **algorithmic bias, loss of accountability, hyper-automation and ethical abdication**, creating a civilizational crisis.

Global AI Ethics Landscape
■ EU Artificial Intelligence Act, 2024 ■ Bletchley Park Declaration (2023) ■ UNESCO Recommendation on the Ethics of AI (2021) ■ Global Partnership on Artificial Intelligence (GPAI)
India’s AI Governance Framework
■ NITI Aayog’s National Strategy for AI (2018) – “AI for All” ■ IndiaAI Mission (2024): Includes a “Safe and Trusted AI” pillar for ethical AI deployment. ■ India AI Governance Guidelines (MeitY)

Bay of Bengal as India’s SHANTI Anchor

India proposed **SHANTI—Securing Holistic Advancement through Norms, Trust and Integrity**—while announcing its candidature for the **UNSC 2028–29 term**.

- The Bay of Bengal serves as the ideal region to operationalise it before expanding it across the wider Indo-Pacific.

SHANTI Framework

- Builds on **SAGAR, 2015** and **MAHASAGAR, 2025**.
- Provides the operational framework for maritime security by promoting **Norms, Trust, and Integrity**.
- Shifts the focus from competition to demand-driven cooperation, influence to institution-building, and crisis management to long-term resilience.
- Seeks to make India a preferred security partner and first responder in the Indo-Pacific.
- Supports inclusive and rules-based maritime governance, strengthens BIMSTEC, enhances maritime resilience and advances the Blue Economy.

India's Maritime Security Evolution		
Initiative	Core Focus	Key Contribution
SAGAR (<i>Security and Growth for All in the Region</i>)	Maritime security and inclusive regional development	Articulated India's vision for security and growth in the Indian Ocean Region .
MAHASAGAR (<i>Mutual and Holistic Advancement for Security and Growth Across Regions</i>)	Wider Indo-Pacific and Global South	Expanded the scope of SAGAR by recognising the interconnectedness of regional and global security.
SHANTI (<i>Securing Holistic Advancement through Norms, Trust and Integrity</i>)	Norms-based maritime governance	Provides the operational framework for governing shared maritime commons through Norms, Trust, and Integrity , positioning India as a preferred security partner and first responder .

Why the Bay of Bengal?

- **Cooperative Maritime Commons:** Unlike the Western Indian Ocean which is plagued by intense militarization, piracy, chokepoint blockades (e.g., **Red Sea/Houthis**), and direct great-power confrontation.
 - The Bay of Bengal is relatively stable, with strong historical, cultural and economic ties among littoral states without the baggage of **active, insurmountable maritime border disputes**.
 - This makes it highly **conducive for establishing a trust-based, rules-bound governance architecture** before attempting to apply it to more volatile regions.
- **Existing Cooperation:** India engages with littoral states through the BIMSTEC Master Plan, Colombo Security Conclave, Lines of Credit, and bilateral logistics, defence and HADR cooperation with Bangladesh, Sri Lanka, Myanmar and Thailand.
- **Malacca Factor:** The Bay links South and Southeast Asia, advances Act East Policy and SAGAR vision, and guards the western approach to the Strait of Malacca.
 - China's push into the Bay through Myanmar's Kyaukphyu port to overcome its "Malacca Dilemma" makes rapid SHANTI implementation vital for preserving a free, open and multipolar region.
- **Shared Non-Traditional Threats:** Cyclones, sea-level rise, coastal erosion, climate migration, IUU fishing, drug trafficking (maritime routes overlap with the "Golden Triangle") and gun-running require coordinated HADR, climate resilience and intelligence-sharing.
- **Normative Support to BIMSTEC:** SHANTI can embed Norms, Trust and Integrity into existing mechanisms, strengthening maritime governance, the Blue Economy and Maritime Domain Awareness without creating new institutions.
- **Regional Stewardship:** SHANTI enables India to act as a non-coercive security partner and first responder, avoiding the "big brother" image and offering a model for wider Indo-Pacific governance. The US's evolving Indo-Pacific approach and military actions against Iran, which disrupted regional sea lanes, highlight the urgency for India to **assert itself and strengthen its claim as the region's preferred security provider**.

Key Challenges to Operationalising SHANTI in Bay of Bengal		
Category	Challenges	Suggestions
Geopolitical & Traditional Security	China's presence and instability in Myanmar threatens regional connectivity and maritime security, including projects such as India-Myanmar-Thailand (IMT) Highway .	Strengthen Maritime Domain Awareness through IFC-IOR and NSA-level coordination among littoral states.
Economic & Infrastructure Connectivity	Fragmented trade procedures, weak ports and financing gaps	Implement the BIMSTEC Maritime Transport Cooperation Agreement and harmonise port regulations.
Institutional Gaps & Trust Deficit	Despite institutions such as BIMSTEC, regional fragmentation and trust deficit persist.	Promote cooperative regionalism through inclusive corridor governance and transparent decision-making through multilateral cooperation.
Environmental & Non-Traditional Threats	Climate change, cyclones, IUU fishing, pollution and trafficking	Develop a coordinated HADR framework with India as a non-coercive first responder , while promoting the Blue Economy , sustainable marine resource management, pollution control, and technology-driven coastal monitoring.

"India's maritime strategy has evolved from SAGAR to MAHASAGAR and the proposed SHANTI." Critically analyse this evolution in the context of changing geopolitical realities.

Drishiti Mains Question

Environment & Ecology

World Elephant Day 2026

India reaffirmed its commitment to elephant conservation during the national **World Elephant Day** 2026 celebrations in Visakhapatnam.

Facts about World Elephant Day

- **Observed:** Annually on **12 August** to raise awareness about conserving **Asian and African elephants**.
- **Origin:** Established on **12 August 2012** by **Patricia Sims of Canada** and the **Elephant Reintroduction Foundation of Thailand**.
- **Theme 2026:** "Bringing the world together to help elephants."
- **Objective:** Highlight threats such as habitat loss and fragmentation, poaching and illegal ivory trade, human-elephant conflict, loss of migration routes and mistreatment in captivity, while promoting conservation and coexistence.

SC ruled that no State can block **elephant corridors** and directed the **Central Government to conduct a nationwide survey** to ensure elephants' **right of passage through traditional corridors across State borders**.

- Elephant corridors are **not State-specific** as herds naturally cross State boundaries; conservation requires **inter-State coordination**.
 - Examples: **Nepal → West Bengal → Jharkhand**; **Yamuna Nagar, Haryana → Uttarakhand → Himachal Pradesh**.

Ban on Barbaric Practices

- SC ordered strict implementation of its **2018 directions** to remove **spikes** and stop using **fireballs** to drive elephants.
- Elephants are **Schedule I animals under the Wildlife (Protection) Act, 1972**, receiving the highest statutory protection.
- **Asian Elephant (*Elephas maximus*)**: Endangered, IUCN Red List; habitat and migratory-corridor protection is crucial for survival.

Key Initiatives Announced on World Elephant Day 2026

- **Regional Action Plan for Southern India:** ₹500-crore plan covering Karnataka, Kerala, Andhra Pradesh and Tamil Nadu for corridor conservation, habitat restoration, early warning, conflict mitigation and community participation.
- **Nilgiri Plan:** Model Elephant Conservation Plan for the **Nilgiri Elephant Reserve** to guide scientific management.
- **Captive Elephant Welfare:** Publication "**Captive Elephant Husbandry: Science, Welfare and Practice**" released to strengthen captive-care standards.
- **Gaj Gaurav Awards 2026:** Recognised exemplary contributions to elephant conservation and management.

- **"30 Giant Steps":** Documents 30 major Project Elephant milestones and initiatives during **2020–2026**.
- **HEC Research:** Phase II of the human-elephant conflict study launched across Kerala, Karnataka, Tamil Nadu, Odisha and West Bengal.

Steps India has Taken to Strengthen Elephant Conservation Project Elephant

- **About:** The **Ministry of Environment, Forest and Climate Change** implements **Project Elephant**, providing financial and technical support for habitat protection, captive-elephant conservation, human-elephant conflict mitigation, AI-enabled monitoring, radio-collaring, genetic databases and early-warning systems.
- **DNA-Based Census:** **SAIEE 2021–25** used India's first DNA-based synchronised elephant census, analysing dung samples through genetic fingerprinting alongside block counts and line-transect surveys.

Elephant Corridor Network

- **Purpose:** Corridors connect fragmented habitats, maintain movement and genetic diversity, and reduce human-elephant conflict.
- **National Network:** **150 corridors across 14 elephant-range States**.
- **Regional Distribution:** **East-Central 52, North-East 48, Southern 32, Northern 18**; **West Bengal has 26**, the highest.
- **Connectivity:** **19 inter-State** and **6 India-Nepal transboundary corridors**.
- **Use Status:** Increased use in **59 corridors (40%)**, stable in **29 (19%)**, declining in **29 (19%)**, and **15 impaired**.
- **Measures:** Scientific monitoring, boundary delineation and integration into State working and management plans.

Growing the Reserve Network

- Elephant Reserves cover around **80,777 km² across 14 major elephant-range States**.
- Since 2014, about **8,610 km²** additional area has been brought under the reserve network.
- Expansion strengthens habitat connectivity, scientific management and long-term landscape-level conservation.

Management Effectiveness Evaluation

- The **Management Effectiveness Evaluation (MEE) framework** for Elephant Reserves was launched in **2023** using measurable indicators.

- It was piloted in **Shivalik, Kaziranga–Karbi Anglong, Mayurbhanj and Nilgiri Elephant Reserves** and is being expanded nationally with **CAMPA funding**.
- The framework links conservation actions to measurable outcomes and supports adaptive management.
- **Regional Action Plans** for all elephant-bearing regions provide a common framework for inter-State coordination on habitat connectivity, conflict mitigation and landscape management.

Railway Collision Mitigation

- Joint surveys covered **127 sensitive railway stretches (3,452.4 km)**.
- Specific interventions recommended on **77 stretches across 14 States (1,965.2 km)**.
- Proposed **705 wildlife-friendly structures** include underpasses, overpasses, ramps, level crossings and bridge modifications.
- Monitoring uses optical-fibre intrusion detection, seismic sensors, thermal cameras and **AI-enabled Distributed Acoustic Sensing** for real-time alerts.
- Every collision is jointly investigated with the State Forest Department.
- Train-hit elephant deaths declined from **26 in 2013 to 12 in 2024**, a reduction of more than **50%**.

Genetic Monitoring of Captive Elephants

- Genetic identification, digital records and regulated transfers are being used to curb illegal trade and movement of captive elephants.
- Under the **Wildlife (Protection) Act, 1972**, such cases are referred to State Chief Wildlife Wardens.
- **Gajah Suchana**: National genetic database containing DNA profiles, morphological characteristics and ownership details of registered captive elephants; supports identity verification and forensic evidence.
- **Captive Elephant (Transfer or Transport) Rules, 2024**: Regulate transfer and transport; a High-Powered Committee constituted pursuant to the Supreme Court's order examines inter-State transfers.

Community Compensation

- Financial assistance under **Project Tiger and Elephant and Development of Wildlife Habitats** supports people affected by human-wildlife conflict.
- **Human casualty compensation**: **₹10 lakh** for death/permanent incapacitation, **₹2 lakh** for grievous injury and treatment costs up to **₹25,000** for minor injuries.
- Crop and property losses are compensated according to State/UT norms.

- The **Human-Wildlife Conflict Advisory** recommends hotspot mapping, rapid-response teams, review committees and faster compensation.
- Elephant-specific conflict-mitigation guidelines issued in **2023** aim to improve coordination and management.

Thinning-Induced Deceleration of Zaskar Glaciers

A study in *The Cryosphere* analysing **12 Zaskar Basin glaciers (1992–2023)** found that glaciers are **thinning and slowing due to climate change**, threatening long-term water availability in the **Zaskar and the wider Upper Indus Basin**.

Key Findings

- **Glacier Slowdown**: Glaciers in the Zaskar region have slowed down by an average of **2.4 m/year per decade** over the last 30 years.
- **Accelerated Thinning**: Surface thinning increased from about **0.22 m/year (2000–05)** to **0.57 m/year (2015–20)**.
- **Mechanism**: Global Warming → glacier mass loss → thinner ice → **lower driving stress** → reduced ability to transport ice from accumulation zones to lower elevations.
- **Non-Uniform Response**: Glacier behaviour varies with **geometry, debris cover and terminus/snout conditions**.

Impact on Upper Indus Water Security

- **Meltwater Dependence**: Glacier melt sustains Indus river flows during dry summer months.
- **Peak Water Phenomenon**: Accelerated melting may initially increase runoff, but after substantial ice loss, **meltwater contribution declines sharply**.
 - Earlier projection cited by the study: **peak water around 2028**.
- **Long-Term Risks**: Declining glacier storage threatens **water security, agriculture, hydropower and downstream ecosystems**.
- **GLOF Risk**: Retreating glaciers can create moraine-dammed lakes; dam failure may cause catastrophic **Glacial Lake Outburst Floods (GLOFs)**.
 - Example: **South Lhonak, Sikkim (2023)**.

Zaskar Valley

Aspect	Detail
Location	Ladakh; northern flank of the Great Himalayan Range
Climate	High-altitude semi-desert; cold and dry due to Himalayan rain shadow
Area	About 5,000 sq km
Culture	Preserves traditional Ladakhi and Tibetan practices
Main Town	Padum , administrative headquarters and tourism/trekking centre

Key Geographical Features

- **Penzi La Pass:** About **4,400 m**; highest point on the Kargil-Zaskar road, broad tableland with high-altitude lakes; gateway linking Zaskar with **Suru Valley and Kargil**.
 - Heavy snowfall historically cuts off the valley for roughly **8 months annually**.
- **Chadar Trek:** Winter route over the **frozen Zaskar River**, historically used when mountain passes were blocked.
- **Drang-Drung Glacier:** Visible from Penzi La; among Ladakh's largest glaciers outside the **Karakoram** and an important freshwater source.
- **Doda (Stod) River:** Originates from **Drang-Drung Glacier near Pensi La** and flows southeast towards Padum.
- **Zaskar River:** Formed by the **Doda and Lungnak rivers near Padum**; flows northeast through a deep gorge before joining the **Indus at Nimmu near Leh**.



Cultural Significance

- **Microcosm of Tibet:** Isolation helped preserve traditional Tibetan socio-cultural practices.
- **Social Practices:** **Polyandry** survives in some remote areas.
- **Padum:** Also known for nearby **Gyalwarina rock-carved statues**.

Major Monasteries	Key Fact
Sani Gonpa	Near Padum; built on level ground; associated with Kanishka through the Kanishka Stupa
Karsha Gonpa	Largest monastery in Zaskar ; founded in 14 th century; eight temples, scrolls and Thangkas
Phugtal Gonpa	Cave monastery in Lungnak Valley; founded by Phagspa Sherab Zangpo in 11 th century
Stongde Monastery	Founded by Tibetan Yogi Marpa ; also called Marpa Ling
Burdan Monastery	17 th -century monastery in Lungnak Valley; prominent seat of Droga order
Zangkul Gonpa	Meditation retreat of Indian Yogi Naropa

Measures for Himalayan Cryosphere Resilience

- **Cryosphere Monitoring:** Expand high-altitude weather stations and measurements of **ice thickness, mass balance and subglacial hydrology**.
- **Reduce Black Carbon:** Regulate **brick kilns and diesel transport** and promote clean cooking fuels.
- **Glacier-to-Basin Management:** Adopt **Integrated River Basin Management** linking glacier melt, snow, groundwater, hydropower, agriculture and downstream demand.
- **GLOF Risk Reduction:** Map vulnerable lakes, deploy **Early Warning Systems** and use measures such as controlled siphoning.
- **Climate Adaptation:** Promote **climate-resilient agriculture** and local innovations such as **Ladakh's Ice Stupas** for storing winter water for spring use.

The Himalayan cryosphere is both an ecological regulator and a strategic water reservoir. Examine the implications of its rapid transformation for the Indian subcontinent.

Drishiti Mains Question

Agasthyamalai Evictions and the Forest Rights Act, 2006

The Forest Department issued eviction notices to thousands of households in the **Agasthyamalai Biosphere Reserve (ABR)**, particularly in Tamil Nadu's **Theni district**, following a SC directive based on Central Empowered Committee (CEC) findings.

- The controversy concerns whether the eviction process adequately protects rights guaranteed under FRA, 2006.

Background of the Eviction Orders

- **Survey Direction:** In March 2025, the Court asked the CEC to survey the Agasthyamalai landscape for non-forestry activities violating the Forest Conservation Act, 1980 and Wildlife (Protection) Act, 1972.
- **Eviction & Restoration:** Based on CEC findings, the Court ordered removal of non-forestry activities, commercial infrastructure and encroachments from protected areas such as **Kalakkad-Mundanthurai and Srivilliputhur-Megamalai Tiger Reserves**.
 - Tamil Nadu was directed to prepare a time-bound, division-wise eviction plan, including rehabilitation **"wherever applicable"** and ecological restoration.
- **Core Conflict:** Conservation and ecological restoration measures are criticised for allegedly bypassing FRA-mandated safeguards against arbitrary eviction.

Key Concerns Regarding Eviction

- **Statutory Verification Bypassed:** FRA requires forest-right claims to be verified through the Gram Sabha and multi-

departmental committees; critics argue occupants are being treated as encroachers without completing this process.

- **Bar on Eviction:** No forest dweller can be evicted until recognition and verification of rights are completed.
 - Wrongful dispossession of SC/ST persons or interference with forest rights may constitute an offence under the SC/ST (Prevention of Atrocities) Act, 1989.
- **Tenure Misinterpretation:** Other Traditional Forest Dwellers (OTFDs) are often required to prove 75 years of occupation of the exact land claimed before 2005, although the FRA does not impose this rigid requirement.
 - FRA does not prohibit all commercial activities; “**bona fide livelihood needs**” include sale of surplus produce and grazing.
- **Nodal Agency Concern:** Tribal-rights advocates have criticised the limited intervention of State tribal-welfare authorities.

Forest Rights Act, 2006

Aspect	Detail
Beneficiaries	Forest Dwelling Scheduled Tribes (FDST) and Other Traditional Forest Dwellers (OTFD)
Eligibility	Residence in forest land for bona fide livelihood needs for at least 3 generations/75 years before 13 Dec 2005
Purpose	Recognises forest rights while strengthening conservation and livelihood/food security
Gram Sabha	Initiates determination of Individual Forest Rights (IFR) and Community Forest Rights (CFR)
Title Rights	Ownership over actually cultivated land, up to 4 hectares ; no new land granted
Use Rights	Minor Forest Produce, grazing areas, etc.
Relief & Development Rights	Rehabilitation after illegal eviction/forced displacement and access to basic amenities, subject to forest-protection restrictions
Forest Management Rights	Right to protect, regenerate, conserve or manage traditionally protected community forest resources

Agasthyamalai Biosphere Reserve

- **Geographical Spread:** About **3,500 sq km**, at the southern end of the Western Ghats, spanning Tamil Nadu and Kerala.
- **Tamil Nadu Protected Areas:** Kalakad-Mundanthurai Tiger Reserve, Srivilliputhur-Megamalai Tiger Reserve, Nellai Wildlife Sanctuary and Kanyakumari Wildlife Sanctuary.
- **Kerala Protected Areas:** Peppara and Neyyar Wildlife Sanctuaries.

- As per the CEC report submitted to SC, the **Agasthyamalai Ecological Landscape includes the Periyar Tiger Reserve in Kerala.**

- **Communities:** Home to indigenous communities including the **Kani tribe**, dependent on forests for livelihood and culture.
- **UNESCO Status:** Included in the **UNESCO World Network of Biosphere Reserves in 2016.**

World Lion Day

On World Lion Day 2026, the Prime Minister highlighted India's Asiatic lion recovery: population rose from 523 (2015) to 891 (2025), supported by scientific conservation, habitat expansion and community participation.

- **World Lion Day:** Observance on 10 August annually; established in 2013 by conservationists Dereck and Beverly Joubert to raise awareness about lion conservation and declining wild populations. They are co-founders of the Big Cats Initiative with National Geographic.

Significance of Asiatic Lion Conservation in India

- **Historical Decline:** Once widely distributed across Asia and the Indian subcontinent; by late 1890s, fewer than 50 lions survived in Gir forests of Gujarat.
 - Junagadh State played an early protection role; Gir Wildlife Sanctuary established in 1965; conservation expanded to Gir NP, Girnar, Mitiyala, Pania and Barda.
- **Recovery:** Population increased from 523 (2015) to 891 (2025), supported by legal protection, habitat expansion, prey availability and community participation.
- **Project Lion:** Launched 15 August 2020; landscape-level approach using habitat improvement, disease monitoring, scientific tracking, radio-collaring, camera traps, prey management and conflict mitigation; Gujarat Forest Department leads implementation and censuses.
- **Landscape Expansion:** Conservation extends beyond protected areas to forests, grasslands, coastal and multi-use landscapes. Distribution range expanded ~59% during 2015-25 to ~35,000 sq km across 11 Saurashtra districts; Barda, without lions since 1879, recorded 17 lions by 2025.
- **Community-Based Conservation:** Gir Gram Samiti (2019) links forest officials and villagers, supported by honorary wildlife wardens, community leaders, eco-guides and eco-development committees.
- **Technology:** GPS tracking monitors lions/vehicles; SIMBA enables AI-based individual identification; radio-collaring and GIS-based real-time monitoring support surveillance, analysis and management.

Key Facts About Asiatic Lion

Parameter	Key Fact
Scientific name	<i>Panthera leo persica</i>
Historical range	West Asia to eastern India
Current core habitat	India's only wild population: Saurashtra, Gujarat - Greater Gir Landscape incl. Gir, Girnar, Pania, Mitiyala, Barda, plus riverine, coastal and multi-use landscapes.
"Second Home"	Natural migration to Barda Wildlife Sanctuary near Porbandar due to carrying-capacity limits in Gir.
Habitat	Dry deciduous forests, scrublands, grasslands and open landscapes.
Social structure	Males generally solitary/small coalitions; females form prides of related females and cubs.
Activity & Diet	Mainly nocturnal/crepuscular; carnivorous with notable reliance on scavenging in Gir.
Niche partitioning	Coexists with Indian leopards and striped hyenas through differences in habitat use, prey selection and activity.
Human coexistence	Maldhari pastoral community has historically coexisted with lions in Gir.

Major Threats

- **Genetic Bottleneck:** Small founder population -> low genetic diversity and inbreeding -> higher disease vulnerability.
- **Disease Risk:** Past Canine Distemper Virus (CDV) and Babesiosis outbreaks caused lion mortalities.
- **Habitat Saturation:** Population pressure pushes lions into human-dominated areas, increasing conflict and livestock depredation.
- **Infrastructure:** Open wells, railway lines and roads in Greater Gir pose accidental-death risks.

SC Extends 10 km Mining Buffer Rule to Wetland Reserves Across India

SC clarified that its mining restrictions near the **Asan Wetland Conservation Reserve** may also apply to other **wetland and community conservation reserves** across India.

- **Asan Mining Restriction:** Under the Feb 2024 interim order, no mining can occur within 10 km of the Asan reserve without **prior permission from the Standing Committee of the National Board** for Wildlife and/or the Environment Ministry.
- **Himachal Pradesh's Plea:** The restriction should not apply because the reserve lies in Uttarakhand and wetland reserves do not necessarily require external buffer zones.

- **SC Direction:** The Standing Committee or Environment Ministry must determine whether the wetland extends into Himachal Pradesh. If it does, the 2024 directions apply; otherwise, mining proposals will be decided under the applicable legal framework.
- **Asan Wetland:** Located at the confluence of the Asan and Yamuna rivers in Uttarakhand; supports migratory waterbirds and aquatic biodiversity; designated a Ramsar site in 2020.

Key Legal Position

- **No Fixed Ramsar Buffer:** Neither the Ramsar Convention nor the Wetlands (Conservation and Management) Rules, 2017 prescribes a fixed buffer or expressly prohibits mining within a specified distance.
- **Wetlands Rules, 2017:** Introduced a decentralised system led by State Wetland Authorities, removed the detailed prohibited-activities list in the 2010 Rules and gave States greater responsibility; their constitutional validity is under challenge before SC.
- **Comparison with Other Protected Areas:** Mining is prohibited within national parks and wildlife sanctuaries and, under SC directions, within 1 km of their boundaries unless a larger notified Eco-Sensitive Zone applies.
 - Mining on forest land requires prior Central approval under the **Forest Conservation Act, 1980** and is generally subject to environmental clearance under the EIA Notification, 2006.

India Launches Clouded Leopard Conservation Action Plan (CAP)

On **International Clouded Leopard Day (4th Aug)**, India reaffirmed conservation of the species through the newly launched **CAP**.

- The **day was established in 2018** by The Aspinall Foundation and is an advocacy observance, not a UN- or treaty-based observance.

Key Features of the Action Plan

- **Institutional Origin:** Developed under the **Govt. of India-GEF-UNDP initiative** and released at the 91st Standing Committee of the National Board for Wild Life meeting in July 2026 at Coimbatore.
- **Core Focus:** Habitat protection, scientific monitoring, landscape connectivity and community participation.
- **Priority Landscapes:** Identifies **14 conservation landscapes across Northeast India** for habitat restoration and wildlife corridors.
- **Scientific Monitoring:** Uses camera traps, occupancy surveys, genetics and environmental DNA (eDNA).

- **Protection Technology:** Strengthens anti-poaching through the **M-StripES smart application**.
- **Inclusive Approach:** Promotes climate-resilient conservation, sustainable livelihoods and transboundary cooperation.
- **Species Recovery Programme:** The Clouded Leopard has been included under the **Integrated Development of Wildlife Habitats (IDWH)** scheme.

Clouded Leopard: Key Facts

Aspect	Fact
Scientific Name	Mainland leopard: <i>Neofelis nebulosa</i> ; Sunda clouded leopard (<i>Neofelis diardi</i>) is a separate species of Sumatra and Borneo.
Range	Himalayan foothills, Southeast Asia and southern China; in India mainly Northeast evergreen forests, from mangroves to Himalayan forests up to about 6,000–9,000 feet.
Size	Smallest member of Pantherinae subfamily ; weighs 11–20 kg.
Special Trait	Largest canines in proportion to skull/body size among living cats (similar in size to tiger's).
Vocalisation	Hyoid bone allows purring but prevents roaring.
Behaviour	More nocturnal and arboreal where tigers/common leopards occur; more daytime activity where larger competitors are absent.

- **Conservation Status and Important Locations**
 - **IUCN:** Vulnerable.
 - **CITES:** Appendix I.
 - **Wildlife (Protection) Act, 1972:** Schedule I.
 - **State Animal:** Meghalaya.
- **Dedicated Park:** Clouded Leopard National Park in Tripura, within Sepahijala Wildlife Sanctuary—**India's first dedicated wildlife park for the species**.
- **Other Key Sites:** Manas Tiger Reserve (Assam), Dampa Tiger Reserve (Mizoram) and Pakke Tiger Reserve (Arunachal Pradesh).

Ken-Betwa River Linking Project

Tribal communities in **MP** protested against the **Ken-Betwa River Linking Project** over concerns related to **displacement, rehabilitation, compensation and environmental impacts**.

Ken-Betwa Link Project

- **About:** The **Ken-Betwa River Linking Project** is India's first river-linking project and the first of **30 links** proposed under the **National Perspective Plan, 1980**.

- **Implementing body:** **Ken-Betwa Link Project Authority**, a Special Purpose Vehicle.
- **Objective:** Transfer water from the **Ken River in MP** to the **Betwa River in UP**. Both are tributaries of the **Yamuna**.

Phase	Major Components
Phase I	Daudhan Dam , low- and high-level tunnels, Ken-Betwa Link Canal and powerhouses
Phase II	Lower Orr Dam , Bina Complex Multipurpose Project and Kotha Barrage

- **Power generation:** **103 MW hydropower** and **27 MW solar power**.
- **Significance:** Seeks to improve **drinking-water supply**, irrigation and regional development in drought-prone **Bundelkhand**, while reducing distress migration.
- **Environmental and Rehabilitation Concerns:**
 - The **Daudhan reservoir** may submerge about **4,141 hectares of the core area** and **1,314 hectares of the buffer area** of the **Panna Tiger Reserve**.
 - Concerns include **displacement, compensation, ecological damage and inadequate rehabilitation**.
 - MP government approved an additional **₹300 crore** for rehabilitation under the **Majhgaon and Runjh irrigation projects**.
 - A fresh survey identified around **750 additional families** for inclusion in rehabilitation plans.

SC Verdict on Ex Post Facto Environmental Clearances

SC has **quashed the Centre's 2021 Office Memorandum on ex post facto Environmental Clearances (ECs)**, reaffirming that **prior ECs under the Environmental Impact Assessment (EIA) Notification, 2006** is mandatory.

- Retrospective environmental regularisation must be granted **only through a statutory notification under the Environment (Protection) Act, 1986**, not an administrative order, as such orders **cannot create a perpetual, parallel amnesty regime**.
 - The 2021 Office Memorandum promoted a **"pollute and then pay"** approach, failing the **tests of proportionality and reasonable classification**, thereby violating **Article 14 (Right to Equality)** and **Article 21 (Right to Life and a Clean Environment)**.
 - ECs already granted under the **2017 Notification** and **2021 OM** remain valid unless individually challenged. Any future amnesties require a valid statutory notification under **Section 3 of the Environment (Protection) Act, 1986**.
- **Supervening Public Interest:** Retrospective clearances are not entirely barred but must be restricted to a **"narrow, time-bound, reasoned, and supervised window"**

applicable only to projects of immense public utility (e.g., hospitals, airports, strategic infrastructure) **under Section 3 of the Environment (Protection) Act, 1986.**

- **Ex Post Facto EC:** Refers to **retrospective environmental clearance** granted to projects that started without prior approval.
- **Shift in Environmental Jurisprudence:** The judgment marks a nuanced transition from an absolutist application of the **precautionary principle** to balancing it with the **polluter pays principle** and **sustainable development**, acknowledging the heavy socioeconomic and environmental costs of demolishing massive, integrated public infrastructure.
- **Mandate of the 2006 EIA Notification:** The overarching legal principle remains that “**prior environmental clearance**” is **mandatory** before the commencement of construction or operations to evaluate ecological impacts, hear public objections, and prescribe safeguards.

Cloudbursts in India

IMD clarified that the **Upper Assam floods were caused by prolonged heavy rainfall, not cloudbursts** in Assam or Nagaland.

Cloudburst

- **Cloudburst:** IMD defines a **cloudburst** as rainfall of **100 mm or more within one hour** over an area of about **20–30 sq. km.**
 - It is a **highly localised and rare** event, more intense than ordinary heavy monsoon rainfall. Such rainfall can trigger **flash floods, landslides and debris flows.**
 - Some scientists use the term “**mini-cloudburst**” for rainfall of about **50 mm in one hour**, which may also cause severe damage.

Cloudburst	Heavy Rainfall
≥ 100 mm rainfall in one hour over 20–30 sq. km	High rainfall spread over a larger area and longer duration
Highly localized	Regional or widespread
Causes flash floods within minutes	Generally causes riverine or urban flooding
Rare event	Common during monsoon

- **Mechanism of Formation:** Moist air is forced upward by mountain slopes through **orographic lifting.**
 - Rapid cooling and condensation form tall **cumulonimbus clouds**, sometimes extending up to **15 km** vertically.
 - Strong **updrafts** hold water droplets inside the cloud.
 - When the droplets become too heavy or the updraft weakens, a large volume of rain falls suddenly over a small area.

- **Geographical Susceptibility in India:** IMD recorded about **30 cloudburst incidents between 1970 and 2016.** Their frequency may rise as **warmer air holds more moisture.**

Region	Reason
Himalayan states and Northeast India	Topography (orographic effect) makes these regions highly prone during southwest monsoon
Western Ghats	Elevated relief obstructs moisture-laden monsoon winds

Forecasting Challenges:

- **Hyper-localised nature:** Cloudbursts affect only **20–30 sq. km**, often below the resolution of numerical weather models.
- **Rapid formation:** They develop within a few hours, providing very limited warning time.
- **Radar limitations:** Mountainous terrain can block **Doppler Weather Radar (DWR)** signals.
- **Sparse observation network:** Remote Himalayan regions have limited **Automatic Weather Stations (AWS)** and rain gauges.
- **Data gaps:** Many observatories record rainfall at **3-hour or 24-hour intervals**, whereas IMD’s cloudburst threshold is **100 mm in one hour.**
- **High computational requirement:** Hyperlocal forecasting needs high-resolution models and advanced computing capacity.

India’s Initiatives to Address Cloudbursts:

- **Nowcasting:** IMD issues short-duration forecasts and frequent alerts for localised extreme weather.
- **Mission Mausam:** India expanded its DWR network from **14 in 2014 to 50**, covering over **87% of the country; 50 more radars** are planned.
 - ❖ AI is being integrated to improve **hyperlocal forecasting** and prediction of extreme weather events.

Misuse of the Term ‘Cloudburst’

The term **cloudburst** is often used for any intense rainfall, even when the **IMD threshold of 100 mm rainfall in one hour over 20–30 sq. km** is not met.

- Such misuse can shift attention from governance failures, including **poor drainage, unplanned construction, deforestation and weak early-warning systems.**
- **IMD** uses a scientific, rainfall-based definition, while **State Disaster Management Authorities (SDMAs)** may rely on field observations due to inadequate real-time monitoring.
- This difference leads to **inconsistent reporting, damage assessment and accountability.**

Event	Assessment
2022 Telangana floods	Cloudburst claims were disputed; experts pointed to issues linked with the Kaleshwaram irrigation project .
2025 Dharali floods, Uttarakhand	Rainfall was below the cloudburst threshold; riverbed construction, deforestation and poor drainage were cited.
2026 Upper Assam floods	IMD attributed the floods to prolonged heavy rainfall , not cloudbursts.

Western Ghats Ecologically

Sensitive Area: 7th Draft Notification

The MoEFCC has **reissued the draft notification (7th)** proposing an **Eco-Sensitive Area** across the **Western Ghats**.

Highlights of the 7th Draft

Notification on Western Ghats ESZ

- **Geographical Spread:**
 - Proposes an **Eco-Sensitive Area (ESA)** of **56,825.7 sq. km** across **Karnataka (highest at 20,668 sq. km), Maharashtra, Kerala, Tamil Nadu, Goa and Gujarat (lowest at 449 sq. km).**
- **Prohibited Activities:** Bans commercial mining, quarrying, sand mining, new thermal power plants, red-category industries and large construction projects.
- **Regulated and Exempted Activities:** **Hydropower projects** will be regulated, while **agriculture, plantations and local livelihoods** will remain unaffected.
- **Phased Notification:** Allows state-wise or combined notification without waiting for consensus from all **six states**.
- **Expert Committee:** The **Sanjay Kumar-led committee**, constituted in **2022** to examine state objections and village-demarcation issues, has been extended until **July 2027**.

Western Ghats ESA Proposal

The proposal to notify the **Western Ghats as an ESA** has remained pending since **August 2011**.

- **Gadgil Committee (2011):** Recommended declaring the entire **Western Ghats** as an ESA and dividing it into **ESZ-I, ESZ-II and ESZ-III** based on ecological sensitivity. States opposed it as excessively restrictive on development.
- **Kasturirangan Committee (2013):** Recommended notifying about **37% of the Western Ghats**, nearly **60,000 sq. km**, as ESA while permitting sustainable development elsewhere.

Challenges in Notifying the Western Ghats ESA

- **Centre–State Deadlock:** Disagreement persists over balancing conservation with development.
- **Boundary Demarcation:** Disputes over satellite mapping, village boundaries and ESA identification delay finalisation.
- **Lack of Conservation Incentives:** States face economic costs without adequate compensation or ecological fiscal transfers.
- **Regulatory Uncertainty:** Repeated draft notifications have created a policy vacuum, enabling illegal mining, quarrying, deforestation and unsustainable development.

Key Facts About the Western Ghats

- **Geographical Expanse:** Extend about **1,490 km** from **Gujarat to Tamil Nadu and Kerala**, covering nearly **1.4 lakh sq. km** across **six states:** Gujarat, Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu.
- **Geological Origin:** Older than the Himalayas; formed along the faulted edge of the **Deccan Plateau** after the break-up of **Gondwana**.
- **Structural Divisions:** Divided into the **Northern, Central and Southern Western Ghats** by the **Goa Gap, Palghat Gap and Sencottah Gap**.
- **UNESCO Status:** Declared a **World Heritage Site in 2012** under Criteria **(ix) and (x)**. It comprises **39 protected components** grouped into **seven sub-clusters**.
- **Divisions of the Western Ghats**
 - **Northern Western Ghats:** Include the **Sahyadri Range** and basaltic Deccan Trap plateaus; major peaks include **Kalsubai, Salher and Mahabaleshwar**.
 - **Central Western Ghats:** Extend across Karnataka and are marked by heavy rainfall, evergreen forests and rivers such as **Sharavathi and Netravati**.
 - **Southern Western Ghats:** Include the **Nilgiri, Anamalai and Cardamom Hills**. **Anamudi (2,695 m)** is the highest peak of the Western Ghats and of India outside the Himalayas.
- **Major Hill Stations:** Include **Matheran, Mahabaleshwar, Panchgani, Lonavala-Khandala, Amboli, Kodagu and Kudremukh**.
- **River Systems:**
 - **West-flowing Rivers:** **Periyar, Netravati, Sharavathi, Mandovi and Bharathappuzha** are short, swift and hydroelectricity-rich rivers draining into the Arabian Sea.
 - **East-flowing Rivers:** **Godavari, Krishna and Kaveri** are longer rivers supporting irrigation, agriculture and drinking-water supply before entering the Bay of Bengal.

- **Ecological and Economic Significance:** One of the world's **eight hottest biodiversity hotspots** and a major barrier influencing the **southwest monsoon, climate and regional hydrology**.
 - Functions as the **water tower of Peninsular India** and contains major tropical evergreen forests.
 - Sequesters nearly **4 million tonnes of carbon annually**, around **10% of India's forest-based carbon sequestration**.
 - Supports mineral-based industries, non-timber forest produce and livelihoods of indigenous and forest-dependent communities.
- **Biodiversity and Conservation:** Contains over **5,000 flowering plant species**; about **65% of amphibians** and **53% of freshwater fishes** are endemic.
 - Supports at least **325 globally threatened species**, including the **Asian Elephant, Tiger, Gaur, Lion-tailed Macaque, Nilgiri Tahr and Nilgiri Langur**.
 - The **IUCN World Heritage Outlook** classifies its conservation status as of **"significant concern"** due to habitat fragmentation, mining, development and linear infrastructure.
 - The continuing debate over its **Eco-Sensitive Area demarcation** reflects the challenge of balancing conservation with regional development.
- **Anaerobic Conditions:** Waterlogged valleys and depressions can create oxygen-deficient conditions where microorganisms release methane while decomposing organic material.
- **Migration Through Faults:** **Faults, cracks, and fissures** allow methane from coal, natural gas, or geothermal deposits to rise and collect in enclosed spaces.
- **Waste-Related Sources:** Poorly managed waste dumps may generate methane through decomposition in oxygen-deficient layers.
- **Sikkim's Geological Context:** Southern Sikkim, including **Namchi**, has small coal deposits and lies within the geologically complex **Rangit Tectonic Window**.
- **Detection Challenge:** Small, localised methane pockets may escape standard geological surveys, which mainly assess rock stability, water conditions, and structural risks.

Vulnerability of Infrastructure in Sikkim

- **Teesta-VI Incident:** The **Teesta-VI tunnel disaster** is Sikkim's first recorded hydropower accident involving a suspected **methane-triggered explosion**.
- **Earlier Project Damage:** The **October 2023 Glacial Lake Outburst Flood (GLOF)** destroyed the **1,200 MW Chungthang dam** and damaged the **510 MW Teesta-V Hydropower Project**, which was again affected by a landslide in **August 2024**.
- **Fragile Himalayan Geology:** Sikkim lies in a **young, tectonically active, and erosion-prone Himalayan region**, making it vulnerable to earthquakes, landslides, and slope failures.
- **Impact of Construction:** Blasting and tunnelling for dams, highways, and railways can destabilise slopes and increase landslide and mudslide risks.
- **Human Pressures:** **Unregulated urbanisation, heavy tourism, deforestation, and poor waste management** intensify environmental and geological vulnerabilities.
- **Climate-Related Threats:** **Cloudbursts, glacier melting, and expanding glacial lakes** are increasing the frequency and intensity of disasters affecting hydropower infrastructure.

Why Has Tunnelling Been Disaster-Prone in India?

- **High Geological Risk:** Tunnelling in the **Himalayan and northeastern regions** is hazardous due to unstable rocks, active faults, heavy rainfall, and frequent landslides.
- **Systemic Failures:** Disasters also arise from poor planning, weak construction quality, inadequate oversight, and limited accountability.

Teesta Tunnel Disaster: Methane Risks and Himalayan Infrastructure Vulnerability

A suspected **methane and toxic gas explosion** inside the under-construction **Head Race Tunnel** of the **Teesta Stage VI Hydropower Project** at **Samardung, Namchi, Sikkim**, killed several workers and trapped others.

How Is Methane Trapped in Hilly Terrain?

- **About Methane:** It is a colourless, odourless, and highly flammable gas that can accumulate in underground rocks and confined spaces such as tunnels.
- **Explosion Mechanism:** Methane mixed with air can ignite from sparks produced by drilling or construction machinery. Tunnel excavation may disturb methane-bearing rocks and trigger a sudden gas release.
- **Methane in Rocks:** It may remain trapped in **porous rocks** or isolated pockets within young and fragile geological formations.
- **Organic Origin:** In **coal-bearing or carbon-rich strata**, methane forms through the long-term decomposition of organic matter.

- **Inadequate Risk Assessment:** Incomplete surveys and weak predictive modelling may miss **fault zones, groundwater, weak rocks, and gas pockets**.
- **Poor Safety Compliance:** Inadequate ventilation, fire protection, escape routes, and real-time monitoring increase underground risks.
- **Worker Vulnerability:** Limited training, insufficient protective gear, and unclear contractor responsibility heighten danger.
- **Need for Reforms:** India needs stricter enforcement, continuous geological monitoring, clear contractor liability, and specialised rescue teams with modern equipment.

Soil Management for Sustainable Agriculture

The Government promotes soil conservation through initiatives such as the **Soil Health Card Scheme, National Mission on Natural Farming, Khet Bachao Abhiyan, geospatial soil mapping and digital agriculture platforms**.

Soil as a Strategic National Resource

- **Foundation of Agriculture:** Soil supports crop production, ecosystems and rural livelihoods. With **46%+ workforce dependent on agriculture**, soil health affects farm incomes, food security and economic stability.
- **Extent of Agricultural Land:** As per **Land Use Statistics 2022–23**, India's reported area is **306.65 million hectares**, with nearly **59% under agriculture**.
- **Food and Climate Security:** Healthy soil improves yields, water and nutrient retention, and reduces external input dependence, supporting **SDG 2 (Zero Hunger)** and **SDG 13 (Climate Action)**.
- **Sustainable Soil Management:**
 - India focuses on **climate-resilient agriculture, natural farming, soil mapping, land restoration and digital agricultural technologies**.
- **Government Initiatives:** **National Mission for Sustainable Agriculture** and **National Mission on Natural Farming** promote resource-efficient and sustainable practices.
- **Soil Health Card Scheme:** Provides soil-testing results and crop-specific nutrient recommendations for balanced fertiliser use.
- **Public Awareness:** **Khet Bachao Abhiyan** promotes soil conservation, balanced fertiliser use and sustainable farming.

How Does Soil Diversity Strengthen Indian Agriculture?

- **Alluvial Soils:** Found in Indo-Gangetic and Brahmaputra plains and river deltas; support intensive agriculture in

Punjab, Haryana, UP, Bihar, Assam, West Bengal, Odisha and Andhra Pradesh.

- **Black Soils:** Found mainly in **Maharashtra, Madhya Pradesh and Gujarat**; clayey, moisture-retentive and rich in lime and calcium.
- **Red Soils:** Spread across **Tamil Nadu, Karnataka, Andhra Pradesh, Madhya Pradesh, Bihar, West Bengal and Rajasthan**; rich in iron but nutrient deficient.
- **Laterite Soils:**
 - Found in high rainfall regions of **Andhra Pradesh, Karnataka, Kerala, Madhya Pradesh, Odisha, Assam and Maharashtra**; formed due to intense leaching.
- **Desert Soils:** Found in **Rajasthan, Gujarat and Leh-Ladakh**; sandy with low moisture retention and soluble salts.
- **Forest and Hill Soils:** Found in Himalayan and North-Eastern regions; vary with altitude, rainfall, temperature, vegetation and slope.
- **Saline and Alkaline Soils:** Found in arid, semi-arid, coastal and waterlogged regions; affected by poor drainage.
- **Marshy Soils:** Found in waterlogged areas of **Tamil Nadu, Bihar and Uttar Pradesh**; rich in organic matter.

Role of Soil in Climate Action and Resilience

- **Soil as a Carbon Sink:** Soil stores atmospheric carbon dioxide as soil organic carbon through vegetation, roots and microorganisms.
- **Benefits of Soil Organic Carbon:** Improves soil structure, fertility, nutrient availability, moisture retention and drought resilience.
- **Carbon Farming:** Promotes practices that remove and store carbon in soil, supporting climate-smart agriculture.
- **Government Support:** **Green India Mission, National Mission for Sustainable Agriculture, National Mission on Natural Farming and Carbon Credit Trading Scheme** promote carbon storage and climate resilience.

National Mission for a Green India

- One of the **eight missions under NAPCC**.
- Enhances forest and tree cover on forest, non-forest and degraded lands, improving ecosystem services, soil carbon and climate resilience.
- Supports India's NDC target of creating an additional carbon sink of **2.5–3.0 billion tonnes CO₂ equivalent by 2030**.
- India created an additional carbon sink of **2.29 billion tonnes CO₂ equivalent (2005–2021)**.

National Mission for Sustainable Agriculture

- Launched in **2014–15** under **NAPCC**.
- Promotes climate-resilient agriculture through **soil health improvement, efficient water use, integrated farming and sustainable resource management**.
- Since 2014–15, **₹2,119.84 crore** released under **Rainfed Area Development**, covering **8.50 lakh hectares** and benefiting **14.35 lakh farmers**.

National Mission on Natural Farming

- Launched in **2024** to promote nature-based farming, improve soil health, increase soil organic carbon and reduce input costs.
- **As of March 2026:**
 - Covered **18,786 clusters across 8.80 lakh hectares**.
 - Enrolled **18.19 lakh+ farmers**.
 - Supported by **33,676 trained Community Resource Persons**.
- Bio-input Resource Centres provide inputs like **Beejamrit and Jeevamrit**.

Carbon Credit Trading Scheme

- Adopted in **2023** to incentivise activities reducing, avoiding or removing greenhouse gas emissions.
- Under voluntary carbon markets, verified activities generate tradable credits, with each credit representing **one metric tonne of CO₂ equivalent**.
- Farmers can sell verified carbon credits for additional income.

Integrated Initiatives for Strengthening Soil Health and Fertiliser Management

- **Soil Health and Fertility Scheme:** Provides Soil Health Cards with nutrient status and crop-specific recommendations for balanced fertiliser use.
- **Training and Grassroots Support:** ATMA, Krishi Vigyan Kendras and **70,002 trained Krishi Sakhis** promote judicious fertiliser use, organic manure and bio-fertilisers.
- **Expansion of Soil Health Cards:** By **13 March 2026**, **25.89 crore Soil Health Cards** generated and **7.17 lakh demonstrations** conducted.
- **School Soil Health Programme:** During **2025–26**, involved **4,430 schools and 1,29,167 students**.

Nutrient-Based Subsidy Scheme

- Links fertiliser subsidies with nutrient content and covers **28 grades of phosphatic and potassic fertilisers**.

- **Direct Benefit Transfer in Fertilisers:** Aadhaar-linked PoS devices ensure authentication and real-time sales tracking; subsidies of **₹6.77 lakh crore (2022–23 to July 2025)** released.
- **Neem-Coated Urea:** Improves nitrogen-use efficiency and prevents diversion.
- **Integrated Fertiliser Management System:** Improves monitoring, logistics and transparency.
- **Khet Bachao Abhiyan:** Promotes soil test-based nutrient management and sustainable farming practices.

Building a Digital Soil Ecosystem in India

- **Nationwide Soil Resource Mapping:** Soil and Land Use Survey of India uses satellite imagery and field observations for village-level soil maps.
 - By September 2024, **29 million hectares** mapped against the target of **142 million hectares**.
 - ₹1,076 crore provided to **UP, MP, Rajasthan, Maharashtra, Tamil Nadu and Andhra Pradesh**.
- **Digital Agriculture Mission:** Integrates soil, farmer, land, weather and crop datasets.
- **AgriStack:** Provides location-specific advisory services by integrating farmer, land, crop and soil data; **7.63 crore Farmer IDs** generated by November 2025.
- **Digital Crop Survey:** Surveyed **23.5 crore crop plots during Rabi 2024–25**.
- **Krishi Decision Support System:** Uses soil, weather and crop data for crop planning and nutrient management.
- **AI-Based Soil Diagnostics:** Uses AI, satellite imagery and remote sensing to identify soil stress and nutrient deficiencies.
- **Precision Resource Application:** Enables site-specific use of water and fertilisers, reducing costs and environmental damage.
- **Farmer Advisory Systems:** Assist in crop selection, sowing periods and nutrient management.
- **Agri-Robotics:** GPS, sensors and AI-based robots support soil sampling, planting and field management.
- **Institutional Innovation:** ICAR–Indian Agricultural Research Institute develops robotic technologies for precision soil analysis and agricultural operations.

Soil health is central to food security, climate resilience and rural livelihoods in India. Discuss the role of natural farming in securing India's agricultural future.

Drishti Mains Question

Science & Technology

Generative AI in Synthetic Biology

Researchers from **Stanford University** and the **Arc Institute** used **generative AI models (Evo 1 and Evo 2)** to design complete, functional genomes of **bacteriophages**.

- Out of **285 AI-designed genomes** synthesised and tested, **16 produced viable phages** capable of infecting *E. coli* and overcoming bacterial resistance.
- This marks a shift from AI **analysing biological data** to **designing functional biological entities**.

Generative Biology

- **Definition: Generative biology** combines **AI and synthetic biology**, where AI models learn patterns in **DNA, RNA and proteins** to design new functional biological entities **de novo (from scratch)**.
- **The Genomic Evolution:**

Stage	Period	Development
Reading	1970s	Sequencing natural genomes (e.g., ΦX174 bacteriophage)
Writing	2000s	Chemical synthesis of known DNA sequences
Modifying	2010s	Gene editing and targeted modifications
Designing	Present	AI-generated genetic blueprints for biological functions

- **Benefits & Opportunities:**
 - **Antimicrobial Resistance (AMR):** Enables customised **bacteriophage therapies** against multidrug-resistant bacteria ('superbugs').
 - **Targeted Gene Delivery:** Helps design synthetic viral vectors for delivering gene-editing tools like **CRISPR**.
 - **Vaccine and Drug Discovery:** Accelerates development of **antigens, monoclonal antibodies and therapeutic proteins**.
 - **Precision Oncology:** Supports engineering of **oncolytic viruses** targeting cancer cells.
- **Implications for India:** AI-driven biological design can support drug discovery, genomics, vaccine development, AMR research
 - India needs stronger biomedical AI, secure computing infrastructure, biological datasets, biosecurity frameworks
 - Initiatives like the **IndiaAI Mission** must balance **AI sovereignty with biosafety safeguards**.

Major Concerns Associated with Generative AI in Synthetic Biology

- **Dual-Use and Bioterrorism Risks:** The same AI capabilities useful for vaccines, gene therapy and AMR research could be misused to design harmful biological agents.
 - Main concern is **capability amplification**, where AI enhances the ability of existing experts or laboratories rather than independently creating pathogens.
- **Ecological Unpredictability:** AI-designed organisms released intentionally or accidentally may have unknown ecological impacts, including **species barriers and zoonotic risks**.
- **Acceleration of Biological Design:** AI can analyse vast genetic possibilities and speed up **Literature review, Biological modelling, Experimental design**
 - Combined with automated laboratories, it can shorten the time between **design and biological testing**.
- **Limitations of Existing Biosecurity Screening:** Current DNA synthesis screening mainly detects similarity with **known pathogens or toxins**.
 - AI-generated designs may be novel and escape existing screening.
 - **Future systems must assess:** Biological function, Potential effects
 - ❖ **Entire innovation chain including:** DNA synthesis providers, Research laboratories, Biological databases, Computing infrastructure
- **Regulatory and Patent Ambiguity:** Existing intellectual property laws do not clearly define ownership of **AI-generated biological entities**.

Way Forward

- **Mandatory Screening of DNA Synthesis:** Implement strict **Know Your Customer (KYC)** norms and sequence screening for gene-synthesis providers.
- **Updating Biological Weapons Convention (BWC), 1972:** International frameworks should explicitly address **AI-generated pathogens and synthetic biology risks**.
- **Hardware and Compute Governance:** Consider oversight of high computing power required for advanced biological AI models.
- **Red Teaming AI Models:** AI systems should undergo simulated **cyber-biological attack testing** to ensure they prevent harmful biological design requests.

The convergence of Artificial Intelligence and synthetic biology presents both transformative opportunities and unprecedented security challenges. Analyse.

Drishti Mains Question

Transforming India's Space Ecosystem

On India's 80th Independence Day, the Government highlighted the country's expanding **space capabilities**, **private-sector ecosystem**, **human-spaceflight ambitions**, **planetary exploration** and **advanced launch technologies**.

Strategic Reforms Driving Growth in India's Space Sector

- **Space Start-ups:** Increased from 1 in 2014 to around 440 by August 2026.
- **Space Economy:** Valued at about **USD 9 billion** in August 2026; projected to reach **USD 40–45 billion** over the next decade.
- **Private Investment:** Rose from **USD 100.5 million** in 2021–22 to **USD 618.5 million** by March 2026.
- **Non-Government Entities:** 113 authorisations granted to 52 NGEs, including 18 start-ups, by mid-2026.
- **Earth Observation Public-Private Partnership:** IN-SPaCe framework for India's **first privately owned Earth-observation satellite constellation**; attracted about **₹1,200 crore** in private commitments.
- **Foreign Satellites Launched:** Increased from 35 before 2014 to 399 by January 2026.
- **International Cooperation:** Over 300 agreements with 61 countries and 5 multilateral organisations.

Strategic Reforms Catalysing Growth in India's Space Sector

- **2020 Reforms:** Opened the space sector to private participation in satellites, launch systems and applications.
- **Indian Space Policy 2023:** Expanded private participation and defined roles of NSIL and IN-SPaCe.
- **NewSpace India Limited (NSIL):** ISRO's commercial arm; revenue rose from **₹321.77 crore** in FY 2021–22 to over **₹3,000 crore** in FY 2024–25.
 - Signed 118 Technology Transfer Agreements covering 83 technologies.
- **IN-SPaCe:** Single-window authorisation body for private space activities.
 - Facilitated 71 ISRO technology transfers by 31 January 2026.
- **FDI Liberalisation, 2024:** 74% automatic route in satellite manufacturing/operations.
 - 49% in launch vehicles and spaceports.
 - 100% in satellite and ground-segment components.

- **Norms, Guidelines and Procedures (NGP) 2024:** Improved authorisation clarity and Ease of Doing Business.
- **Funding Support:** ₹1,000 crore Venture Capital Fund, ₹500 crore Technology Adoption Fund, and ₹75 crore Satellite Bus as a Service Scheme.

Expanding India's Space Capabilities Beyond Earth

Self-Reliant Space Transportation

- India operates PSLV, GSLV, GSLV Mk III and SSLV.
- Between July 2023 and June 2026, India completed 12 launch missions, carrying 11 national and 9 international satellites.
 - Sriharikota: Two operational launch pads; third launch pad under development.
 - Kulasekarapattinam, Tamil Nadu: Emerging as India's second launch site, mainly for SSLV and private launches.

Lunar Exploration through the Chandrayaan Programme

- **Chandrayaan-1 (2008):** Detected water molecules and hydroxyl on the lunar surface.
- **Chandrayaan-2 (2019):** Captured lunar images up to 30 cm resolution.
- **Chandrayaan-3 (2023):** Made India the first country to soft-land near the lunar south polar region and fourth to soft-land on the Moon.
- **Chandrayaan-4 (2027):** Planned for lunar sample collection and return.
- **Chandrayaan-5/LUPEX:** Joint ISRO–JAXA mission to study lunar volatiles, including water, near the south pole.

Beyond the Moon: Mars, Sun and Deep Space

- **Mars Orbiter Mission:** Entered Mars orbit on 24 September 2014; operated for over 8 years.
- **AstroSat:** India's first dedicated astronomy mission covering X-ray, optical and ultraviolet bands.
- **Aditya-L1:** India's first space-based solar mission; operates near Sun–Earth L1, about 1.5 million km from Earth.
- **XPoSat:** Launched in January 2024 for X-ray polarisation astronomy.
- **SpaDeX:** In January 2025, India became the 4th country to demonstrate autonomous space docking and undocking.

Axiom-4 and Human Spaceflight Capability

- **Group Captain Shubhanshu Shukla** became the first Indian astronaut to visit the ISS through Axiom-4, launched on 25 June 2025.
- Mission concluded on 15 July 2025 and provided experience relevant to Gaganyaan, crewed operations and microgravity research.

NavIC — India's Indigenous Navigation System

- **Navigation with Indian Constellation (NavIC):** Provides navigation services across India and up to **1,500 km beyond its boundaries**.
- **NVS-01 and NVS-02:** Launched in **May 2023** and **January 2025**.
- **NVS-03:** Ready for launch; **NVS-04 and NVS-05** under advanced realisation.
- Applications include **power-grid synchronisation, train tracking, Aadhaar geo-tagging and mobile chipsets**.

Key Priorities in India's Future Space Programme

Human Spaceflight and Orbital Infrastructure

- **Gaganyaan:** Expanded in **September 2024** to **₹20,193 crore**, covering **8 missions**.
- **Objective:** Send **3 astronauts** to around **400 km orbit** for **3 days** and return safely.
- **First Uncrewed Mission:** Targeted for **Q4 2026**, including **Vyommitra** validation.
- **Crewed Gaganyaan:** Targeted for **2027**.
- **Bharatiya Antariksh Station (BAS):** Planned as a **5-module station by 2035**. **BAS-01** targeted for **2028**.
 - Intended for **long-duration human spaceflight, microgravity research and orbital operations**.

Advanced Orbital Docking Capabilities

- **SPADEX-2 and SPADEX-3:** Proposed to develop capabilities for:
 - **Spacecraft docking.**
 - **Sample and crew transfer.**
 - **Transfer of power, fluids and propellants.**
- Relevant to **Chandrayaan-4, Gaganyaan and BAS**.

Venus Orbiter Mission

- Targeted for **March 2028**.
- Will study Venus's **geology, atmosphere and ionosphere**.
- Will demonstrate **aerobraking and thermal-management technologies** for future deep-space missions.

India's space sector reforms are transforming the country from a government-led space programme into a broad-based space economy. Discuss.

Drishti Mains Question

Rotating Detonation Engines

- India-based defence start-up **D-Propulse** demonstrated **India's first indigenous 5 kN air-breathing Rotating Detonation Engine (RDE)** at a **DRDO facility in Hyderabad**, achieving **Technology Readiness Level-5 (TRL-5)**.
- It strengthens India's indigenous **next-generation air-breathing propulsion capability**.

- **RDE:** Uses continuous **supersonic detonation waves**, offering potentially **higher thermodynamic efficiency** and a **more compact combustor** than conventional engines.
- **Deflagration vs. Detonation:**

Feature	Deflagration	Detonation
Flame speed	Subsonic	Supersonic
Combustion	At constant pressure	At constant volume
Effect	Greater heat loss due to expansion	Shock wave compresses mixture before combustion, generating greater pressure and thrust

- **RDE Working Mechanism:** Uses an **annular combustion chamber** formed between two concentric cylinders.
 - Fuel and oxidiser are continuously injected ahead of the detonation wave, sustaining a **continuous circular detonation** and producing sustained thrust.
 - Unlike RDEs, **Pulsed Detonation Engines (PDEs)** burn fuel intermittently in a straight tube.
- **Thermodynamic Efficiency & Payload Advantage:** RDEs may provide **10-25% higher thermodynamic efficiency** than traditional rocket engines.
 - A **20% efficiency gain** can require about **17% less fuel**, reducing launch costs and allowing larger **satellite payloads or missile warheads**.
- **Key Strategic Applications:**
 - **Hypersonic systems:** Potential use in hypersonic cruise missiles such as **BrahMos-II**, high-speed aircraft/UAVs and **ramjet-type propulsion**.
 - **Space Launch Vehicles & In-Space Thrusters:** Can improve the **payload-to-fuel mass ratio** for orbital and deep-space missions.
- **Engineering and Material Hurdles:** Must withstand temperatures above **2,000°C**, pressures of **10-100 atmospheres** and detonation velocities above **1,500 m/s**.
 - Requires **microsecond-level fuel-injection precision** and advanced computing to control combustion instability.
- **Global R&D Status:** RDE technology remains in the **R&D stage globally**; **Lockheed Martin, GE Aerospace, Astrobotic and SpaceWorks** are conducting hot-fire tests for missile and space applications.

Total Solar Eclipse

On **12 August 2026**, a **total solar eclipse** was visible across **eastern Greenland, western Iceland, northern Russia, northern Spain and a small part of Portugal**. Scientists used it to study the Sun's faint **corona**, normally obscured by the bright solar surface.

Total Solar Eclipse

- **Alignment:** It occurs when the **Sun, Moon and Earth align nearly in a straight line (syzygy)**.
- **Shadow:** The Moon passes between Earth and the Sun and completely blocks the Sun for observers within the **umbra**.
 - **Umbra:** Region of complete shadow where totality occurs.
 - **Penumbra:** Region of partial shadow producing a partial eclipse.
 - As Earth rotates and the Moon moves, the shadow sweeps across Earth, creating the **eclipse path**; the narrow umbral region is the **path of totality**.
 - Since the umbra is narrow, totality is visible over a small area and usually lasts only **a few minutes or less at one location**.
- **Totality:** The period when the Sun is completely covered; the sky darkens and the **corona** becomes visible, enabling study of solar activity.

Solar Eclipse and its Types

- **Solar Eclipse:** Occurs when the **Moon passes between the Sun and Earth**, blocking some or all sunlight.
 - It occurs only during a **New Moon** with suitable alignment.
 - Eclipses do not occur every New Moon because the Moon's orbit is inclined about **5°** to the **ecliptic plane**. They occur during **two eclipse seasons each year**, when the New Moon crosses this plane.

Types of Solar Eclipse

- **Total:** Moon completely covers the Sun's disc within the **path of totality**, revealing the corona.
- **Annular:** Moon is near its farthest point from Earth and appears smaller, leaving a bright **"ring of fire."** It involves the **antumbra**, the continuation of the shadow beyond the umbra's tip.
- **Partial:** Sun, Moon and Earth are not perfectly aligned, so only part of the Sun is covered.
- **Hybrid:** Appears total from some locations and annular from others due to Earth's curvature and the changing position of the Moon's shadow.

Significance of a Total Solar

Eclipse in Studying the Solar Corona

- **Inner Corona:** During totality, the Moon blocks the bright **photosphere**, making the faint inner corona directly visible—an area difficult for coronagraphs because of scattered light and diffraction.

- **Coronal Heating Problem:** The corona exceeds **1 million°C**, while the photosphere is about **5,500°C**. Eclipse observations help study how energy reaches the outer atmosphere.
- **Magnetic Structures:** Coronal plasma, streamers and related structures reveal the Sun's magnetic field and help explain solar activity and the **11-year solar cycle**.
- **Space Weather:** The corona produces the **solar wind** and is the source region of **Coronal Mass Ejections (CMEs)**, which can disrupt satellites, GPS, communications and power grids.
- **Solar Wind:** Observations help explain how charged particles such as electrons and protons are accelerated into space.

Historical Discoveries During Total Solar Eclipses

- **Helium (1868):** During the **1868 eclipse in India**, French astronomer **Pierre Jules Janssen** detected an unknown yellow spectral line in the solar chromosphere. English astronomer **Norman Lockyer** later named the proposed element **helium**, after *Helios*, the Greek word for Sun.
- **General Relativity (1919):** **Arthur Eddington's** eclipse observations provided early observational evidence for Einstein's prediction that gravity bends light, supporting the **General Theory of Relativity**.

Physicist Deepak Dhar

Awarded the 2026 Dirac Medal

Theoretical physicist **Prof. Deepak Dhar** has been named co-recipient of the **2026 Dirac Medal** of the **Abdus Salam International Centre for Theoretical Physics (ICTP)**, Trieste for contributions to **statistical physics**, especially the **Abelian Sandpile Model**.

- **Co-recipients:** **Bernard Derrida, Marc Mézard and Haim Sompolsky.**
- **Significance:** Dhar is the **2nd scientist working in India** to receive the medal, after **Ashoke Sen (2014)**.

Abelian Sandpile Model

- **Developed:** By Deepak Dhar in **1990**, building on the **Self-Organized Criticality** model proposed in 1987 by **Per Bak, Chao Tang and Kurt Wiesenfeld**.
- **Basic Idea:** Sand grains are added until a **critical state** is reached; an additional grain may trigger an avalanche of varying size.
- **Abelian Property:** The **final stable state is independent of the order in which grains topple**, making cascading behaviour mathematically tractable.
- **Self-Organized Criticality:** Shows how systems can naturally evolve to a critical state where small disturbances may trigger minor or very large events.

- **Applications:** Used to understand cascading behaviour in earthquakes, forest fires, traffic jams, neural activity and financial-market fluctuations.

Deepak Dhar

- Born in 1951, Pratapgarh, UP. Studied at Allahabad University and IIT Kanpur; PhD from Caltech (1978). Served as a teaching assistant to Richard Feynman at Caltech.
- **Boltzmann Medal (2022):** First Indian recipient; co-recipient with John Hopfield. Among only 10 scientists globally to have received both the Boltzmann Medal and Dirac Medal.
- **Other honours:** Padma Bhushan (2023), Shanti Swarup Bhatnagar Prize (1991), The World Academy of Sciences (TWAS) Prize (2002).

Dirac Medal

- **Institution & History:** Instituted in 1985 by ICTP, Trieste (Italy) in honour of Paul A.M. Dirac, Nobel laureate and quantum mechanics pioneer.
 - **ICTP:** Founded in 1964; international institute for physical and mathematical sciences under Italy, UNESCO and IAEA; a UNESCO Category 1 Institute.
- **Selection & Criteria:** Announced annually on 8th Aug (Dirac's birthday) for outstanding contributions to theoretical physics.
- **Exclusion Rule:** Traditionally not awarded to existing Nobel Prize, Fields Medal or Wolf Prize recipients; some medallists later won the Nobel Prize.
- **Founding Vision:** ICTP was founded in 1964 by Abdus Salam to promote scientific collaboration between the Global South and the rest of the world.

NASA Invites ISRO to Join Moon Base Programme

NASA invited ISRO to join its Moon Base programme to establish a permanent human outpost near the Moon's South Pole.

- **India-US CSJWG:** Invitation came at the 9th India-US Civil Space Joint Working Group meeting at ISRO Headquarters, Bengaluru.
- **TRUST Initiative:** Meeting advanced civil and commercial space cooperation under US-India TRUST (Transforming the Relationship Utilising Strategic Technology), reviewed NISAR (NASA-ISRO Synthetic Aperture Radar) mission and reaffirmed UN COPUOS long-term sustainability guidelines.
- **Artemis Link:** Builds on India-US cooperation under the Artemis Accords, which India joined in June 2023.

Moon Base Programme & Artemis Framework

- **About Artemis Programme:** NASA-led programme to return humans to the Moon for the first time since Apollo

(1972), establish a permanent lunar base and prepare for future crewed Mars missions.

- **Moon Base Project:** Three-phase plan for humanity's first outpost near the lunar South Pole; region is important due to water ice in shadowed areas.
 - **Phase I (Through 2029):** Robotic South Pole missions, technology demonstrations and groundwork.
 - **Phase II (2029-2032):** Building systems and critical infrastructure for long-term presence.
 - **Phase III:** Assemble permanent outpost for astronauts to live, work and conduct extensive research.
- **Artemis Accords:** Established in 2020 by NASA, US State Department and seven initial signatories; non-binding principles for peaceful, safe and sustainable civil space exploration. India became 27th signatory in 2023.
- **Aligns with Space Vision 2047:** Indian Moon landing by 2040; Bharatiya Antariksh Station first module by 2028 and full station by 2035.
 - Would move India from independent missions such as Chandrayaan-3 towards complex multilateral human-spaceflight projects and strengthen its role as a global space power.

Lab-Grown Diamonds

LGDs are gaining attention as a sustainable alternative amid declining natural reserves and mining-related environmental and ethical concerns. India is promoting indigenous production through research and technological innovation.

Key Features and Production

- **About:** LGDs are synthetic diamonds produced in laboratories and are chemically, physically, and optically identical to mined diamonds, with the same crystalline carbon structure.
- **Properties:** They have the same rarity, scintillation, lustre, and durability as natural diamonds.
- **Traceability and Ethics:** LGDs offer full supply-chain traceability and avoid links with environmental destruction, human rights abuses, and conflict financing associated with mined diamonds.
- **Cost-Effectiveness:** Improved equipment and manufacturing efficiency have reduced LGD prices, making them more affordable.
- **Production Methods:**
 - **HPHT (High Pressure, High Temperature) (1954):** Replicates conditions in the Earth's mantle by subjecting carbon to extreme pressure and temperature.

- **CVD (Chemical Vapour Deposition) (1962):** Places a **diamond seed** in carbon-rich gases such as methane; ionised plasma enables carbon to deposit layer by layer.
- **Impact of Adopting Lab-Grown Diamonds:**
 - **Environmental Sustainability:** Avoids deforestation, soil erosion, high water use, and ecological displacement caused by mining.
 - **Carbon Footprint Reduction:** Renewable-energy-based production lowers emissions and aligns with the **UN SDGs** and circular economy principles.
 - **Industrial Advancements:** Used in cutting, drilling, semiconductors, quantum computing, and protective coatings due to their durability and thermal conductivity.

India's Push for Indigenous Production

- **Global Role:** India processes about 90% of the world's diamonds and accounts for nearly 75% of global diamond trade by value.
- **LGD Production:** India manufactured over 3 million LGDs in 2023, more than 15% of global production.
- **Budget Support:** Union Budget 2023-24 supported research in HPHT and CVD technologies.
- **InCent-LGD:** Commerce Ministry granted about ₹243 crore for five years to IIT Madras to develop indigenous machinery, processes and diamond seeds.

Concerns Regarding Natural Diamond Mining

- **Dwindling Supplies:** Mines accounting for **90% of global production** may be exhausted within **50 years**; major mines such as **Argyle** and **Diavik** are nearing exhaustion.
- **Occupational Health Hazards:** Asbestos exposure can cause **asbestosis, pleural plaques, and mesothelioma**.
- **"Blood Diamonds" and Conflict:** Illicit diamond sales fund military action and civil violence; gaps in the **Kimberley Process** enable smuggling.
 - India is the **Kimberley Process Chair for 2026** and a **founding member of KPCS**, leading efforts against conflict diamonds and for transparent trade.

- **Majhgawan Diamond Mine, Panna, Madhya Pradesh,** operated by **NMDC**, is India's only active mechanised commercial diamond mine.
- Major historical regions– **Panna Diamond Belt, Krishna River gravels, Mahanadi River gravels, and Wairagarh Conglomerates.**
- Major kimberlite deposits– **Majhgawan Pipe and Wajrakarur Pipe.**

World Hepatitis Day (WHD) 2026

World Hepatitis Day was observed on **28 July 2026** to raise awareness about viral hepatitis.

- Theme 2026 - **"Hepatitis: Let's Break It Down."**

Key Facts About Hepatitis

- **About:** Hepatitis is liver inflammation caused by **viral infections, alcohol, certain drugs or autoimmune disorders**.
- **Major Types:** The five main viruses are **hepatitis A, B, C, D and E**.
 - Chronic **hepatitis B and C** cause over **90% of hepatitis-related deaths and disability**, mainly through cirrhosis, liver failure and liver cancer.
- **Global Burden:** Viral hepatitis causes nearly **1.34 million deaths** and **2 million new infections annually**. Around **287 million people** live with chronic hepatitis B or C, mostly undiagnosed.
- **India's Burden:** India has about **29 million hepatitis B** and **5.5 million hepatitis C** cases (second highest-burden country).
 - It accounts for **10–15%** of the global hepatitis B carrier pool.
- **Seroprevalence:** In **2021**, hepatitis B seroprevalence was **0.85%** and hepatitis C was **0.29%** in India.
- **Diagnosis Gap:** Only about **2.4%** of hepatitis B cases in India are diagnosed, as early infection is often asymptomatic.
- **Vaccination Gap:** Hepatitis B birth-dose coverage is around **63%**, compared with **93%** third-dose coverage, increasing the risk of mother-to-child transmission.
- **Symptoms:** Possible signs include fatigue, loss of appetite, nausea, abdominal discomfort, dark urine and jaundice.
- **Major Barriers:** Low awareness, stigma, misinformation, weak primary-care integration and high treatment costs restrict testing and treatment.
- **WHO Targets for 2030:** Reduce new hepatitis B infections by **90%**, hepatitis-related deaths by **65%**, and achieve **90% diagnosis coverage**.

National Viral Hepatitis Control Programme (NVHCP)

- **Launch:** Started in **2018** to provide hepatitis screening, diagnosis and treatment and reduce related illness and deaths.
- **Coverage:** Addresses **hepatitis A, B, C, D and E** through preventive, promotive and curative measures, supporting the **2030 Sustainable Development Goal** on communicable diseases.
- **Surveillance and Treatment:** **Hepatitis A and E** outbreaks are monitored through the **Integrated Disease Surveillance Programme**; free diagnosis and medicines are provided for **hepatitis B and C**.
- **Digital Monitoring:** The **NVHCP Management Information System portal** enables paperless registration, reporting and treatment monitoring.

Supporting National Health Programmes

- **Reproductive, Maternal, Newborn, Child, Adolescent Health and Nutrition programme:** Provides universal screening of pregnant women for hepatitis B.
- **Universal Immunisation Programme:** Provides the **hepatitis B birth dose** to newborns; babies of infected mothers also receive **hepatitis B immunoglobulin within 24 hours**.
- **National AIDS Control Programme:** Screens high-risk groups for hepatitis B and C and vaccinates eligible healthcare workers and high-risk individuals.
- **National Tuberculosis Elimination Programme:** Supports hepatitis diagnosis using existing **molecular-testing machines**.
- **Blood Transfusion Services:** Link hepatitis-positive blood donors with healthcare facilities for evaluation and treatment.
- **Helpline: 1800-11-6666** provides information on prevention, transmission, treatment and service centres.

Mega Data Centres and the Sustainability Challenge

Protests in Andhra Pradesh and Maharashtra raised concerns over the water, electricity and environmental impact of mega data centres.

- **Google:** 1 GW AI data centre hub in Visakhapatnam–Anakapalli.
- **Amazon:** 422 MW hyperscale data centre in Thane.

Data Centres

- **About:**
 - Data centres are specialised facilities that securely store, manage and process electronic data using servers, storage systems, networks, cooling, power and security infrastructure. They support digital services such as cloud computing and artificial intelligence. They house **critical Information Technology (IT) infrastructure**.
- **Geographical Distribution:**
 - India has around **150 data centres**, led by AWS, Microsoft Azure and Google Cloud. **Mumbai accounts for over 50%** of the capacity, while Ahmedabad, Pune and Visakhapatnam are emerging hubs.
- **Growth Drivers:**
 - **Digitalisation:** India generates nearly **20% of global data** but has only **3% of global data-centre capacity**.
 - **AI and Cloud:** AI capacity may rise from **275 MW to 6,546 MW by 2030**; total capacity may reach **12 GW**.

- **Digital Users:** Over **1.03 billion internet users** and about **22 billion monthly UPI transactions**.
- **Government Support:** Subsidised land and power, tax benefits and faster approvals.
 - E.g. Gujarat's Data Centre Policy 2026–29 targets ₹6 lakh crore in investment and 7.5 GW of capacity, while Andhra Pradesh plans to develop about 5 GW of capacity in Visakhapatnam.
- **Private Investment:** Google, Reliance–Meta, Yotta and TCS HyperVault.
- **Data Localisation:** Promotes domestic and secure data storage.
- **Government Initiatives:** Dedicated State data-centre policies (Gujarat, Uttar Pradesh, Tamil Nadu, Telangana, Maharashtra and Andhra Pradesh).
 - Fiscal incentives and concessional land.
 - Data-centre parks, substations and single-window clearances.
 - **Digital India** and **National Informatics Centre Data Centres**.

Major Concerns	Way Forward
Water Stress: A 1 MW facility may use 25.5 million litres annually.	Treated wastewater, closed-loop cooling and rainwater harvesting. Avoid water-stressed, densely populated and ecologically sensitive areas.
Electricity Grid Pressure: AI data centres may add 26.3 GW demand by 2031–32.	Renewable energy, efficient servers and battery backup.
Environmental Impact: Data centres may consume 7% of India's electricity by 2040.	Public consultation and transparent environmental assessments.
Limited Employment: High resource use but few permanent jobs.	Ensure local employment, skill development and benefit-sharing.
Weak Transparency: Unclear resource-use data and inadequate public consultation.	Mandatory disclosure of water, power, emissions and waste.
Strategic Risks: Excessive subsidies, stranded assets and dependence on global hyperscalers.	Real-time monitoring, public reporting and penalties for violations.

Mega data centres are critical for India's digital economy and AI ambitions, but they also raise concerns over water use, power demand and environmental governance. Discuss.

Drishti Mains Question

History, Art & Culture

Royal Indian Navy Mutiny of 1946

As India marked its **80th Independence Day (15th August 2026)**, the **Royal Indian Navy (RIN) Mutiny of 1946** is being remembered for its role in **bringing India's independence closer**. Although the mutiny ended with the sailors' surrender, it **shook British confidence in the loyalty of Indian armed forces**, a crucial pillar of colonial rule.

What is the Royal Indian Navy (RIN) Mutiny of 1946?

■ About:

- The Royal Indian Navy (RIN) Mutiny of 1946 was a major uprising by Indian naval ratings against British authority, which received widespread support from workers, students and civilians and generated sympathy within other branches of the armed forces.
- Starting as a strike on **18th February 1946**, in Bombay (now Mumbai), the revolt rapidly expanded into a broader anti-imperial movement across British India, accelerating the nation's path to independence.

■ Causes and Triggers:

- **Post-War Demobilization:** During World War II, the **Royal Indian Navy** expanded from under **2,000 to nearly 30,000 personnel**. After the war, Britain planned to reduce it to around **11,000**, creating **job insecurity and loss of status** among Indian sailors.
- **Racial Discrimination and Poor Conditions:** Indian ratings faced **racial discrimination, poor living conditions and inferior food** compared to British personnel. Between **1942 and 1945**, **nine smaller mutinies** had already occurred over morale and management issues.
- **Immediate Spark:** Anti-British slogans such as **"Quit India"** and **"Down with the British"** were painted at HMIS Talwar in late 1945 and again in early February 1946
 - ❖ In February 1946, Commander **Arthur Frederick King's racial abuse** and refusal to address their complaints triggered the full-scale strike on **18th February 1946**.
- **Nationalist Fervour:** The **Indian National Army (INA) trials** and the influence of **Subhas Chandra Bose** intensified nationalist sentiment among the sailors, turning service grievances into a broader **anti-colonial protest**.

- The revolt **spread from Bombay to naval bases in Karachi, Visakhapatnam, Jamnagar and Kolkata**, involving around **78 ships, 20 shore establishments and nearly 20,000 personnel**.
- The rebels formed a **Naval Strike Committee**, called themselves the **Indian National Navy and Azad Hindis** (drawing inspiration from the INA), removed British flags and hoisted **Congress and Muslim League flags**.
- The revolt also received widespread **civilian support**, particularly from dockyard and mill workers, railway workers and students, disrupting **public transport and commercial activity** in Bombay.
- **Demands of the Mutineers:** Better food, rations and living conditions, parity in pay and allowances with the Royal Navy, fair employment prospects after demobilisation, protection from victimisation, and access to **NAAFI (Navy, Army and Air Force Institutes) stores**.
 - **Release and fair trials for INA detainees**, withdrawal of British Indian troops from **Indonesia**, and opposition to using Indian personnel for **British imperial expansion**.
- **British Crackdown:** British forces used **disproportionate force**, killing at least **400 civilians**.
- **Resolution and Aftermath:** **Sardar Vallabhbhai Patel** and **Muhammad Ali Jinnah** persuaded the sailors to end the mutiny to stop the spreading violence.
 - The mutiny eventually ended with the surrender of the rebel sailors, limiting its immediate political success. Unlike the INA, it lacked a charismatic national leader who could become its enduring public symbol.
 - The RIN Mutiny, often described as the **"last war of independence,"** shook British confidence by demonstrating that they could no longer **rely unconditionally on Indian armed forces to sustain colonial rule**, thereby **hastening British withdrawal from India**.

Indian Independence Act, 1947

The **80th Independence Day** brings attention to the **Indian Independence Act, 1947**, the landmark legislation that **ended British rule in India and provided for the creation of India and Pakistan as independent dominions**.

- It was passed on **4th July 1947**, received **Royal Assent on 18th July 1947**, and came into effect on **15th August 1947**.

Key Features of the Act:

- **Partition:** Created **India and Pakistan** as independent dominions.
- **British Sovereignty:** Ended British legal authority and the monarch's title of **Emperor of India**.
- **Viceroy:** Replaced the Viceroy with a **Governor-General for each dominion**.
- **Constituent Assemblies:** Empowered them to **frame constitutions and repeal/modify existing laws**.
- **Princely States:** Ended **British paramountcy**, allowing them to join India, Pakistan or remain independent.
- **Interim Framework:** Continued the **Government of India Act, 1935**, with modifications, until new constitutions were framed.
- **Boundary Demarcation:** Punjab and Bengal boundaries were determined by **Boundary Commissions chaired by Sir Cyril Radcliffe**.
- **Secretary of State:** Abolished the office of **Secretary of State for India**.
- **Background:**
 - **Freedom Struggle:** Mass movements such as the **Non-Cooperation Movement** and **Quit India Movement** intensified the demand for independence.
 - **Muslim League's Demand:** The **Lahore Resolution, 1940** strengthened the demand for a separate Muslim state.
 - **Failure of Cabinet Mission Plan, 1946:** Its attempt to preserve a **united India with provincial autonomy** failed amid political differences and communal tensions.
 - **Attlee's Announcement:** In **February 1947**, British PM **Clement Attlee** announced transfer of power to India by **June 1948 at the latest**.
 - **Mountbatten Plan:** The **3rd June Plan** accelerated the transfer of power, was accepted by the **Congress and Muslim League**, and became the basis of the **Indian Independence Act, 1947**.

Veeragallu Discovery Sheds Light on Ruler Vishnuvardhana's Death

A **12th-century Veeragallu (hero stone)** bearing an **Old Kannada inscription** unearthed in **Neeragunda village, Karnataka**, may help resolve the long-standing historical debate over the **death of Hoysala ruler Vishnuvardhana**, making it an important archaeological discovery.

- **Discovery Site:** The **Veeragallu** was discovered during restoration work at the **Hoysala-era Lakshmi Narayana Swamy Temple in Neeragunda, Karnataka**.

- **Inscription Details:** The inscription states that Maravve erected the Veeragallu in **memory of her husband, Boppa, a Hoysala soldier**, who died shortly after the **death of King Vishnuvardhana at Bankapura** (present-day Haveri district).
 - It records the date as **Ekadashi of Pushya Masa, Durmathi Samvatsara, Year 1064**, corresponding to **11th December 1141 CE (Gregorian Calendar)**.
 - If authenticated, this inscription could resolve the long-standing dispute regarding Vishnuvardhana's death, previously believed by different scholars to have occurred in 1149 CE or 1152 CE.
- **About Veeragallu (Hero Stones):** **Memorial stones** erected to honour individuals who died defending their **king, village, or cattle**.
 - Usually feature **three panels** depicting the **battle scene, ascent to heaven, and worship before a deity**, symbolising **Moksha (salvation)**.
 - Reflect the Hoysala **Garuda tradition**, where elite royal bodyguards displayed absolute loyalty and, in some cases, sacrificed their lives following the king's death.
- **King Vishnuvardhana:** Originally known as **Bitti Deva**, he was one of the **greatest rulers of the Hoysala dynasty**.
 - **Defeated the Cholas in the Battle of Talakad (1116 CE)**, consolidating Hoysala power in southern India.
 - The Chennakeshava Temple at Belur, Karnataka, is one of the finest masterpieces of Hoysala architecture. It was consecrated by the Hoysala ruler **Vishnuvardhana** in **1117 CE** to commemorate his **1116 CE victory over the Cholas** and was originally known as the **Vijaya Narayana Temple**.
 - Influenced by **Ramanujacharya**, he embraced **Srivaishnavism** while continuing to patronise **Jainism and Shaivism**.
- **About the Hoysala Empire:** Ruled parts of present-day Karnataka, Tamil Nadu, and Andhra Pradesh between the **11th and 14th centuries CE**.
 - Capitals were **Belur** and later **Halebidu (Dwarasamudra)**.
 - **Hoysala temples** are renowned for their **distinctive Vesara-style architecture**, characterized by **complex stellate (star-shaped) plans, raised circumambulatory platforms (Jagati), tiered friezes, and intricate soapstone carvings**.
 - The Sacred Ensembles of the Hoysalas (Belur, Halebidu, and Somanathapura) were inscribed as a **UNESCO World Heritage Site in 2023**.

Social Issues

Centre's Opposition to Creamy Layer for SC/STs

The Union Government opposed applying the creamy layer principle to SCs/STs before the SC

- This followed the 7-judge Constitution Bench judgment in *State of Punjab v. Davinder Singh (2024)*, which allowed sub-classification within SC/ST quotas and suggested considering creamy-layer exclusion to prevent “elite capture” of reservation benefits.

‘Creamy Layer’ Concept

- **Creamy Layer:** Identifies and excludes relatively socially, economically and educationally advanced individuals within a reserved category from reservation benefits.
- **Origin:** *Indra Sawhney v. Union of India (1992)* upheld 27% OBC reservation while requiring exclusion of the creamy layer.
- **Current Application:**
 - **OBCs:** Only Non-Creamy Layer (NCL) candidates receive 27% reservation in public employment and education.
 - **SCs/STs:** Creamy layer has traditionally not applied because their backwardness stems from untouchability, social stigma and geographical isolation, not merely economic deprivation.

Sub-categorisation and Creamy Layer

Concept	Meaning
Sub-categorisation	Divides an existing quota among different groups within the same reserved category.
Creamy Layer	Excludes relatively advanced individuals from reservation benefits altogether.

- States such as Telangana, Haryana, Punjab and Andhra Pradesh have sub-classified SC quotas; Mizoram has sub-categorised ST quotas.
- At the Centre, ST sub-categorisation is used in Eklavya Model Residential School admissions to ensure minimum representation of PVTGs.
- EWS reservation (introduced in 2019) also uses income and asset limits and is available only to persons outside SC, ST and OBC reservations.

Centre's Opposition to the Creamy Layer for SC/STs

- **Historical and Structural vs. Economic Backwardness:** SC/ST disadvantage arises from centuries of oppression, untouchability (Article 17), social stigma and geographical isolation, and cannot be reduced to economic deprivation.

- **Persistence of Caste Stigma:** Economic advancement does not necessarily remove caste discrimination or deficits in social and cultural capital.
- **Fallacy of Economic Determinism:** SC/ST backwardness is structurally different from OBC (identified under Article 340) social and educational backwardness, making income an inadequate proxy.
- **Parliamentary Supremacy and Separation of Powers:**
 - Articles 341(2) and 342(2) provide that inclusion or exclusion from SC/ST Presidential Lists can only be done by Parliament by law.
 - *State of Maharashtra v. Milind (2001)* held that courts cannot alter these lists.
 - However, *Jarnail Singh (2018)* held that creamy-layer exclusion does not itself alter Presidential Lists.
- **Constitutional Protection: Article 16(4A)**, inserted by the 77th Constitutional Amendment, 1995, specifically enables reservation in promotion for SCs/STs.
- **Requirement of Empirical Data:** The Centre argues that creamy-layer exclusion without a contemporary nationwide socio-economic caste database would be premature and arbitrary.
 - **Census 2027** will include caste enumeration, but its results are not yet available.

Measures for Equitable SC/ST Reservation

- **Prioritizing Sub-Categorization over Economic Exclusion:** Use the *Davinder Singh* ruling for evidence-based sub-classification and dedicated sub-quotas for the most disadvantaged groups.
- **Institutionalizing Pre-Legislative Socio-Economic Audits:** Any restructuring of reservation should follow a comprehensive socio-economic census and empirical assessment of intra-group disparities.
- **De-linking Social Capital from Financial Capital:** Treat reservation as an instrument of representation and dignity, not merely poverty alleviation.
- **Deploying Targeted Means Tests Outside Quotas:** Use economic criteria in scholarships, housing and financial grants to target poorer sections without restricting constitutional reservation.

Distinguish between sub-categorisation and creamy layer exclusion. Why may sub-categorisation be considered a more suitable approach for addressing intra-category disparities among SCs/STs?

Drishiti Mains Question

Debate on Alcohol Prohibition

Bihar's **2016 alcohol prohibition** is under debate after the 'Macro Perspective of Bihar's Development Achievements, Unfinished Agenda, and the Way Forward' report at the **India Policy Forum 2026** recommended lifting the ban due to **limited impact on violence against women and significant revenue losses**.

- Reliance on **NCRB data** is contested because higher reported crime may reflect **better reporting**. **Rape cases declined or remained relatively stable after the ban**, highlighting the need for **specific, causally relevant indicators**.

Constitutional and Legal Provisions Regarding Alcohol Regulation in India

Provision/SC Judgements	Key Point
Article 47 (DPSP)	Directs the State to endeavour to prohibit consumption of intoxicating drinks and drugs injurious to health , except for medicinal purposes.
Seventh Schedule, State List, Entry 8	States regulate production, manufacture, possession, transport, purchase and sale of intoxicating liquor .
Khoday Distilleries Ltd. v. State of Karnataka (1994)	No fundamental right to trade in liquor ; States may regulate, prohibit or create a monopoly.
State of Tamil Nadu v. K. Balu (2016)	Article 19(1)(g) does not protect liquor trade; restrictions can be imposed for public safety .
State of U.P. v. Lalta Prasad Vaish & Sons (2024)	State regulatory competence under Entry 8 extends to certain non-potable/industrial alcohol .

- **Regulations Related to Alcohol Usage in India:**
 - **National Action Plan for Drug Demand Reduction (NAPDDR) 2021-22:** Prevention, treatment, counselling, rehabilitation and demand reduction for alcoholism and substance dependence.
 - **National Mental Health Policy 2014:** Links alcohol use with mental illness.
 - **National Health Policy 2017 and National Suicide Prevention Strategy 2022:** Recommend alcohol-control measures.
 - **Control of Noncommunicable Diseases (NMAP) 2017-2022:** Advocates a **national alcohol policy**.

NOTE: NFHS-6 (2023-24) reports alcohol use among **18.9% of men and 1.1% of women aged 15+**, against **18.7% and 1.3% respectively in NFHS-5**.

Arguments Regarding Alcohol Prohibition

Arguments in Favour of Prohibition

- **Gender-based violence:** Studies found reduced **spousal violence** in Bihar; a **2024 Lancet Regional Health study** estimated the ban prevented **over 21 lakh cases of violence**.
- **Household expenditure:** Reduced alcohol spending can redirect income towards **education, nutrition and durable assets**.
- **Public health:** Can reduce alcohol-related diseases and injuries; the **2024 Lancet study** estimated Bihar's ban prevented **18 lakh cases of overweight/obesity among men**.
- **Economic costs:** A **2019 study** estimated alcohol consumption costs India around **1.45% of GDP annually**; social costs are estimated at **over three times** the economic gains from alcohol excise taxes.

Arguments Against Complete Prohibition

- **Parallel economy:** Encourages **illegal markets, liquor mafias, cross-border smuggling and corruption**.
- **Hooch tragedies:** Unregulated liquor may cause mass poisoning; a **2025 Amritsar hooch tragedy** killed **21 people**, with suspected **methanol contamination**.
- **Inequitable enforcement:** Poorer groups may face greater risk of arrest, while affluent consumers can access smuggled liquor.
- **Revenue loss:** Alcohol excise duties often contribute **15%-25% of states' total tax revenue**, affecting spending on infrastructure, welfare and de-addiction.
- **Judiciary and prisons:** Stringent prohibition laws can cause **mass arrests, prison overcrowding and judicial burden**.
- **Limited long-term effectiveness:** Experiences such as **US Prohibition (1920-33)** and **Haryana's 1990s ban** show that complete bans may fail to eliminate demand.

Measures to Make Alcohol-control Policy more Effective

- **Applying 'Nudge' Theory:** Use **strict age-gating, prohibition of surrogate advertising and graphic health warnings** instead of relying only on blanket bans.
- **Community-Led De-addiction:** Involve **PRIs and women's SHGs**, including models such as **Kudumbashree**, in identifying vulnerable households and counselling.
- **Technological Intervention:** Consider **state-monopolised, biometric-linked rationing/smart liquor cards** to limit purchases.
- **Ring-Fencing Funds:** Statutorily earmark part of alcohol excise revenue, such as **20%**, for **psychiatric healthcare, de-addiction centres and nutrition schemes**.

- **Prioritise Rehabilitation over Criminalisation:** Treat alcohol dependence mainly as a **public-health and social issue**, strengthen treatment and rehabilitation, while strictly targeting **illegal supply networks**.

Article 47 provides a constitutional basis for regulating intoxicating substances, but its implementation raises questions of State capacity, individual liberty and fiscal sustainability. Discuss.

Drishti Mains Question

The State of Food Security and Nutrition in the World 2026

A joint **UN report SOFI 2026** revealed that Africa has surpassed Asia for the first time as the region with the highest absolute population facing hunger.

Key Findings of SOFI 2026

- **Global Hunger:** Undernourishment declined for the third consecutive year, from 8.6% (2022) to 7.8% (2025); however, 645 million people remained undernourished, above pre-pandemic levels.
- **Africa as Hunger Epicentre:** Africa recorded 309 million hungry people (20% of its population), surpassing Asia (292 million; 6%); it may host 56% of the world's undernourished population by 2030.
 - Drivers include climate change, conflicts (Sudan and the Democratic Republic of the Congo), poverty, high food and energy prices, fertiliser costs and geopolitical tensions.
- **Food Insecurity:** About 2.1 billion people (25.8%) faced moderate or severe food insecurity in 2025, still above pre-Covid levels.
- **Cost of a Healthy Diet:** CoHD rose to 4.28 PPP dollars per person per day in 2025; 2.69 billion people (32.7%) could not afford a healthy diet, with Africa recording 66.6% unaffordability.

Nutrition Indicators and Programmes

Indicator / Programme	Status	Key Point
Stunting (under five)	35.5% -> 29.3%	Declined b/w NFHS-5 (2019-21) and NFHS-6 (2023-24).
Severe wasting	7.7% -> 5.2%	Declined over the same period.
Underweight	32.1% -> 31.8%	—
Complementary feeding, 6-8 months	45.9% -> 59.5%	Improved infant feeding; malnutrition nevertheless remains high.
PM POSHAN	Bal Vatika-Class VIII	Hot cooked meals in eligible government and aided schools; supports nutrition, enrolment, attendance and learning.
Anaemia	Women (15–49 years) 57%; children (6–59 months) 67.1%	NFHS-5 levels remain high; Anemia Mukh Bharat and Rice Fortification address hidden hunger.

- **SDG 2 Off Track:** In March 2025, Minimum Dietary Diversity for Children (MDD-C) and Women (MDD-W) became official indicators for SDG 2.2 diet quality dimension. Only 30.8% of children aged 6-23 months achieved MDD-C (29.4% a decade earlier), while 62.7% of women aged 15-49 achieved MDD-W.

- Progress on stunting, wasting, anaemia, low birthweight and obesity remains inadequate. Low farm productivity, weak value chains, high input costs, poor logistics and staple-crop bias keep nutritious food expensive.

India and the SOFI 2026 Report

- **Regional Progress:** India contributed significantly to Southern Asia's reduction in undernourishment from 13.9% (2021) to 9.3% (2025), along with lower food insecurity.
- **Diet Affordability:** Excluding India would reverse the trend among lower-middle-income countries from a decline to an increase in the population unable to afford a healthy diet.
- **Nutrition Outcomes:** Southern Asia remains on track for reducing child stunting and halting child overweight, but maternal anaemia remains a major concern.
- **Price Sensitivity:** Rising fruit and vegetable prices disproportionately increase healthy-diet costs across South Asia.

State of Food Security and Nutrition in India

- **Food Safety Net:** NFSA, 2013 and universalised **PM Garib Kalyan Anna Yojana (PMGKAY)** provide free foodgrains to over 81 crore AAY and Priority Households (PHH) beneficiaries.
- **Production and Distribution:** National Food Security & Nutrition Mission (NFSNM) supports foodgrain production; NFSA distributes it through TPDS, covering 75% of the rural and 50% of the urban population.

Draft National Food Security (Amendment) Bill, 2026

- **Proposed Change:** AAY entitlement from fixed 35 kg per household to 7 kg per beneficiary per month, capped at 35 kg per household.
- **Rationale:** Remove per-capita disparities, rationalise distribution and align entitlements with household size and nutritional needs.
- **Concern:** AAY households with fewer than five members may receive less foodgrain, affecting vulnerable families and State allocations.

Structural Bottlenecks

- **Cereal-Centric Policy:** Rice-wheat procurement supports self-sufficiency but discourages pulses, oilseeds and nutritious millets (Shree Anna), while contributing to groundwater depletion and ecological stress in north-western India.
- **Post-Harvest Gaps:** Weak cold chains, storage and logistics cause losses in fruits, vegetables, dairy and other perishables, affecting farmer incomes and nutrition.
- **Limited Dietary Diversity:** NFSA, PM POSHAN, ICDS and Rice Fortification strengthen calorie and micronutrient security, but regular access to proteins, pulses, eggs, milk, fruits and leafy vegetables remains uneven.
- **Targeting Gaps:** Despite digitised ration cards, 99.9% Aadhaar seeding and One Nation One Ration Card (ONORC), NFSA coverage based on the 2011 Census, urbanisation, migration and inclusion-exclusion errors leave vulnerable households out.
- **Climate Risk:** Over 50% of India’s net sown area remains rain-fed and vulnerable to erratic monsoons, heatwaves, floods and droughts.

Way Forward

- **Expand Nutritious Food Supply:** Redirect agricultural R&D and incentives towards fruits, vegetables, pulses and legumes; reduce food loss and improve trade integration.
- **Reduce Food Costs:** Invest in cold chains, storage, transport and post-harvest infrastructure; post-farmgate activities account for nearly 70-75% of consumer food costs, especially for perishables.
- **Region-Specific Strategies:** Strengthen rural-urban market links and post-harvest management, tailored to regional food-system challenges.
- **Nutrition-Sensitive Protection:** Reorient subsidies and welfare towards diverse, nutrient-dense foods rather than mainly staple cereals.
- **Data-Driven Governance:** Develop granular sub-national food-price and diet-affordability monitoring for timely interventions.
- **Inclusive and Sustainable Reform:** Sequence demand- and supply-side measures, support smallholders, women and

youth, and use true-cost accounting without compromising the environment or rural livelihoods.

The affordability of a healthy diet has emerged as the biggest challenge to achieving SDG-2. Discuss.

Drishti Mains Question

NITI Aayog Report on India’s Caregiving Ecosystem

NITI Aayog’s report “Reimagining Care: Strategies for Empowering Caregivers in Viksit Bharat@2047” proposes a professional, inclusive and future-ready caregiving ecosystem and seeks to position India as a global hub for skilled caregivers.

Need for Organised Caregiving Ecosystem in India	
Demographic Transition	Elderly population (60+) may rise from 149 million in 2022 to 347 million (20.8%) by 2050; 26.8 million Indians live with disabilities, while chronic disease, dementia and age-related illness increase long-term, geriatric, palliative and home-care needs.
Family Care Burden	Nearly 10% of the population are family caregivers, mostly untrained and unpaid; professionalisation can meet care demand, create jobs and strengthen Viksit Bharat@2047.
Global Care Deficit	WHO projects a shortage of nearly 11 million health workers by 2030; OECD countries may need 13.5 million additional care workers by 2040, creating an opportunity for India.
Economic Potential	Care economy accounts for 16% of global employment; unpaid care contributes nearly USD 11 trillion annually. India’s care-services market may grow from USD 29.62 billion in 2023 to USD 72.31 billion by 2030 (CAGR 13.76%).
Women’s Empowerment	NSO Time Use Survey 2024: 41% of women aged 15-59 performed caregiving versus 21.4% of men; women spent 140 minutes daily versus 74 minutes for men. Formalisation can reduce unpaid work and raise female labour-force participation.
Global Competitiveness	Standardised training and G2G mobility with ageing societies (Japan, Germany, the UK, and other OECD countries) can generate employment, remittances and soft power.
Inclusive Growth	Improves care for the elderly, PwDs and other vulnerable groups and advances the Purple Economy by enabling PwDs to work as trained caregivers.

- Article 41 directs the State to provide public assistance in old age, sickness, disability and other vulnerable conditions, while professional caregiving promotes the right to work.

Parameter	Care Economy	Monetized Economy
Valuation & Measurement	Largely unpaid, invisible and excluded from GDP.	Formally measured and directly contributes to GDP.
Nature of Work	Caregiving and domestic services sustaining well-being and human capital.	Market-oriented goods and services for economic exchange.
Workforce Composition	Predominantly women, reflecting gendered norms.	More diversified formal or informal workforce.
Regulation	Limited legal recognition, contracts, social security and labour protection.	May have contracts, labour protection and social security, though gaps remain.
Economic Recognition	Often treated as family or moral responsibility.	Recognised as income-generating, productive and taxable activity.

Challenges in India's Caregiving Ecosystem	
Regulatory Gaps	Not yet a regulated profession; lacks dedicated law, accreditation and quality assurance, limiting professionalisation.
Unpaid Care Burden	Mostly unpaid, undervalued and borne by women. Despite RPwD Act, 2016 recognition for high-support family caregivers, comprehensive financial recognition, long-term care finance, insurance and social security remain absent, reducing female labour-force participation and deepening inequality.
Skills and Workforce	Training is not fully aligned with global standards; specialised dementia, stroke and palliative-care skills and certified caregivers remain limited, causing turnover and reliance on informal care.
Infrastructure and Delivery	Respite care is scarce; formal services are inadequate, especially in rural areas affected by migration; social acceptance is low.
Technology and Safeguarding	Weak background checks, accreditation and grievance systems undermine trust; limited digital health records, telemedicine and technology-based training restrict scale.

NITI Aayog Strategies

- **National Policy on Caregiving:** Formalise the sector, improve Centre-State coordination, and expand capacity building.
- **National Caregiver Council (NCC):** Apex body for regulation, accreditation, certification, registry, grievance redressal, and standards.
- **Professionalise the Caregiving Workforce:** Introduce NSQF-aligned certification, standard curricula, specialised geriatric, palliative, and dementia-care cadres, and caregiving modules in education and digital platforms.
- **Promote Global Caregiver Mobility:** Build G2G partnerships with Japan, Germany, the UK, and Australia, supported by ethical recruitment, language training, and reintegration.
- **Strengthen Institutional Ecosystem:** Strengthen RCI, NISD, Sector Skill Councils, National Centre for Ageing, and Regional Geriatric Centres, and standardise training.
- **Extend Social Security:** Provide health and accident insurance, coverage under the **Unorganised Workers' Social Security Act, 2008**, and support care entrepreneurship through Mudra and Startup India.
- **Support Family and Community Care:** Reduce women's unpaid care burden by strengthening family caregivers, SHGs, Ayushman Arogya Mandirs, and CSR-supported community care.
- **Provide Caregiver Welfare:** Introduce statutory care leave, respite care, workplace sensitisation, and counselling.
- **Leverage Technology:** Create a **National Digital Caregiver Portal** for registry, certified directory, background verification, multilingual training, and technology-enabled services.
- **Societal Care Model:** Sweden and Denmark follow shared responsibility among the State, families, communities, and the private sector.

Examine the role of the care economy in promoting inclusive growth, gender equality, and employment generation in India.

Drishiti Mains Question

FACTS FOR PRELIMS

REPORT & INDICES

Oldest Dated Biosignature Evidence in Indian Rock

A study published in the **Proceedings of the National Academy of Sciences (PNAS)** reported evidence of **microbial life** around **3.5 billion years ago** in the **Singhbhum Craton**, eastern India.

- **Discovery Site:** Evidence was found in a **carbon-bearing banded chert rock from Bhitardari** in the Singhbhum Craton, which spans **Jharkhand and Odisha**.
 - **Chert** is a silica-rich rock capable of preserving organic material for billions of years.
- **Singhbhum Craton:** One of India's oldest geological formations, preserving records of crust formation from **over 4 billion years ago**.
- **Zircon Dating:** Four out of eight analysed zircons indicated an age of approximately **3.5 billion years**, helping establish the rock's age.
- **Carbon Isotope Evidence:** Carbon isotope composition matched patterns linked to **carbon fixation by living organisms**, indicating biological origin.
- **Organic Matter Analysis:** The carbon was identified as **degraded organic matter** rather than later inorganic contamination, supporting evidence of ancient microbial activity.
- **Significance:** The discovery is among the **oldest directly dated biosignature records of microbial life**, based on multiple chemical and geological indicators. It provides insights into **early evolution of life on Earth** and the search for life on other habitable worlds.

India Becomes World's 2nd Largest Supplier of Seafarers

According to the **BIMCO-ICS Seafarer Workforce Report 2026**, India has become the **world's second-largest supplier of seafarers** to the global shipping industry.

- **Global seafaring workforce:** ~**2.57 million** against demand of nearly **2.55 million**, indicating shortage of qualified maritime professionals, especially officers.
- **Factors Behind Growth:** Expansion of **maritime training institutes**.

- Modernisation of maritime infrastructure.
- Improved **skill development programmes**.
- Alignment with international maritime standards.
- Digitisation of certification and regulatory services by the **Directorate General of Shipping (DGS)**.
- Compliance with the **International Maritime Organization (IMO) Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW)** enhanced global recognition of Indian seafarers.

Indicator	Data
Total Indian seafarers	311,936
Global share	12.16%
Global rank	2 nd -largest supplier
Largest supplier	Philippines
Indian officers	140,718 (13.41% of global officers)
Indian ratings	171,218 (11.29% of global ratings)

- **Maritime Legislative Reforms:** Inland Vessels Act, 2021, Merchant Shipping Act, 2025, Indian Ports Act, 2025, Coastal Shipping Act, 2025, Bills of Lading Act, 2025 and Carriage of Goods by Sea Act, 2025
- **Mission 20%:** Aims to increase India's share of global seafarers from **12.16% to 20%**. Objective—ensure **one out of every five global seafarers is Indian**.

India's 1st Comprehensive Catalogue of Lepidoptera

The **Zoological Survey of India** has completed **India's first comprehensive Catalogue of Lepidoptera**, covering butterflies and moths, after more than 12 years of research.

- **Publication:** A **22-volume** authoritative inventory; final volume published in *Zootaxa*.
- **Species Diversity:** India has **13,703 species** of butterflies and moths across **3,705 genera, 102 families and 31 superfamilies**.
- **Global Share:** Indian Lepidoptera represent about **8.25%** of the world's known diversity.
- **Geometroidea:** India hosts **2,205 species**, including **2,100 Geometridae**, 95 Uraniidae, nine Epicopeiidae and one Pseudobistonidae; Sematuridae is absent.

- **Ecological Role:** Act as pollinators, herbivores and detritivores and form an important food source for birds, bats and other insectivores.
- **Significance:** Provides a baseline for biodiversity assessment, climate studies, ecological monitoring and conservation planning.

Yale Environmental Performance Index 2026

India's Rank: 176th of 177 countries, with a score of 22.46; above only Laos and 7th among 8 South Asian countries.

- **Top Rank:** Estonia, with 74.79.

Key Points

- **About EPI:** Biennial index initiated in 2000.
 - Published by **Yale Center for Environmental Law & Policy** and **CIESIN, Columbia University**.
 - Tracks progress towards the **SDGs, Paris Agreement and Kunming-Montreal Biodiversity Framework**.
- **India's Ranking Trend:** 141st (2016) → 177th (2018) → 168th (2020) → 180th (2022) → 176th (2024 and 2026).
 - India has remained among the **5 worst-performing countries since 2022**.
- **Concerns:** Poor performance in environmental health, air quality and biodiversity; decline in fisheries, tree cover and pesticide-pollution control.
- **Relative Progress:** Better performance in climate mitigation, waste management and sanitation.
- **India's Criticism:** EPI overlooks development level, population, historical emissions and **CBDR (Common But Differentiated Responsibilities)**, and relies on extrapolations and simplified cross-country comparisons.
- **Yale's View:** It measures present environmental conditions, not future promises or historical responsibility.

India Improves Ranking in Structural and Pro-Competitive Reforms

The **Competere Foundation** released the report "India's Next Growth Frontier: Reducing Anti-Competitive Market Distortions to Build on India's 2010–2023 Reform Progress".

- **About:** India rose **25 places** in the **Market Distortions Performance Index**, from **82nd** in 2010 to **57th** in 2023.
- **Assessment Framework:** Measures progress in reducing **market distortions** and improving **competitiveness**.
- **Key Pillars:** Property-rights protection, Domestic competition, International competition
- **Areas Covered:** Competition policy, investment conditions, digital markets, and external regulatory barriers.
- **Major Reform Drivers:** **GST** unified indirect taxation and reduced interstate barriers.

- **IBC** enabled time-bound insolvency resolution and improved credit discipline and capital allocation.
- **Key Recommendations:** Adopt **evidence-based competition policy**.
 - Review **sector-specific investment restrictions**.
 - Strengthen cooperation with **like-minded trading partners**.
- **Significance:** Continued reforms are needed to improve **productivity, competitiveness, investment climate**, and **global integration**.

AWARDS & HONOURS

Gallantry Awards 2026

On 80th Independence Day, President Droupadi Murmu approved **78 Gallantry Awards**.

- Awards included **9 Kirti Chakras (7 posthumous)**, **19 Shaurya Chakras (1 posthumous)**, **36 Sena Medals for Gallantry (5 posthumous)**, **5 Bar to Sena Medals (Gallantry)**, **3 Nao Sena Medals for Gallantry** and **5 Vayu Sena Medals for Gallantry**.
- **Gallantry Awards:** Recognise exceptional **bravery, courage and self-sacrifice** by military personnel and civilians.
 - **Wartime Gallantry Awards:** **Param Vir Chakra** (highest military decoration), **Maha Vir Chakra** and **Vir Chakra**.
 - **Peacetime Gallantry Awards:** **Ashoka Chakra**, **Kirti Chakra** and **Shaurya Chakra**.
 - ❖ Instituted in **1952** as **Ashoka Chakra Class-I, II and III** and renamed in **1967**.
- **Order of Precedence:** **Param Vir Chakra** > **Ashoka Chakra** > **Maha Vir Chakra** > **Kirti Chakra** > **Vir Chakra** > **Shaurya Chakra**.
- **Recommendation and Conferment Process:**
 - Recommendations originate from **military units, CAPFs or police organisations** and pass through the chain of command.
 - Scrutinised by the **Central Honours and Awards Committee** and finally approved by the **President of India**.
 - Announced twice yearly on **Republic Day and Independence Day**.
- **Additional Recognition:** Repeated acts of similar gallantry are recognised through a **Bar to the Chakra**; both the **original award and Bar can be conferred posthumously**.

Fields Medal 2026

The **2026 Fields Medals**, among mathematics' highest honours, were awarded to **Yu Deng, John Pardon, Jacob Tsimerman and Hong Wang** for outstanding mathematical research.

- **About:** Established in 1936 through **John Charles Fields**, it is awarded by the **International Mathematical Union** for outstanding achievements and future promise in mathematics.
- **Eligibility:** Awarded every four years to 2–4 mathematicians below 40 years.
- **Funding:** Financed through the **J.C. Fields Trust** at the **University of Toronto**; the **Fields Institute** also supported the **2026 prize fund**.
- **2026 Awards:** **Hong Wang** became the **third woman** recipient after **Maryam Mirzakhani (2014)** and **Maryna Viazovska (2022)**.
 - **Yu Deng** and **Hong Wang** became the first Chinese nationals to win since **Shing-Tung Yau (1982)**.
- **India's Position:** Indian-origin mathematicians **Manjul Bhargava (2014)** and **Akshay Venkatesh (2018)** are previous recipients.
- **Significance:** Mathematical excellence supports advances in **artificial intelligence, physics and engineering**, reflecting growing scientific and technological capability.

IMPORTANT DAYS

154th Birth Anniversary of Aurobindo Ghose

PM paid tribute to Sri Aurobindo on 15th August 2026.

- **About:** Born **15 August 1872, Calcutta**; died **5 December 1950, Pondicherry**. He was a **yogi, philosopher, poet, seer and nationalist**.
 - Passed the **ICS examination** but was not appointed after failing to appear for the compulsory **horse-riding test**.
- **Contributions:**
 - **Revolutionary Career:** Advocated **radical nationalism and mass mobilisation** before Gandhi's leadership.
 - ❖ **New Lamps for Old** criticised the Congress' moderate policies.
 - ❖ Arrested in the **Alipore Bomb Case (1908)**; defended by **Chittaranjan Das**.
 - **Spiritual and Philosophical:** Established **Sri Aurobindo Ashram (1926), Pondicherry**.
 - ❖ Collaborated with **Mirra Alfassa (The Mother)**, who later founded **Auroville**.
 - **Literary:**
 - ❖ **Major works:** *The Life Divine, Savitri, Essays on the Gita, The Synthesis of Yoga* and *Defense of Indian Culture*.
 - ❖ Wrote for **Bande Mataram** and founded **Karmayogin**; associated with **Jugantar** and **Anushilan Samiti**.

❖ Nominated for **Nobel Prize in Literature (1943)** and **Nobel Peace Prize (1950)**.

- **Legacy and Influence:** Envisioned India as a leading **spiritual and civilisational force**, stressing **cultural self-confidence and spiritual regeneration**.

150th Birth Anniversary of Pingali Venkayya

PM paid tribute to Pingali Venkayya on his 150th birth anniversary (2 August), recalling his contribution as the **designer of India's National Flag** and its role in patriotism and nation-building.

- **Early Life:** Born on 2 August 1876 in Krishna District, Andhra Pradesh; joined the British Army at 19, fought in the Second Boer War (1899-1902) in South Africa where he first met and formed a lasting association with Mahatma Gandhi.
- **National Flag Design:** Presented the Swaraj Flag on khadi to Gandhi at the 1921 INC session in Vijayawada. The **original red-green flag represented Hindu and Muslim communities and carried the Gandhi Charkha**; on Gandhi's suggestion, a white band for peace was added, while the charkha symbolised Swaraj and self-reliance.
 - The INC adopted the design in 1931, making it the basis of the modern Tricolour. In 1947, a Constituent Assembly ad hoc committee retained the 1931 tricolour but replaced the Charkha with the Ashoka Chakra. The present flag was adopted on 22 July 1947 and became the National Flag on 15 August 1947.
- **National Movement:** Active in the freedom struggle from 1906 to 1922, including the Vande Mataram and Home Rule movements.
- **Polymath:**
 - **'Japan Venkayya'** - delivered a full-length speech in Japanese at Bapatla (Andhra Pradesh) in 1913.
 - **'Patti (Cotton) Venkayya'** - known for agricultural research specifically on Cambodia Cotton.
 - **'Diamond Venkayya'** - geology graduate; set records in diamond mining in Andhra Pradesh, wrote *The Mother of Diamonds* (1955), and served as an advisor to India's Mineral Research Department after Independence.
- **Legacy:** Worked as a plague inspector in Madras and Bellary, established an educational institution in Machilipatnam, received a commemorative postage stamp in 2009, and had AIR Vijayawada renamed in his honour in 2015.

Global Tiger Day

Observed every year on 29 July.

- **Origin:** Established at **St. Petersburg Tiger Summit (2010)**, where 13 Tiger Range Countries (TRCs) adopted **St. Petersburg Declaration**, launched **Global Tiger Recovery**

Programme, and committed to the **TX2 Goal** of doubling wild tigers by 2022; the global goal was not fully achieved.

- **13 TRCs:** Bangladesh, Bhutan, Cambodia, China, India, Indonesia, Laos, Malaysia, Myanmar, Nepal, Russia, Thailand, and Vietnam.
- **Theme 2026:** “Coexistence in Action: Scaling Solutions for People and Tigers”
- **IUCN Initiative: Integrated Tiger Habitat Conservation Programme** supports habitat restoration, anti-poaching and community-led coexistence across Asia.
- **NTCA Eco-tourism Webpage (2026):** NTCA launched an integrated platform for **authentic safari and accommodation bookings**, promoting **responsible eco-tourism**.
- **India’s Status:** Tiger population reached **3,682** and Tiger Reserves increased to **58**.
- **Major Indian Initiatives:**
 - **Project Tiger, 1973:** Protects wild tigers and their habitats.
 - **NTCA:** Statutory body under the **Wild Life (Protection) Act, 1972**.
 - **All India Tiger Estimation:** Conducted every four years using camera traps, GIS and **M-STRIPES**.
 - **International Big Cat Alliance:** India-led initiative for conserving seven big-cat species.

Tiger

- **Habitat:** Rainforests, grasslands, savannas, mangrove swamps
- **Protection Status:**
 - **IUCN Red List:** Endangered; **CITES:** Appendix I; **WPA 1972:** Schedule I

DEFENCE & SECURITY

Exercise Udara Shakti 2026

IAF participated in **Exercise Udara Shakti 2026**, a bilateral air exercise with the **Royal Malaysian Air Force (RMAF)** at **Subang Airbase, Malaysia**.

- Reflects India’s growing defence cooperation with Southeast Asian partners under the **Act East Policy** and **Indo-Pacific strategy** - following the IAF’s participation in **Exercise Pitch Black 2026 in Australia**.
- **Other India–Malaysia Military Exercises:**
 - **Samudra Laksamana** – Indian Navy
 - **Exercise Harimau Shakti** – Indian Army

MH-60R Seahawk Helicopter

Indian Navy received another **MH-60R Seahawk** from the **US**, taking deliveries to **21 of 24 helicopters ordered**.

MH-60R Seahawk Helicopter

- **About:** Also called “**Romeo**”; an all-weather, **shipborne multi-role naval helicopter** manufactured by **Lockheed Martin**.
- **Primary Role:** Designed mainly for **Anti-Submarine Warfare (ASW)** and **Anti-Surface Warfare (ASuW)**.
- **Advanced Sensors:** Equipped with **multi-mode radar**, **dipping sonar**, **sonobuoys**, **electro-optical/infrared sensors**, **electronic-support measures**, **datalinks** and **survivability systems**.
- **Weapons:** Mk-54 lightweight torpedoes, AGM-114 Hellfire missiles, rockets and machine guns.
- **Multi-Mission Capability:** Can perform **search and rescue**, **medical evacuation**, **surveillance** and **logistics-support missions**.
- **India’s Procurement:** India signed a deal with the US in **2020** for **24 MH-60Rs** to replace the ageing **Sea King fleet**.
 - **INAS 334 ‘Seahawks’** commissioned at **Kochi** in **March 2024**.
 - **INAS 335 ‘Ospreys’** raised at **Goa** in **December 2025**.
 - India also signed a **5-year sustainment-support package** under the **US Foreign Military Sales programme**.
- **Significance:** Strengthens India’s **anti-submarine warfare**, **maritime-domain awareness** and **shipborne aviation capability**, while deepening **India-US defence cooperation**.

Exercise MAITREE-XV

Indian Army contingent departed for **Surat Thani, Thailand**, for the **15th India–Thailand Joint Military Exercise MAITREE-XV (2026)**.

- **Bilateral military exercise** between the Indian Army and Royal Thai Army.
- **Origin:** Began in 2006 as **TICAFE**; renamed **MAITREE** (“Friendship”) in 2013.
- **Hosting:** Conducted alternately in India and Thailand; resumed in 2024 after a five-year gap.
- **Significance:** Enhances interoperability, military cooperation and mutual trust; 2026 marks 20 years since its launch.

India Successfully Test-fires Agni-4 Ballistic Missile

Recently, India successfully conducted the test launch of the **Agni-4 medium-range ballistic missile** from the **Integrated Test Range (ITR), Chandipur, Odisha**.

- The launch was carried out under the aegis of the **Strategic Forces Command**, which manages **India’s strategic nuclear forces**.
- **About:** Agni-4 is an indigenously developed, **intermediate-range, surface-to-surface ballistic missile** belonging to India’s Agni missile series.

- **Features:** Developed by the **DRDO**, Agni-IV is approximately 20 metres long, has a strike range of about 4,000 km and can carry a payload of around 1,000 kg.
 - The missile uses **advanced avionics, onboard computers, inertial navigation and guidance systems** to achieve high accuracy and maintain stability during flight.

INS Malvan Commissioned into the Indian Navy

Indian Navy commissioned INS Malvan, the second indigenously designed and built **Mahe-class Anti-Submarine Warfare Shallow Water Craft (ASW-SWC)**, at Karwar, Karnataka.

- **Induction:** Second of eight Mahe-class vessels being built by Cochin Shipyard Limited, Kochi, and the sixth of 16 ASW-SWCs being inducted into the Navy.
- **Legacy:** Named after Malvan, Maharashtra, linked to **Chhatrapati Shivaji Maharaj's maritime legacy**; continues the name of a naval minesweeper that served until 2003.
- **Capabilities:** Detects, tracks and engages underwater threats in shallow waters; supports anti-submarine warfare, underwater surveillance, mine warfare, low-intensity operations, coastal defence and littoral security.
- **Strategic Role:** Protects harbour approaches, sea lanes, naval bases, ports and offshore installations, where larger warships may be less effective.
- **Maritime Awareness:** Operates with coastal surveillance systems, patrol aircraft, submarines and larger surface combatants to safeguard sea lines of communication and strategic waters.

SUMMITS & CONFERENCES

GI & Beyond 2.0 Summit (2026)

The summit was organised by office of the Development Commissioner (Handlooms), Ministry of Textiles in association with Handloom Export Promotion Council (HEPC).

- **Venue:** Dr. Ambedkar International Centre, New Delhi.
- **Objective:** To promote GI-tagged handloom and handicraft products, strengthen the GI ecosystem, and expand market opportunities for artisans and weavers.

Key Highlights

- Digital Handloom Atlas – **“One Platform. Endless Markets”** launched
 - A digital platform to showcase India's handloom products, including GI-tagged products.
 - Aims to improve visibility, connect weavers with domestic and international markets, and enhance market access.

- **New GI Certificates presented for:** Ladakh Nambu Textile, Ladakh Pashmina Textile, Himachal Gudma, Himachal Lingchay, Jharkhand Bamboo Craft, Jharkhand Tussar Silk Saree & Fabrics, Munda Jewellery of Jharkhand, Bareilly Terracotta
- **Exhibition:** Displayed 100+ GI-registered handloom and handicraft products from across India.
 - Included live demonstrations by artisans showcasing traditional craftsmanship.

PLACES IN NEWS

Hara Mangrove Forests

Oil residues have reached parts of the **Hara mangrove forests around Iran's Qeshm Island**, threatening the protected ecosystem amid conflict and shipping-related spills in the **Persian Gulf**.

Hara Forests

- **Location:** Around **Qeshm Island and the Khuran Strait**, near the **Strait of Hormuz**.
- **Flora:** Dominated by **grey mangrove (*Avicennia marina*)**, adapted to saline waters through specialised salt-filtering roots.
- **Biodiversity:** Important breeding, nursery and feeding habitat for **fish, shrimp, crabs and other marine species**; also supports migratory birds, marine mammals and **critically endangered hawksbill sea turtles**.
- **UNESCO Status:** Part of the **UNESCO-designated Hara Biosphere Reserve**.
- **Importance:** Supports **local fishing communities** and **nature-based tourism**, including mangrove boat tours.

China's Antelope Reef Base

Satellite imagery shows China has completed the **first phase of land reclamation and infrastructure development at Antelope Reef in the disputed Paracels archipelago**, indicating further military consolidation in the **South China Sea**.

- **Infrastructure Developed:** The reclaimed island spans nearly **6 km** and includes:
 - Helicopter pad.
 - **680-metre wharf** adjoining a deep-water harbour.
 - **3-km straight coastline**, assessed as a potential **military runway**.
- **China's Position:** China has not acknowledged a new military base and says development serves **civilian purposes**, including **weather forecasting and scientific research**.
- **The 'Bastion' Strategy:** Antelope Reef could help create a protected maritime **“bastion” for China's nuclear-armed**

ballistic missile submarines, shielding them from US and Vietnamese naval operations.

- **The Territorial Dispute:** Vietnam claims the entire **Paracel Islands**, while China has exercised **de facto control** since 1974, following the **Battle of the Paracel Islands**.
 - The South China Sea has overlapping claims involving **China, Vietnam, the Philippines, Malaysia, Brunei and Taiwan**.
 - China's **Nine-Dash Line** claims were rejected by the **2016 South China Sea Arbitral Tribunal**, which held that claims to historic rights beyond **UNCLOS maritime entitlements** have no legal basis.
- **Potential Militarization Impact on India:** More than **55% of India's maritime trade** passes through the **South China Sea and Malacca Strait**; increased militarisation can threaten **Sea Lines of Communication (SLOCs)**.
 - China's naval consolidation supports its **Maritime Silk Road** and enables greater power projection into the **Indian Ocean**, challenging India's role as a **net security provider in the Indian Ocean Region**.

Black Sea

Russian attacks on **Ukraine's Black Sea ports and shipping infrastructure** have disrupted agricultural exports during peak wheat harvest season. Around **90% of Ukraine's wheat, corn and sunflower seed exports** move through the **Black Sea**.

Black Sea

- **About:** Major inland sea between **Eastern Europe and Western Asia**, connected to the Atlantic Ocean through the Mediterranean Sea, historically called the **Euxine Sea (Pontus Euxinus)**.
- **Bordering Countries (BURGeR-T):** Bulgaria, Ukraine, Russia, Georgia, Romania, Türkiye.
- **Connection with the Mediterranean:** Black Sea → Bosphorus Strait → Sea of Marmara → Dardanelles Strait → Aegean Sea → Mediterranean Sea.
- **Turkish Straits System:** Bosphorus Strait + Sea of Marmara + Dardanelles Strait.
- **Connection with the Sea of Azov:** Through the **Strait of Kerch**.
- **Surrounding Mountains:** **Pontic Mountains** in the south, **Caucasus Mountains** in the east and **Crimean Mountains** in the north.
- **Major Inflowing Rivers:** **Danube, Dnieper and Dniester**. The **Danube**, Europe's second-longest river after the Volga, enters through the **Danube Delta**.
- **Economic Importance:** Major route for **grain, fertilisers, energy products and other commodities**, making it important for **global trade and food security**.

- **Role in the Russia–Ukraine War:** Russia seeks to restrict **Ukrainian maritime access**, while Ukraine has challenged Russian naval power using **anti-ship missiles, drones and other asymmetric methods**.



India's First Telecom Manufacturing Zone in Gwalior

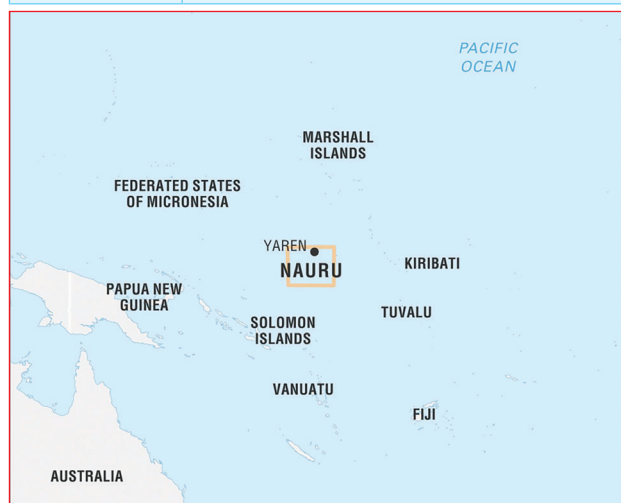
Gwalior's **Telecom Manufacturing Zone (TMZ)** secured **fresh commitments of ₹2,000 crore**, taking total committed investment to ₹5,500 crore.

- **About:** India's first dedicated telecom manufacturing cluster, planned as a globally competitive, end-to-end telecom manufacturing ecosystem in Gwalior.
- **Objective:** Transform India from a telecom services-led economy into a products and manufacturing hub under **"Make in India, Make for the World"** through domestic value addition, indigenous intellectual property and exports.
- **Integrated Ecosystem:** Covers design, R&D, testing, manufacturing, semiconductors, optical fibre, telecom equipment, electronic components and allied industries, supported by common plug-and-play testing facilities.
- **Employment:** Expected to generate more than 18,000 jobs.
- **Strategic Advantages:** Central location, connectivity, skilled workforce, competitive costs, industrial base and improved logistics, including the upcoming Agra-Gwalior Greenfield Expressway.
- **Implementation:** A Special Purpose Vehicle comprising the Department of Telecommunications and MP Government, with State support for land, capital, skills, employment-linked incentives and logistics.

Nauru Renamed as Republic of Naoero

The **Pacific island nation Nauru** adopted its **indigenous name, the Republic of Naoero**, to reaffirm cultural identity, linguistic heritage and national pride.

Location & Size	Northeast of Australia; coral-limestone island of about 21 sq km. World's smallest island country; population about 12,000; third-smallest country by population and land area after Vatican City and Monaco .
Identity Changes	International code: NRU → NRO ; citizens: dei-Naoero instead of Nauruans. Aircraft, ships and official records will adopt the revised name.
Colonial History	Known as Pleasant Island; annexed by Germany in 1888; administered by Australia, Britain and New Zealand; occupied by Japan during World War II; later a UN trusteeship; independent since 1968.
Origin of "Nauru"	Naoero was altered because colonisers found the indigenous pronunciation difficult; "Nauru" entered official use during German rule and continued after independence.
Language & Recognition	Renaming seeks to protect dorerin Naoero, classified by UNESCO as severely endangered . Countries and international organisations were notified; the UN and several governments have begun using Naoero.
Significance	Reflects decolonisation through reclaiming indigenous names, similar to Eswatini, Türkiye and the Micronesian State of Chuuk.



Glaw Lake: India's 101st Ramsar Site

Glaw Lake in Arunachal Pradesh became India's 101st Ramsar Site and the State's first wetland of international importance, increasing **India's Ramsar network from 26 sites in 2014 to 101 in 2026**.

- **Location:** Pristine freshwater lake within Kamlang Tiger Reserve and Wildlife Sanctuary in the Eastern Himalayas.
- **Physical Features:** Fed by perennial mountain streams and surrounded by dense natural vegetation.

- **Biodiversity:** More than 150 tree species and 49 orchid species occur in the wetland and its catchment.
- **Significance:** Supports biodiversity conservation, climate resilience, water conservation and sustainable livelihoods.

Ceuta

Spain's autonomous city of Ceuta faced a major migration crisis in July 2026, when nearly 60,000 migrants attempted to enter the Spanish enclave from Morocco.

- **Location:** Spanish autonomous city on Africa's northern Mediterranean coast, bordering Morocco and facing the Strait of Gibraltar; area about 18.5 sq. km.
- **European Territories in Africa:** Ceuta, Melilla and the Plazas de Soberanía—Chafarinas Islands, Alhucemas Islands and Peñón de Vélez de la Gomera—are the only European territories on the African mainland.
- **Sovereignty Dispute:** Spain exercises sovereignty, while Morocco has sought their return since its independence in 1956.
- **Geographical Significance:** Located at the eastern entrance of the Strait of Gibraltar, a major chokepoint connecting the Atlantic Ocean and Mediterranean Sea.
 - It forms one of the EU's only two land borders with Africa; the other is Melilla.



- Important for maritime security, trade and migration management, and among the shortest entry routes into Europe for migrants from Africa and West Asia.
- **Key Issues:** Hotspot for irregular migration, trafficking and humanitarian crises; also fuels Spain–Morocco tensions and raises the challenge of balancing border security with refugee and human-rights obligations.

Shipki La Pass

Cross-border trade between India and China through Shipki La resumed on 1 August 2026 after a six-year Covid-19-related suspension.

- **Location:** High-altitude pass above 4,000 m in Kinnaur district, Himachal Pradesh, on the India–China border with the Tibet Autonomous Region.



- **Earlier Name:** Known as Pema La or Shared Gate; renamed Shipki La by the Indo-Tibetan Border Police after 1962.
- **Terrain:** The Sutlej (Langqen Zangbo in Tibet) enters India through this region. The 28-km Namgia–Shipki La stretch is barren, and heavy winter snowfall closes the pass.
- **Strategic Importance:** Indian positions lie at a higher elevation than those on the Chinese side; the area also contains buildings used by Indian security agencies.
- **Historical Significance:** Part of the ancient Silk Route. Before Independence, Tibetan traders used Shipki La and the old Hindustan–Tibet Road to reach Rampur Bushahr, especially for the Lavi Fair.
- **Trade History:** Closed after the 1962 India–China War and reopened in the early 1990s under a bilateral agreement. Trade generally operates for a few months from June after the snow recedes.
- **Trade Items:** Imports include wool, livestock, yak products, religious items and minerals; exports include grains, spices, tobacco, timber and metal tools.
- **Distinctive Feature:** Trade traditionally follows a barter system, unlike Lipulekh and Nathu La, where transactions use currency.

Stone Age Tool, Sangam

Workshops Unearthed in Tamil Nadu

Excavations at **Malayadipatti**, near Karivalamvanthanallur in Tenkasi district, revealed a long-lived settlement along the **Vaippar River**.

- **Excavation:** TN State Department of Archaeology has opened **18 trenches** since April as part of its **wider archaeological programme**, which also includes **Keeladi**.
- **Cultural Sequence:** Continuous occupation from the **Microlithic period through the Iron Age and into Early Historic and Historic periods**, with no clearly identifiable Neolithic or Chalcolithic phases.

- **Major Finds:** Microlithic tools, iron implements, hematite, glass and coral beads, shell bangles, terracotta figurines, spindle whorls, gaming pieces, copper coins, graffiti-marked pottery and black-and-red ware.
- **Structural Evidence:** An **8.7 × 4.2 metre** brick structure with steps may have been an underground granary or rainwater-harvesting tank - reflecting advanced construction and water-management practices.
- **Burial Evidence:** A **red-slipped burial pot decorated with painted peacocks, along with human teeth and a black-and-red ware vessel**, suggests ritual burial practices that may also have links with references to peacocks in **Sangam literature**.
- **Craft and Trade:** Iron-tool making, bead and shell-bangle production, and weaving suggest an organised production centre connected with east–west coast trade routes.
- **Significance:** Highlights Tamil Nadu's antiquity, advanced technology and economic vitality and the settlement developed from the Stone Age to the Sangam period.

Qeshm Island

The **US struck IRGC military infrastructure on Iran's Qeshm Island**, citing threats to American forces and maritime security.



- **Location:** Largest island in the Persian Gulf, at the mouth of the **Strait of Hormuz**, in Iran's Hormozgan Province near Bandar Abbas; separated from mainland Iran by the **Khuran/Clarence Strait**.
- **Key Ecological Features:**
 - **Qeshm Geopark:** Middle East's first UNESCO-recognised geopark; includes the Valley of Stars, Chahkuh Gorge and Namakdan Salt Cave.

- **Hara Mangroves:** Important breeding ground for migratory birds.
- **Namakdan Salt Cave:** Among the world's longest salt caves, extending over 6 km.
- **Strategic Importance:** Helps Iran monitor and project power across the Strait of Hormuz, through which around **20% of global oil and LNG trade** passes.
 - Hosts **IRGC radar systems, drone facilities, missile batteries, naval assets,** and coastal surveillance infrastructure.

Pickaxe Mountain

The **US threatened military strikes on Iran's Pickaxe Mountain** amid reports that it may house **deep-underground nuclear facilities** linked to Iran's **nuclear programme**.

- The development revived debate over **India's decision not to launch a pre-emptive strike on Pakistan's Kahuta nuclear facility in the 1980s**.

Pickaxe Mountain

- **Geographical Setting:** Located about **220 km south of Tehran**, near **Natanz**, in the **Karkas Range**, which extends over **100 km** between **Kashan and Ardestan**.
- **Tectonic Origin:** Part of the **Urumieh-Dokhtar magmatic belt**, formed by the subduction of the **Neo-Tethyan Ocean** before the **Arabia-Eurasia collision**.
- **Rock Composition:** Dominated by hard volcanic rocks such as **andesite, rhyolite and dacite**, along with **pyroclastic deposits and volcanic tuffs**.
- **Structural Complexity:** Faults, fractures and varying rock densities disperse shock waves, making **blast modelling and military penetration difficult**.
- **Strategic Advantage:** Deep underground shielding and strong volcanic rock increase the difficulty of kinetic strikes, strengthening Iran's strategic position near the **Strait of Hormuz**.

India's Kahuta Dilemma and Nuclear Restraint

- **Pokhran-I (1974)**, code-named **Smiling Buddha**, prompted Pakistan to accelerate its nuclear programme centred on the **Kahuta uranium enrichment facility**.
- In **1981**, the **Indian Air Force** examined the feasibility of a pre-emptive strike on Kahuta, but no official strike plan was confirmed.
- India chose **strategic restraint**, while Pakistan acquired nuclear capability by the **late 1980s**, altering South Asia's strategic balance.
- India and Pakistan signed the **Agreement on the Prohibition of Attack against Nuclear Installations and Facilities, 1988**.

- Since **1992**, both countries have annually exchanged lists of their nuclear installations.
- India currently follows **credible minimum deterrence** and **strategic restraint**, while strengthening air and missile defence through **Project Kusha** and **Sudarshan Chakra**.

PORTALS & APPS

Centralised Public Grievance Redress and Monitoring System (CPGRAMS)

The **CPGRAMS** has emerged as the **Government's flagship digital grievance-redressal platform** through citizen-centric service delivery.

- **About:** CPGRAMS is the Government's **flagship 24x7 digital grievance-redressal platform**, connecting **Central Ministries/Departments, States and UTs** for lodging and tracking public-service grievances.
- **Evolution:** Since 2014, it has shifted from a largely manual to a **technology-driven system**:
 - Annual grievances: **3.01 lakh (2014) → ~27 lakh (2024)**.
 - Grievance Redressal Officers: **10,232 → over 1.11 lakh (2025)**.
- **Access Channels:** CPGRAMS portal, mobile app, **UMANG**, post and **5+ lakh Common Service Centres**, supported by **~2.5 lakh Village Level Entrepreneurs**.
- **Technology:** Uses **auto-routing, grievance categorisation, unique registration IDs, escalation, citizen feedback and appellate review** for greater accountability and transparency.
- **Samadhan Didi:** Launched **May 2026**; an **AI-enabled voice chatbot integrated with BHASHINI** that enables speech-based grievance registration in preferred languages and automatically identifies the relevant Ministry/Department/category.
- **NextGen CPGRAMS:** Being developed by **DARPG** with **AI-based routing, support for 22 Scheduled Languages, omni-channel registration, automated escalation, accessibility features and real-time analytics**.

SPECIES IN NEWS

India's 'Peacock Diplomacy' at the United Nations

India donated **5 Indian male peafowls to the United Nations Office at Geneva (UNOG)**, highlighting biodiversity heritage, multilateralism and international cooperation.

- **Gift:** Four blue peafowls + one white-feathered male, reflecting the UN's blue and white colours.
 - Second such Indian gift to UNOG; in 1981, PM Indira Gandhi sent a breeding pair of peafowls from Delhi Zoo.
- **Ariana Park:** 46-hectare park around UN Geneva; has hosted peacocks for over a century. It was the estate of

Swiss art collector Gustave Revilliod, bequeathed to Geneva in 1890.

- **Soft Power:** The gesture is described as “**Peacock Diplomacy**”, comparable to China’s “Panda Diplomacy”, reinforcing India’s peaceful and multilateral image.
- **Cultural Significance:** Indian peacock is India’s National Bird, symbolising grace, beauty and resilience; the gift also highlights biodiversity conservation and environmental stewardship.

Indian Peafowl

Aspect	Detail
Taxonomy	<i>Pavo cristatus</i> ; family Phasianidae. Male = peacock; female = peahen.
Sexual Dimorphism	Male: metallic-blue plumage and long train with ocelli; female: muted brown plumage aiding camouflage during ground nesting.
Habitat	Native to Indian subcontinent; dry deciduous forests, scrublands, semi-desert, agricultural and human-dominated landscapes.
Behaviour	Mainly ground-dwelling by day; roosts in trees at night.
Diet/Ecology	Omnivorous: seeds, fruits, insects, small mammals and reptiles including small snakes; contributes to natural pest control.
Conservation	IUCN: Least Concern; Wildlife (Protection) Act, 1972: Schedule I; CITES: Appendix III.

Scientists Develop IoT-Based Health Tracker for Yaks

An IoT-based smart collar has been developed to monitor the health, stress levels, and movement of **high-altitude yaks in remote Himalayan grazing areas** near the international border.

- **About:** The Indian yak (*Bos grunniens*), “**ship of the Himalayas**”, is a high-altitude bovine valued for its ability to survive extreme cold and difficult terrain.
- **Distribution:** ~58,000 in India; ~50% in Ladakh; also in Arunachal Pradesh, Sikkim, Himachal Pradesh, Uttarakhand. Wild yaks mainly on Tibetan Plateau.
- **Habitat:** High-altitude **alpine meadows, alpine steppes, and desert steppes, generally 3,200-5,400m**. They prefer cold plateaus and may migrate seasonally.
- **Physical Features:** Wild male weigh up to **1,000 kg, while females weigh ~300 kg**. They have long blackish-brown hair, curved horns, broad hooves, large lungs, a high red blood cell count, and high haemoglobin concentration, enabling them to survive in low-oxygen environments.
- **Behaviour:** Daytime grazers; females and calves form large herds, males solitary.

- **Diet:** Grasses, shrubs, mosses, lichens.
- **Reproduction and Lifespan:** Mating season begins in September, & calves are born in June after a gestation period of ~9 months. Females generally produce **one calf every two years**, lifespan ~25 years.
- **Conservation Status (Wild Yak)(*Bos mutus*):**
 - **IUCN Red List:** Vulnerable
 - **CITES:** Appendix I
 - **Wildlife Protection Act, 1972:** Schedule I

MISCELLANEOUS

Helicase: The DNA Separator

Recent research has clarified how **replicative helicases initiate DNA replication**, particularly the activation of **MCM2–7 complexes into CMG helicases**.

- **About Helicase:** Enzyme that **unwinds and separates the two DNA strands**, enabling replication.
- **Need for DNA Unwinding:** DNA strands in the **double helix** are held together by **hydrogen bonds** and must separate before copying.
- **ATP-Driven Activity:** Replicative helicases are loaded at or near **replication origins** and move along DNA using energy from **nucleotide hydrolysis**.
- **Breaking Base-Pair Interactions:** Helicase disrupts interactions between complementary bases, exposing the individual DNA strands.
- **Formation of Replication Fork:** Unwinding creates a **Y-shaped replication fork**, where parental strands become available to replication machinery.
- **Role of Topoisomerases:** Helicase activity creates **torsional stress and supercoiling** ahead of the fork; **topoisomerases** relieve this stress and prevent excessive twisting or DNA damage.
- **Template for New DNA:** Each separated parental strand acts as a **template for synthesis of a complementary strand**.
- **Regulation of DNA Replication:** Replication is tightly controlled during the cell cycle.
 - Recent research shows how **inactive MCM2–7 double hexamers are remodelled into active CMG helicases**, linking helicase activation with **initial DNA melting and unwinding**.

District Collectors to Grant Citizenship Under Section 6B

MHA has amended the CAA citizenship-processing framework, transferring pending applications from **Empowered Committees and District-Level Committees to District Collectors** in specified States and UTs.

- **Citizenship (Third Amendment) Rules, 2026:** Authorise **District Collectors** to receive, scrutinise and dispose of applications for registration or naturalisation under **Section 6B of the Citizenship Act, 1955**.
- **CAA Eligibility:** Covers **Hindus, Sikhs, Buddhists, Jains, Parsis and Christians** from **Afghanistan, Bangladesh and Pakistan** who entered India **on or before 31 December 2014** and fulfil prescribed conditions.
- **Areas Covered:** Gujarat, Rajasthan, Punjab, West Bengal, Assam, Tripura, J&K, and Ladakh.
- **Exclusion of Protected Areas:** **Sixth Schedule** tribal areas of Assam, Meghalaya, Mizoram and Tripura.
 - Areas under the **Inner Line**.
 - Thus, tribal areas of **Assam and Tripura** are excluded from the new arrangement.
- **Powers of Collector:** Receive applications **electronically**.
 - Verify documents and conduct enquiries.
 - Assess eligibility under **Section 6B**.
 - Administer the **Oath of Allegiance**.
 - Grant citizenship by **registration or naturalisation** if the applicant is eligible and a **“fit and proper person.”**
- **Grounds for Rejection:** Application may be rejected if the applicant, despite reasonable opportunities, fails to **appear personally, subscribe to the application or take the Oath of Allegiance**.
- **Significance:** Makes the **District Collector the key competent authority** for CAA applications in notified regions, replacing the earlier multi-agency mechanism and aiming for **more direct and efficient processing**.

India Under US Scrutiny Over Alleged Chinese Transshipment and Tariff Evasion

A White House report, ‘**The Great Transshipment Scam**’, alleges that **India and other countries are enabling Chinese exporters to evade US tariffs** by routing goods through third countries.

- India has been placed in **Tier 1**, alongside **Canada, EU, Japan and Mexico**, among over **40 jurisdictions** identified as having elevated transshipment risk.
- **Context of US-China Tariff War:** The US imposed tariffs of up to **100% on Chinese goods under Section 301 of the Trade Act, 1974**.
 - Chinese exporters allegedly began routing goods through third countries to **conceal origin and reduce tariff liability**.
 - The report highlights the **Pune-Gujarat-Chennai production belt** for absorbing Chinese industrial goods.
- **Transshipment:** Goods are routed through a third country where **limited assembly, repackaging, relabelling or**

documentation changes may make them appear to originate there.

- Chinese exporters allegedly use **“screwdriver factories”** for minimal assembly to exploit **tariff differentials** and alter the apparent **country of origin**.
- **Scale and Impact:** Around **USD 67 billion** of US-bound goods were estimated to have been transshipped through **Mexico, India and Vietnam in 2025**.
 - Estimated loss of US tariff revenue: **USD 28 billion**.
- **Compounding Trade Frictions:** The issue adds pressure to **India-US trade relations**.
 - It follows a **10% US tariff on India citing forced-labour compliance gaps**.
 - It also comes amid proposed US legislation threatening tariffs of up to **100% on countries importing Russian oil**.

India’s 1st Guide to Grasslands and Other Open Natural Ecosystems

India’s first-ever Guide to Grasslands and Other Open Natural Ecosystems (ONEs) of India, launched at **UNCCD COP17, Ulaanbaatar, Mongolia**.

- **Coverage:** Documents **grasslands, savannas, deserts, scrublands, ravines and rocky plateaus** across India.
- **Key Landscapes:** Includes Terai and Himalayan grasslands, Thar, Banni, Deccan savannas, Aravalli scrublands, Yamuna-Chambal ravines and Western Ghats rocky plateaus.
- **Ecological Role:** ONEs support biodiversity, carbon sequestration, soil-water processes and climate resilience; habitat for Great Indian Bustard, floricans, Indian wolf and blackbuck.
- **Livelihood Role:** Sustain **pastoral and livestock-dependent communities**, including traditional **Orans and gauchars**.
- **Conservation:** Promotes **science-based restoration, landscape planning and India’s Land Degradation Neutrality (LDN) goals**.
- **Prepared By:** Developed under **MoEFCC**, with partners including **ATREE and IUCN**.

BRICS Condemns EU Carbon Tax

At the **12th BRICS Environment Ministers’ Meeting, New Delhi**, members opposed the **EU’s Carbon Border Adjustment Mechanism (CBAM)** and called for urgent scaling-up of **adaptation finance** by developed countries ahead of **COP31**.

- BRICS termed **CBAM** a **“unilateral, punitive, discriminatory and protectionist”** measure that could undermine developing countries’ climate-resilience efforts.
- **CBAM Scope:** Entered definitive phase on **1st January 2026**.
 - Covers **iron & steel, aluminium, cement, fertilisers, hydrogen and electricity**; importers must account and pay for their embedded **carbon emissions**.

- EU says CBAM prevents **carbon leakage**.
- Part of EU's "**Fit for 55**" package, targeting at least **55% GHG-emission reduction by 2030 from 1990 levels**.
- **Impact on India:** Iron and steel constitute **~90% of India's exports to the EU** covered under CBAM.
 - **June 2026 Nature Climate Change analysis:** High-emission Indian steel firms saw declines in **EU export quantities and revenues**, while **lower-emission firms maintained exports**.
- **Demand for Adaptation Finance:** BRICS demanded finance that is "**new, additional, predictable, adequate and accessible**."
 - Favoured **grants and concessional finance** to prevent greater debt burdens on developing countries.
 - Called on developed countries to honour the climate commitment to **triple adaptation finance to developing countries by 2035**.
 - Adaptation finance supports **water security, climate-resilient agriculture, infrastructure, disaster preparedness and resilient livelihoods**; mitigation finance focuses on **reducing GHG emissions**.

Gyan Bharatam Mission

Parliamentary Standing Committee on Transport, Tourism and Culture flagged the **slow pace of manuscript digitisation** under the **Gyan Bharatam Mission**.

- **Current Progress:** Only about **7.5 lakh manuscripts** digitised so far; interim target is **20 lakh by 31 March 2027**.
- **About:** Announced in **Union Budget 2025-26** to **preserve, digitise and disseminate India's manuscript heritage**.
 - Approved as a **Central Sector Scheme for 2025-31** with an outlay of **₹491.66 crore**.
- **Background:** Builds on the **National Mission for Manuscripts (2003)**, which documented **44.07 lakh+ manuscripts** in the **Kriti Sampada repository**.
- **Digital Repository & Target:** Seeks to create a **National Digital Repository (NDR)** and digitise **1 crore manuscripts by 2030**.
- **Significance:** Links manuscript heritage with **education, research, digital access and global knowledge exchange**, supporting "**Virasat aur Vikas**" and **Viksit Bharat @2047**.

West Nile Fever

Recently, health authorities issued a caution following suspected cases of **West Nile fever (WNF)** in **Ernakulam district, Kerala**.

- **Cause:** Caused by **West Nile virus (WNV)**, a positive-sense single-stranded RNA virus of **Flaviviridae family and Orthoflavivirus genus**.

- **Transmission:** Mainly through bites of infected mosquitoes. **Casual human-to-human transmission is not documented**; rare transmission occurs through **blood transfusion, organ transplantation, laboratory exposure and mother-to-child transmission**.
- **Vector and Reservoir:** **Birds are principal reservoir hosts**; **Culex mosquitoes** are the main vectors and are most active at night.
- **Symptoms:** Around **80% asymptomatic, 20% symptomatic and less than 1% develop neuroinvasive disease**. Incubation period is usually **2–6 days**, ranging up to **14 days**.
- **Geographical Distribution:** Found mainly in **Africa, Europe, Middle East, North America and West Asia**.
 - First isolated in **West Nile district, Uganda, in 1937**; detected in birds in the **Nile Delta in 1953**.
- **Diagnosis and Treatment:** Confirmed through **laboratory testing** with assessment of symptoms and mosquito exposure. **No licensed human vaccine or specific antiviral treatment**.
- **Prevention:** Prevent mosquito bites and eliminate mosquito breeding sites.

Centre Relaxes Construction Rules Along Ganga Floodplains

The Government issued the **River Ganga Authorities Order, 2026**, amending the **2016 framework** notified under the **Environment Act, 1986**, which supports the **Namami Gange Programme**.

- **Removal of Blanket Ban:** The earlier blanket "**construction-free zone**" has been replaced with a **stratified, approval-based framework**, easing restrictions while retaining safeguards.
- **Floodplain Zones:** Based on **Central Water Commission (CWC)** guidelines:
 - **Active Flood Plain:** Submerged at least **once every 5 years**; strict construction prohibition continues, with limited exceptions for temporary structures such as religious events.
 - **Regulatory Zone:** Flooded **once every 5–25 years**; construction permitted with official clearance.
 - **Warning Zone:** Flooded **once every 25–100 years**; construction permitted with clearances.
- **Approval Mechanism:** Riverfront infrastructure, bridges, roads and similar works still require prior approval from the **NMCG** and regional authorities.
- **Government View:** The amendment clarifies the previously undefined term "**active floodplain**" and, according to the government, does not dilute environmental safeguards.

- **Concern:** The 2016 blanket ban had often been invoked before the **National Green Tribunal (NGT)** to stop potentially harmful riverfront works and embankments, raising concerns about future ecological impacts.

Government Adds 11 More International Ports for E-Visa Holders

The Government notified **11 additional international entry points** for foreign nationals holding valid **e-Visas**, expanding India's electronic visa network.

- **New Entry Points: 9 land ports**—Agartala, Darranga, Gede, Ghojadanga, Haridaspur, Jaigaon, Dawki, Moreh and Attari (Road)—and **2 airports at Bhopal and Tirupati**.
- **Total Network:** Authorised e-Visa entry points increased to **88**—37 airports, 38 seaports and 13 land ports.
- **Objective:** Liberalise, streamline and simplify the visa regime, facilitate legitimate foreign travel and improve e-Visa accessibility.
- **Background:** India introduced the **e-Visa with Electronic Travel Authorisation (ETA) in 2014** for 43 countries and later expanded it to citizens of **172 countries**.
 - Eligible foreign nationals receive an **ETA before arrival**, simplifying pre-entry approval.
- **Coverage:** The e-Visa facility covers **17 sub-categories**, including tourist, business, medical, student, mountaineering, transit, miscellaneous and group visitor visas for cruise travellers.
- **Digital Efficiency:** Around **95% of e-Visa applications are processed within 72 hours**, while about **78% of all Indian visas issued are electronic**.
- **Significance:** Improves access through airports and land borders, supports tourism and business travel, and strengthens technology-based border management with security safeguards.

CBDC-Based DBT for Food Subsidies

Recently, the Government of India launched a **CBDC-based DBT system** for food subsidies under the **Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY)**.

- **About:** **Chandigarh and Dadra & Nagar Haveli** became the first UTs to provide food subsidy to all eligible PMGKAY beneficiaries through **programmable ₹ tokens**.
- **Mechanism:** Eligible beneficiaries receive the subsidy directly in their CBDC wallets instead of conventional bank-account transfers.
- **Background:** It builds on earlier CBDC-based welfare pilots in **Puducherry and Gujarat** and marks complete digital adoption among eligible beneficiaries in the two UTs.
- **Key Features:** Enables **secure, real-time and end-to-end traceable transactions**, allowing better monitoring of subsidy transfers and utilisation.

- **Objectives:** Reduce leakages, diversion and cash handling while enabling faster, transparent and secure welfare delivery.
- **Significance:** Promotes financial inclusion through simple wallet-based transactions and demonstrates the Digital Rupee's potential for large-scale welfare delivery and future payment systems.

SC Clarifies the Scope of Police Remand Under the BNSS

SC, in *State of Andhra Pradesh v. Suda Suresh Veera Venkata Naga Raju*, clarified that **Section 187(2) of the BNSS, 2023** permits police custody in parts during the **first 40 or 60 days of detention**, subject to an overall limit of **15 days**, depending on the applicable **60-day or 90-day detention period**.

- **Change from CrPC:** Under **Section 167 of the CrPC, 1973**, police custody was restricted to the **first 15 days of remand**.
- **Purpose of Extended Window:** It allows police custody when **fresh facts, discoveries or investigative leads** emerge later during investigation.
- **No Absolute 15-Day Cut-off:** A Magistrate or court cannot impose an **absolute, non-extendable cut-off** preventing further police custody within the statutory **40/60-day window**.
- **Detention Period (Sections 58 & 187 BNSS):** A person ordinarily cannot be kept in police custody beyond **24 hours without a Magistrate's authorisation**.
 - If investigation cannot be completed, a Magistrate may authorise detention up to **90 days for offences punishable with death, life imprisonment or imprisonment of 10 years or more**, and **60 days for other offences**.
 - After expiry of the applicable period, the accused is entitled to **default bail**, if prepared to and does furnish bail.
- **Right to an Advocate (Section 38 BNSS):** An arrested person may meet an advocate during interrogation, but this does **not require continuous physical presence throughout interrogation**; the lawyer may remain within sight but cannot intervene.
- **Audio-Visual Recording:** Continuous videography during transit is **not required**; recording of **actual interrogation sessions and discovery/recovery proceedings** is sufficient.

Government Approves Polymer Banknotes for Field Trials

Government approved RBI's proposal under **Section 25 of the RBI Act, 1934** to introduce **1 billion pieces each of ₹10 and ₹20 polymer notes** for field trials.

- **Parallel Circulation:** Polymer notes will circulate alongside existing paper notes, not immediately replace them.
- **Purpose of Trials:** To assess **cost, durability, security, environmental impact and suitability** for Indian climatic conditions.
- **Material:** Made from **biaxially oriented polypropylene (BOPP)** instead of cotton-pulp paper.
- **Advantages:** More **durable, water-resistant and tear-resistant**, with transparent windows and embedded security features that make counterfeiting harder.
 - Polymer notes generally last **2.5–4 times longer** than paper notes.
- **Need:** India reportedly spends nearly **₹5,000 crore annually** on replenishing and maintaining banknotes and destroys around **20–24 billion soiled notes each year** under the RBI’s **Clean Note Policy**.
- **Why ₹10 and ₹20 First:** Lower denominations deteriorate faster due to frequent circulation; polymer notes could reduce **replacement frequency and lifecycle costs**.

Light-Activated Nanobots
for Targeted Breast Cancer Therapy

INST, Mohali developed **light-activated multifunctional nanobots** for more precise and targeted breast-cancer therapy.

- **Working: upconversion nanoparticle (UCNP)-based nanobots** respond to near-infrared light and show phototaxis, enabling directional movement toward the light source for localised action.
- **Cancer Targeting:** Folic-acid functionalisation targets breast-cancer cells overexpressing folate receptors; a 980-nm near-infrared laser produces localised heating to destroy cancer cells.
- **Combined Therapy:** A photosensitizer **generates reactive oxygen species (ROS)**; combined photothermal and photodynamic effects inhibit tumours more than either therapy alone.
- **Smart Control:** Therapeutic action can be influenced by laser intensity, pH and glutathione concentration.
- **Advantage & Testing:** Fuel-free, externally controlled and localised treatment may reduce healthy-tissue damage and systemic toxicity; efficacy was demonstrated in cellular models and breast tumour-bearing mice, showing potential for precise, targeted and minimally invasive therapy.

Supreme Court Directs States to
Declare Cancer as Notifiable Disease

SC directed the **19 States/UTs** that have not yet done so to consider declaring cancer a **notifiable disease**.

- **Current Status:** **17 of 36 States/UTs** have already notified cancer, following recommendations of a **Parliamentary Standing Committee**.
- **Reason:** PIL highlighted gaps in the **ICMR National Cancer Registry Programme**, which reportedly covers only about **10% of India’s population**, with rural coverage as low as **1%**.
- **Significance:** Notification would make cancer reporting mandatory, enabling **systematic data collection, surveillance, early detection and better planning for treatment and care**.

Notifiable Disease

Aspect	Detail
Meaning	■ Disease legally required to be reported to public-health authorities under laws such as the Epidemic Diseases Act, 1897
Surveillance Framework	■ Integrated Disease Surveillance Programme (IDSP) , managed by NCDC
Reporting Requirement	■ Public/private healthcare providers must report suspected, probable and confirmed cases
Common Examples	■ Tuberculosis, Dengue, Malaria, Cholera, Hepatitis, Measles, Leptospirosis and Polio ■ Covid-19 was universally declared notifiable ■ Recent examples- Snakebite (2024) and human rabies (2025)
Federal Character	■ India has no uniform national list ; notification largely rests with State/UT governments, causing variation
Global Framework	■ International Health Regulations (2005) under WHO are a legally binding international law that require member countries to notify WHO about events with possible international public-health implications

EVEREST: India’s 1st Private
800 kN FFSC LOX-Methane Engine

Bengaluru-based Astrobase Space Technologies unveiled EVEREST, India’s first privately built **800 kN Full-Flow Staged Combustion (FFSC) LOX-Methane rocket engine**.

- **Key Specifications:** **800 kN thrust**, specific impulse of about **340 seconds**, and **50%–110% throttle range**; designed for **reusable medium-lift launch vehicles**.
- **FFSC Technology:** Fuel and oxidiser pass through **separate pre-burners** before entering the main combustion chamber; generated gases drive turbopumps and then enter the main chamber, enabling **better propellant utilisation and higher efficiency**.
- **LOX-Methane Advantage:** Relatively clean combustion and multiple-cycle FFSC architecture make it suitable for **reusable launch systems**.

- **Launch Capability:** Proposed vehicle architecture targets about **3 tonnes to 500-km LEO**, with **return-to-launch-site recovery**, aiming to improve turnaround time, precision control and launch economics.
- **Development Roadmap:** Development began in **2024**; subsystem testing completed. **Full-scale hot-fire testing at Anantapur, Andhra Pradesh, in 2026** is the next major milestone, followed by stage testing and vehicle integration.
 - Around **20 engines** are planned to be manufactured/ tested before the first orbital flight, targeted for **2028**.
- **Government Support:** Recipient of **IN-SPACe's Technology Adoption Fund** for development of advanced private-sector space technologies.
- **Global Significance:** According to Astrobases, India is the **4th country after Russia, the US and China** to develop FFSC technology; Astrobases is also the **4th commercial company globally** developing a high-thrust FFSC engine.
- **Strategic Significance:** Could strengthen India's capabilities in **reusable launch vehicles, high-cadence launches, indigenous propulsion and commercial access to space**, enhancing participation in the global space economy.

Nasha Mukh Yuva for Viksit Bharat Sankalp Abhiyan

The PM launched the **Nasha Mukh Yuva for Viksit Bharat Sankalp Abhiyan** with over 1 crore youth participating virtually from more than 28,000 locations across India.

- **Objective:** Build a **drug-free, physically healthy, mentally resilient and confident youth population**, linked with **Viksit Bharat@2047**.
- **Origin:** Evolved from a **pilot project in Kashi (U.P.)**, supported by **125+ spiritual organisations and NGOs** under the **Kashi Declaration**.
- **100-Week Action Plan:** To run for **100 Sundays** through **sports, art, culture, meditation and community engagement** against substance abuse.
- **Youth-Led Jan Bhagidari:** Involves **students, NSS members, MY Bharat volunteers, colleges and communities** through anti-drug pledges and promotion of drug-free lifestyles.
- **Prevention & Early Intervention:** Emphasises **parent-child communication, early identification of addiction risks, campus awareness** and rejection of the social normalisation of substance abuse.
- **Rehabilitation:** Promotes **medical treatment, rehabilitation centres, counselling, yoga and meditation**, along with ending stigma against recovering individuals.
- **Enforcement & Security:** Calls for stronger action against **drug trafficking and smuggling**, recognising links with **organised crime and national-security threats**, alongside increased seizures and action against traffickers.

Glueball

Scientists have found **some of the strongest evidence** yet for the long-predicted glueball, with **X(2370)** emerging as a leading candidate.

- **Strong Force & QCD:** The strong force binds quarks into protons/neutrons and helps bind protons and neutrons in nuclei; it is described by Quantum Chromodynamics (QCD).
- **Gluons:** Gluons mediate the strong force and, unlike photons, carry a type of charge that allows them to interact with one another.
- **Glueball:** QCD predicts short-lived composite particles made predominantly of gluons rather than the quark-antiquark pairs of conventional mesons.
- **Detection Challenge:** Glueballs decay rapidly into lighter particles and can mix with ordinary hadrons having similar quantum properties, making identification difficult.
- **X(2370):** First observed in J/ψ meson decays in 2011; its mass and properties broadly match theoretical predictions for a pseudoscalar glueball.
 - **Recent Evidence—** further analysis of J/ψ decay data strengthens the possibility that X(2370) is a glueball candidate.
- **Significance:** Confirmation would provide landmark experimental evidence for hadronic matter composed predominantly of gluons, the force carriers of the strong interaction.

Money Bill Route to Increase Supreme Court Judges' Strength

Parliament passed the Supreme Court (Number of Judges) Amendment Bill, 2026, raising the sanctioned strength from 34 to 38 judges, including the CJI.

- **Purpose:** Creates four additional judicial posts to address SC's workload; appointments also require supporting staff, residences, vehicles and security.
- **Money Bill Basis:** The government linked the legislation's expenditure to the Consolidated Fund of India.
- **Criticism:** Use of Article 110(1)(g) on "incidental matters" may bypass effective Rajya Sabha scrutiny; Justice D.Y. Chandrachud's 2018 Aadhaar dissent called such use a "fraud on the Constitution" and a "subterfuge".
- **Speaker's Role:** The scope of the Lok Sabha Speaker's power to certify a Money Bill is pending before a seven-judge Constitution Bench.
- **Article 110:** A Money Bill primarily concerns taxation, borrowing, appropriation of public funds or incidental matters. The Rajya Sabha cannot reject or amend it and may only recommend changes.

Research, Development and Innovation Fund

The ₹1 lakh crore RDIF is in focus over concerns about company selection for concessional loans under the deep-tech financing initiative.

Aspect	Description
About & Outlay	DST flagship initiative and Special Purpose Fund under ANRF; approved in 2025 with ₹1 lakh crore over six years to catalyse private-sector R&D.
Financing	Collateral-free, low-cost, long-term finance for high-risk, high-impact projects at TRL 4 (Technology Readiness Level and above in AI, quantum, semiconductors, space, defence, robotics, clean energy and digital healthcare.
Objectives	Promote research-driven entrepreneurship, technology self-reliance and critical-technology acquisition; create a Deep-Tech Fund of Funds; strengthen strategic autonomy and sustainable growth.
Funding Route	Funds flow through AIFs, DFIs, NBFCs and FROs to Indian RDI-intensive firms/startups; loans are routed through Second-Level Fund Managers (SLFMs), which select Eligible Technology Entities (ETEs).
Current SLFMs	Technology Development Board (DST) and Biotechnology Industry Research Assistance Council (Department of Biotechnology).

Indigenous African Swine Fever Vaccine Developed by ICAR

ICAR-NIHSAD, Bhopal, developed India’s first indigenous live attenuated African Swine Fever (ASF) vaccine using an attenuated Genotype II virus with specific gene deletions and the MA-104 cell line, enabling scalable and cost-effective production.

- **About ASF:** A highly contagious viral disease of domestic and wild pigs, causing high fever, internal bleeding and mortality of up to 100%.
- **WOAH Status:** A listed disease under the World Organisation for Animal Health Terrestrial Animal Health Code; member countries must notify outbreaks.
- **Transmission:** Direct contact with infected pigs or carcasses; contaminated feed, food waste, equipment and materials; and soft ticks in some regions.
- **Human Health Risk:** Not zoonotic and **does not infect humans**, but people can indirectly spread it through contaminated clothing, footwear, vehicles, equipment and pork products.
- **Global Spread:** First described in Kenya in 1921; after emerging in Georgia in 2007, spread across Europe and later Asia, Oceania and parts of the Caribbean, while continuing in Africa.

- **India:** First detected in 2020; **spread across several States and Union Territories**, especially the North-East, causing pig deaths, movement restrictions and disruption of pork production and trade.
- **Significance:** Can reduce pig mortality and farmers’ losses, strengthen livestock and food biosecurity, and support India as a supplier of affordable veterinary vaccines.

Heatwaves and Lightning Included as Notified Natural Calamities

The Union Government added heatwaves and lightning to the notified calamities eligible under SDRF and NDRF guidelines for the **2026-31 award period**.

- **Basis:** The decision follows the 16th Finance Commission’s recommendation, citing the rising frequency and intensity of climate-related extreme weather.
- **Financial Assistance:** States may use SDRF funds for relief; NDRF assistance may supplement State resources, and heatwave-mitigation projects are eligible under Disaster Mitigation Funds.
- **Expanded List:** The notified categories increased from 12 to 14, adding heatwaves and lightning to cyclone, drought, earthquake, fire, flood, tsunami, hailstorm, landslide, avalanche, cloudburst, pest attack, and frost and cold wave.
- **Significance:** Strengthens disaster-risk financing and aligns disaster management with growing climate-change impacts.

Concept	Definition / Key Facts
Heatwave	A prolonged period of abnormally high temperatures relative to a region’s normal climate; occurs mainly from March to June, sometimes July , with May as the peak month. IMD uses region-specific thresholds.
Lightning	A rapid atmospheric electrical discharge caused by separation and accumulation of opposite charges; may occur within or between clouds, cloud-air, or cloud-ground, usually during thunderstorms. Lightning may also occur in volcanic ash plumes, wildfire-generated storm clouds and intense snowstorms. Lightning rod – a pointed copper conductor installed on buildings to intercept strikes and safely conduct current into the ground through earthing.

Varsha Ashok Aglawe: 1st Woman DG of Geological Survey of India

Varsha Ashok Aglawe became the **54th Director General of GSI** and the first woman to head it in its 176-year history. She succeeded Asit Saha.

- **Experience:** Joined GSI in 1992 after serving with the Indian Bureau of Mines; has over three decades of scientific and administrative experience.
- **Expertise:** Specialist in palaeontology and palynology; researched Late Cretaceous–Palaeocene sediments, the K–T boundary (**Cretaceous–Tertiary boundary**) and the Satpura–Gondwana basins.
- **Contribution:** Identified the Salbardi–Ghorpend area in Central India as a new dinosaur nesting site and participated in international research, including collaboration with the University of Michigan.
- **Priorities:** Critical and strategic minerals, deep-seated and concealed deposits, advanced geophysical surveys, offshore mineral exploration and stronger laboratory infrastructure.

GSI

- **Establishment:** Established in 1851 to locate coal deposits for railways; presently an attached office of the Ministry of Mines.
- **Organisation:** Headquartered in Kolkata; six regional offices at Kolkata, Lucknow, Jaipur, Nagpur, Hyderabad and Shillong, with state units across most states.
- **Functions:** Geological mapping, mineral-resource assessment, geophysical, airborne and marine surveys, natural-hazard studies, glaciology, seismotectonics, geoinformatics, research and maintenance of the national geoscientific database.

Leptospirosis in Kerala

Kerala faces a high disease and mortality burden from leptospirosis, especially during monsoon and post-monsoon periods.

- **Cause:** A zoonotic disease caused by pathogenic spiral-shaped bacteria (spirochaetes) of the genus *Leptospira*.
- **Reservoirs:** Wild and domestic mammals such as rodents, cattle, buffaloes, pigs, dogs, horses, sheep and goats. The bacteria can persist in renal tubules and sometimes the reproductive tract.
- Carrier animals may be asymptomatic or mildly ill but shed bacteria through their urine for long periods. Rodents are key reservoirs because their urine contaminates soil and water.
- **Transmission:** Direct contact with infected animal urine or indirect contact with contaminated water, soil or food. Entry occurs through cuts, abrasions, mucous membranes or macerated skin; human-to-human spread is rare.
- **Symptoms:** Range from mild flu-like illness to Weil's syndrome, meningitis and pulmonary haemorrhage. Incubation is usually 5-14 days; common symptoms are

fever, headache and muscle pain, while jaundice generally indicates severe disease.

- **Diagnosis:** Often underdiagnosed because symptoms resemble other febrile illnesses and access to suitable diagnostic tests may be limited.
- **Treatment:** Antibiotics such as doxycycline, penicillin G or ceftriaxone, depending on disease severity.
- **Prevention:** Control animal reservoirs, maintain sanitation, avoid contaminated water or soil and use protective clothing. Serovar-specific vaccines exist for animals such as dogs, pigs and cattle, but may not completely prevent renal carriage.
- **Distribution:** Occurs worldwide but is prevalent in South-East Asia; India, Indonesia, Thailand and Sri Lanka report most cases, particularly during the rainy season.

Registration of Births and Deaths (Amendment) Bill, 2026

The Lok Sabha passed the Bill by voice vote without debate to amend the **Registration of Births and Deaths Act, 1969**.

Key Provisions

- **Delay beyond 2 years:** Requires an order from a **Judicial Magistrate First Class (JMFC)**.
- **Delay of 1–2 years:** Requires approval from a **DM, SDM or authorised Executive Magistrate**.
- Replaces references to the **CrPC, 1973** with the **BNSS, 2023** and updates the definition of "Executive Magistrate."
- Aims to promote timely registration and improve the reliability of civil records.
- **RBD Act, 1969 (Amended in 2023):**
 - Mandates registration of all **births, deaths and stillbirths** at their place of occurrence.
 - **RGI** oversees the national system; Chief, District and Local Registrars manage registration and certificates.
 - Registrations have been digitally maintained through the **Civil Registration System (CRS)** since October 2023.
 - Events must be reported within **21 days**; delayed registration attracts a prescribed late fee.

Samudra Manthan Scheme (National Offshore Exploration Scheme)

The Union Cabinet approved the scheme to unlock India's offshore energy potential.

- **About:** Central Sector Scheme of the Ministry of Petroleum and Natural Gas, with an outlay of **₹84,084 crore up to FY 2030–31**.

- **Focus:** Deepwater, ultra-deepwater and frontier offshore hydrocarbon exploration.
- **Need:** Offshore exploration is risky, capital-intensive and technologically complex, with a **5–10-year gestation period**. A deepwater well costs **\$125–150 million**, while existing fields face a **6–7% annual natural production decline**.
- **Objectives:** De-risk exploration, attract private investment, increase domestic production, reduce imports and strengthen energy security (**Viksit Bharat, Make in India and Atmanirbhar Bharat**).
- **Allocation:** **₹43,200 crore** for drilling 60 deepwater exploratory wells; **₹28,534 crore** for seismic surveys; **₹10,000 crore** for common offshore infrastructure hubs; and **₹2,000 crore** for oil and gas manufacturing and services zones.
- **Risk Sharing:** Government support of up to **50% of drilling costs**, capped at **₹675 crore per well** to encourage sustained public and private investment in offshore basins such as **Krishna-Godavari, Cauvery, Mahanadi and the Andaman region**.
- **Progress:** ONGC plans **150 wells in seven years** to assess **5,600 MMTOE**; Oil India is exploring Andaman region and has reported gas in **Vijayapuram-2 and 3**.
- **Targets:** Add over **600 MMTOE** reserves, raise oil and gas production from **62 to 80 MMTOE**, and expand resources from **1.6 to 2.2 billion tonnes of oil equivalent**.
- **Significance:** Could reduce crude imports by nearly **₹1 lakh crore annually**, generate employment and strengthen indigenous offshore manufacturing and services.

First Possible Exomoon Around a Brown Dwarf

Using ESO's **Very Large Telescope (VLT)** in Chile, astronomers found strong evidence of a large body orbiting the brown dwarf **CD-35 2722 B**.

- The brown dwarf itself orbits a star located about **72 light-years from Earth**. Since the newly detected candidate orbits a body that itself revolves around a star, it has been described as an **exosatellite** and a potential exomoon.
- It represents the **first plausible detection of an exosatellite**.
- **Detection Method:** Radial velocity method, which detects a body's wobble caused by an orbiting companion.
- **Characteristics of the Exosatellite:** A giant gaseous body with a minimum estimated mass of about **90% of Jupiter's mass**, placing it in the planetary-mass category.
 - It follows an apparently eccentric orbit around the brown dwarf **CD-35 2722 B** and completes one revolution in approximately **170 days**.

Exomoon

A natural satellite orbiting a planet or other celestial body outside the Solar System. No universally accepted formal definition exists.

Brown Dwarfs

- Objects between giant planets and small stars.
- Form like stars but cannot sustain hydrogen fusion; hence called "failed stars."
- Generally **13–80 times that of Jupiter masses**. **CD-35 2722 B** is about **30–33 times as massive as Jupiter**.

Disabling BitChat in India

The **Indian Cyber Crime Coordination Centre (I4C)** under the **Ministry of Home Affairs** directed GitHub to remove BitChat repositories, citing possible misuse for unlawful assemblies and evasion of lawful surveillance.

- Notice was issued under **Section 79(3)(b), IT Act, 2000**, read with **Rule 3(1)(d), IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021**.
- **Key Legal Concern:** Section 79 of the IT Act deals with "safe harbour" protection for intermediaries, not blocking powers.
 - Lawful blocking of public access typically falls under **Section 69A of the IT Act** and the **Blocking Rules, 2009**, which mandate a formal hearing and written reasons.
 - In **Shreya Singhal v. Union of India (2015)**, the **Supreme Court** clarified that intermediaries are required to remove content **only upon a court order or a valid government direction issued through due process**, while reaffirming **Article 19(1)(a)** protections for online speech.
- **Civil Society Argument:** Referring to **Anuradha Bhasin v. Union of India (2020)**, IFF (Internet Freedom Foundation) contended that restrictions affecting digital communication must satisfy the **principle of proportionality**, requiring measures to be **necessary, proportionate, and limited in scope and duration**.

DAANVEER Initiative

The **Ministry of Panchayati Raj (MoPR)**, with the **National Informatics Centre (NIC)** and **DigiHaat**, launched DAANVEER to strengthen Gram Panchayat digital infrastructure through voluntary citizen participation.

Key Features

- **About:** Transparent, verifiable, end-to-end digital platform for citizens and organisations to support identified technology and infrastructure needs of eligible Gram Panchayats.

- **Objective:** Bridge the digital infrastructure gap and promote efficient, transparent and citizen-centric local governance.
- **Tagline:** “Give Back to Your Village”, encouraging citizens and the Indian diaspora to support native villages.
- **Mechanism:** Available through the **Meri Panchayat App** for donating standardised, quality-verified computer bundles.
- **DigiHaat Integration:** Enables selection of State-specific, pre-approved computer bundles with end-to-end digital tracking.
- **Transparency:** Tracks donations from dispatch to installation and provides a digital certificate of appreciation.
- **Coverage:** Nearly **2.5 lakh Gram Panchayats**.
- **Digital Ecosystem:** Builds on **e-GramSwaraj, Meri Panchayat, Sabha Saar, AuditOnline and Gram Manchitra**.
- **Significance:** Strengthens voluntary public participation, Digital India, Panchayati Raj institutions and grassroots digital governance.

Bank of Baroda Data Breach on Dark Web

BoB has initiated a forensic investigation following an alleged leak of **customer-related data onto the dark web**.

- Cyber incidents, such as the **Tata Electronics ransomware attack (2026)**, **BSNL data breach (2024)**, **WazirX crypto heist (2024)**, **ICMR data leak (Dark Web, 2023)**, **Air India cyberattack (2021)**, and the latest BoB data breach, have raised serious concerns over India’s cybersecurity preparedness.
- **Layers of the World Wide Web**
 - **Surface Web:** Publicly accessible websites indexed by search engines.
 - **Deep Web:** Legitimate but unindexed content requiring authentication, such as banking portals, emails and private databases.
 - **Dark Web:** An intentionally hidden part of the **Deep Web**, accessible through specialised software such as **Tor**. It conceals users’ identities and IP addresses and is often misused for **illegal trade, cybercrime and data trafficking**.

India’s Cybersecurity and Data-Protection Framework

- **CERT-In:** India’s nodal agency for cyber threats; severe incidents must be reported within **6 hours**.
- **RBI Cybersecurity Framework:** Requires banks to adopt a **board-approved cybersecurity policy**, establish a **Cyber Crisis Management Plan**, and promptly report unusual incidents.

- **DPDP Act, 2023:** Requires **Data Fiduciaries**, including banks, to protect personal data and imposes penalties for inadequate safeguards.
- **IT Act, 2000: Sections 43A and 66** address failure to protect sensitive data and unauthorised computer access.

BHAVYA Rasayan Scheme

The **Union Cabinet** approved the **Bharat Audyogik Vikas Yojana Rasayan (BHAVYA Rasayan) Scheme** to establish **three dedicated Chemical Parks** across India.

BHAVYA Rasayan Scheme

- **About:** Announced in the **Union Budget 2026–27** to establish **three integrated Chemical Parks** for chemical, petrochemical, downstream and ancillary industries.
- **Funding:** Total outlay of **₹3,030 crore**, including **₹3,000 crore** for infrastructure and utilities.
- **Centre–State Contribution:** Centre will provide up to **₹1,000 crore per park**, while the State must contribute at least **₹500 crore**.
- **Implementation:** To be implemented from **FY 2026–27 to FY 2030–31** by State Governments through the **Challenge Route**.
- **Land Requirement:** Each park must have at least **8 sq. km or 2,000 acres** of contiguous, encumbrance-free land.
- **Infrastructure:** Includes effluent treatment, hazardous-waste management, water supply, solvent recovery, pipelines, logistics and warehousing.
- **Benefits:** Will reduce production and logistics costs, attract investment, promote exports and import substitution, generate employment and support **Atmanirbhar Bharat and Viksit Bharat @ 2047**.

PRAXIS Mission

NASA has selected **PRAXIS (Planetary Rings Autonomous EXploration with In-situ Sampling)** for **Innovative Advanced Concepts (NIAC) 2026 Phase I funding**. **PRAXIS** proposes the first mission to directly collect particles from **planetary ring systems** & helps in improving understanding of rings around **Saturn, Uranus, Neptune**, and smaller bodies such as **Chariklo and Chiron**.

- **Nature of Saturn’s Rings:** Saturn’s rings consist mainly of **water ice**, with particles ranging from **microscopic grains to house-sized boulders** that continuously merge, and break apart.
- **Background:** Despite the **Cassini mission’s** discoveries, questions remain about how planetary rings **form, evolve, move, and disappear**.
- **About:** **PRAXIS** would make the first close-up observations of **millimetre- to centimetre-sized ring particles**.

- **Objective:** It would measure particle **size, porosity, structure, and chemical composition** to explain features such as **density waves, gaps, propellers, and self-gravity wakes**.
- **AI-Powered Robotic System:** The mission would use an **AI-driven, bio-inspired robot** for autonomous navigation, collision avoidance, target selection, sampling, and real-time analysis.
- **Sampling Method:** The spacecraft would graze the ring plane and use a **long, soft, deployable boom** for **touch-and-go sampling**, avoiding direct passage through hazardous ring regions.
- **Exploration of Multiple Regions:** PRAXIS would sample and compare particles from **different ring sections and gaps**.
- **Future Potential:** **NIAC Phase I** will focus on simulations and system design. **Phase II**, if approved, would support prototypes, laboratory tests, feasibility demonstrations, and operational planning.

Maharashtra to Get India's 1st AI-Powered Bird Sanctuary

A ₹45-crore Eco-Nature Park at the **Thane Creek Flamingo Sanctuary**, featuring India's 1st **AI-powered bird detection & monitoring centre**.

- **AI-Based Monitoring System:** AI-enabled cameras will use automated image and sound recognition to identify bird species, track migration and analyse activity across a **1.5-km park stretch**.
- **Thane Creek Flamingo Sanctuary:** Located between **Mumbai and Navi Mumbai**, it was declared a wildlife sanctuary in **2015** & is **India's 1st flamingo sanctuary**. This coastal wetland supports thousands of flamingos and over **200 bird species**, including greater and lesser flamingos, herons, egrets, shorebirds and kingfishers.
- **Migratory Importance:** Part of **Central Asian Flyway**, an important migratory route connecting Europe, Central Asia and the Indian subcontinent.

Flamingos

- **About:** Tall water birds with long, S-shaped necks & stick-like legs.
 - Highly social, often in large flocks inhabiting shallow, eutrophic water bodies and alkaline lakes.
- **Species:** 6 species found globally in **tropical and subtropical regions**, with only **2 found in India (Greater & Lesser Flamingo)**.
 - **Other Species:** **Chilean Flamingo** (*Phoenicopterus chilensis*), **American/Caribbean Flamingo**

(*Phoenicopterus ruber*), **Andean Flamingo** (*Phoenicoparrus andinus*) & **James's or Puna Flamingo** (*Phoenicoparrus jamesi*).

- **Diet & Colouration:** Feed on **algae, molluscs, and crustaceans**. Their **plumage colour**, ranging from **white to pink** to orange, is determined by **carotenoid pigments** in their diet.
- **Habitat:** Shallow wetlands, mudflats and coastal lagoons.
- **Threats:** Habitat degradation, pollution, coastal development, human disturbance and changes in water quality.

Supreme Court Clarifies Scope of Special Intensive Revision

Recently, the **Supreme Court** observed that the **Special Intensive Revision (SIR) process** is limited to electoral purposes and cannot be used for other objectives, including determining citizenship or affecting access to welfare benefits.

- The Court was hearing a petition alleging that the **West Bengal government** was using SIR data to remove names from welfare schemes such as **food security programmes, women's welfare schemes, and Backward Caste certifications**.
- The Court observed that, as per its earlier judgment in the **2025 Bihar SIR case**, SIR outcomes cannot be used for any purpose other than elections. The **Election Commission** must refer citizenship-related issues to the government for adjudication under the **Citizenship Act, 1955**.

Special Intensive Revision

- **SIR:** The SIR is a comprehensive exercise conducted under the supervision of the **Chief Election Commissioner** to update and purify electoral rolls.
- **Legal Basis:** The SIR is anchored in **Article 324 of the Indian Constitution**, which grants the ECI superintendence over elections, and **Section 21(3) of the Representation of the People Act (RPA), 1950**, which specifically empowers the commission to order a special revision.
- **Objective:** Its primary objective is to **remove duplicate, deceased, shifted, and ineligible voters, including "ghost voters,"** to ensure that electoral rolls accurately reflect the eligible voting population and uphold the democratic principle of **"one person, one vote."**
- **Background:** The Special Intensive Revision (SIR) of electoral rolls has been conducted **eight times between 1951 and 2004**, with the last SIR carried out during 2002–2004, over two decades ago.

TOP 200 PRACTICE MCQs

August 2026 Special Edition

Coverage: 25 July – 24 August 2026

Q1. Under the modified Supreme Court ruling regarding entry-level judicial services, what is the transitional relief window during which fresh law graduates can appear without prior bar practice?

- A) Up to 31 December 2025
- B) 20 May 2025 to 31 March 2027
- C) 1 January 2026 to 1 January 2028
- D) Up to 15 August 2027

Q2. What is the full form of the SHANTI Act introduced in 2025 to replace the CLND Act 2010?

- A) Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India Act
- B) Strategic Handling and Atomic Navigation for Technological India Act
- C) Safety Harnessing and Atomic Network for Transforming India Act
- D) Sovereign Handling and Atomic Nuclear Transformation of India Act

Q3. Under the SHANTI Act, 2025, which body receives statutory status to regulate nuclear safety?

- A) BARC
- B) AERB (Atomic Energy Regulatory Board)
- C) NPCIL
- D) Department of Atomic Energy

Q4. Under the SHANTI Act 2025, what is the operator liability cap for nuclear installations with thermal power above 3,600 MW?

- A) Rs. 1,500 crore
- B) Rs. 3,000 crore
- C) Rs. 750 crore
- D) Rs. 5,000 crore

Q5. Which constitutional provision empowers the Parliament to alter state boundaries or change the name of a state (such as renaming Kerala to Keralam)?

- A) Article 1
- B) Article 2
- C) Article 3
- D) Article 4

Q6. On India's 80th Independence Day, the Prime Minister articulated which conceptual framework to guide the journey toward Viksit Bharat 2047?

- A) Navratna Vision
- B) Sapt Dhara Framework
- C) Panchamrit Pledge
- D) Ashtalakshmi Path

Q7. In the context of the SHANTI Act, 2025, what is the time limit for filing claims regarding personal injury arising from nuclear damage?

- A) 5 years
- B) 10 years
- C) 20 years
- D) 30 years

Q8. Which committee in 1997 recommended amending the Atomic Energy Act 1962 to establish an independent Nuclear Safety Regulatory Authority?

- A) Homi Bhabha Committee
- B) Raja Ramanna Committee
- C) Kasturirangan Committee
- D) K. Subrahmanyam Committee

Q9. The Foreign Contribution (Regulation) Amendment Bill, 2026 primarily aims to enhance oversight on foreign funding of which entities?

- A) Political parties
- B) NGOs and voluntary associations
- C) Private educational trusts only
- D) Multinational corporations

Q10. Which Article of the Constitution guarantees the Right to Freedom of Speech and Expression, under which the Right to Protest peacefully is protected?

- A) Article 14
- B) Article 19(1)(a) and 19(1)(b)
- C) Article 21
- D) Article 25

Q11. Under the Pradhan Mantri Viksit Bharat Rojgar Yojana, what is the primary incentive provided to employers?

- A) 100% corporate tax exemption
- B) Subsidy on Employees' Provident Fund (EPF) contributions for new hires
- C) Free land allocation in SEZs
- D) Interest-free working capital loans

Q12. The Supreme Court recently extended protection under Section 498A (cruelty) of the Indian Penal Code/BNS to which category of relationships?

- A) Live-in relationships having characteristics of marriage
- B) Workplace colleagues
- C) Business partnerships
- D) Distant commercial relations

Q13. What is the primary objective of the GOBARdhan initiative under the Department of Drinking Water and Sanitation?

- A) Milk yield enhancement
- B) Converting cattle dung and organic waste into biogas and bio-CNG
- C) Desalination of seawater
- D) Cattle insurance coverage

Q14. The Amrit Bharat Station Scheme aims to modernize how many railway stations across India?

- A) 500 stations
- B) 1,000 stations
- C) 1,321 stations
- D) 2,500 stations

Q15. What is the primary focus of the Pradhan Mantri Surya Sarovar Yojana?

- A) Deep sea mining
- B) Installing floating solar panels on Amrit Sarovar water bodies
- C) Subsidizing desert safari tourism
- D) Rainwater harvesting in urban schools

Q16. The Public Examinations (Prevention of Unfair Means) Amendment Bill, 2026 proposes the establishment of which fast-track mechanism?

- A) Special Fast-Track Courts for paper leak cases
- B) Military tribunals
- C) Online AI re-examinations within 24 hours
- D) Complete cancellation of national testing agencies

Q17. Which landmark 1987 Supreme Court case established the principle of 'Absolute Liability' in India?

- A) Kesavananda Bharati case
- B) M.C. Mehta v. Union of India (Oleum Gas Leak case)
- C) Maneka Gandhi case
- D) Vishaka v. State of Rajasthan

Q18. The Government e-Marketplace (GeM) completed how many years of operation in 2026?

- A) 5 Years
- B) 8 Years
- C) 10 Years
- D) 15 Years

Q19. Under the Supreme Court guidelines on ordering DNA/paternity tests in matrimonial disputes, when should courts direct such tests?

- A) In all divorce petitions automatically
- B) Only as an exceptional last resort when strong prima facie case exists
- C) On oral demand by either spouse
- D) Paternity tests are completely prohibited

Q20. What is 'Digital Arrest', which was recently flagged by the Supreme Court and cyber agencies?

- A) Legitimate electronic surveillance under IT Act
- B) A cyber fraud where scammers pose as law enforcement via video calls to extort money
- C) Biometric attendance for undertrials
- D) Judicial custody inside digital lockups

Q21. Which Article of the Constitution provides for recruitment of persons other than District Judges to the judicial service by the Governor?

- A) Article 214
- B) Article 233
- C) Article 234
- D) Article 235

Q22. The Supreme Court's newly released Handbook on Gender Sensitivity specifically advises judges and lawyers against using which of the following?

- A) Hindi terms in judgments
- B) Gender-stereotypical language and patriarchal clichés
- C) Quotations from international conventions
- D) Dissenting opinions

Q23. What is 'Non-Revenue Water' (NRW), a major challenge in urban water governance?

- A) Water used exclusively for agriculture
- B) Water produced and lost before reaching customers due to leaks, theft or unmetered consumption
- C) Bottled mineral water exempt from GST
- D) River water flowing across state borders

Q24. Under the PM-KISAN scheme, what is the annual financial support provided to eligible farmer families?

- A) Rs. 4,000 in two installments
- B) Rs. 6,000 in three equal installments
- C) Rs. 8,000 in four installments
- D) Rs. 12,000 in monthly installments

Q25. The Tribunals Reforms Bill, 2026 aims to accomplish which major structural change?

- A) Abolishing all high courts
- B) Rationalizing and merging redundant tribunals into regular judicial bodies
- C) Creating separate tribunals for each district
- D) Replacing Supreme Court with a National Tribunal

Q26. Which mission was launched to provide affordable nutraceuticals and dietary supplements through Jan Aushadhi Kendras?

- A) Jan Aushadhi Poshan Abhiyan
- B) Jan Aushadhi Nutraceuticals Initiative
- C) Ayushman Bharat Bharat Poshan
- D) Swasthya Suposhan Mission

Q27. Under the Smart Border Architecture, which advanced technology is deployed along India's international borders?

- A) Comprehensive Integrated Border Management System (CIBMS) with radars, sensors and thermal cameras
- B) Continuous physical brick walls across all borders
- C) Subterranean moat network
- D) Automated electrified wire grids only

Q28. The Indian Statistical Institute (ISI) Bill, 2026 aims to strengthen which aspect of the institute?

- A) Privatizing statistical data gathering
- B) Modernizing governance, research autonomy, and expanding data science programs
- C) Merging ISI into NITI Aayog
- D) Shutting down all regional centres

Q29. The Revamped Khelo India Scheme primarily focuses on which tier of sports development?

- A) Only Olympic medalists' overseas training
- B) Grassroots talent identification, state sports academies, and rural competitions
- C) IPL franchise ownership
- D) Construction of Formula 1 tracks

Q30. In the Supreme Court ruling on Electoral Black Money, which aspect of election expenditure was emphasized?

- A) Mandating cryptocurrency donations
- B) Real-time tracking of candidate spending and stringent compliance with expenditure caps
- C) Removing all spending limits for national parties
- D) Cash donations up to Rs. 1 crore made anonymous

Q31. The Prevention of Insults to National Honour (Amendment) Act, 2026 introduces specific provisions regarding:

- A) Mandatory singing of anthem in cinemas
- B) Digital misuse, AI-generated desecration and strict penalties for disrespect to the National Flag and Anthem
- C) Compulsory flag hoisting on private vehicles
- D) Abolition of state flags

Q32. Under the PM CARES Fund legal debate, which statutory body's audit was demanded by transparency activists?

- A) Comptroller and Auditor General (CAG) of India
- B) Law Commission of India
- C) Election Commission
- D) CBI Financial Cell

Q33. Which Article of the Indian Constitution vests administrative, disciplinary and operational control over subordinate courts in the High Court?

- A) Article 226
- B) Article 235
- C) Article 311
- D) Article 356

Q34. What is the primary role of the e-Courts Integrated Mission Mode Project in Indian judiciary?

- A) Replacing judges with AI bots
- B) Digital filing, case information systems, virtual hearings, and national judicial data grid
- C) Conducting televised trials
- D) Eliminating written evidence

Q35. Under Article 233 of the Constitution, who appoints District Judges in consultation with the High Court?

- A) President of India
- B) Governor of the State
- C) Chief Minister
- D) Chief Justice of India

Q36. What was the core issue addressed in the debate over 'Relevance of Regional Parties in India'?

- A) Regional parties must be banned
- B) Regional parties articulate sub-national aspirations, deepening cooperative federalism and coalition accountability
- C) Only two-party system is allowed constitutionally
- D) Regional parties cannot participate in Rajya Sabha elections

Q37. Under the Convention on Supplementary Compensation for Nuclear Damage (CSC), what is the international tier of compensation funded by?

- A) Private insurance companies only
- B) Public contributions by Contracting Parties based on installed nuclear capacity and UN assessment rate
- C) United Nations General Fund
- D) World Bank loans

Q38. What percentage of research scholars supported under the National Fellowship for ST Students in 2025–26 were women?

- A) 25%
- B) 33%
- C) 56%
- D) 75%

Q39. Under the Pre-Matric Scholarship scheme for tribal students, how many beneficiaries were reached between April 2024 and August 2026?

- A) 5.5 lakh
- B) 12.2 lakh
- C) 28.99+ lakh
- D) 50 lakh

Q40. What is the primary function of the Atomic Energy Redressal Advisory Council created under SHANTI Act 2025?

- A) Nuclear weapon testing oversight
- B) Reviewing nuclear liability claims and providing grievance dispute redressal
- C) Exporting enriched uranium
- D) Setting electricity consumer tariffs

Q41. The Kerala Legislative Assembly passed a unanimous resolution to rename the state as 'Keralam' primarily reflecting which linguistic heritage?

- A) Tamil roots
- B) Malayalam pronunciation and historical identity
- C) Sanskrit literature
- D) Tulu heritage

Q42. Under the new National Security Guidelines for Renewable Energy Projects near International Borders, what clearance is mandatory?

- A) Approval from the Ministry of Defence and strict vetting of equipment origin/SCADA systems
- B) Total ban on all solar projects within 500 km of borders
- C) Compulsory 100% state government equity
- D) No security clearances are required

Q43. India formally standardized and released native names for 27 places in Arunachal Pradesh in response to renaming attempts by which country?

- A) Myanmar
- B) China
- C) Bhutan
- D) Nepal

Q44. What is the deadline set by the Union Ministry of Home Affairs for achieving a 'Naxal-Free India' through integrated security and development?

- A) March 2026
- B) December 2028
- C) August 2030
- D) October 2027

Q45. The Cauvery Water Dispute involves which group of states/UTs?

- A) Maharashtra, Karnataka, Telangana
- B) Karnataka, Tamil Nadu, Kerala and Puducherry
- C) Andhra Pradesh, Odisha, Tamil Nadu
- D) Kerala, Goa, Karnataka

Q46. Where are the Commonwealth Games 2026 scheduled to be hosted after Glasgow took over host responsibilities?

- A) Birmingham, UK
- B) Glasgow, Scotland
- C) Gold Coast, Australia
- D) Victoria, Canada

Q47. Kargil Vijay Diwas is observed annually on which date to commemorate the success of Operation Vijay in 1999?

- A) 15th July
- B) 26th July
- C) 10th August
- D) 15th August

Q48. What is the primary objective of the Positive Indigenisation Lists notified by the Ministry of Defence?

- A) Banning private defence manufacturing
- B) Prohibiting imports of specified military equipment to promote domestic manufacturing
- C) Importing 100% of fighter aircraft
- D) Privatizing defence ordnance factories

Q49. Which newly raised border guarding formation focuses on technology-driven drone and sensor surveillance along India's northern borders?

- A) Border Electronic Wing
- B) Smart Surveillance Battalions under ITBP/BSF
- C) Digital Task Force
- D) Cyber Border Corps

Q50. The 'Operation Vijay' in 1999 was executed to recapture high-altitude military posts in which sector of Jammu & Kashmir/Ladakh?

- A) Siachen only
- B) Kargil, Dras, Batalik, and Mushkoh Valley
- C) Chushul and Galwan
- D) Baramulla and Uri

Q51. What is the name of India's indigenous Light Combat Aircraft (LCA) developed by HAL and ADA?

- A) Marut
- B) Tejas
- C) Rafale
- D) Sukhoi Su-30MKI

Q52. Which state's assembly passed a resolution opposing the creation of a 'Frontier Nagaland Territory' out of its eastern districts?

- A) Assam
- B) Nagaland
- C) Manipur
- D) Arunachal Pradesh

Q53. Under the inter-state river water disputes framework, which Article of the Constitution empowers Parliament to adjudicate river valley disputes?

- A) Article 260
- B) Article 262
- C) Article 280
- D) Article 301

Q54. What is the central strategy of the 'SAMADHAN' doctrine used by security forces in Left-Wing Extremism (LWE) affected areas?

- A) Unilateral ceasefires
- B) Smart leadership, aggressive strategy, motivation, actionable intelligence, and technology
- C) Complete surrender of weapons without rehabilitation
- D) Relocation of entire tribal villages

Q55. Which Indian state has developed the 'Surakshit Bachpan' campaign to combat cyber exploitation and social media radicalization among youth?

- A) Uttar Pradesh
- B) Kerala
- C) Karnataka
- D) Maharashtra

Q56. What is the purpose of the 'Vibrant Villages Programme' along India's northern border areas?

- A) Creating exclusive industrial parks
- B) Comprehensive development of border villages to prevent out-migration and enhance border security
- C) Relocating border residents to metropolitan cities
- D) Nuclear missile silo placement

Q57. In which union territory/state was the world's highest fighter airfield constructed at Nyoma by the Border Roads Organisation (BRO)?

- A) Ladakh
- B) Uttarakhand
- C) Sikkim
- D) Arunachal Pradesh

Q58. Which river basin dispute involves the states of Goa, Karnataka, and Maharashtra?

- A) Mahadayi (Mandovi) River
- B) Godavari River
- C) Narmada River
- D) Krishna River

Q59. What was the name of the naval operation conducted during the 1999 Kargil conflict by the Indian Navy to blockade Pakistani ports?

- A) Operation Trident
- B) Operation Talwar
- C) Operation Cactus
- D) Operation Python

Q60. Which constitutional schedule governs the administration of tribal areas in Assam, Meghalaya, Tripura, and Mizoram?

- A) Fifth Schedule
- B) Sixth Schedule
- C) Seventh Schedule
- D) Eighth Schedule

Q61. How many strategic critical minerals were officially identified by the Ministry of Mines to secure India's supply chains?

- A) 12
- B) 20
- C) 30
- D) 50

Q62. What does the term 'Greenium' refer to in sovereign and corporate debt markets?

- A) Penalty for carbon emissions
- B) The pricing benefit (lower yield/interest rate) that issuers receive when selling green bonds compared to conventional bonds
- C) Green insurance premium for farmers
- D) Subsidized electricity tariff for solar parks

Q63. Under the revamped Foreign Assets of Small Taxpayers Disclosure Scheme, 2026, who are the primary beneficiaries?

- A) Large multinational conglomerates
- B) Small retail taxpayers and professionals who inadvertently failed to report small foreign shares/ESOPs/bank accounts
- C) Shell companies registered in tax havens
- D) Public sector banks

Q64. Which state in India accounts for over 85% of national production of Makhana (fox nuts), which recently received major export support?

- A) Uttar Pradesh
- B) Bihar (Mithila region)
- C) West Bengal
- D) Assam

Q65. What is the primary objective of the Electronics Component Manufacturing Scheme (SPECS 2.0)?

- A) Importing finished smartphones from China
- B) Subsidizing domestic capital expenditure for passive components, sub-assemblies, and semiconductor packaging
- C) Selling subsidized televisions in rural areas
- D) Replacing 5G with 2G spectrum

Q66. Which international body adopted Convention No. 193 regarding Decent Work in the Platform and Gig Economy?

- A) WTO
- B) ILO (International Labour Organization)
- C) OECD
- D) World Bank

Q67. The Reserve Bank of India (RBI) decided to resume licensing of Urban Cooperative Banks (UCBs) after a hiatus since which year?

- A) 1991
- B) 2004
- C) 2014
- D) 2020

Q68. India recently allowed 100% Foreign Direct Investment (FDI) under the automatic route for which model in e-commerce meant strictly for exports?

- A) Marketplace model for domestic grocery
- B) Inventory-based model for cross-border export sales
- C) Multi-brand domestic retail stores
- D) Peer-to-peer cryptocurrency exchanges

Q69. The World Bank's 'B-READY' (Business Ready) index replaces which erstwhile flagship report?

- A) Human Development Index
- B) Ease of Doing Business Report
- C) Global Competitiveness Report
- D) World Development Indicators

Q70. Under the Minimum Support Price (MSP) regime, the government fixes MSP for how many mandated crops based on CACP recommendations?

- A) 14 crops
- B) 22 mandated crops (plus Fair and Remunerative Price for Sugarcane)
- C) 35 crops
- D) All agricultural produce

Q71. What is the primary role of the National Cooperative Development Corporation (NCDC)?

- A) Privatizing agricultural cooperatives
- B) Planning, promoting, and financing production, processing, and marketing through cooperatives
- C) Collecting corporate taxes from cooperatives
- D) Managing currency mints in India

Q72. According to the NITI Aayog Report on India's Regulatory Regime in Professional Services, what is the main hurdle identified?

- A) Too few professionals in India
- B) Rigid entry barriers, fragmented state licensing, and lack of mutual recognition agreements with global trading partners
- C) Zero taxation on professional fees
- D) Complete lack of internet access

Q73. Under the Geographical Indications of Goods (Registration and Protection) Act, 1999, a GI registration is valid for how many years before requiring renewal?

- A) 5 years
- B) 10 years
- C) 15 years
- D) 20 years

Q74. What is the significance of the Democratic Republic of Congo (DRC) in the global critical mineral supply chain?

- A) World's largest producer of Lithium
- B) World's largest producer of Cobalt (over 70% of global supply)
- C) Exclusive source of enriched Uranium
- D) Largest exporter of crude petroleum

Q75. What is the Real Effective Exchange Rate (REER) used to measure?

- A) Inflation rate of food commodities only
- B) The value of a currency against a weighted basket of foreign currencies, adjusted for inflation differentials
- C) Gold reserves held by commercial banks
- D) Government's fiscal deficit as percentage of GDP

Q76. Which scheme was launched to position India as a global MICE (Meetings, Incentives, Conferences, and Exhibitions) tourism destination?

- A) Meet in India Initiative
- B) Bharat Parv Tourism Portal
- C) Swadesh Darshan 2.0 (MICE segment)
- D) Global Bharat Yatra

Q77. What was the main driver of the surge in India's merchandise exports in mid-2026?

- A) Only raw cotton shipments
- B) Engineering goods, electronic goods, and refined petroleum products
- C) Export of unrefined mineral ores
- D) Gold bar exports

Q78. Under the PM Surya Ghar: Muft Bijli Yojana, what is the maximum capacity of rooftop solar subsidized for residential households?

- A) 1 kW
- B) 3 kW
- C) 10 kW
- D) 25 kW

Q79. The National Clean Energy Fund (NCEF) in India was initially funded through which levy?

- A) Corporate CSR tax
- B) Clean Energy Cess (GST Compensation Cess on Coal)
- C) Petroleum road cess
- D) Airline carbon surcharge

Q80. In the context of the platform economy, what is the defining feature of 'gig workers' under the Code on Social Security, 2020?

- A) Full-time permanent government employment
- B) Persons who perform work or participate in a work arrangement and earn from such activities outside of traditional employer-employee relationships
- C) Factory workers entitled to pension only
- D) Agricultural tenants

Q81. Which agency serves as the nodal organisation for conducting competitive reverse bidding for utility-scale solar and wind projects in India?

- A) NTPC only
- B) SECI (Solar Energy Corporation of India)
- C) BHEL
- D) Power Finance Corporation

Q82. What is the target for India's non-fossil fuel installed electric capacity by 2030 as per its updated Nationally Determined Contributions (NDCs)?

- A) 300 GW
- B) 400 GW
- C) 500 GW (50% cumulative electric power capacity)
- D) 1,000 GW

Q83. Under the GST framework, what is the role of the Goods and Services Tax Network (GSTN)?

- A) Judicial adjudication of tax disputes
- B) Providing shared IT infrastructure, return filing portal, and registration services to taxpayers and governments
- C) Setting GST rates unilaterally
- D) Printing physical tax stamps

Q84. What is 'Angel Tax' which was recently rationalized/abolished under income tax amendments to support the startup ecosystem?

- A) Tax on charity trusts
- B) Tax levied under Section 56(2)(viib) on excess share premium received by unlisted companies above fair market value
- C) Customs duty on imported drones
- D) Capital gains tax on listed stocks

Q85. Which index published by the Reserve Bank of India tracks the extent of digital payments adoption across the country?

- A) Digital Financial Inclusion Index (DFII)
- B) RBI Digital Payments Index (RBI-DPI)
- C) Unified Payments Scale (UPS)
- D) FinTech Growth Index

Q86. What is the primary purpose of the 'Trade Receivables Discounting System' (TReDS)?

- A) Equity crowdfunding for NGOs
- B) Electronic platform for facilitating financing/discounting of trade receivables of MSMEs from corporate buyers
- C) Subsidizing export freight charges
- D) Retail mortgage lending

Q87. Under the Insolvency and Bankruptcy Code (IBC), what is the maximum statutory timeline for completion of the Corporate Insolvency Resolution Process (CIRP)?

- A) 90 days
- B) 180 days (extendable up to 330 days including litigation)
- C) 5 years
- D) No time limit exists

Q88. What is the key objective of the Production Linked Incentive (PLI) scheme for Semiconductor and Display Fabrication?

- A) Subsidizing imported silicon chips
- B) Providing 50% fiscal support on capital expenditure for setting up silicon fabs and ATMP/OSAT facilities in India
- C) Banning all domestic chip design
- D) Manufacturing cathode ray televisions

Q89. Which category of banks was mandated by the RBI to maintain a Priority Sector Lending (PSL) target of 75% of Adjusted Net Bank Credit (ANBC)?

- A) Foreign Banks with >20 branches
- B) Urban Cooperative Banks (UCBs) and Regional Rural Banks (RRBs)
- C) Public Sector Commercial Banks (PSBs)
- D) Non-Banking Financial Companies (NBFCs)

Q90. What is 'Sovereign Rating' assigned by global agencies like Moody's, S&P, and Fitch?

- A) Military strength assessment
- B) Independent assessment of an obligor's (country's) creditworthiness and ability to repay debt without default
- C) Democratic freedom index
- D) Olympic medal potential score

Q91. Under the Foreign Trade Policy 2023, what is India's merchandise and services export target by the year 2030?

- A) USD 1 Trillion
- B) USD 2 Trillion
- C) USD 5 Trillion
- D) USD 500 Billion

Q92. What is the primary function of the National Agriculture Market (e-NAM) portal?

- A) Selling seeds on retail credit
- B) Pan-India electronic trading portal networking existing APMC mandis to create a unified national market for agricultural commodities
- C) Regulating tractor manufacturing
- D) Imposing state border tolls on crops

Q93. Which regulatory body governs credit rating agencies and merchant bankers in India's capital markets?

- A) RBI
- B) SEBI (Securities and Exchange Board of India)
- C) IRDAI
- D) PFRDA

Q94. What is the purpose of the 'Bharat Kosh' portal developed by the Controller General of Accounts (CGA)?

- A) Filing income tax returns
- B) Non-tax receipt portal facilitating digital payment of fees, fines, royalties and spectrum charges to the Government of India
- C) Direct gold purchasing app
- D) Rural loan application kiosk

Q95. What is the meaning of 'Core Inflation' in macroeconomic indicators?

- A) Total CPI inflation including all items
- B) Headline inflation excluding volatile components like food and fuel
- C) Inflation measured exclusively at wholesale petrol pumps
- D) Annual gold price increase

Q96. Which committee recommended the constitution of the Monetary Policy Committee (MPC) in India?

- A) Urjit Patel Committee
- B) Raghuram Rajan Committee
- C) Bimal Jalan Committee
- D) Y.V. Reddy Committee

Q97. What is the target inflation band mandated for the RBI under the flexible inflation targeting framework?

- A) 2% with a tolerance band of +/- 1%
- B) 4% with a tolerance band of +/- 2% (i.e. 2% to 6%)
- C) Zero inflation
- D) 6% to 10%

Q98. Under the PM Gati Shakti National Master Plan, how many infrastructure pillars (engines) drive multi-modal connectivity?

- A) 3 Engines
- B) 5 Engines
- C) 7 Engines (Roads, Railways, Airports, Ports, Mass Transport, Waterways, Logistics)
- D) 12 Engines

Q99. What is 'Prompt Corrective Action' (PCA) framework implemented by the Reserve Bank of India?

- A) Disciplinary action against retail customers for late payments
- B) Supervisory mechanism triggered when a commercial bank's capital, asset quality (NPA), or profitability breaches specified risk thresholds
- C) Automated cancellation of banking licenses
- D) Direct police raid on defaulting borrowers

Q100. Which public sector institution acts as the national coordinator for retail electronic payments including UPI, IMPS, and RuPay?

- A) State Bank of India
- B) National Payments Corporation of India (NPCI)
- C) Reserve Bank Information Technology (ReBIT)
- D) NITI Aayog Digital Cell

Q101. The landmark India–Japan Maritime Security Pact signed recently focuses on which strategic ocean region?

- A) Baltic Sea
- B) Indo-Pacific and the Malacca Strait approaches
- C) Arctic Ocean
- D) Caribbean Sea

Q102. What is the significance of the Makkah Joint Defence Agreement signed by regional powers?

- A) Joint space exploration treaty
- B) Collective regional security and counter-terror maritime cooperation centered around the Red Sea and Gulf of Aden
- C) Exclusive oil pricing cartel
- D) Mutual visa-free tourism treaty

Q103. The President of India recently undertook an official State Visit to which Eastern European nation to strengthen bilateral high-tech and defence ties?

- A) Romania
- B) Belarus
- C) Albania
- D) Moldova

Q104. What is the core message of the 'Rome Declaration for an Unarmed and Disarming Peace' endorsed by global religious and civil society leaders?

- A) Promoting unilateral nuclear proliferation
- B) Moral appeal for mutual de-escalation, phasing out nuclear arsenals, and investing in climate resilience rather than armaments
- C) Mandatory conscription in European armies
- D) Privatizing UN peace missions

Q105. Which international agreement requires India and Pakistan to annually exchange lists of their nuclear installations on the 1st of January since 1992?

- A) Tashkent Declaration
- B) Agreement on the Prohibition of Attack against Nuclear Installations and Facilities (1988)
- C) Simla Agreement
- D) Lahore Declaration

Q106. Ariana Park, where peacocks have been hosted for over a century, is located adjacent to which international headquarters?

- A) United Nations Office at Geneva (Palais des Nations)
- B) International Court of Justice, The Hague
- C) UNESCO Headquarters, Paris
- D) NATO Headquarters, Brussels

Q107. In diplomatic strategy, why is the Bay of Bengal described as India's 'SHANTI Anchor'?

- A) It has no geopolitical significance
- B) It bridges India's Act East Policy with Neighborhood First, anchoring maritime stability and BIMSTEC connectivity
- C) It is declared an international demilitarized zone
- D) It is an exclusive domestic lake

Q108. The proposed US–Saudi Arabia Nuclear Cooperation Agreement is negotiated under which section of the US Atomic Energy Act?

- A) Section 123 (123 Agreement)
- B) Section 301
- C) Section 232
- D) Section 504

Q109. What is the primary trade dynamic in 'India's Trade Balancing Act' navigating relations between the US and China?

- A) Joining a complete trade embargo against all global economies
- B) Reducing critical import dependencies on China while expanding value-added manufacturing exports to the West and upholding strategic autonomy
- C) Pegging the Indian Rupee to the Chinese Yuan
- D) Exiting the World Trade Organization

Q110. Which multilateral grouping consists of Bangladesh, Bhutan, India, Myanmar, Nepal, Sri Lanka, and Thailand?

- A) ASEAN
- B) BIMSTEC
- C) SAARC
- D) SCO

Q111. What was the focal theme of the bilateral discussion on 'Climate Resilience Pathway' between India and China?

- A) Space colonization
- B) Glacial melt monitoring, sharing transboundary hydrological data in the Himalayas, and joint disaster early warning systems
- C) Subsidizing private coal mining in Africa
- D) Merging the two countries' electrical grids

Q112. The International North-South Transport Corridor (INSTC) connects India with Russia via which intermediate nation?

- A) Iran
- B) Myanmar
- C) Japan
- D) Egypt

Q113. Where is the permanent Secretariat of BIMSTEC located?

- A) New Delhi, India
- B) Dhaka, Bangladesh
- C) Colombo, Sri Lanka
- D) Bangkok, Thailand

Q114. India recently signed a Comprehensive Economic Partnership Agreement (CEPA) with which Middle Eastern nation that came into force in 2022?

- A) United Arab Emirates (UAE)
- B) Jordan
- C) Yemen
- D) Syria

Q115. What is the 'Quad' diplomatic grouping composed of?

- A) India, Russia, China, and South Africa
- B) India, United States, Japan, and Australia
- C) India, UK, France, and Germany
- D) US, UK, Australia, and New Zealand

Q116. Which strait connects the Indian Ocean to the South China Sea, serving as one of the world's most critical maritime chokepoints?

- A) Strait of Hormuz
- B) Strait of Malacca
- C) Bab-el-Mandeb
- D) Bosphorus Strait

Q117. What is the primary purpose of the 'Colombo Security Conclave' (CSC)?

- A) Currency union negotiations
- B) Regional maritime security, counter-terrorism, and cyber security cooperation in the Indian Ocean
- C) Free trade zone in tea and spices
- D) Submarine arms sales

Q118. Which African nation recently joined the BRICS group as a new member?

- A) Egypt and Ethiopia
- B) Nigeria and Kenya
- C) Ghana and Senegal
- D) Sudan and Mali

Q119. What is the name of the strategic port in Iran developed with Indian assistance to provide access to Afghanistan and Central Asia?

- A) Gwadar Port
- B) Chabahar Port (Shahid Beheshti terminal)
- C) Bandar Abbas
- D) Duqm Port

Q120. Which international convention governs the rights of nations over territorial waters, EEZs, and high seas?

- A) UNCLOS (UN Convention on the Law of the Sea)
- B) Geneva Convention 1949
- C) Kyoto Protocol
- D) Chicago Convention

Q121. The 'Supply Chain Resilience Initiative' (SCRI) was launched jointly by India, Japan, and which other country?

- A) Australia
- B) South Korea
- C) United Kingdom
- D) Canada

Q122. Where is the headquarters of the International Solar Alliance (ISA) located?

- A) Paris, France
- B) Gurugram, India
- C) Geneva, Switzerland
- D) Nairobi, Kenya

Q123. What is the primary role of the 'Global Biofuels Alliance' (GBA) launched during India's G20 Presidency?

- A) Phasing out ethanol blending
- B) Accelerating global adoption of sustainable biofuels, setting technical standards, and facilitating international trade in biofuels
- C) Subsidizing petroleum drilling
- D) Banning electric mobility

Q124. Which country is the host of the United Nations Peacekeeping mission UNIFIL in which Indian peacekeepers have served for decades?

- A) Lebanon
- B) South Sudan
- C) Cyprus
- D) Democratic Republic of Congo

Q125. What is the main objective of the India–Middle East–Europe Economic Corridor (IMEC) announced at the New Delhi G20 Summit?

- A) Creating a space station
- B) Multi-modal ship-to-rail transit network connecting India, the Arabian Gulf, and Europe to stimulate economic development and green energy connectivity
- C) Military alliance against ASEAN
- D) Submarine telecom cable only

Q126. World Elephant Day is celebrated globally each year on which date?

- A) 29th July
- B) 12th August
- C) 16th September
- D) 5th June

Q127. World Lion Day is observed annually on which date to raise awareness about the conservation of Asiatic and African lions?

- A) 10th August
- B) 15th August
- C) 3rd March
- D) 4th October

Q128. Recent glaciological research revealed 'thinning-induced deceleration' in the glaciers of which Himalayan region?

- A) Zaskar Range (Ladakh)
- B) Nilgiri Hills
- C) Satpura Range
- D) Aravalli Range

Q129. What is the primary objective of India's newly launched 'Clouded Leopard Conservation Action Plan' (CAP)?

- A) Relocating clouded leopards to Thar Desert
- B) Targeted habitat protection, genetic corridors, and community-led anti-poaching monitoring across North-East India's rainforests
- C) Keeping clouded leopards in urban pet shelters
- D) Banning national parks in the Himalayas

Q130. The Supreme Court extended a mandatory 10 km mining buffer rule around which ecological category across India?

- A) Urban cricket stadiums
- B) Designated Wetland Reserves and Ramsar Sites
- C) Private agricultural farms
- D) Commercial SEZs

Q131. The Ken-Betwa River Linking Project, India's first river interlinking project under the National Perspective Plan, is situated in which geographical region?

- A) Thar Desert
- B) Bundelkhand region (Madhya Pradesh & Uttar Pradesh)
- C) Deccan Plateau
- D) Konkan Coast

Q132. The Ministry of Environment released the 7th Draft Notification to declare Ecologically Sensitive Areas (ESA) covering nearly 56,800 sq km in which mountain range?

- A) Eastern Ghats
- B) Western Ghats
- C) Vindhya Range
- D) Shivalik Hills

Q133. What was the primary hazard highlighted during the Teesta Tunnel Disaster in the Eastern Himalayas?

- A) Underground methane gas pockets, high geostress, and intense seismic vulnerability in fragile Himalayan rock formations
- B) Tsunami waves
- C) Volcanic lava flows
- D) Desert sandstorm accumulation

Q134. Under the Forest Rights Act (FRA), 2006, what was the legal issue raised during the Agasthyamalai forest evictions in South India?

- A) Evicting indigenous Kani tribal communities without recognizing their statutory Community Forest Resource (CFR) rights
- B) Converting forests into shopping malls
- C) Building an international airport inside the biosphere
- D) Granting private mining leases to foreign firms

Q135. What meteorologically defines a 'Cloudburst' as per the India Meteorological Department (IMD) criteria?

- A) Any rainfall exceeding 10 mm in 24 hours
- B) Rainfall of 100 mm or more per hour over a localized geographical area of approximately 20 to 30 sq km
- C) Continuous drizzle lasting for 7 days
- D) Hailstorms occurring at sub-zero temperatures

Q136. What is the primary ecological risk associated with 'Ex Post Facto' (retrospective) Environmental Clearances according to Supreme Court rulings?

- A) It promotes environmental sustainability
- B) It undermines the Precautionary Principle and encourages irreversible ecological degradation before assessment
- C) It lowers government tax revenues
- D) It delays tree plantation drives

Q137. Hara Mangroves, which serve as an indispensable breeding ground for migratory birds and marine life, are situated along the Persian Gulf coast of which nation?

- A) Iran
- B) Oman
- C) Kuwait
- D) Qatar

Q138. Which national park in Madhya Pradesh serves as the exclusive habitat submerged partially by the Daudhan Dam under the Ken-Betwa Link?

- A) Kanha National Park
- B) Panna Tiger Reserve
- C) Bandhavgarh National Park
- D) Pench National Park

Q139. The Asiatic Lion (*Panthera leo persica*) is found in the wild exclusively in which protected area in India?

- A) Jim Corbett National Park, Uttarakhand
- B) Gir National Park and Wildlife Sanctuary, Gujarat
- C) Kaziranga National Park, Assam
- D) Ranthambore National Park, Rajasthan

Q140. Under Project Elephant launched in 1992, how many officially notified Elephant Reserves currently exist in India?

- A) 12
- B) 20
- C) 33
- D) 55

Q141. What is the primary conservation objective of 'Project Dolphin' launched by the Government of India?

- A) Establishing ocean circus entertainment
- B) Conserving both riverine (Gangetic & Indus) and oceanic dolphins through habitat restoration and anti-poaching measures
- C) Exporting dolphin oil for pharmaceuticals
- D) Dredging river beds for deep navigation

Q142. Which wetland in Kerala, designated as a Ramsar site, is India's longest lake and home to traditional Vembanad fish sanctuaries?

- A) Ashtamudi Lake
- B) Vembanad-Kol Wetland
- C) Sasthamkotta Lake
- D) Chilika Lake

Q143. What is 'Soil Organic Carbon' (SOC) and why is it critical for sustainable agriculture?

- A) Synthetic chemical pesticide residue
- B) The carbon component of organic matter in soil that determines fertility, moisture retention, and agricultural resilience against droughts
- C) Coal particles mixed with compost
- D) Radioactive tracer element

Q144. The Great Indian Bustard (GIB), critically endangered under IUCN Red List, faces its most acute mortality threat from:

- A) Collisions with overhead high-voltage power transmission lines in Gujarat and Rajasthan
- B) Deep sea trawling nets
- C) Freezing winter blizzards
- D) Excessive pesticide use in coffee plantations

Q145. Which flamingos species is classified as 'Vulnerable' under IUCN and native to the high-altitude Andean salt lakes of South America?

- A) Greater Flamingo
- B) Andean Flamingo (*Phoenicoparrus andinus*)
- C) Lesser Flamingo
- D) American Flamingo

Q146. Under the Ramsar Convention, which was the first wetland site designated in India in 1981 alongside Keoladeo National Park?

- A) Chilika Lake (Odisha)
- B) Loktak Lake (Manipur)
- C) Sambhar Lake (Rajasthan)
- D) Wular Lake (J&K;)

Q147. What is the primary ecological hazard of 'Methane Leaks' during tunnelling and infrastructure blasting in deep mountain strata?

- A) Rapid formation of acid rain
- B) Catastrophic subterranean explosions, toxic asphyxiation of workers, and flash fires
- C) Soil salinization
- D) Groundwater freezing

Q148. The 'Kasturirangan Committee' report on the Western Ghats recommended declaring what percentage of the Western Ghats as an Ecologically Sensitive Area?

- A) 10%
- B) 37%
- C) 64% (as recommended by Madhav Gadgil)
- D) 100%

Q149. What is 'Bio-char' used in regenerative agriculture and carbon sequestration?

- A) Liquid petroleum fuel
- B) Charcoal produced by pyrolysis of biomass in the absence of oxygen, used as a permanent soil enhancer and carbon sink
- C) Chemical urea fertilizer
- D) Plastic packaging additive

Q150. The concept of 'Ecological Footprint' measures:

- A) Total length of hiking trails in national parks
- B) The amount of biologically productive land and water area required to produce the resources an individual or population consumes and absorb its wastes
- C) Number of wild animals photographed per hectare
- D) Weight of synthetic shoes manufactured annually

Q151. Under the National Green Tribunal (NGT) Act, 2010, the tribunal is mandated to dispose of applications or appeals within how many months of filing?

- A) 3 months
- B) 6 months
- C) 12 months
- D) 24 months

Q152. Which international environmental treaty regulates the transboundary movement of hazardous wastes and their disposal?

- A) Basel Convention (1989)
- B) Stockholm Convention
- C) Rotterdam Convention
- D) Minamata Convention

Q153. What is the primary cause of 'Coral Bleaching' across tropical reef ecosystems?

- A) Excessive snowfall
- B) Prolonged sea surface temperature rise causing corals to expel symbiotic photosynthetic zooxanthellae algae
- C) Deep sea oil exploration noise
- D) Plastic micro-bead ingestion only

Q154. Under the Wildlife Protection Act (WPA), 1972 as amended in 2022, how many schedules of protected wildlife now exist?

- A) 6 Schedules
- B) 4 Schedules (two for specially protected fauna, one for plants, and one for CITES specimens)
- C) 2 Schedules
- D) 10 Schedules

Q155. Which Indian state has the highest percentage of its geographical area under forest cover as per the India State of Forest Report (ISFR)?

- A) Madhya Pradesh (highest by total area)
- B) Mizoram (highest by percentage >84%)
- C) Arunachal Pradesh
- D) Kerala

Q156. Indian theoretical physicist Deepak Dhar was co-awarded the prestigious 2026 Dirac Medal for his seminal contributions to which field?

- A) Quantum cryptography
- B) Statistical physics, exact solutions of self-organized criticality, and abelian sandpile models
- C) Dark matter detection
- D) Gravitational wave astronomy

Q157. What is the key propulsion breakthrough in 'Rotating Detonation Engines' (RDE) currently under advanced aerospace testing?

- A) Using steam power in jet engines
- B) Generating thrust through continuous supersonic detonation waves propagating circumferentially around an annular channel, offering up to 25% higher thermal efficiency
- C) Using solid wood pellets as jet propellant
- D) Completely eliminating liquid fuel combustion

Q158. NASA extended an invitation to ISRO to join which historic international exploration framework focused on permanent lunar surface operations?

- A) Apollo 2.0 Taskforce
- B) Artemis Moon Base Programme and Gateway Lunar Architecture
- C) Soviet Lunar Alliance
- D) Mars Pathfinder Consortium

Q159. In the field of Synthetic Biology, how is Generative AI transforming protein engineering?

- A) Generating animated biology videos
- B) De novo design of novel functional enzymes, therapeutic antibodies, and biocatalysts with customized molecular folding structures
- C) Replacing physical petri dishes with video games
- D) Manufacturing plastic polymers directly from text prompts

Q160. What are 'Lab-Grown Diamonds' (LGD) produced through the Chemical Vapour Deposition (CVD) method?

- A) Fake plastic crystals
- B) Diamonds created by depositing carbon atoms from a hydrocarbon gas mixture onto a seed crystal, having identical optical, chemical, and physical properties as mined diamonds
- C) Glass beads coated with synthetic polish
- D) Silicon carbide cubic zirconia

Q161. World Hepatitis Day (WHD) is observed annually on 28th July to mark the birthday of which Nobel laureate who discovered the Hepatitis B virus?

- A) Dr. Baruch Blumberg
- B) Dr. Alexander Fleming
- C) Dr. Robert Koch
- D) Dr. Edward Jenner

Q162. What is the primary sustainability challenge posed by modern 'Mega Data Centres' powering generative AI infrastructure?

- A) Shortage of computer keyboards
- B) Enormous electricity consumption, immense water usage for evaporative cooling towers, and heavy carbon footprints
- C) Physical lack of building cement
- D) Excessive fiber optic cable weight

Q163. What astronomical phenomenon occurs during a 'Total Solar Eclipse' when the Moon passes directly between the Sun and Earth?

- A) The Sun's faint outer atmosphere (Corona) becomes visible as the Moon's umbral shadow falls on Earth
- B) The Earth ceases rotation for several minutes
- C) The Moon glows with bright blue light
- D) All solar radiation permanently vanishes

Q164. Under India's National Quantum Mission (NQM), which tier of quantum technology aims at establishing secure communication networks over 2,000 km?

- A) Quantum Computing only
- B) Quantum Communication via satellite-based and ground fiber-optic Quantum Key Distribution (QKD)
- C) Quantum internet for household streaming
- D) Classical radio encryption

Q165. Which virus causes Hepatitis B, an infectious disease primarily attacking which human organ?

- A) Lungs
- B) Liver
- C) Heart
- D) Kidneys

Q166. What is the core mandate of IN-SPACe (Indian National Space Promotion and Authorization Centre)?

- A) Direct control of military satellites only
- B) Single-window autonomous nodal agency to authorize, promote, and hand-hold Non-Government Entities (NGEs) and private space startups in India
- C) Conducting civilian airline flights
- D) Regulating automobile GPS units

Q167. Which propulsion technology uses solar radiation pressure exerted on large reflective membranes to propel spacecraft without onboard propellant?

- A) Solar Sails (Photon propulsion)
- B) Nuclear thermal rocket
- C) Ion thruster
- D) Cryogenic upper stage

Q168. What is 'CRISPR-Cas9' technology primarily utilized for in modern biotechnology?

- A) 3D printing metal tools
- B) Targeted molecular editing of specific DNA sequences in genomes to correct genetic disorders or engineer resilient crops
- C) Designing computer anti-virus software
- D) Purifying distilled water

Q169. The James Webb Space Telescope (JWST) primarily operates in which portion of the electromagnetic spectrum to observe the earliest galaxies?

- A) Gamma rays
- B) Infrared (Near-Infrared and Mid-Infrared)
- C) Ultraviolet
- D) X-rays

Q170. What is the term 'Hallucination' in the context of Large Language Models (LLMs) and Artificial Intelligence?

- A) Physical overheating of CPU chips
- B) Generation of confident, factually incorrect, or fabricated statements that are not grounded in training data or reality
- C) Screen flickering on computer monitors
- D) Cyber attack involving malware

Q171. Under the National Supercomputing Mission (NSM), which supercomputer was installed at the Indian Institute of Science (IISc), Bengaluru?

- A) Param Pravega
- B) Param Siddhi-AI
- C) Param Shivay
- D) Pratyush

Q172. What is 'Solid-State Battery' technology and why is it considered superior to conventional Lithium-ion batteries?

- A) It uses liquid acid that never evaporates
- B) It replaces flammable liquid electrolytes with solid electrolytes, offering higher energy density, faster charging, and eliminated risk of thermal runaway fires
- C) It generates electricity from coal
- D) It requires no charging ever

Q173. What is the primary function of an 'Air-Breathing Scramjet' engine developed by ISRO?

- A) Flying at subsonic passenger speeds
- B) Utilizing atmospheric oxygen from the air as oxidizer during supersonic combustion (Mach 6+), drastically reducing launch vehicle liftoff weight
- C) Underwater submarine propulsion
- D) Powering household air conditioners

Q174. Which mission was launched by ISRO in collaboration with NASA featuring a dual-frequency L-band and S-band Synthetic Aperture Radar (SAR)?

- A) NISAR Mission
- B) Chandrayaan-3
- C) Aditya-L1
- D) Gaganyaan

Q175. What is 'Green Hydrogen' as defined under the National Green Hydrogen Mission?

- A) Hydrogen painted with green dye
- B) Hydrogen produced through electrolysis of water powered entirely by renewable electricity (solar, wind)
- C) Hydrogen derived from burning plastic waste
- D) Naturally occurring gas found in dense forests

Q176. What is 'Nuclear Fusion', the process powering stars that experimental reactors like ITER are attempting to harness on Earth?

- A) Splitting heavy uranium atoms into lighter nuclei
- B) Fusing light atomic nuclei (e.g. Deuterium and Tritium) under extreme temperature and pressure to form helium, releasing colossal energy without long-lived radioactive waste
- C) Burning chemical hydrogen gas in air
- D) Mining thorium from coastal sands

Q177. Which type of orbit is typically located approximately 35,786 km directly above Earth's equator, allowing satellites to match Earth's rotation period?

- A) Low Earth Orbit (LEO)
- B) Geostationary Earth Orbit (GEO)
- C) Polar Sun-Synchronous Orbit (SSO)
- D) Molniya Orbit

Q178. What is the disease vector responsible for transmitting the Dengue, Chikungunya, and Zika viruses to humans?

- A) Anopheles mosquito
- B) Aedes aegypti mosquito
- C) Culex mosquito
- D) Tsetse fly

Q179. What is 'Quantum Supremacy' in computing benchmarks?

- A) The political rule of scientists
- B) The demonstrated ability of a programmable quantum computer to solve a specialized computational problem that is practically impossible for any classical supercomputer to solve in feasible time
- C) Replacing all personal laptops with quantum chips
- D) Connecting 10 billion smartphones

Q180. Which Indian mission placed a solar observatory craft around the Sun–Earth Lagrange Point 1 (L1) to study coronal mass ejections and solar winds continuously?

- A) Mangalyaan
- B) Aditya-L1
- C) AstroSat
- D) XPoSat

Q181. The historic Royal Indian Navy (RIN) Mutiny of February 1946, which dealt a decisive blow to British colonial authority, began aboard which naval vessel in Bombay?

- A) HMIS Vikrant
- B) HMIS Talwar
- C) HMIS Mysore
- D) HMIS Delhi

Q182. The Indian Independence Act, 1947 received Royal Assent on which date, formalizing the partition and independence of India and Pakistan?

- A) 3rd June 1947
- B) 18th July 1947
- C) 15th August 1947
- D) 26th January 1950

Q183. A recent discovery of a rare 12th-century 'Veeragallu' (hero stone) provided new historical insights into the death of King Vishnuvardhana of which dynasty?

- A) Chola Dynasty
- B) Hoysala Dynasty (Karnataka)
- C) Rashtrakuta Dynasty
- D) Pallava Dynasty

Q184. According to the State of Food Security and Nutrition in the World (SOFI) 2026 report, published jointly by FAO, IFAD, UNICEF, WFP, and WHO, what is the major global challenge?

- A) Surplus grain in all developing nations
- B) Persistent global hunger affecting over 730 million people, aggravated by geopolitical conflicts, climate extremes, and economic downturns
- C) Complete eradication of undernutrition
- D) Global drop in meat consumption

Q185. The NITI Aayog Report on 'India's Caregiving Ecosystem' highlighted the critical economic contribution of which demographic?

- A) Foreign corporate consultants
- B) Unpaid and informal domestic caregivers, predominantly women, and the urgent need for institutional support, elder care infrastructure, and childcare subsidies
- C) Automated nursing robots only
- D) Commercial hospital CEOs

Q186. Which Indian city was chosen as the host of the International Olympic Committee (IOC) Session that approved cricket's inclusion in the 2028 Los Angeles Olympics?

- A) New Delhi
- B) Mumbai
- C) Ahmedabad
- D) Bengaluru

Q187. The National Sports Day in India is celebrated annually on 29th August to honor the birth anniversary of which legendary sports icon?

- A) Milkha Singh
- B) Major Dhyan Chand
- C) K.D. Jadhav
- D) Kapil Dev

Q188. Which state government launched the 'Mukhyamantri Vayu Swasthya Seva' (Air Health Service) to provide super-specialist medical services to remote and tribal districts via aircraft?

- A) Odisha
- B) Himachal Pradesh
- C) Sikkim
- D) Jharkhand

Q189. What is the official mascot of the 38th National Games hosted by Uttarakhand?

- A) Veera
- B) Himali (Himalayan Monal)
- C) Asha
- D) Moga

Q190. Which Indian airport became the first in the world to operate entirely on solar energy, earning the Champions of the Earth award from the UNEP?

- A) Indira Gandhi International Airport, Delhi
- B) Cochin International Airport (CIAL), Kerala
- C) Kempegowda International Airport, Bengaluru
- D) Chhatrapati Shivaji Maharaj International Airport, Mumbai

Q191. What is the primary objective of the 'One Nation, One Registration' portal under the Digital India Land Records Modernization Programme?

- A) Mandatory vehicle pooling
- B) Anywhere-anytime national digital registration of deeds, land parcels, and real estate transactions with unique ULPIN integration
- C) Registering voters online only
- D) Issuing passport clearance within 1 hour

Q192. The 'DIRAC Medal', one of the highest honors in theoretical physics, is awarded annually by which renowned scientific institute located in Trieste, Italy?

- A) CERN
- B) ICTP (Abdus Salam International Centre for Theoretical Physics)
- C) Max Planck Institute
- D) Cavendish Laboratory

Q193. Under the 'Heritage City Development and Augmentation Yojana' (HRIDAY), which ministry spearheads the holistic preservation of heritage cities?

- A) Ministry of Culture
- B) Ministry of Housing and Urban Affairs (MoHUA)
- C) Ministry of Tourism
- D) Archaeological Survey of India

Q194. Which biosphere reserve located at the tri-junction of Kerala, Karnataka, and Tamil Nadu was the first biosphere reserve established in India in 1986?

- A) Gulf of Mannar
- B) Nilgiri Biosphere Reserve
- C) Sundarbans
- D) Nanda Devi

Q195. What is the unique feature of the 'PM SVANidhi' scheme for street vendors launched by the Ministry of Housing and Urban Affairs?

- A) Micro-credit collateral-free working capital loan of up to Rs. 50,000 with 7% interest subsidy and cashback incentives on digital transactions
- B) Free four-wheeler commercial vehicles
- C) Exemption from all state municipal taxes forever
- D) Relocation of street vendors to air-conditioned malls

Q196. Which international financial institution headquartered in Manila, Philippines approved a multi-million-dollar loan for sustainable urban transport in India?

- A) World Bank
- B) Asian Development Bank (ADB)
- C) New Development Bank (NDB)
- D) Asian Infrastructure Investment Bank (AIIB)

Q197. In Indian constitutional history, which landmark 1973 verdict established the inviolable 'Basic Structure Doctrine'?

- A) Golaknath case
- B) Kesavananda Bharati v. State of Kerala
- C) Minerva Mills case
- D) S.R. Bommai case

Q198. What is the primary role of the Indian National Centre for Ocean Information Services (INCOIS) located in Hyderabad?

- A) Manufacturing commercial cargo ships
- B) Providing ocean data, potential fishing zone (PFZ) advisories, tsunami early warnings, and storm surge alerts to coastal communities
- C) Drilling deep sea oil wells
- D) Issuing merchant navy passports

Q199. The 'Global Innovation Index' (GII), where India has steadily ascended in global rankings, is published annually by which specialized agency of the United Nations?

- A) UNCTAD
- B) WIPO (World Intellectual Property Organization)
- C) World Bank
- D) ITU

Q200. What is the overarching goal of the 'Sapt Dhara' strategic vision unveiled on India's 80th Independence Day?

- A) Developing 7 international space shuttles
- B) Seven foundational streams of national transformation (including self-reliance, inclusive social security, green transition, youth empowerment, digital infrastructure, and good governance) to achieve Viksit Bharat by 2047
- C) Constructing 7 mega river dams across northern India
- D) Formulating a 7-nation defence treaty in the Indian Ocean

OFFICIAL ANSWER KEY — PART 1 (Q.1 – Q.100)

Comprehensive Verified Answer Key for Practice MCQs 1 to 100 · sarkaripath.onl

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	26	B	51	B	76	A
2	A	27	A	52	B	77	B
3	B	28	B	53	B	78	B
4	B	29	B	54	B	79	B
5	C	30	B	55	B	80	B
6	B	31	B	56	B	81	B
7	C	32	A	57	A	82	C
8	B	33	B	58	A	83	B
9	B	34	B	59	B	84	B
10	B	35	B	60	B	85	B
11	B	36	B	61	C	86	B
12	A	37	B	62	B	87	B
13	B	38	C	63	B	88	B
14	C	39	C	64	B	89	B
15	B	40	B	65	B	90	B
16	A	41	B	66	B	91	B
17	B	42	A	67	B	92	B
18	C	43	B	68	B	93	B
19	B	44	A	69	B	94	B
20	B	45	B	70	B	95	B
21	C	46	B	71	B	96	A
22	B	47	B	72	B	97	B
23	B	48	B	73	B	98	C
24	B	49	B	74	B	99	B
25	B	50	B	75	B	100	B

OFFICIAL ANSWER KEY — PART 2 (Q.101 – Q.200)

Comprehensive Verified Answer Key for Practice MCQs 101 to 200 · sarkaripath.onl

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
101	B	126	B	151	B	176	B
102	B	127	A	152	A	177	B
103	A	128	A	153	B	178	B
104	B	129	B	154	B	179	B
105	B	130	B	155	B	180	B
106	A	131	B	156	B	181	B
107	B	132	B	157	B	182	B
108	A	133	A	158	B	183	B
109	B	134	A	159	B	184	B
110	B	135	B	160	B	185	B
111	B	136	B	161	A	186	B
112	A	137	A	162	B	187	B
113	B	138	B	163	A	188	A
114	A	139	B	164	B	189	B
115	B	140	C	165	B	190	B
116	B	141	B	166	B	191	B
117	B	142	B	167	A	192	B
118	A	143	B	168	B	193	B
119	B	144	A	169	B	194	B
120	A	145	B	170	B	195	A
121	A	146	A	171	A	196	B
122	B	147	B	172	B	197	B
123	B	148	B	173	B	198	B
124	A	149	B	174	A	199	B
125	B	150	B	175	B	200	B